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ISLAMIC REPUBLIC OF PAKISTAN

WATER & POWER DEVELOPMENT AUTHORITY

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

Background Papers

PAKISTAN
DRAINAGE SECTOR ENVIRONMENTAL ASSESSMENT
- NATIONAL DRAINAGE PROGRAMME

VOLUME 4: DATA

(WATER, SOIL AND AGRICULTURE)

JUNE 1993

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NATIONAL ENGINEERING SERVICES PAKISTAN (PVT) LIMITED

Mott MacDonald International Limited

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PAKISTAN
DRAINAGE SECTOR ENVIRONMENTAL ASSESSMENT
-NATIONAL DRAINAGE PROGRAMME

(LIST OF VOLUMES)

VOLUME 1 - PAKISTAN DRAINAGE SECTOR ENVIRONMENTAL ASSESSMENT

VOLUME 2 - CONCEPT FRAMEWORK
NATIONAL DRAINAGE PROGRAMME

VOLUME 3 - SUPPLEMENTARY REPORTS

VOLUME 4 - DATA (WATER, SOIL & AGRICULTURE)

ACKNOWLEDGEMENT

The data given in Annex I would not have been possible without the untiring effort of Mr. Muhammad Shafiq, Research Officer, Planning Division, WAPDA. NESPAK gratefully acknowledges his contribution.

PAKISTAN
DRAINAGE SECTOR ENVIRONMENTAL ASSESSMENT
- NATIONAL DRAINAGE PROGRAMME

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CONVERSION TABLE

Non-SI Units		SI Units
1 inch (in)	=	25.4 millimeters (mm)
1 foot (ft)	=	30.5 centimeters (cm)
1 yard (yd)	=	0.915 meters (m)
1 mile (mi)	=	1.609 kilometers (km)
1 acre (ac)	=	0.405 hectares (ha)
1 square mile (sq.mi)	=	259 ha
1 pound (lb)	=	0.454 kilograms (kg)
1 long ton (lg ton)	=	1.016 metric tons (t)
1 cubic foot/second	=	0.0283 cubic meters/second (m^3/sec)
	=	28.32 litres/second (l/sec)
1 acre foot (AF)	=	1233.5 cubic meters (m^3)
1 parts per million (ppm)	=	1 milli gram/litre (mg/l)
	=	0.00156 milli mho/cm (mmho/cm)
	=	0.00156 decisemen/meter (dS/m)

ANNEX - I

A. DETAIL OF GROSS AREA IN COMPLETED SCARPS, ON-GOING SCARPS AND IN CANAL COMMANDS OUTSIDE SCARPS HAVING GROUND WATER TABLE DEPTH LESS THAN 5-FEET.

B. SCARPS-WISE DEPTH TO WATERTABLE.

C. CANAL COMMAND-WISE DEPTH TO WATERTABLE.

A. DETAIL OF GROSS AREA IN COMPLETED SCARPS, ON-GOING SCARPS AND IN CANAL COMMANDS OUTSIDE SCARPS HAVING GROUND WATER TABLE DEPTH LESS THAN 5-FEET

TABLE–1

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1982)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	5.568	5.007	0.427	0.427	0.018	0.101	0.973
NWFP	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Sindh/Balochistan	2.035	1.746	0.162	0.314	0.277	0.220	0.973
i. Non–Rice	1.648	1.418	0.162	0.130	0.277	0.022	0.591
ii. Rice	0.387	0.328	0.000	0.184	0.000	0.198	0.382
Total:	7.829	6.951	0.635	0.741	0.295	0.321	1.992
i. Non–Rice	7.442	6.623	0.630	0.557	0.295	0.123	1.605
ii. Rice	0.387	0.328	0.000	0.184	0.000	0.198	0.382
II. On–Going SCARPs							
Punjab	2.903	2.502	0.180	0.421	0.326	0.016	0.943
NWFP	0.220	0.180	0.045	0.000	0.000	0.000	0.045
Sindh/Balochistan	4.757	3.927	0.007	0.397	0.871	2.214	3.489
i. Non–Rice	2.298	1.807	0.007	0.379	0.871	0.265	1.522
ii. Rice	2.459	2.120	0.000	0.018	0.000	1.949	1.967
Total:	7.880	6.609	0.232	0.818	1.197	2.230	4.477
i. Non–Rice	5.421	4.489	0.232	0.800	1.197	0.281	2.510
ii. Rice	2.459	2.120	0.000	0.018	0.000	1.949	1.967
III. Out Side SCARPs							
Punjab	16.149	12.760	0.532	0.500	0.316	0.081	1.429
NWFP	0.887	0.597	0.034	0.000	0.000	0.000	0.034
Sindh/Balochistan	7.755	7.583	0.014	0.534	1.562	1.115	3.225
i. Non–Rice	6.510	6.418	0.014	0.064	1.562	0.340	1.980
ii. Rice	1.245	1.165	0.000	0.470	0.000	0.775	1.245
Total:	24.791	20.940	0.580	1.034	1.878	1.196	4.688
i. Non–Rice	7.442	19.775	0.580	0.564	1.878	0.421	3.443
ii. Rice	0.387	1.165	0.000	0.470	0.000	0.775	1.245
Gross Total (I TO III) :	40.500	34.500	1.447	2.593	3.370	3.747	11.157

Source: SCARP Monitoring, WAPDA

TABLE–2

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1982)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.075	0.005	0.000	0.000	0.080
SCARP II							
2. Non Saline	1.649	1.490	0.244	0.014	0.011	0.007	0.276
3. Shahpur	0.112	0.090	0.000	0.027	0.000	0.000	0.027
SCARP III							
4. Non Saline	1.138	0.952	0.000	0.270	0.000	0.018	0.288
5. Saline	0.142	0.098	0.000	0.049	0.000	0.058	0.107
SCARP IV							
6. Muridke Mangtanwala	0.558	0.544	0.000	0.001	0.000	0.000	0.001
SCARP V							
7. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
8. Satiana (P)	0.118	0.098	0.070	0.000	0.007	0.000	0.077
SCARP VI							
9. Allahabad (P)	0.216	0.202	0.008	0.046	0.000	0.007	0.061
SCARP VIII							
10. Minchinabad (P)	0.078	0.073	0.000	0.015	0.000	0.000	0.015
11. AWL TP Link	0.020	0.020	0.008	0.000	0.000	0.000	0.008
12. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.011	0.011
Sub–Total (a):	5.471	4.912	0.407	0.427	0.018	0.101	0.953
(b) Surface Drainage Projects							
1. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.020	0.000	0.000	0.000	0.020
Sub–Total (b):	0.097	0.095	0.020	0.000	0.000	0.000	0.020
Total (I):	5.568	5.007	0.427	0.427	0.018	0.101	0.973
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.082	0.000	0.066	0.000	0.148
2. North Rohri	0.793	0.686	0.037	0.000	0.000	0.000	0.037
3. Kandhkot (P)	0.013	0.012	0.000	0.009	0.000	0.000	0.009
Sub–Total (a):	1.247	1.078	0.119	0.009	0.066	0.000	0.194
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.043	0.305	0.211	0.133	0.692
Sub–Total (b):	0.701	0.594	0.043	0.305	0.211	0.133	0.692
Total (II):	1.948	1.672	0.162	0.314	0.277	0.133	0.886
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
IV. N.W.F.P.							
(a) Tubewell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.043	0.000	0.000	0.000	0.043
2. Bannu	0.090	0.080	0.003	0.000	0.000	0.000	0.003
Sub–Total (a):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Total (IV):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Gross Total (I to IV) :	7.829	6.951	0.635	0.741	0.295	0.321	1.992

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–3

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1982)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP II							
1. Saline	0.690	0.550	0.025	0.121	0.121	0.008	0.275
SCARP V							
2. Khairwala Unit	0.333	0.301	0.000	0.000	0.020	0.000	0.020
SCARP VI							
3. Panjnad Abbasia (Rem.)	1.460	1.267	0.050	0.300	0.057	0.008	0.415
SCARP VII							
4. Pandoki Unit	0.196	0.174	0.093	0.000	0.004	0.000	0.097
SCARP VIII							
5. Fordwah Sadiqia II	0.111	0.100	0.012	0.000	0.094	0.000	0.106
6. AWL TSMB Link	0.090	0.090	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):	2.880	2.482	0.180	0.421	0.296	0.016	0.913
(b) Surface Drainage Project							
7. AWL CJ Link	0.023	0.020	0.000	0.000	0.045	0.000	0.045
Sub–Total (b):	0.023	0.020	0.000	0.000	0.045	0.000	0.045
Total (I):	2.903	2.502	0.180	0.421	0.341	0.016	0.958
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.336	0.000	0.007	0.343
2. South Rohri (FGW)	0.541	0.375	0.007	0.000	0.000	0.000	0.007
Sub–Total (a):	0.982	0.775	0.007	0.336	0.000	0.007	0.350
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.061	0.000	0.451	0.512
4. LBOD Core Programme	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.842	1.756	2.598
Sub–Total (b):	3.730	3.116	0.000	0.061	0.842	2.207	3.110
(c) Tile Drainage Projects							
6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.029	0.000	0.029
Sub–Total (c):	0.045	0.036	0.000	0.000	0.029	0.000	0.029
Total (II):	4.757	3.927	0.007	0.397	0.871	2.214	3.489
III. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.045	0.000	0.000	0.000	0.045
Sub–Total (a):	0.220	0.180	0.045	0.000	0.000	0.000	0.045
Total (III):	0.220	0.180	0.045	0.000	0.000	0.000	0.045
Gross Total (I to III):	7.880	6.609	0.232	0.818	1.212	2.230	4.492

Source: SCARP Monitoring, WAPDA

TABLE–4

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1982)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.149	12.760	0.532	0.500	0.316	0.081	1.429
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.486	0.000	0.757	1.243
Gudu L. Bank	0.519		0.000	0.038	0.000	0.295	0.333
Sukkur R. Bank	1.461		0.000	0.010	1.048	0.061	1.119
Sukkur L. Bank	4.532		0.014	0.000	0.514	0.002	0.530
Sub–Total (II):	7.755	7.583	0.014	0.534	1.562	1.115	3.225
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.034	0.000	0.000	0.000	0.034
Gross Total (I TO III) :	24.791	20.940	0.580	1.034	1.878	1.196	4.688

Source: SCARP Monitoring, WAPDA

TABLE–5

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1982)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.175	0.000	0.111	0.286
Kandhkot (Pilot)	0.013	0.012	0.000	0.009	0.000	0.000	0.009
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.184	0.000	0.198	0.382
II. On–Going							
North Dadu	0.497	0.430	0.000	0.018	0.000	0.451	0.469
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	1.498	1.498
Sub–Total (II):	2.459	2.120	0.000	0.018	0.000	1.949	1.967
III. Remaining Area	1.245	1.165	0.000	0.470	0.000	0.775	1.245
Gross Total (I TO III) :	4.091	3.613	0.000	0.672	0.000	2.922	3.594

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–6

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	5.568	5.007	0.306	0.268	0.025	0.078	0.677
NWFP	0.226	0.198	0.048	0.000	0.000	0.000	0.048
Sindh/Balochistan	2.035	1.746	0.118	0.047	0.222	0.134	0.521
i. Non–Rice	1.648	1.418	0.118	0.039	0.222	0.100	0.479
ii. Rice	0.387	0.328	0.000	0.008	0.000	0.034	0.042
Total:	7.829	6.951	0.472	0.315	0.247	0.212	1.246
i. Non–Rice	7.442	6.623	0.472	0.307	0.247	0.178	1.204
ii. Rice	0.387	0.328	0.000	0.008	0.000	0.034	0.042
II. On–Going SCARPs							
Punjab	2.903	2.502	0.169	0.295	0.285	0.000	0.749
NWFP	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Sindh/Balochistan	4.757	3.927	0.008	0.028	0.228	0.243	0.507
i. Non–Rice	2.298	1.807	0.008	0.022	0.228	0.164	0.422
ii. Rice	2.459	2.120	0.000	0.006	0.000	0.079	0.085
Total:	7.880	6.609	0.214	0.323	0.513	0.243	1.293
i. Non–Rice	5.421	4.489	0.214	0.317	0.513	0.164	1.208
ii. Rice	2.459	2.120	0.000	0.006	0.000	0.079	0.085
III. Out Side SCARPs							
Punjab	16.149	12.760	0.589	0.220	0.318	0.064	1.191
NWFP	0.887	0.597	0.043	0.000	0.000	0.000	0.043
Sindh/Balochistan	7.755	7.583	0.036	0.016	1.383	0.272	1.707
i. Non–Rice	6.510	6.418	0.036	0.000	1.383	0.199	1.618
ii. Rice	1.245	1.165	0.000	0.016	0.000	0.073	0.089
Total:	24.791	20.940	0.668	0.236	1.701	0.336	2.941
i. Non–Rice	23.546	19.775	0.668	0.220	1.701	0.263	2.852
ii. Rice	1.245	1.165	0.000	0.016	0.000	0.073	0.089
Gross Total (I TO III) :	40.500	34.500	1.354	0.874	2.461	0.791	5.480

Source: SCARP Monitoring, WAPDA

TABLE–7

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.064	0.000	0.000	0.000	0.064
SCARP II							
2. Non Saline	1.649	1.490	0.208	0.021	0.009	0.009	0.247
3. Shahpur	0.112	0.090	0.000	0.010	0.000	0.000	0.010
SCARP III							
4. Non Saline	1.138	0.952	0.000	0.155	0.000	0.011	0.166
5. Saline	0.142	0.098	0.000	0.058	0.000	0.036	0.094
SCARP IV							
6. Muridke Mangtanwala	0.558	0.544	0.000	0.002	0.000	0.000	0.002
SCARP V							
7. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
8. Satiana (P)	0.118	0.098	0.006	0.000	0.004	0.000	0.010
SCARP VI							
9. Allahabad (P)	0.216	0.202	0.005	0.034	0.000	0.013	0.052
SCARP VIII							
10. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
11. AWL TP Link	0.020	0.020	0.007	0.000	0.000	0.000	0.007
12. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.009	0.009
Sub–Total (a):	5.471	4.912	0.292	0.280	0.013	0.078	0.663
(b) Surface Drainage Projects							
1. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.014	0.000	0.000	0.000	0.014
Sub–Total (b):	0.097	0.095	0.014	0.000	0.000	0.000	0.014
Total (I):	5.568	5.007	0.306	0.280	0.013	0.078	0.677
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.093	0.000	0.077	0.000	0.170
2. North Rohri	0.793	0.686	0.025	0.000	0.000	0.000	0.025
3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):			0.118	0.000	0.077	0.000	0.195
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.000	0.047	0.145	0.098	0.290
Sub–Total (b):			0.000	0.047	0.145	0.098	0.290
Total (II):			0.118	0.047	0.222	0.098	0.485
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.036	0.036
Sub–Total (a):			0.000	0.000	0.000	0.036	0.036
Total (III):			0.000	0.000	0.000	0.036	0.036
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.038	0.000	0.000	0.000	0.038
2. Bannu	0.090	0.080	0.010	0.000	0.000	0.000	0.010
Sub–Total (a):			0.048	0.000	0.000	0.000	0.048
Total (IV):			0.048	0.000	0.000	0.000	0.048
Gross Total (I to IV) :			0.472	0.327	0.235	0.212	1.246

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–8

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
(a) Tubewell Drainage Project							
SCARP II							
1. Saline	0.690	0.550	0.000	0.167	0.104	0.000	0.271
SCARP V							
2. Khairwala Unit	0.333	0.301	0.005	0.000	0.001	0.000	0.006
SCARP VI							
3. Panjnad Abbasia (Rem.)	1.460	1.267	0.094	0.128	0.069	0.000	0.291
SCARP VII							
4. Pandoki Unit	0.196	0.174	0.060	0.000	0.002	0.000	0.062
SCARP VIII							
5. Fordwah Sadiqia II	0.111	0.100	0.010	0.000	0.088	0.000	0.098
6. AWL TSMB Link	0.090	0.090	0.000	0.000	0.015	0.000	0.015
Sub–Total (a):	2.880	2.482	0.169	0.295	0.279	0.000	0.743
(b) Surface Drainage Project							
7. AWL CJ Link	0.023	0.020	0.000	0.000	0.006	0.000	0.006
Sub–Total (b):	0.023	0.020	0.000	0.000	0.006	0.000	0.006
Total (I):	2.903	2.502	0.169	0.295	0.285	0.000	0.749
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.022	0.000	0.000	0.022
2. South Rohri (FGW)	0.541	0.375	0.008	0.000	0.000	0.000	0.008
Sub–Total (a):	0.982	0.775	0.008	0.022	0.000	0.000	0.030
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.006	0.000	0.045	0.051
4. LBOD Core Programme (Spinal Drain)	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.207	0.198	0.405
Sub–Total (b):	3.730	3.116	0.000	0.006	0.207	0.243	0.456
(c) Tile Drainage Projects							
6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.021	0.000	0.021
Sub–Total (c):	0.045	0.036	0.000	0.000	0.021	0.000	0.021
Total (II):	4.757	3.927	0.008	0.028	0.228	0.243	0.507
III. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Sub–Total (a):	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Total (III):	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Gross Total (I to III) :	7.880	6.609	0.214	0.323	0.513	0.243	1.293

Source: SCARP Monitoring, WAPDA.

TABLE–9

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.149	12.760	0.589	0.220	0.318	0.064	1.191
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.016	0.000	0.056	0.072
Gudu L. Bank	0.519		0.000	0.000	0.000	0.020	0.020
Sukkur R. Bank	1.461		0.000	0.010	0.485	0.196	0.691
Sukkur L. Bank	4.532		0.036	0.000	0.898	0.000	0.934
Sub–Total (II):	7.755	7.583	0.036	0.026	1.383	0.272	1.717
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.043	0.000	0.000	0.000	0.043
Gross Total (I to III) :	24.791	20.940	0.668	0.246	1.701	0.336	2.951

Source: SCARP Monitoring, WAPDA.

TABLE–10

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.008	0.000	0.000	0.008
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.034	0.034
Sub–Total (I):	0.387	0.328	0.000	0.008	0.000	0.034	0.042
II. On–Going							
North Dadu	0.497	0.430	0.000	0.006	0.000	0.039	0.045
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	0.040	0.040
Sub–Total (II):	2.459	2.120	0.000	0.006	0.000	0.079	0.085
III. Remaining Area	1.245	1.165	0.000	0.016	0.000	0.073	0.089
Gross Total (I to III) :	4.091	3.613	0.000	0.030	0.000	0.186	0.216

Source: SCARP Monitoring, WAPDA.

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–11

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	5.568	5.007	0.747	0.467	0.023	0.190	1.427
NWFP	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Sindh/Balochistan	2.035	1.746	0.169	0.272	0.227	0.205	0.873
i. Non–Rice	1.648	1.418	0.169	0.074	0.227	0.019	0.489
ii. .Rice	0.387	0.328	0.000	0.198	0.000	0.186	0.384
Total:	7.829	6.951	0.970	0.739	0.250	0.395	2.354
i. Non–Rice	7.442	6.623	0.970	0.541	0.250	0.209	1.970
ii. Rice	0.387	0.328	0.000	0.198	0.000	0.186	0.384
II. On–Going SCARPs							
Punjab	2.903	2.502	0.207	0.515	0.474	0.023	1.219
NWFP	0.220	0.180	0.064	0.000	0.000	0.000	0.064
Sindh/Balochistan	4.757	3.927	0.017	0.446	0.799	2.577	3.839
i. Non–Rice	2.298	1.807	0.017	0.428	0.799	0.354	1.598
ii. Rice	2.459	2.120	0.000	0.018	0.000	2.223	2.241
Total:	7.880	6.609	0.288	0.961	1.273	2.600	5.122
i. Non–Rice	5.421	4.489	0.288	0.943	1.273	0.377	2.881
ii. Rice	2.459	2.120	0.000	0.018	0.000	2.223	2.241
III. Out Side SCARPs							
Punjab	16.149	12.760	0.716	0.461	0.623	0.296	2.096
NWFP	0.887	0.597	0.055	0.000	0.000	0.000	0.055
Sindh/Balochistan	7.755	7.583	0.046	0.624	2.137	1.082	3.889
i. Non–Rice	6.510	6.418	0.046	0.153	2.137	0.308	2.644
ii. Rice	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Total:	24.791	20.940	0.817	1.085	2.760	1.378	6.040
i. Non–Rice	23.546	19.775	0.817	0.614	2.760	0.604	4.795
ii. Rice	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	40.500	34.500	2.075	2.785	4.283	4.373	13.516

Source: SCARP Monitoring, WAPDA

TABLE–12

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.112	0.012	0.000	0.000	0.124
SCARP II							
2. Non Saline	1.649	1.490	0.423	0.027	0.021	0.013	0.484
3. Shahpur	0.112	0.090	0.000	0.000	0.000	0.089	0.089
SCARP III							
4. Non Saline	1.138	0.952	0.000	0.342	0.000	0.016	0.358
5. Saline	0.142	0.098	0.000	0.043	0.000	0.053	0.096
SCARP IV							
6. Muridke Mangtanwala	0.558	0.544	0.000	0.010	0.000	0.000	0.010
SCARP V							
7. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
8. Satiana (P)	0.118	0.098	0.104	0.000	0.002	0.000	0.106
SCARP VI							
9. Allahabad (P)	0.216	0.202	0.007	0.033	0.000	0.004	0.044
SCARP VIII							
10. Minchinabad (P)	0.078	0.073	0.049	0.000	0.000	0.000	0.049
11. AWL TP Link	0.020	0.020	0.010	0.000	0.000	0.000	0.010
12. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.015	0.015
Sub–Total (a):	5.471	4.912	0.707	0.467	0.023	0.190	1.387
(b) Surface Drainage Projects							
1. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.040	0.000	0.000	0.000	0.040
Sub–Total (b):	0.097	0.095	0.040	0.000	0.000	0.000	0.040
Total (I):	5.568	5.007	0.747	0.467	0.023	0.190	1.427
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.093	0.000	0.043	0.000	0.136
2. North Rohri	0.793	0.686	0.040	0.000	0.000	0.000	0.040
3. Kandhkot (P)	0.013	0.012	0.000	0.008	0.000	0.000	0.008
Sub–Total (a):	1.247	1.078	0.133	0.008	0.043	0.000	0.184
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.036	0.264	0.184	0.118	0.602
Sub–Total (b):	0.701	0.594	0.036	0.264	0.184	0.118	0.602
Total (II):	1.948	1.672	0.169	0.272	0.227	0.118	0.786
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.047	0.000	0.000	0.000	0.047
2. Bannu	0.090	0.080	0.007	0.000	0.000	0.000	0.007
Sub–Total (a):	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Total (IV):	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Gross Total (I to IV) :	7.829	6.951	0.970	0.739	0.250	0.395	2.354

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–13

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP II							
1. Saline	0.690	0.550	0.027	0.175	0.217	0.008	0.427
SCARP V							
2. Khairwala Unit	0.333	0.301	0.016	0.028	0.000	0.000	0.044
SCARP VI							
3. Panjnad Abbasia (Rem.)	1.460	1.267	0.056	0.312	0.131	0.015	0.514
SCARP VII							
4. Pandoki Unit	0.196	0.174	0.095	0.000	0.003	0.000	0.098
SCARP VIII							
5. Fordwah Sadiqia II	0.111	0.100	0.013	0.000	0.093	0.000	0.106
6. AWL TSMB Link	0.090	0.090	0.000	0.000	0.015	0.000	0.015
Sub–Total (a):	2.880	2.482	0.207	0.515	0.459	0.023	1.204
(b) Surface Drainage Project							
7. AWL CJ Link	0.023	0.020	0.000	0.000	0.015	0.000	0.015
Sub–Total (b):	0.023	0.020	0.000	0.000	0.015	0.000	0.015
Total (I):	2.903	2.502	0.207	0.515	0.474	0.023	1.219
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.393	0.000	0.007	0.400
2. South Rohri (FGW)	0.541	0.375	0.015	0.000	0.000	0.000	0.015
Sub–Total (a):	0.982	0.775	0.015	0.393	0.000	0.007	0.415
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.002	0.053	0.000	0.459	0.514
4. LBOD Core Programme	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.757	2.111	2.868
Sub–Total (b):	3.730	3.116	0.002	0.053	0.757	2.570	3.382
(c) Tile Drainage Projects							
6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Sub–Total (c):	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Total (II):	4.757	3.927	0.017	0.446	0.799	2.577	3.839
III. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.064	0.000	0.000	0.000	0.064
Sub–Total (a):	0.220	0.180	0.064	0.000	0.000	0.000	0.064
Total (III):	0.220	0.180	0.064	0.000	0.000	0.000	0.064
Gross Total (I to III):	7.880	6.609	0.288	0.961	1.273	2.600	5.122

Source: SCARP Monitoring, WAPDA

TABLE–14

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.149	12.760	0.716	0.461	0.623	0.296	2.096
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.487	0.000	0.756	1.243
Gudu L. Bank	0.519		0.000	0.125	0.000	0.251	0.376
Sukkur R. Bank	1.461		0.000	0.012	1.119	0.069	1.200
Sukkur L. Bank	4.532		0.046	0.000	1.018	0.006	1.070
Sub–Total (II):	7.755	7.583	0.046	0.624	2.137	1.082	3.889
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.055	0.000	0.000	0.000	0.055
Gross Total (I to III):	24.791	20.940	0.817	1.085	2.760	1.378	6.040

Source: SCARP Monitoring, WAPDA

TABLE–15

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1983)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.186	0.000	0.099	0.285
Kandhkot (Pilot)	0.013	0.012	0.000	0.012	0.000	0.000	0.012
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.198	0.000	0.186	0.384
II. On–Going							
North Dadu	0.497	0.430	0.000	0.018	0.000	0.474	0.492
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	1.749	1.749
Sub–Total (II):	2.459	2.120	0.000	0.018	0.000	2.223	2.241
III. Remaining Area	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	4.091	3.613	0.000	0.687	0.000	3.183	3.870

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals

TABLE–16

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	5.568	5.007	0.123	0.143	0.004	0.085	0.355
NWFP	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Sindh/Balochistan	2.035	1.746	0.194	0.037	0.228	0.097	0.556
i. Non–Rice	1.648	1.418	0.194	1.010	0.228	0.024	1.456
ii. Rice	0.387	0.328	0.000	0.027	0.000	0.073	0.100
Total:	7.829	6.951	0.361	0.180	0.232	0.182	0.955
i. Non–Rice	7.442	6.623	0.361	0.153	0.232	0.109	0.855
ii. Rice	0.387	0.328	0.000	0.027	0.000	0.073	0.100
II. On–Going SCARPs							
Punjab	2.903	2.502	0.190	0.149	0.169	0.072	0.580
NWFP	0.220	0.180	0.030	0.000	0.000	0.000	0.030
Sindh/Balochistan	4.757	3.927	0.009	0.024	0.393	0.298	0.724
i. Non–Rice	2.298	1.807	0.009	0.016	0.393	0.044	0.462
ii. Rice	2.459	2.120	0.000	0.008	0.000	0.254	0.262
Total:	7.880	6.609	0.229	0.173	0.562	0.370	1.334
i. Non–Rice	5.421	4.489	0.229	0.165	0.562	0.116	1.072
ii. Rice	2.459	2.120	0.000	0.008	0.000	0.254	0.262
III. Out Side SCARPs							
Punjab	16.149	12.760	0.420	0.307	0.255	0.028	1.010
NWFP	0.887	0.597	0.053	0.000	0.000	0.000	0.053
Sindh/Balochistan	7.755	7.583	0.065	0.005	1.513	0.127	1.710
i. Non–Rice	6.510	6.418	0.065	0.002	1.513	0.057	1.637
ii. Rice	1.245	1.165	0.000	0.003	0.000	0.070	0.073
Total:	24.791	20.940	0.538	0.312	1.768	0.155	2.773
i. Non–Rice	23.546	19.775	0.538	0.309	1.768	0.085	2.700
ii. Rice	1.245	1.165	0.000	0.003	0.000	0.070	0.073
Gross Total (I TO III) :	40.500	34.500	1.128	0.665	2.562	0.707	5.062

Source: SCARP Monitoring, WAPDA

TABLE–17

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1994)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.009	0.001	0.000	0.000	0.010
SCARP II							
2. Non Saline	1.649	1.490	0.090	0.004	0.002	0.000	0.096
3. Shahpur	0.112	0.090	0.002	0.000	0.000	0.000	0.002
SCARP III							
4. Non Saline	1.138	0.952	0.000	0.069	0.000	0.032	0.101
5. Saline	0.142	0.098	0.000	0.039	0.000	0.036	0.075
SCARP IV							
6. Muridke Mangtanwala	0.558	0.544	0.002	0.000	0.000	0.000	0.002
SCARP V							
7. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
8. Satiana (P)	0.118	0.098	0.002	0.000	0.000	0.000	0.002
SCARP VI							
9. Allahabad (P)	0.216	0.202	0.006	0.032	0.002	0.009	0.049
SCARP VIII							
10. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
11. AWL TP Link	0.020	0.020	0.007	0.000	0.000	0.000	0.007
12. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.009	0.009
Sub–Total (a):	5.471	4.912	0.120	0.145	0.004	0.086	0.355
(b) Surface Drainage Projects							
1. Paharang Drain (Out Side Scarp–I)	0.097	0.093	0.005	0.000	0.000	0.000	0.005
Sub–Total (b):	0.097	0.093	0.005	0.000	0.000	0.000	0.005
Total (I):	5.568	5.005	0.125	0.145	0.004	0.086	0.360
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.136	0.000	0.072	0.000	0.208
2. North Rohri	0.793	0.686	0.049	0.000	0.000	0.000	0.049
3. Kandhkot (P)	0.013	0.012	0.000	0.007	0.000	0.000	0.007
Sub–Total (a):	1.247	1.078	0.185	0.007	0.072	0.000	0.264
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.009	0.030	0.156	0.025	0.220
Sub–Total (b):	0.701	0.594	0.009	0.030	0.156	0.025	0.220
Total (II):	1.948	1.672	0.194	0.037	0.228	0.025	0.484
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.072	0.072
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.072	0.072
Total (III):	0.087	0.074	0.000	0.000	0.000	0.072	0.072
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 1/	0.136	0.118	0.043	0.000	0.000	0.000	0.043
2. Bannu	0.090	0.080	0.003	0.000	0.000	0.000	0.003
3. Mardan (Yar Hussain Sharif)	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Total (IV):	0.136	0.118	0.043	0.000	0.000	0.000	0.043
Gross Total (I to IV) :	7.739	6.869	0.362	0.182	0.232	0.183	0.959

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–18

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
(a) Tubewell Drainage Project							
SCARP II							
1. Saline	0.690	0.550	0.030	0.023	0.029	0.000	0.082
SCARP V							
2. Khairwala Unit	0.333	0.301	0.014	0.000	0.001	0.000	0.015
SCARP VI							
3. Panjnad Abbasia (Rem.)	1.460	1.267	0.070	0.125	0.114	0.002	0.311
SCARP VII							
4. Pandoki Unit	0.196	0.174	0.066	0.000	0.003	0.000	0.069
SCARP VIII							
5. Fordwah Sadiqia II	0.111	0.100	0.000	0.000	0.009	0.070	0.079
6. AWL TSMB Link	0.090	0.090	0.010	0.001	0.004	0.015	0.030
Sub–Total (a):	2.880	2.482	0.190	0.149	0.160	0.087	0.586
(b) Surface Drainage Project							
7. AWL CJ Link	0.023	0.020	0.000	0.000	0.009	0.000	0.009
Sub–Total (b):	0.023	0.020	0.000	0.000	0.009	0.000	0.009
Total (I):	2.903	2.502	0.190	0.149	0.169	0.087	0.595
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.018	0.000	0.002	0.020
2. South Rohri (FGW)	0.541	0.375	0.009	0.000	0.001	0.000	0.010
Sub–Total (a):	0.982	0.775	0.009	0.018	0.001	0.002	0.030
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.006	0.005	0.056	0.067
4. LBOD Core Programme	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.345	0.240	0.585
Sub–Total (b):	3.730	3.116	0.000	0.006	0.350	0.296	0.652
(c) Tile Drainage Projects							
6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Sub–Total (c):	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Total (II):	4.757	3.927	0.009	0.024	0.393	0.298	0.724
III. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.030	0.000	0.000	0.000	0.030
Sub–Total (a):	0.220	0.180	0.030	0.000	0.000	0.000	0.030
Total (III):	0.220	0.180	0.030	0.000	0.000	0.000	0.030
Gross Total (I to III) :	7.880	6.609	0.229	0.173	0.562	0.385	1.349

Source: SCARP Monitoring, WAPDA.

TABLE–19

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.149	12.760	0.420	0.307	0.255	0.028	1.010
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.003	0.000	0.070	0.073
Gudu L. Bank	0.519		0.000	0.002	0.000	0.050	0.052
Sukkur R. Bank	1.461		0.021	0.000	0.588	0.007	0.616
Sukkur L. Bank	4.532		0.044	0.000	0.925	0.000	0.969
Sub–Total (II):	7.755	7.583	0.065	0.005	1.513	0.127	1.710
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.053	0.000	0.000	0.000	0.053
Gross Total (I to III) :	24.791	20.940	0.538	0.312	1.768	0.155	2.773

Source: SCARP Monitoring, WAPDA.

TABLE–20

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.020	0.000	0.001	0.021
Kandhkot (Pilot)	0.013	0.012	0.000	0.007	0.000	0.000	0.007
Hairdin	0.087	0.074	0.000	0.000	0.000	0.072	0.072
Sub–Total (I):	0.387	0.328	0.000	0.027	0.000	0.073	0.100
II. On–Going							
North Dadu	0.497	0.430	0.000	0.008	0.000	0.054	0.062
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	0.200	0.200
Sub–Total (II):	2.459	2.120	0.000	0.008	0.000	0.254	0.262
III. Remaining Area	1.245	1.165	0.000	0.003	0.000	0.070	0.073
Gross Total (I to III) :	4.091	3.613	0.000	0.038	0.000	0.397	0.435

Source: SCARP Monitoring, WAPDA.

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–21

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Completed SCARPs							
Punjab	5.568	5.007	0.571	0.519	0.052	0.080	1.222
NWFP	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Sindh/Balochistan	2.035	1.746	0.248	0.341	0.253	0.176	1.018
i. Non–Rice	1.648	1.418	0.248	0.107	0.253	0.026	0.634
ii. Rice	0.387	0.328	0.000	0.234	0.000	0.150	0.384
Total:	7.829	6.951	0.873	0.860	0.305	0.256	2.294
i. Non–Rice	7.442	6.623	0.873	0.623	0.308	0.116	1.920
ii. Rice	0.387	0.328	0.000	0.234	0.000	0.150	0.384
II. On–Going SCARPs							
Punjab	3.033	2.611	0.340	0.404	0.322	0.070	1.136
NWFP	0.220	0.180	0.057	0.000	0.000	0.000	0.057
Sindh/Balochistan	4.757	3.927	0.018	0.316	0.926	2.577	3.837
i. Non–Rice	2.298	1.807	0.018	0.297	0.926	0.308	1.549
ii. Rice	2.456	2.120	0.000	0.019	0.000	2.269	2.288
Total:	8.010	6.718	0.415	0.720	1.248	2.647	5.030
i. Non–Rice	5.551	4.598	0.415	0.701	1.248	0.378	2.742
ii. Rice	2.459	2.120	0.000	0.019	0.000	2.269	2.288
III. Out Side SCARPs							
Punjab	16.019	12.651	0.529	0.840	0.496	0.022	1.887
NWFP	0.887	0.597	0.064	0.000	0.000	0.000	0.064
Sindh/Balochistan	7.755	7.583	0.260	0.564	1.918	0.996	3.738
i. Non–Rice	6.510	6.418	0.260	0.093	1.918	0.222	2.493
ii. Rice	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Total:	24.661	20.831	0.853	1.404	2.414	1.018	5.689
i. Non–Rice	23.416	19.666	0.853	0.933	2.414	0.244	4.444
ii. Rice	1.240	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	40.500	34.500	2.141	2.984	3.967	3.921	13.013

Source: SCARP Monitoring, WAPDA

TABLE–22

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I SCARP II	1.217	1.141	0.121	0.003	0.000	0.000	0.124
2. Non Saline	1.649	1.490	0.302	0.043	0.043	0.000	0.388
3. Shahpur SCARP III	0.112	0.090	0.000	0.069	0.000	0.000	0.069
4. Non Saline	1.138	0.952	0.000	0.275	0.000	0.034	0.309
5. Saline SCARP IV	0.142	0.098	0.000	0.051	0.000	0.038	0.089
6. Muridke Mangtanwala SCARP V	0.558	0.544	0.005	0.012	0.000	0.000	0.017
7. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
8. Satiana (P) SCARP VI	0.118	0.098	0.084	0.000	0.007	0.000	0.091
9. Allahabad (P) SCARP VIII	0.216	0.202	0.008	0.029	0.002	0.003	0.042
10. Minchinabad (P)	0.078	0.073	0.000	0.037	0.000	0.000	0.037
11. AWL TP Link	0.020	0.020	0.010	0.000	0.000	0.000	0.010
12. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.005	0.005
Sub–Total (a):	5.471	4.912	0.532	0.519	0.052	0.080	1.183
(b) Surface Drainage Projects							
1. Paharang Drain (Out Side Scarp–1)	0.097	0.095	0.039	0.000	0.000	0.000	0.039
Sub–Total (b):	0.097	0.095	0.039	0.000	0.000	0.000	0.039
Total (I):	5.568	5.007	0.571	0.519	0.052	0.080	1.222
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.116	0.000	0.067	0.000	0.183
2. North Rohri	0.793	0.686	0.049	0.000	0.000	0.000	0.049
3. Kandhkot (P)	0.013	0.012	0.000	0.011	0.000	0.000	0.011
Sub–Total (a):	1.247	1.078	0.165	0.011	0.067	0.000	0.243
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.083	0.330	0.186	0.089	0.688
Sub–Total (b):	0.701	0.594	0.083	0.330	0.186	0.089	0.688
Total (II):	1.948	1.672	0.248	0.341	0.253	0.089	0.931
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.047	0.000	0.000	0.000	0.047
2. Bannu	0.090	0.080	0.007	0.000	0.000	0.000	0.007
Sub–Total (a):	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Total (IV):	0.226	0.198	0.054	0.000	0.000	0.000	0.054
Gross Total (I to IV) :	7.829	6.951	0.873	0.860	0.305	0.256	2.294

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–23

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1984)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	SCARP II							
	1. Saline	0.690	0.550	0.075	0.116	0.140	0.000	0.331
	SCARP V							
	2. Khairwala Unit	0.333	0.301	0.022	0.000	0.003	0.000	0.025
	SCARP VI							
	3. Panjnad Abbasia (Rem.)	1.460	1.267	0.076	0.266	0.125	0.000	0.467
	SCARP VII							
	4. Pandoki Unit	0.196	0.174	0.093	0.000	0.009	0.000	0.102
	SCARP VIII							
	5. Fordwah Sadiqia II	0.111	0.100	0.002	0.022	0.015	0.070	0.109
	6. AWL TSMB Link	0.090	0.090	0.000	0.000	0.015	0.000	0.015
	Sub–Total (a):	2.880	2.482	0.268	0.404	0.307	0.070	1.049
(b)	Surface Drainage Project							
	7. AWL CJ Link	0.023	0.020	0.000	0.000	0.015	0.000	0.015
	Sub–Total (b):	0.023	0.020	0.000	0.000	0.015	0.000	0.015
(c)	Tile Drainage Projects							
	SCARP V							
	8. Drainage IV 1/	0.130	0.109	0.072	0.000	0.000	0.000	0.072
	Sub–Total (b):	0.130	0.109	0.072	0.000	0.000	0.000	0.072
	Total (I):	3.033	2.611	0.340	0.404	0.322	0.070	1.136
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Ghotki (FGW)	0.441	0.400	0.000	0.283	0.000	0.013	0.296
	2. South Rohri (FGW)	0.541	0.375	0.018	0.000	0.009	0.000	0.027
	Sub–Total (a):	0.982	0.775	0.018	0.283	0.009	0.013	0.323
(b)	Surface Drainage Project							
	3. North Dadu	0.514	0.445	0.000	0.033	0.000	0.481	0.514
	4. LBOD Core Programme (Spinal Drain)	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.880	2.083	2.963
	Sub–Total (b):	3.730	3.116	0.000	0.033	0.880	2.564	3.477
(c)	Tile Drainage Projects							
	6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.037	0.000	0.037
	Sub–Total (c):	0.045	0.036	0.000	0.000	0.037	0.000	0.037
	Total (II):	4.757	3.927	0.018	0.316	0.926	2.577	3.837
III.	**N.W.F.P.**							
(a)	Tile Drainage Project							
	1. Mardan	0.220	0.180	0.057	0.000	0.000	0.000	0.057
	Sub–Total (a):	0.220	0.180	0.057	0.000	0.000	0.000	0.057
	Total (III):	0.220	0.180	0.057	0.000	0.000	0.000	0.057
	Gross Total (I to III):	8.010	6.718	0.415	0.720	1.248	2.647	5.030

Source: SCARP Monitoring, WAPDA

TABLE–24

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.651	0.529	0.840	0.496	0.022	1.887
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.487	0.000	0.756	1.243
Gudu L. Bank	0.519		0.000	0.075	0.000	0.229	0.304
Sukkur R. Bank	1.461		0.240	0.002	0.932	0.005	1.179
Sukkur L. Bank	4.532		0.020	0.000	0.986	0.006	1.012
Sub–Total (II):	7.755	7.583	0.260	0.564	1.918	0.996	3.738
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.064	0.000	0.000	0.000	0.064
Gross Total (I TO III) :	24.661	20.831	0.853	1.404	2.414	1.018	5.689

Source: SCARP Monitoring, WAPDA

TABLE–25

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1984)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.223	0.000	0.063	0.286
Kandhkot (Pilot)	0.013	0.012	0.000	0.011	0.000	0.000	0.011
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.234	0.000	0.150	0.384
II. On–Going							
North Dadu	0.497	0.430	0.000	0.019	0.000	0.478	0.497
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	1.791	1.791
Sub–Total (II):	2.459	2.120	0.000	0.019	0.000	2.269	2.288
III. Remaining Area	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	4.091	3.613	0.000	0.724	0.000	3.193	3.917

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–26

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.281	5.577	0.116	0.092	0.075	0.039	0.322
NWFP	0.226	0.198	0.034	0.000	0.000	0.000	0.034
Sindh/Balochistan	2.035	1.746	0.181	0.025	0.270	1.100	1.576
i. Non–Rice	1.648	1.418	0.181	0.008	0.270	0.024	0.483
ii. Rice	0.387	0.328	0.000	0.017	0.000	0.076	0.093
Total:	8.542	7.521	0.331	0.117	0.345	1.139	1.932
i. Non–Rice	8.155	7.193	0.331	0.100	0.345	0.063	0.839
ii. Rice	0.387	0.328	0.000	0.017	0.000	0.076	0.093
II. On–Going SCARPs							
Punjab	2.326	2.046	0.082	0.060	0.109	0.056	0.307
NWFP	0.220	0.180	0.025	0.000	0.000	0.000	0.025
Sindh/Balochistan	4.757	3.927	0.007	0.026	0.344	0.061	0.438
i. Non–Rice	2.298	1.807	0.007	0.024	0.344	0.001	0.376
ii. Rice	2.459	2.120	0.000	0.002	0.000	0.060	0.062
Total:	7.303	6.153	0.114	0.086	0.453	0.117	0.770
i. Non–Rice	4.844	3.853	0.114	0.084	0.453	0.057	0.708
ii. Rice	2.459	2.120	0.000	0.002	0.000	0.060	0.062
III. Out Side SCARPs							
Punjab	16.019	12.651	0.245	0.123	0.207	0.013	0.588
NWFP	0.887	0.597	0.037	0.000	0.000	0.000	0.037
Sindh/Balochistan	7.755	7.583	0.065	0.007	1.725	0.330	2.127
i. Non–Rice	6.510	6.418	0.065	0.006	1.725	0.173	1.969
ii. Rice	1.245	1.165	0.000	0.001	0.000	0.157	0.158
Total:	24.661	20.831	0.347	0.130	1.932	0.343	2.752
i. Non–Rice	23.416	19.666	0.347	0.129	1.932	0.186	2.594
ii. Rice	1.245	1.165	0.000	0.001	0.000	0.157	0.158
Gross Total (I TO III) :	40.506	34.505	0.792	0.333	2.730	1.599	5.454

Source: SCARP Monitoring, WAPDA

TABLE–27

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.010	0.000	0.000	0.000	0.010
SCARP II							
2. Non Saline	1.649	1.490	0.067	0.008	0.008	0.000	0.083
3. Shahpur	0.690	0.550	0.050	0.027	0.059	0.000	0.136
4. Shahpur	0.112	0.090	0.000	0.003	0.000	0.000	0.003
SCARP III							
5. Non Saline	1.138	0.952	0.000	0.013	0.000	0.004	0.017
6. Saline	0.142	0.098	0.000	0.027	0.000	0.032	0.059
SCARP IV							
7. Muridke Mangtanwala	0.558	0.544	0.000	0.000	0.000	0.000	0.000
SCARP V							
8. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
9. Satiana (P)	0.118	0.098	0.002	0.000	0.000	0.000	0.002
SCARP VI							
10. Allahabad (P)	0.216	0.202	0.017	0.009	0.001	0.003	0.030
SCARP VIII							
11. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
12. AWL TP Link	0.020	0.020	0.008	0.000	0.000	0.000	0.008
13. AWL RQB Link	0.054	0.050	0.000	0.005	0.000	0.000	0.005
Sub–Total (a):	6.161	5.462	0.156	0.092	0.068	0.039	0.355
(b) Surface Drainage Projects							
14. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.005	0.000	0.000	0.000	0.005
15. AWL CJ Link	0.023	0.020	0.000	0.000	0.007	0.000	0.007
Sub–Total (b):	0.120	0.115	0.005	0.000	0.007	0.000	0.012
Total (I):	6.281	5.577	0.161	0.092	0.075	0.039	0.367
II. Sind							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.111	0.000	0.077	0.000	0.188
2. North Rohri	0.793	0.686	0.035	0.000	0.000	0.000	0.035
3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):	1.247	1.078	0.146	0.000	0.077	0.000	0.223
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.035	0.025	0.193	0.026	0.279
Sub–Total (b):	0.701	0.594	0.035	0.025	0.193	0.026	0.279
Total (II):	1.948	1.672	0.181	0.025	0.270	0.026	0.502
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.074	0.074
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.074	0.074
Total (III):	0.087	0.074	0.000	0.000	0.000	0.074	0.074
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.029	0.000	0.000	0.000	0.029
2. Bannu	0.090	0.080	0.005	0.000	0.000	0.000	0.005
Sub–Total (a):	0.226	0.198	0.034	0.000	0.000	0.000	0.034
Total (IV):	0.226	0.198	0.034	0.000	0.000	0.600	0.034
Gross Total (I to IV) :	8.542	7.521	0.376	0.117	0.345	0.139	0.977

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–28

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Khairwala Unit	0.333	0.301	0.005	0.000	0.000	0.000	0.005
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.043	0.056	0.101	0.000	0.200
SCARP VII							
3. Pandoki Unit	0.196	0.174	0.007	0.000	0.000	0.000	0.007
SCARP VIII							
4. Fordwah Sadiqia II	0.111	0.100	0.000	0.003	0.006	0.056	0.065
5. AWL TSMB Link	0.090	0.090	0.018	0.001	0.002	0.000	0.021
Sub–Total (a):	2.190	1.932	0.073	0.060	0.109	0.056	0.298
(b) Surface Drainage Project							
6. AWL CJ Link	0.006	0.005	0.000	0.000	0.000	0.000	0.000
Sub–Total (b):	0.006	0.005	0.000	0.000	0.000	0.000	0.000
(c) Tile Drainage Projects							
7. Drainage IV	0.130	0.109	0.009	0.000	0.000	0.000	0.009
Sub–Total (a):	0.130	0.109	0.009	0.000	0.000	0.000	0.009
Total (I):	2.326	2.046	0.082	0.060	0.109	0.056	0.307
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.023	0.000	0.002	0.025
2. South Rohri (FGW)	0.541	0.375	0.007	0.000	0.003	0.000	0.010
Sub–Total (a):	0.982	0.775	0.007	0.023	0.003	0.002	0.035
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.003	0.000	0.029	0.032
4. LBOD Core Programme	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.299	0.030	0.329
Sub–Total (b):	3.730	3.116	0.000	0.003	0.299	0.059	0.361
(c) Tile Drainage Projects							
6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Sub–Total (c):	0.045	0.036	0.000	0.000	0.042	0.000	0.042
Total (II):	4.757	3.927	0.007	0.026	0.344	0.061	0.438
III. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.025	0.000	0.000	0.000	0.025
Sub–Total (a):	0.220	0.180	0.025	0.000	0.000	0.000	0.025
Total (III):	0.220	0.180	0.025	0.000	0.000	0.000	0.025
Gross Total (I to III) :	7.303	6.153	0.114	0.086	0.453	0.117	0.770

Source: SCARP Monitoring, WAPDA.

TABLE–29

Detail of Gross Area of Canal Commands Outside SCARPs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.651	0.245	0.123	0.207	0.013	0.588
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.001	0.000	0.157	0.158
Gudu L. Bank	0.519		0.000	0.005	0.000	0.163	0.168
Sukkur R. Bank	1.461		0.026	0.001	0.653	0.005	0.685
Sukkur L. Bank	4.532		0.039	0.000	1.072	0.005	1.116
Sub–Total (II):	7.755	7.583	0.065	0.007	1.725	0.330	2.127
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.037	0.000	0.000	0.000	0.037
Gross Total (I to III) :	24.661	20.831	0.347	0.130	1.932	0.343	2.752

Source: SCARP Monitoring, WAPDA.

TABLE–30

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.017	0.000	0.002	0.019
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.074	0.074
Sub–Total (I):	0.387	0.328	0.000	0.017	0.000	0.076	0.093
II. On–Going							
North Dadu	0.497	0.430	0.000	0.002	0.000	0.030	0.032
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	0.030	0.030
Sub–Total (II):	2.459	2.120	0.000	0.002	0.000	0.060	0.062
III. Remaining Area	1.245	1.165	0.000	0.001	0.000	0.157	0.158
Gross Total (I to III) :	4.091	3.613	0.000	0.020	0.000	0.293	0.313

Source: SCARP Monitoring, WAPDA.

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–31

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	6.281	5.577	0.406	0.445	0.097	0.094	1.042
NWFP	0.226	0.198	0.040	0.000	0.000	0.000	0.040
Sindh/Balochistan	2.035	1.746	0.251	0.330	0.271	0.170	1.022
i. Non–Rice	1.648	1.418	0.251	0.097	0.271	0.020	0.639
ii. Rice	0.387	0.328	0.000	0.233	0.000	0.150	0.383
Total:	8.542	7.521	0.697	0.775	0.368	0.264	2.104
i. Non–Rice	8.155	7.193	0.697	0.542	0.368	0.114	1.721
ii. Rice	0.387	0.328	0.000	0.233	0.000	0.150	0.383
II. On–Going SCARPs							
Punjab	2.326	2.046	0.211	0.204	0.095	0.053	0.563
NWFP	0.220	0.180	0.069	0.000	0.000	0.000	0.069
Sindh/Balochistan	4.757	3.927	0.023	0.315	1.046	2.498	3.882
i. Non–Rice	2.298	1.807	0.023	0.297	1.046	0.186	1.552
ii. Rice	2.459	2.120	0.000	0.018	0.000	2.312	2.330
Total:	7.303	6.153	0.303	0.519	1.141	2.551	4.514
i. Non–Rice	4.844	4.033	0.303	0.501	1.141	0.239	2.184
ii. Rice	2.459	2.120	0.000	0.018	0.000	2.312	2.330
III. Out Side SCARPs							
Punjab	16.019	12.651	0.526	0.516	0.323	0.012	1.377
NWFP	0.887	0.597	0.030	0.000	0.000	0.000	0.030
Sindh/Balochistan	7.755	7.583	0.205	0.583	1.905	1.062	3.755
i. Non–Rice	6.510	6.418	0.205	0.112	1.905	0.288	2.510
ii. Rice	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Total:	24.661	20.831	0.761	1.099	2.228	1.074	5.162
i. Non–Rice	23.416	19.666	0.761	0.628	2.228	0.300	3.917
ii. Rice	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	40.506	34.505	1.761	2.393	3.737	3.889	11.780

Source: SCARP Monitoring, WAPDA

TABLE–32

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1985)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
				Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
		GA	CCA	P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	1. SCARP I SCARP II	1.217	1.141	0.041	0.003	0.000	0.000	0.044
	2. Non Saline	1.649	1.490	0.243	0.012	0.014	0.000	0.269
	3. Saline	0.690	0.550	0.059	0.108	0.083	0.000	0.250
	4. Shahpur SCARP III	0.112	0.090	0.000	0.027	0.000	0.000	0.027
	5. Non Saline	1.138	0.952	0.000	0.191	0.000	0.031	0.222
	6. Saline	0.142	0.098	0.000	0.063	0.000	0.033	0.096
	SCARP IV							
	7. Muridke Mangtanwala SCARP V	0.558	0.544	0.004	0.016	0.000	0.000	0.020
	8. Shorkot Kamalia (P)	0.169	0.154	0.001	0.000	0.000	0.000	0.001
	9. Satiana (P) SCARP VI	0.118	0.098	0.015	0.000	0.000	0.000	0.015
	10. Allahabad (P) SCARP VIII	0.216	0.202	0.017	0.010	0.000	0.000	0.027
	11. Minchinabad (P)	0.078	0.073	0.000	0.015	0.000	0.000	0.015
	12. AWL TP Link	0.020	0.020	0.010	0.000	0.000	0.000	0.010
	13. AWL RQB Link	0.054	0.050	0.000	0.000	0.000	0.015	0.015
	Sub–Total (a):	6.161	5.462	0.390	0.445	0.097	0.079	1.011
(b)	Surface Drainage Projects							
	14. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.016	0.000	0.000	0.000	0.016
	15. AWL CJ Link	0.023	0.020	0.000	0.000	0.000	0.015	0.015
	Sub–Total (b):	0.120	0.115	0.016	0.000	0.000	0.015	0.031
	Total (I):	6.281	5.577	0.406	0.445	0.097	0.094	1.042
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.123	0.000	0.070	0.000	0.193
	2. North Rohri	0.793	0.686	0.044	0.000	0.000	0.000	0.044
	3. Kandhkot (P)	0.013	0.012	0.000	0.009	0.000	0.000	0.009
	Sub–Total (a):	1.247	1.078	0.167	0.009	0.070	0.000	0.246
(b)	Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.084	0.321	0.201	0.083	0.689
	Sub–Total (b):	0.701	0.594	0.084	0.321	0.201	0.083	0.689
	Total (II):	1.948	1.672	0.251	0.330	0.271	0.083	0.935
III.	**Balochistan**							
(a)	Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
IV.	**N.W.F.P.**							
(a)	Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.035	0.000	0.000	0.000	0.035
	2. Bannu	0.090	0.080	0.005	0.000	0.000	0.000	0.005
	Sub–Total (a):	0.226	0.198	0.040	0.000	0.000	0.000	0.040
	Total (IV):	0.226	0.198	0.040	0.000	0.000	0.000	0.040
	Gross Total (I to IV) :	8.542	7.521	0.697	0.775	0.368	0.264	2.104

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–33

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1985)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
				Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
		GA	CCA	P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	SCARP V							
	1. Khairwala Unit	0.333	0.301	0.003	0.000	0.000	0.000	0.003
	SCARP VI							
	2. Panjnad Abbasia (Rem.)	1.460	1.267	0.069	0.177	0.048	0.000	0.294
	SCARP VII							
	3. Pandoki Unit	0.196	0.174	0.133	0.000	0.007	0.000	0.140
	SCARP VIII							
	4. Fordwah Sadiqia II	0.111	0.100	0.000	0.027	0.025	0.053	0.105
	5. AWL TSMB Link	0.090	0.090	0.000	0.000	0.015	0.000	0.015
	Sub–Total (a):	2.190	1.932	0.205	0.204	0.095	0.053	0.557
(b)	Surface Drainage Project							
	6. AWL CJ Link	0.006	0.005	0.000	0.000	0.000	0.000	0.000
	Sub–Total (b):	0.006	0.005	0.000	0.000	0.000	0.000	0.000
(c)	Tile Drainage Projects							
	7. Drainage IVC 1/	0.130	0.109	0.006	0.000	0.000	0.000	0.006
	Sub–Total (c):	0.130	0.109	0.006	0.000	0.000	0.000	0.006
	Total (I):	2.326	2.046	0.211	0.204	0.095	0.053	0.563
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Ghotki (FGW)	0.441	0.400	0.000	0.282	0.000	0.029	0.311
	2. South Rohri (FGW)	0.541	0.375	0.023	0.000	0.007	0.000	0.030
	Sub–Total (a):	0.982	0.775	0.023	0.282	0.007	0.029	0.341
(b)	Surface Drainage Project							
	3. North Dadu	0.514	0.445	0.000	0.033	0.000	0.481	0.541
	4. LBOD Core Programme	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.999	1.988	2.987
	Sub–Total (b):	3.730	3.116	0.000	0.033	0.999	2.469	3.528
(c)	Tile Drainage Projects							
	6. East Khairpur Tile	0.045	0.036	0.000	0.000	0.040	0.000	0.040
	Sub–Total (c):	0.045	0.036	0.000	0.000	0.040	0.000	0.040
	Total (II):	4.757	3.927	0.023	0.315	1.046	2.498	3.909
III.	**N.W.F.P.**							
(a)	Tile Drainage Project							
	1. Mardan	0.220	0.180	0.069	0.000	0.000	0.000	0.069
	Sub–Total (a):	0.220	0.180	0.069	0.000	0.000	0.000	0.069
	Total (III):	0.220	0.180	0.069	0.000	0.000	0.000	0.069
	Gross Total (I to III):	7.303	6.153	0.303	0.519	1.141	2.551	4.541

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage IV Project is 0.354 MA and 0.296 MA respectively, which is partially overlapped by SCARP–I, Satiana and Paharang Drainage Project. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–34

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.651	0.526	0.516	0.323	0.012	1.377
II. Sindh/Balochistan							
Gudu R. Bank	1.243		0.000	0.487	0.000	0.756	1.243
Gudu L. Bank	0.519		0.000	0.090	0.000	0.296	0.386
Sukkur R. Bank	1.461		0.162	0.006	0.985	0.007	1.160
Sukkur L. Bank	4.532		0.043	0.000	0.920	0.003	0.966
Sub–Total (II):	7.755	7.583	0.205	0.583	1.905	1.062	3.755
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.030	0.000	0.000	0.000	0.030
Gross Total (I TO III) :	24.661	20.831	0.761	1.099	2.228	1.074	5.162

Source: SCARP Monitoring, WAPDA

TABLE–35

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1985)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.224	0.000	0.063	0.287
Kandhkot (Pilot)	0.013	0.012	0.000	0.009	0.000	0.000	0.009
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.233	0.000	0.150	0.383
II. On–Going							
North Dadu	0.497	0.430	0.000	0.018	0.000	0.479	0.497
Kotri Surface Drainage	1.962	1.690	0.000	0.000	0.000	1.833	1.833
Sub–Total (II):	2.459	2.120	0.000	0.018	0.000	2.312	2.330
III. Remaining Area	1.245	1.165	0.000	0.471	0.000	0.774	1.245
Gross Total (I TO III) :	4.091	3.613	0.000	0.722	0.000	3.236	3.958

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–36

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	6.281	5.577	0.230	0.227	0.090	0.064	0.611
NWFP	0.226	0.198	0.032	0.000	0.000	0.000	0.032
Sindh/Balochistan	2.080	1.782	0.162	0.061	0.226	0.131	0.580
i. Non–Rice	1.648	1.418	0.162	0.041	0.226	0.046	0.475
ii. Rice	0.387	0.328	0.000	0.020	0.000	0.085	0.105
Total:	8.587	7.557	0.424	0.288	0.316	0.195	1.223
i. Non–Rice	8.200	7.229	0.424	0.268	0.316	0.110	1.118
ii. Rice	0.387	0.328	0.000	0.020	0.000	0.085	0.105
II. On–Going SCARPs							
Punjab	2.326	2.046	0.055	0.153	0.092	0.045	0.345
NWFP	0.220	0.180	0.022	0.000	0.000	0.000	0.022
Sindh/Balochistan	6.228	5.254	0.013	0.023	0.720	0.074	0.830
i. Non–Rice	3.679	3.047	0.013	0.021	0.720	0.001	0.755
ii. Rice	2.549	2.207	0.000	0.002	0.000	0.073	0.075
Total:	8.774	7.480	0.090	0.176	0.812	0.119	1.197
i. Non–Rice	6.225	6.273	0.090	0.174	0.812	0.046	1.122
ii. Rice	2.549	2.207	0.000	0.002	0.000	0.073	0.075
III. Out Side SCARPs							
Punjab	16.019	12.651	0.340	0.361	0.232	0.007	0.940
NWFP	0.887	0.597	0.040	0.000	0.000	0.000	0.040
Sindh/Balochistan	6.239	5.720	0.081	0.039	1.436	0.304	1.860
i. Non–Rice	5.084	4.642	0.081	0.039	1.436	0.153	1.709
ii. Rice	1.155	1.078	0.000	0.000	0.000	0.151	0.151
Total:	23.145	18.968	0.461	0.400	1.668	0.311	2.840
i. Non–Rice	21.990	17.890	0.461	0.400	1.668	0.160	2.689
ii. Rice	1.155	1.078	0.000	0.000	0.000	0.151	0.151
Gross Total (I TO III) :	40.506	34.005	0.975	0.864	2.796	0.625	5.260

Source: SCARP Monitoring, WAPDA

TABLE–37
Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth
(April/June 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I SCARP II	1.217	1.141	0.027	0.005	0.000	0.000	0.032
2. Non Saline	1.649	1.490	0.130	0.025	0.011	0.000	0.166
3. Saline	0.690	0.550	0.024	0.021	0.071	0.000	0.116
4. Shahpur SCARP III	0.112	0.090	0.000	0.020	0.000	0.000	0.020
5. Non Saline	1.138	0.952	0.000	0.084	0.000	0.025	0.109
6. Saline	0.142	0.098	0.000	0.063	0.000	0.033	0.096
SCARP IV							
7. Muridke Mangtanwala SCARP V	0.558	0.544	0.000	0.001	0.000	0.000	0.001
8. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
9. Satiana (P) SCARP VI	0.118	0.098	0.005	0.000	0.000	0.000	0.005
10. Allahabad (P) SCARP VIII	0.216	0.202	0.004	0.002	0.000	0.006	0.012
11. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
12. AWL TP Link	0.020	0.020	0.010	0.000	0.000	0.000	0.010
13. AWL RQB Link	0.054	0.050	0.012	0.006	0.000	0.000	0.018
Sub–Total (a):	6.161	5.462	0.214	0.227	0.082	0.064	0.587
(b) Surface Drainage Projects							
14. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.016	0.000	0.000	0.000	0.016
15. AWL CJ Link	0.023	0.020	0.000	0.000	0.008	0.000	0.008
Sub–Total (b):	0.120	0.115	0.016	0.000	0.008	0.000	0.024
Total (I):	6.281	5.577	0.230	0.227	0.090	0.064	0.611
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.133	0.000	0.038	0.000	0.171
2. North Rohri	0.793	0.686	0.020	0.000	0.000	0.000	0.020
3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):	1.247	1.078	0.153	0.000	0.038	0.000	0.191
(b) Surface Drainage Project							
1. Larkana Shikarpur 1/	0.701	0.594	0.009	0.061	0.153	0.047	0.270
2. LBOD Core Programme (Spinal Drain)	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sub–Total (b):	0.701	0.594	0.009	0.061	0.153	0.047	0.270
(c) Tile Drainage Projects							
1. East Khairpur	0.045	0.036	0.000	0.000	0.035	0.000	0.035
Sub–Total (c):	0.045	0.036	0.000	0.000	0.035	0.000	0.035
Total (II):	1.993	1.708	0.162	0.061	0.226	0.047	0.496
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.084	0.084
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.084	0.084
Total (III):	0.087	0.074	0.000	0.000	0.000	0.084	0.084
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.027	0.000	0.000	0.000	0.027
2. Bannu	0.090	0.080	0.005	0.000	0.000	0.000	0.005
Sub–Total (a):	0.226	0.198	0.032	0.000	0.000	0.000	0.032
Total (IV):	0.226	0.198	0.032	0.000	0.000	0.000	0.032
Gross Total (I to IV) :	8.587	7.557	0.424	0.288	0.316	0.195	1.223

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–38

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1986)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	SCARP V							
	1. Khairwala Unit	0.333	0.301	0.003	0.000	0.000	0.000	0.003
	SCARP VI							
	2. Panjnad Abbasia (Rem.)	1.460	1.267	0.007	0.129	0.079	0.000	0.215
	SCARP VII							
	3. Pandoki Unit	0.196	0.174	0.007	0.000	0.000	0.000	0.007
	SCARP VIII							
	4. Fordwah Sadiqia II	0.111	0.100	0.000	0.023	0.007	0.045	0.075
	5. AWL TSMB Link	0.090	0.090	0.027	0.001	0.006	0.000	0.034
	Sub–Total (a):	2.190	1.932	0.044	0.153	0.092	0.045	0.334
(b)	Surface Drainage Project							
	6. AWL CJ Link	0.006	0.005	0.000	0.000	0.000	0.000	0.000
	Sub–Total (b):	0.006	0.005	0.000	0.000	0.000	0.000	0.000
(c)	Tile Drainage Projects							
	7. Drainage IVC 1/	0.130	0.109	0.011	0.000	0.000	0.000	0.011
	Sub–Total (c):	0.130	0.109	0.011	0.000	0.000	0.000	0.011
	Total (I):	2.326	2.046	0.055	0.153	0.092	0.045	0.345
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Ghotki (FGW)	0.441	0.400	0.000	0.021	0.000	0.002	0.023
	2. South Rohri (FGW)	0.541	0.375	0.013	0.000	0.003	0.000	0.016
	Sub–Total (a):	0.982	0.775	0.013	0.021	0.003	0.002	0.039
(b)	Surface Drainage Project							
	3. North Dadu	0.514	0.445	0.000	0.002	0.000	0.028	0.030
	4. LBOD Stage I	1.426	1.275	0.000	0.000	0.600	0.000	0.600
	5. Kotri Surface Drainage	3.216	2.671	0.000	0.000	0.117	0.014	0.131
	Sub–Total (b):	5.156	4.391	0.000	0.002	0.717	0.042	0.761
	Total (II):	6.138	5.166	0.013	0.023	0.720	0.044	0.800
III.	**Balochistan**							
(a)	Surface Drainage Projects							
	1. Hairdin II	0.090	0.087	0.000	0.000	0.000	0.030	0.030
	Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.030	0.030
	Total (III):	0.090	0.087	0.000	0.000	0.000	0.030	0.030
IV.	**N.W.F.P.**							
(a)	Tile Drainage Project							
	1. Mardan	0.220	0.180	0.022	0.000	0.000	0.000	0.022
	Sub–Total (a):	0.220	0.180	0.022	0.000	0.000	0.000	0.022
	Total (IV):	0.220	0.180	0.022	0.000	0.000	0.000	0.022
	Gross Total (I to IV) :	8.774	7.479	0.090	0.176	0.812	0.119	1.197

Source: SCARP Monitoring, WAPDA.

1/ GA and CCA of Drainage IV Project is 0.354 MA and 0.296 MA respectively, which is partially overlapped by SCARP–I, Satiana and Paharang Drainage Project. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–39

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.651	0.340	0.361	0.232	0.007	0.940
II. Sindh/Balochistan							
Gudu R. Bank	1.242		0.000	0.015	0.003	0.127	0.145
Gudu L. Bank	0.519		0.000	0.024	0.000	0.169	0.193
Sukkur R. Bank	1.373		0.032	0.000	0.598	0.003	0.633
Sukkur L. Bank	3.105		0.049	0.000	0.835	0.005	0.889
Sub–Total (II):	6.239	5.720	0.081	0.039	1.436	0.304	1.860
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.040	0.000	0.000	0.000	0.040
Gross Total (I to III) :	23.145	18.968	0.461	0.400	1.668	0.311	2.840

Source: SCARP Monitoring, WAPDA.

TABLE–40

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.020	0.000	0.001	0.021
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.084	0.084
Sub–Total (I):	0.387	0.328	0.000	0.020	0.000	0.085	0.105
II. On–Going							
Hairdin Phase 0.00011	0.090	0.087	0.000	0.000	0.000	0.030	0.030
North Dadu	0.497	0.430	0.000	0.002	0.000	0.028	0.030
Kotri Surface Drainage Drainage	1.962	1.690	0.000	0.000	0.000	0.015	0.015
Sub–Total (II):	2.549	2.207	0.000	0.002	0.000	0.073	0.075
III. Remaining Area	1.155	1.078	0.000	0.000	0.000	0.151	0.151
Gross Total (I TO III) :	4.091	3.613	0.000	0.022	0.000	0.309	0.331

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–41

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.281	5.575	0.329	0.677	0.101	0.079	1.186
NWFP	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Sindh/Balochistan	2.080	1.782	0.697	0.000	0.366	0.000	1.063
i. Non–Rice	1.693	1.454	0.515	0.000	0.170	0.000	0.685
ii. Rice	0.387	0.328	0.182	0.000	0.196	0.000	0.378
Total:	8.587	7.555	1.070	0.677	0.467	0.079	2.293
i. Non–Rice	8.200	7.227	0.888	0.677	0.271	0.079	1.915
ii. Rice	0.387	0.328	0.182	0.000	0.196	0.000	0.378
II. On–Going SCARPs							
Punjab	2.320	2.041	0.208	0.123	0.068	0.079	0.478
NWFP	0.220	0.180	0.047	0.000	0.000	0.000	0.047
Sindh/Balochistan	6.228	5.166	0.943	0.000	3.448	0.000	4.391
i. Non–Rice	3.769	3.046	0.711	0.000	1.303	0.000	2.014
ii. Rice	2.459	2.120	0.232	0.000	2.145	0.000	2.377
Total:	8.768	7.387	1.198	0.123	3.516	0.079	4.916
i. Non–Rice	6.327	5.267	0.966	0.123	1.375	0.079	2.543
ii. Rice	2.459	2.120	0.232	0.000	2.145	0.000	2.377
III. Out Side SCARPs							
Punjab	16.019	12.653	0.348	0.691	0.436	0.055	1.530
NWFP	0.887	0.597	0.054	0.000	0.000	0.000	0.054
Sindh/Balochistan	6.852	6.221	1.821	0.000	1.318	0.000	3.139
i. Non–Rice	5.607	5.143	1.623	0.000	0.266	0.000	1.889
ii. Rice	1.245	1.165	0.198	0.000	1.052	0.000	1.250
Total:	23.758	19.471	2.223	0.691	1.754	0.055	4.723
i. Non–Rice	22.513	18.306	2.025	0.691	0.702	0.055	3.473
ii. Rice	1.245	1.165	0.198	0.000	1.052	0.000	1.250
Gross Total (I TO III) :	41.113	34.413	4.491	1.491	5.737	0.213	11.932

Source: SCARP Monitoring, WAPDA

TABLE–42

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	0.043	0.004	0.000	0.000	0.047
SCARP II							
2. Non Saline	1.649	1.490	0.174	0.024	0.011	0.000	0.209
3. Saline	0.690	0.550	0.049	0.086	0.075	0.000	0.210
4. Shahpur	0.112	0.090	0.000	0.059	0.000	0.000	0.059
SCARP III							
5. Non Saline	1.138	0.950	0.000	0.369	0.000	0.024	0.393
6. Saline	0.142	0.098	0.000	0.046	0.000	0.055	0.101
SCARP IV							
7. Muridke Mangtanwala	0.558	0.544	0.004	0.028	0.000	0.000	0.032
SCARP V							
8. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
9. Satiana (P)	0.118	0.098	0.015	0.000	0.002	0.000	0.017
SCARP VI							
10. Allahabad (P)	0.216	0.202	0.006	0.009	0.000	0.000	0.015
SCARP VIII							
11. Minchinabad (P)	0.078	0.073	0.000	0.040	0.000	0.000	0.040
12. AWL TP Link	0.020	0.020	0.016	0.000	0.000	0.000	0.016
13. AWL RQB Link	0.054	0.050	0.000	0.012	0.000	0.000	0.012
Sub–Total (a):	6.161	5.460	0.309	0.677	0.088	0.079	1.153
(b) Surface Drainage Projects							
14. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.020	0.000	0.000	0.000	0.020
15. AWL CJ Link	0.023	0.020	0.000	0.000	0.013	0.000	0.013
Sub–Total (b):	0.120	0.115	0.020	0.000	0.013	0.000	0.033
Total (I):	6.281	5.575	0.329	0.677	0.101	0.079	1.186
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.179	0.000	0.041	0.000	0.220
2. North Rohri	0.793	0.686	0.052	0.000	0.007	0.000	0.059
3. Kandhkot (P)	0.013	0.012	0.013	0.000	0.000	0.000	0.013
Sub–Total (a):	1.247	1.078	0.245	0.000	0.047	0.000	0.292
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.450	0.000	0.199	0.000	0.649
Sub–Total (b):	0.701	0.594	0.450	0.000	0.199	0.000	0.649
(c) Tile Drainage Project							
1. East Khairpur	0.045	0.036	0.003	0.000	0.033	0.000	0.035
Sub–Total (c):	0.045	0.036	0.003	0.000	0.033	0.000	0.035
Total (II):	1.993	1.708	0.697	0.000	0.279	0.000	0.976
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.087	0.000	0.087
Sub–Total (a):	0.087	0.074	0.000	0.000	0.087	0.000	0.087
Total (III):	0.087	0.074	0.000	0.000	0.087	0.000	0.087
Total (II+III):	2.080	1.782	0.697	0.000	0.366	0.000	1.063
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.038	0.000	0.000	0.000	0.038
2. Bannu	0.090	0.080	0.006	0.000	0.000	0.000	0.006
Sub–Total (a):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Total (IV):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Gross Total (I to IV) :	8.587	7.555	1.070	0.677	0.467	0.079	2.293

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–43

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Khairwala Unit	0.333	0.301	0.002	0.000	0.000	0.000	0.002
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.106	0.117	0.033	0.006	0.262
SCARP VII							
3. Pandoki Unit	0.196	0.174	0.064	0.000	0.003	0.000	0.067
SCARP VIII							
4. Fordwah Sadiqia II	0.111	0.100	0.000	0.006	0.025	0.073	0.104
5. AWL TSMB Link	0.090	0.090	0.016	0.000	0.007	0.000	0.023
Sub–Total (a):	2.190	1.932	0.188	0.123	0.068	0.079	0.458
(Bc) Tile Drainage Projects							
SCARP V							
6. Drainage IV 1/	0.130	0.109	0.020	0.000	0.000	0.000	0.020
Sub–Total (b):	0.130	0.109	0.020	0.000	0.000	0.000	0.020
Total (I):	2.320	2.041	0.208	0.123	0.068	0.079	0.478
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.387	0.000	0.013	0.000	0.400
2. South Rohri (FGW)	0.541	0.375	0.022	0.000	0.000	0.000	0.022
Sub–Total (a):	0.982	0.775	0.409	0.000	0.013	0.000	0.422
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.082	0.000	0.432	0.000	0.514
4. Kotri Surface Drainage	3.216	2.671	0.242	0.000	2.743	0.000	2.985
5. LBOD Stage–1	1.426	1.275	0.210	0.000	0.260	0.000	0.470
Sub–Total (b):	5.156	4.391	0.534	0.000	3.435	0.000	3.969
Total (II):	6.138	5.166	0.943	0.000	3.448	0.000	4.391
III. BALUCHISTAN							
(a) Surface Drainage Project							
1. Hairdin II	0.090	0.087	Data Not Available				
Total (III):	0.090	0.087					
Total (II+III):	6.228	5.253	0.943	0.000	3.448	0.000	4.391
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.047	0.000	0.000	0.000	0.047
Sub–Total (a):	0.220	0.180	0.047	0.000	0.000	0.000	0.047
Total (III):	0.220	0.180	0.047	0.000	0.000	0.000	0.047
Gross Total (I to III):	8.768	7.387	1.198	0.123	3.516	0.079	4.916

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–44

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.653	0.348	0.691	0.436	0.055	1.530
II. Sindh/Balochistan							
Gudu R. Bank	1.855		1.000	0.000	0.263	0.000	1.263
Gudu L. Bank	0.519		0.024	0.000	0.358	0.000	0.382
Sukkur R. Bank	1.373		0.415	0.000	0.600	0.000	1.015
Sukkur L. Bank	3.105		0.382	0.000	0.097	0.000	0.479
Sub–Total (II):	6.852	6.221	1.821	0.000	1.318	0.000	3.139
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.054	0.000	0.000	0.000	0.054
Gross Total (I TO III) :	23.758	19.471	2.223	0.691	1.754	0.055	4.723

Source: SCARP Monitoring, WAPDA

TABLE–45

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1986)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.169	0.000	0.109	0.000	0.278
Kandhkot (Pilot)	0.013	0.012	0.013	0.000	0.000	0.000	0.013
Hairdin	0.087	0.074	0.000	0.000	0.087	0.000	0.087
Sub–Total (I):	0.387	0.328	0.182	0.000	0.196	0.000	0.378
II. On–Going							
North Dadu	0.497	0.430	0.080	0.000	0.417	0.000	0.497
Kotri Surface Drainage	1.962	1.690	0.152	0.000	1.728	0.000	1.880
Sub–Total (II):	2.459	2.120	0.232	0.000	2.145	0.000	2.377
III. Remaining Area	1.245	1.165	0.198	0.000	1.052	0.000	1.250
Gross Total (I TO III) :	4.091	3.613	0.612	0.000	3.393	0.000	4.005

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–46

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	DATA NOT AVAILABLE						
NWFP	0.226	0.198	0.035	0.000	0.000	0.000	0.035
Sindh/Balochistan	2.080	1.782	0.261	0.000	0.190	0.000	0.451
i. Non–Rice	1.693	1.454	0.260	0.000	0.108	0.000	0.368
ii. Rice	0.387	0.328	0.001	0.000	0.082	0.000	0.083
Total:	2.306	1.980	0.296	0.000	0.190	0.000	0.486
i. Non–Rice	1.919	1.652	0.295	0.000	0.108	0.000	0.403
ii. Rice	0.387	0.328	0.001	0.000	0.082	0.000	0.083
II. On–Going SCARPs							
Punjab	DATA NOT AVAILABLE						0.478
NWFP	0.220	0.180	0.020	0.000	0.000	0.000	0.020
Sindh/Balochistan	6.227	5.253	0.545	0.000	0.612	0.000	1.157
i. Non–Rice	3.679	3.046	0.522	0.000	0.484	0.000	1.006
ii. Rice	2.548	2.207	0.023	0.000	0.128	0.000	0.151
Total:	6.447	5.433	0.565	0.000	0.612	0.000	1.655
i. Non–Rice	3.899	3.226	0.542	0.000	0.484	0.000	1.026
ii. Rice	2.548	2.207	0.023	0.000	0.128	0.000	0.151
III. Out Side SCARPs							
Punjab	DATA NOT AVAILABLE						0.000
NWFP	0.887	0.597	0.035	0.000	0.000	0.000	0.035
Sindh/Balochistan	6.852	6.221	0.437	0.000	0.748	0.000	1.185
i. Non–Rice	5.607	5.056	0.430	0.000	0.673	0.000	1.103
ii. Rice	1.245	1.165	0.007	0.000	0.075	0.000	0.082
Total:.	7.739	6.818	0.472	0.000	0.748	0.000	1.220
i. Non–Rice	6.494	5.653	0.465	0.000	0.673	0.000	1.138
ii. Rice	1.245	1.165	0.007	0.000	0.075	0.000	0.082
Gross Total (I TO III) :	16.492	14.231	1.333	0.000	1.550	0.000	3.361

Source: SCARP Monitoring, WAPDA

TABLE–47

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I	1.217	1.141	DATA NOT AVAIALBLE				
SCARP II							
2. Non Saline	1.649	1.490	DATA NOT AVAIALBLE				
3. Saline	0.690	0.550	DATA NOT AVAIALBLE				
4. Shahpur	0.112	0.090	DATA NOT AVAIALBLE				
SCARP III							
5. Non Saline	1.138	0.950	DATA NOT AVAIALBLE				
6. Saline	0.142	0.098	DATA NOT AVAIALBLE				
SCARP IV							
7. Muridke Mangtanwala	0.558	0.544	DATA NOT AVAIALBLE				
SCARP V							
8. Shorkot Kamalia (P)	0.169	0.154	DATA NOT AVAIALBLE				
9. Satiana (P)	0.118	0.098	DATA NOT AVAIALBLE				
SCARP VI							
10. Allahabad (P)	0.216	0.202	DATA NOT AVAIALBLE				
SCARP VIII							
11. Minchinabad (P)	0.078	0.073	DATA NOT AVAIALBLE				
12. AWL TP Link	0.020	0.020	DATA NOT AVAIALBLE				
13. AWL RQB Link	0.054	0.050	DATA NOT AVAIALBLE				
Sub–Total (a):	6.161	5.460	0.000	0.000	0.000	0.000	0.000
(b) Surface Drainage Projects							
14. Paharang Drain (Out Side Scarp–I)	0.097	0.095	DATA NOT AVAIALBLE				0.000
15. AWL CJ Link	0.023	0.020	DATA NOT AVAIALBLE				0.000
Sub–Total (b):	0.120	0.115	0.000	0.000	0.000	0.000	0.000
Total (I):	6.281	5.575	0.000	0.000	0.000	0.000	0.000
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.144	0.000	0.051	0.000	0.195
2. North Rohri	0.793	0.686	0.031	0.000	0.000	0.000	0.031
3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Sub–Total (a):	1.247	1.078	0.175	0.000	0.051	0.000	0.226
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.070	0.000	0.045	0.000	0.115
Sub–Total (b):	0.701	0.594	0.070	0.000	0.045	0.000	0.115
(c) Tile Drainage Project							
5. East Khairpur	0.045	0.036	0.016	0.000	0.022	0.000	0.038
Sub–Total (c):	0.045	0.036	0.016	0.000	0.022	0.000	0.038
Total (II):	1.993	1.708	0.261	0.000	0.118	0.000	0.379
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.072	0.000	0.072
Sub–Total (a):	0.087	0.074	0.000	0.000	0.072	0.000	0.072
Total (III):	0.087	0.074	0.000	0.000	0.072	0.000	0.072
Total (II+III):	2.080	1.782	0.261	0.000	0.190	0.000	0.451
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.032	0.000	0.000	0.000	0.032
2. Bannu	0.090	0.080	0.003	0.000	0.000	0.000	0.003
Sub–Total (a):	0.226	0.198	0.035	0.000	0.000	0.000	0.035
Total (IV):	0.226	0.198	0.035	0.000	0.000	0.000	0.035
Gross Total (I to IV) :	8.587	7.555	0.296	0.000	0.190	0.000	0.486

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–48

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Khairwala Unit	0.333	0.301	DATA NOT AVAIALBLE				0.000
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	DATA NOT AVAIALBLE				0.000
SCARP VII							
3. Pandoki Unit	0.196	0.174	DATA NOT AVAIALBLE				0.000
SCARP VIII							
4. Fordwah Sadiqia II	0.111	0.100	DATA NOT AVAIALBLE				0.000
5. AWL TSMB Link	0.090	0.090	DATA NOT AVAIALBLE				0.000
Sub–Total (a):	2.199	1.932	0.000	0.000	0.000	0.000	0.000
(b) Tile Drainage Projects							
SCARP V							
6. Drainage IV 1/	0.130	0.109	DATA NOT AVAIALBLE				0.000
Sub–Total (b):	0.130	0.109	0.000	0.000	0.000	0.000	0.000
Total (I):	2.320	2.041	0.000	0.000	0.000	0.000	0.000
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.014	0.000	0.003	0.000	0.017
2. South Rohri (FGW)	0.541	0.375	0.036	0.000	0.000	0.000	0.036
Sub–Total (a):	0.982	0.775	0.050	0.000	0.003	0.000	0.053
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.005	0.000	0.033	0.000	0.038
4. Kotri Surface Drainege	3.216	2.671	0.053	0.000	0.229	0.000	0.282
5. LBOD Stage–1	1.426	1.275	0.438	0.000	0.314	0.000	0.752
Sub–Total (b):	5.156	4.391	0.495	0.000	0.577	0.000	1.072
Total (II):	6.138	5.166	0.545	0.000	0.580	0.000	1.125
III. Balochistan							
(a) Surface Drainage Project							
1. Hairdin	0.090	0.087	0.000	0.000	0.032	0.000	0.032
Total III:	0.090	0.087	0.000	0.000	0.032	0.000	0.032
Total (II+III):	6.228	5.253	0.545	0.000	0.612	0.000	1.157
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.020	0.000	0.000	0.000	0.020
Sub–Total (a):	0.220	0.180	0.020	0.000	0.000	0.000	0.020
Total (IV):	0.220	0.180	0.020	0.000	0.000	0.000	0.020
Gross Total (I to IV):	8.768	7.474	0.565	0.000	0.612	0.000	1.177

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–49

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	16.019	12.653					0.000
II. Sindh/Balochistan							
Gudu R. Bank	1.803		0.034	0.000	0.057	0.000	0.091
Gudu L. Bank	0.513		0.012	0.000	0.045	0.000	0.057
Sukkur R. Bank	1.454		0.029	0.000	0.308	0.000	0.337
Sukkur L. Bank	3.082		0.362	0.000	0.338	0.000	0.700
Sub–Total (II):	6.852	6.221	0.437	0.000	0.748	0.000	1.185
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.035	0.000	0.000	0.000	0.035
Gross Total (I TO III) :	23.758	19.471	0.472	0.000	0.748	0.000	1.220

Source: SCARP Monitoring, WAPDA

TABLE–50

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.001	0.000	0.010	0.000	0.011
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.072	0.000	0.072
Sub–Total (I):	0.387	0.328	0.001	0.000	0.082	0.000	0.083
II. On–Going							
Hairdin II	0.089	0.087	0.000	0.000	0.032	0.000	0.032
North Dadu	0.497	0.430	0.003	0.000	0.034	0.000	0.037
Kotri Surface Drainage	1.962	1.690	0.020	0.000	0.062	0.000	0.082
Sub–Total (II):	2.548	2.207	0.023	0.000	0.128	0.000	0.151
III. Remaining Area	1.245	1.165	0.007	0.000	0.075	0.000	0.082
Gross Total (I TO III) :	4.180	3.700	0.032	0.000	0.284	0.000	0.316

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.
Rice, Fuleli and Pinyari.

TABLE–51

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.891	6.122	0.180	0.730	0.029	0.166	1.105
NWFP	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Sindh/Balochistan	2.080	1.782	0.510	0.089	0.189	0.253	1.041
i. Non–Rice	1.693	1.454	0.510	0.004	0.189	0.001	0.704
ii. Rice	0.387	0.328	0.000	0.085	0.000	0.252	0.337
Total:	9.197	8.102	0.734	0.819	0.218	0.419	2.190
i. Non–Rice	8.810	7.774	0.734	0.734	0.218	0.167	1.853
ii. Rice	0.387	0.328	0.000	0.085	0.000	0.252	0.337
II. On–Going SCARPs							
Punjab	1.680	1.466	0.086	0.192	0.038	0.002	0.318
NWFP	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Sindh/Balochistan	6.228	5.253	0.274	0.419	1.093	2.079	3.865
i. Non–Rice	3.680	3.046	0.274	0.260	1.093	0.007	1.634
ii. Rice	2.548	2.207	0.000	0.159	0.000	2.073	2.232
Total:	8.128	6.899	0.397	0.611	1.131	2.081	4.220
i. Non–Rice	5.580	4.692	0.397	0.462	1.121	0.009	1.989
ii. Rice	2.548	2.207	0.000	0.149	0.010	2.073	2.232
III. Out Side SCARPs							
Punjab	16.049	12.681	0.308	0.497	0.290	0.100	1.195
NWFP	0.887	0.597	0.043	0.000	0.000	0.000	0.043
Sindh/Balochistan	6.852	6.221	0.527	1.097	1.059	0.389	3.072
i. Non–Rice	5.090	4.521	0.527	0.203	1.059	0.045	1.834
ii. Rice	1.762	1.700	0.000	0.894	0.000	0.344	1.238
Total:	23.788	19.499	0.878	1.594	1.349	0.489	4.310
i. Non–Rice	22.026	17.799	0.884	0.700	1.349	0.145	3.078
ii. Rice	1.762	1.700	0.000	0.894	0.000	0.344	1.238
Gross Total (I TO III) :	41.113	34.500	2.009	3.024	2.698	2.989	10.720

Source: SCARP Monitoring, WAPDA

TABLE–52

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
1. SCARP I SCARP II	1.217	1.141	0.003	0.009	0.000	0.000	0.012
2. Non Saline	1.649	1.490	0.047	0.024	0.002	0.000	0.074
3. Saline	0.690	0.550	0.033	0.088	0.005	0.000	0.126
4. Shahpur SCARP III	0.112	0.090	0.003	0.056	0.000	0.000	0.059
5. Non Saline	1.138	0.950	0.000	0.416	0.000	0.024	0.440
6. Saline SCARP IV	0.142	0.098	0.000	0.051	0.000	0.069	0.120
7. Muridke Mangtanwala SCARP V	0.558	0.544	0.001	0.002	0.000	0.000	0.003
8. Shorkot Kamalia (P)	0.169	0.154	0.005	0.000	0.000	0.000	0.005
9. Satiana (P)	0.118	0.098	0.005	0.000	0.000	0.000	0.005
10. Khairwala Unit SCARP VI	0.333	0.301	0.001	0.000	0.000	0.000	0.001
11. Allahabad (P) SCARP VII	0.216	0.202	0.018	0.041	0.000	0.000	0.059
12. Pandoki Unit SCARP VIII	0.196	0.174	0.046	0.000	0.002	0.000	0.048
13. Minchinabad (P)	0.078	0.073	0.000	0.022	0.000	0.000	0.022
14. Fordwah Sadiqia II	0.111	0.100	0.000	0.009	0.009	0.073	0.091
15. AWL TP Link	0.020	0.020	0.018	0.000	0.000	0.000	0.018
16. AWL RQB Link	0.024	0.022	0.000	0.012	0.000	0.000	0.012
Sub–Total (a):	6.771	6.007	0.180	0.730	0.019	0.166	1.094
(b) Surface Drainage Projects							
17. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
18. AWL CJ Link	0.023	0.020	0.000	0.000	0.011	0.000	0.011
Sub–Total (b):	0.120	0.115	0.000	0.000	0.011	0.000	0.011
Total (I):	6.891	6.122	0.180	0.730	0.029	0.166	1.105
II. Sindh							
(a) Tubewell Drainage Project							
1. Khairpur	0.441	0.380	0.150	0.000	0.025	0.000	0.175
2. North Rohri	0.793	0.686	0.057	0.000	0.000	0.000	0.057
3. Kandhkot (P)	0.013	0.012	0.000	0.012	0.000	0.000	0.012
Sub–Total (a):	1.247	1.078	0.207	0.012	0.025	0.000	0.244
(b) Surface Drainage Project							
4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.290	0.077	0.139	0.166	0.672
Sub–Total (b):	0.701	0.594	0.290	0.077	0.139	0.166	0.672
(c) Tile Drainage Project							
1. East Khairpur	0.045	0.036	0.013	0.000	0.025	0.000	0.038
Sub–Total (c):	0.045	0.036	0.013	0.000	0.025	0.000	0.038
Total (II):	1.993	1.708	0.510	0.089	0.189	0.166	0.954
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Total (II+III):	2.080	1.782	0.510	0.089	0.189	0.253	1.041
IV. N.W.F.P.							
(a) Tubwell Drainage Project							
1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.038	0.000	0.000	0.000	0.038
2. Bannu	0.090	0.080	0.006	0.000	0.000	0.000	0.006
Sub–Total (a):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Total (IV):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Gross Total (I to IV) :	9.197	8.102	0.734	0.819	0.218	0.419	2.189

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–53

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project SCARP VI							
1. Panjnad Abbasia (Rem.)	1.460	1.267	0.062	0.188	0.028	0.002	0.280
2. AWL TSMB Link	0.090	0.090	0.017	0.004	0.010	0.000	0.031
Sub–Total (a):	1.550	1.357	0.079	0.192	0.038	0.002	0.311
(b) Tile Drainage Projects SCARP V							
3. Drainage IV 1/	0.130	0.109	0.007	0.000	0.000	0.000	0.007
Sub–Total (b):	0.130	0.109	0.007	0.000	0.000	0.000	0.007
Total (I):	1.680	1.466	0.086	0.192	0.038	0.002	0.318
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.260	0.000	0.008	0.268
2. South Rohri (FGW)	0.541	0.375	0.009	0.000	0.000	0.000	0.009
Sub–Total (a):	0.982	0.775	0.009	0.260	0.000	0.008	0.277
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.149	0.000	0.360	0.509
4. LBOD Stage I	1.426	1.275	0.238	0.000	0.152	0.000	0.390
5. Kotri Surface Drainage	3.216	2.671	0.027	0.010	0.941	1.678	2.656
Sub–Total (b):	5.156	4.391	0.265	0.159	1.093	2.038	3.555
Total (II):	6.138	5.166	0.274	0.419	1.093	2.046	3.832
III. Balochistan							
(a) Surface Drainage Projects							
1. Hab Jin II	0.090	0.087	0.000	0.000	0.000	0.033	0.033
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.033	0.033
Total (III):	0.090	0.087	0.000	0.000	0.000	0.033	0.033
Total (I+III):	6.228	5.253	0.274	0.419	1.093	2.079	3.865
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Sub–Total (a):	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Total (IV):	0.220	0.180	0.037	0.000	0.000	0.000	0.037
Gross Total (I to IV):	8.128	6.899	0.397	0.611	1.131	2.082	4.220

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV Project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP I, Sataina and Pahrang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–54

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
Sub–Total (I):	16.049	12.681	0.308	0.497	0.290	0.100	1.195
II. Sindh/Balochistan							
Gudu R. Bank	1.855		0.000	0.838	0.000	0.379	1.217
Gudu L. Bank	0.519		0.000	0.259	0.000	0.010	0.269
Sukkur R. Bank	1.373		0.214	0.000	0.785	0.000	0.999
Sukkur L. Bank	3.105		0.313	0.000	0.274	0.000	0.587
Sub–Total (II):	6.852	6.221	0.527	1.097	1.059	0.389	3.072
III. N.W.F.P.							
Sub–Total (III):	0.887	0.597	0.043	0.000	0.000	0.000	0.043
Gross Total (I TO III) :	23.788	19.499	0.878	1.594	1.349	0.489	4.310

Source: SCARP Monitoring, WAPDA

TABLE–55

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1987)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.073	0.000	0.166	0.239
Kandhkot (Pilot)	0.013	0.012	0.000	0.012	0.000	0.000	0.012
Hairdin	0.087	0.074	0.000	0.000	0.000	0.086	0.086
Sub–Total (I):	0.387	0.328	0.000	0.085	0.000	0.252	0.337
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.033	0.033
North Dadu	0.497	0.430	0.000	0.149	0.000	0.360	0.509
Kotri Surface Drainage	1.962	1.690	0.000	0.010	0.000	1.680	1.690
Sub–Total (II):	2.548	2.207	0.000	0.159	0.000	2.073	2.232
III. Remaining Area	1.762	1.700	0.000	0.894	0.000	0.344	1.238
Gross Total (I TO III) :	4.697	4.235	0.000	1.138	0.000	2.669	3.807

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–56

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	6.891	6.122	0.052	0.288	0.019	0.129	0.487
NWFP	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Sind/Balochistan	2.080	1.782	0.132	0.010	0.102	0.042	0.286
i. Non–Rice	1.693	1.454	0.132	0.000	0.102	0.000	0.234
ii. Rice	0.387	0.328	0.000	0.010	0.000	0.042	0.052
Total:	9.197	8.102	0.230	0.297	0.121	0.171	0.819
i. Non–Rice	8.810	7.774	0.230	0.287	0.121	0.129	0.767
ii. Rice	0.387	0.328	0.000	0.010	0.000	0.042	0.052
II. On–Going SCARPs							
Punjab	1.844	1.595	0.117	0.028	0.048	0.002	0.195
NWFP	0.377	0.331	0.063	0.000	0.000	0.000	0.063
Sind/Balochistan	6.228	5.253	0.283	0.011	0.514	0.065	0.873
i. Non–Rice	3.680	3.046	0.283	0.002	0.514	0.000	0.799
ii. Rice	2.548	2.207	0.000	0.009	0.000	0.065	0.074
Total:	8.449	7.179	0.463	0.039	0.562	0.067	1.131
i. Non–Rice	5.901	4.972	0.463	0.030	0.562	0.002	1.057
ii. Rice	2.548	2.207	0.000	0.009	0.000	0.065	0.074
III. Out Side SCARPs							
Punjab	15.885	12.552	0.133	0.273	0.180	0.066	0.652
NWFP	0.782	0.510	0.032	0.000	0.000	0.000	0.032
Sind/Balochistan	6.852	6.221	0.214	0.044	0.745	0.058	1.061
i. Non–Rice	5.090	4.521	0.214	0.010	0.745	0.035	1.004
ii. Rice	1.762	1.700	0.000	0.034	0.000	0.023	0.057
Total:	23.519	19.283	0.379	0.317	0.925	0.124	1.745
i. Non–Rice	21.757	17.583	0.379	0.283	0.925	0.101	1.688
ii. Rice	1.762	1.700	0.000	0.034	0.000	0.023	0.057
Gross Total (I TO III) :	41.165	34 564	1.072	0.653	1.608	0.362	3.695

Source: SCARP Monitoring, WAPDA

TABLE–57

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1988)

(Million Acres)

SCARPS			Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
					P	NP	P	NP	
I.	Punjab								
	(a)	Tubewell Drainage Project							
		1. SCARP I SCARP II	1.217	1.141	0.003	0.002	0.000	0.000	0.005
		2. Non Saline	1.649	1.490	0.014	0.006	0.0003	0.000	0.020
		3. Saline	0.690	0.550	0.001	0.036	0.000	0.000	0.037
		4. Shahpur SCARP III	0.112	0.090	0.0005	0.022	0.000	0.000	0.022
		5. Non Saline	1.138	0.950	0.000	0.146	0.000	0.024	0.170
		6. Saline	0.142	0.098	0.000	0.039	0.000	0.062	0.101
		SCARP IV							
		7. Muridke Mangtanwala SCARP V	0.558	0.544	0.000	0.000	0.000	0.000	0.000
		8. Shorkot Kamalia (P)	0.169	0.154	0.001	0.000	0.000	0.000	0.001
		9. Satiana (P)	0.118	0.098	0.009	0.000	0.000	0.000	0.009
		10. Khairwala Unit SCARP VI	0.333	0.301	0.000	0.000	0.000	0.000	0.000
		11. Allahabad (P) SCARP VII	0.216	0.202	0.010	0.028	0.000	0.004	0.042
		12. Pandoki (CBDC) Unit SCARP VIII	0.196	0.174	0.000	0.000	0.000	0.000	0.000
		13. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
		14. Fordwah Sadiqia–II	0.111	0.100	0.000	0.007	0.011	0.039	0.057
		15. AWL TP Link	0.020	0.020	0.014	0.000	0.000	0.000	0.014
		16. AWL RQB Link	0.024	0.022	0.000	0.002	0.000	0.000	0.002
		Sub–Total (a):	6.771	6.007	0.052	0.288	0.012	0.129	0.480
	(b)	Surface Drainage Projects							
		17. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
		18. AWL CJ Link	0.023	0.020	0.000	0.000	0.007	0.000	0.007
		Sub–Total (b):	0.120	0.115	0.000	0.000	0.007	0.000	0.007
		Total (I):	6.891	6.122	0.052	0.288	0.019	0.129	0.487
II.	Sind								
	(a)	Tubewell Drainage Project							
		1. Khairpur	0.441	0.380	0.044	0.000	0.050	0.000	0.094
		2. North Rohri	0.793	0.686	0.049	0.000	0.000	0.000	0.049
		3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
		Sub–Total (a):	1.247	1.078	0.093	0.000	0.050	0.000	0.143
	(b)	Surface Drainage Project							
		4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.027	0.010	0.045	0.000	0.081
		Sub–Total (b):	0.701	0.594	0.027	0.010	0.045	0.000	0.081
	(c)	Tile Drainage Project							
		1. East Khairpur	0.045	0.036	0.013	0.000	0.008	0.000	0.020
		Sub–Total (c):	0.045	0.036	0.013	0.000	0.008	0.000	0.020
		Total (II):	1.993	1.708	0.132	0.010	0.102	0.000	0.244
III.	Balochistan								
	(a)	Surface Drainage Projects							
		1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.042	0.042
		Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.042	0.042
		Total (III):	0.087	0.074	0.000	0.000	0.000	0.042	0.042
		Total (II+III):	2.080	1.782	0.132	0.010	0.102	0.042	0.286
IV.	N.W.F.P.								
	(a)	Tubwell Drainage Project							
		1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.039	0.000	0.000	0.000	0.039
		2. Bannu	0.090	0.080	0.007	0.000	0.000	0.000	0.007
		Sub–Total (a):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
		Total (IV):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
		Gross Total (I to IV) :	9.197	8.102	0.230	0.297	0.121	0.171	0.819

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–58

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1988)

(Millions Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Gojra Khewra Phase–I	0.038	0.030	0.001	0.000	0.000	0.000	0.001
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.068	0.028	0.038	0.002	0.136
3. Hadali Sub Unit	0.164	0.137	0.037	0.000	0.007	0.000	0.044
4. AWL TSMB Link	0.052	0.052	0.011	0.000	0.003	0.000	0.014
Sub–Total (a):	1.714	1.486	0.117	0.028	0.048	0.002	0.195
(b) Tile Drainage Projects							
SCARP V							
5. Drainage IV 1/	0.130	0.109	0.000	0.000	0.000	0.000	0.0000
Sub–Total (b):	0.130	0.109	0.000	0.000	0.000	0.000	0.0000
Total (I):	1.844	1.595	0.117	0.028	0.048	0.002	0.195
II. Sind							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.001	0.000	0.000	0.001
2. South Rohri (FGW)	0.541	0.375	0.004	0.000	0.000	0.000	0.004
Sub–Total (a):	0.982	0.775	0.004	0.001	0.000	0.000	0.005
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.002	0.008	0.000	0.015	0.025
4. LBOD Stage I	1.426	1.275	0.269	0.000	0.424	0.000	0.693
5. Kotri Surface Drainage	3.216	2.671	0.007	0.002	0.090	0.035	0.135
Sub–Total (b):	5.156	4.391	0.279	0.010	0.514	0.050	0.853
Total (II):	6.138	5.166	0.283	0.011	0.514	0.050	0.858
III. BALOCHISTAN							
(a) Surface Drainage Project							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.015	0.015
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.015	0.015
Total (III):	0.090	0.087	0.000	0.000	0.000	0.015	0.015
Total (II+III):	6.228	5.253	0.283	0.011	0.514	0.065	0.873
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.041	0.000	0.000	0.000	0.041
2. Chashma Command Area Development	0.157	0.151	0.022	0.000	0.000	0.000	0.022
Sub–Total (a):	0.377	0.331	0.063	0.000	0.000	0.000	0.063
Total (IV):	0.377	0.331	0.063	0.000	0.000	0.000	0.063
Gross Total (I to IV):	8.449	7.179	0.463	0.039	0.562	0.067	1.131

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–59

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
Sub–Total (I):	15.885	12.552	0.133	0.273	0.180	0.066	0.652
II. Sind/Balochistan							
Gudu R. Bank	1.855		0.000	0.044	0.000	0.030	0.074
Gudu L. Bank	0.519		0.000	0.000	0.000	0.028	0.028
Sukkur R. Bank	1.373		0.036	0.000	0.424	0.000	0.460
Sukkur L. Bank	3.105		0.178	0.000	0.321	0.000	0.499
Sub–Total (II):	6.852	6.221	0.214	0.044	0.745	0.058	1.061
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.032	0.000	0.000	0.000	0.032
Gross Total (I TO III) :	23.519	19.283	0.379	0.317	0.925	0.124	1.745

Source: SCARP Monitoring, WAPDA

TABLE–60

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.010	0.000	0.000	0.010
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.042	0.042
Sub–Total (I):	0.387	0.328	0.000	0.010	0.000	0.042	0.052
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.015	0.015
North Dadu	0.497	0.430	0.000	0.007	0.000	0.014	0.021
Kotri Surface Drainage	1.962	1.690	0.000	0.002	0.000	0.036	0.038
Sub–Total (II):	2.548	2.207	0.000	0.009	0.000	0.065	0.074
III. Remaining Area	1.762	1.700	0.000	0.034	0.000	0.023	0.057
Gross Total (I TO III) :	4.697	4.235	0.000	0.054	0.000	0.129	0.183

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–61

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth
(October 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.891	6.122	0.371	0.803	0.075	0.171	1.420
NWFP	0.226	0.198	0.051	0.000	0.000	0.000	0.051
Sindh/Balochistan	2.080	1.782	0.469	0.310	0.114	0.099	0.991
i. Non–Rice	1.693	1.454	0.469	0.064	0.114	0.000	0.646
ii. Rice	0.387	0.328	0.000	0.246	0.000	0.099	0.345
Total:	9.197	8.102	0.891	1.112	0.188	0.270	2.462
i. Non–Rice	8.810	7.774	0.891	0.866	0.188	0.171	2.117
ii. Rice	0.387	0.328	0.000	0.246	0.000	0.099	0.345
II. On–Going SCARPs							
Punjab	1.844	1.595	0.082	0.062	0.049	0.000	0.193
NWFP	0.377	0.331	0.071	0.000	0.000	0.000	0.071
Sindh/Balochistan	6.228	5.253	0.159	0.517	1.666	2.081	4.422
i. Non–Rice	3.680	3.046	0.159	0.245	1.666	0.072	2.141
ii. Rice	2.548	2.207	0.000	0.272	0.000	2.009	2.281
Total:	8.449	7.179	0.311	0.579	1.715	2.081	4.686
i. Non–Rice	5.901	4.972	0.311	0.307	1.715	0.072	2.405
ii. Rice	2.548	2.207	0.000	0.272	0.000	2.009	2.281
III. Out Side SCARPs							
Punjab	15.885	12.552	0.327	0.641	0.250	0.086	1.304
NWFP	0.782	0.510	0.035	0.000	0.000	0.000	0.035
Sindh/Balochistan	6.852	6.221	0.432	0.198	1.268	1.409	3.307
i. Non–Rice	5.090	4.521	0.432	0.198	1.268	1.100	2.998
ii. Rice	1.762	1.700	0.000	0.000	0.000	0.309	1.238
Total:	23.519	19.283	0.794	0.839	1.518	1.495	4.646
i. Non–Rice	21.757	17.583	0.794	0.839	1.518	1.186	4.337
ii. Rice	1.762	1.700	0.000	0.000	0.000	0.309	0.309
Gross Total (I TO III) :	41.165	34.564	1.997	2.531	3.421	3.846	11.794

Source: SCARP Monitoring, WAPDA

TABLE–62

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1988)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	1. SCARP I SCARP II	1.217	1.141	0.040	0.007	0.000	0.000	0.047
	2. Non Saline	1.649	1.490	0.154	0.055	0.010	0.000	0.219
	3. Saline	0.690	0.550	0.061	0.103	0.043	0.000	0.207
	4. Shahpur SCARP III	0.112	0.090	0.003	0.073	0.000	0.000	0.075
	5. Non Saline	1.138	0.950	0.000	0.420	0.000	0.042	0.462
	6. Saline SCARP IV	0.142	0.098	0.000	0.046	0.000	0.065	0.111
	7. Muridke Mangtanwala SCARP V	0.558	0.544	0.000	0.015	0.000	0.000	0.015
	8. Shorkot Kamalia (P)	0.169	0.154	0.006	0.000	0.000	0.000	0.006
	9. Satiana (P)	0.118	0.098	0.020	0.000	0.000	0.000	0.020
	10. Khairwala Unit SCARP VI	0.333	0.301	0.000	0.000	0.000	0.000	0.000
	11. Allahabad (P) SCARP VII	0.216	0.202	0.014	0.028	0.000	0.000	0.042
	12. Pandoki (CBDC) Unit SCARP VIII	0.196	0.174	0.062	0.000	0.000	0.000	0.062
	13. Minchinabad (P)	0.078	0.073	0.000	0.021	0.000	0.000	0.021
	14. Fordwah Sadiqia–II	0.111	0.100	0.000	0.021	0.010	0.065	0.096
	15. AWL TP Link	0.020	0.020	0.012	0.000	0.000	0.000	0.012
	16. AWL RQB Link	0.024	0.022	0.000	0.013	0.000	0.000	0.013
	Sub–Total (a):	6.771	6.007	0.371	0.803	0.063	0.171	1.408
(b)	Surface Drainage Projects							
	17. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
	18. AWL CJ Link	0.023	0.020	0.000	0.000	0.012	0.000	0.012
	Sub–Total (b):	0.120	0.115	0.000	0.000	0.012	0.000	0.012
	Total (I):	6.891	6.122	0.371	0.803	0.075	0.171	1.420
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.114	0.000	0.024	0.000	0.138
	2. North Rohri	0.793	0.686	0.043	0.000	0.000	0.000	0.043
	3. Kandhkot (P)	0.013	0.012	0.000	0.009	0.000	0.000	0.009
	Sub–Total (a):	1.247	1.078	0.157	0.009	0.024	0.000	0.190
(b)	Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.312	0.301	0.056	0.012	0.681
	Sub–Total (b):	0.701	0.594	0.312	0.301	0.056	0.012	0.681
(c)	Tile Drainage Project							
	1. East Khairpur	0.045	0.036	0.000	0.000	0.033	0.000	0.033
	Sub–Total (c):	0.045	0.036	0.000	.0.000	0.033	0.000	0.033
	Total (II):	1.993	1.708	0.469	0.310	0.114	0.012	0.904
III.	**Balochistan**							
(a)	Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (II+III):	2.080	1.782	0.469	0.310	0.114	0.099	0.991
IV.	**N.W.F.P.**							
(a)	Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.042	0.000	0.000	0.000	0.042
	2. Bannu	0.090	0.080	0.009	0.000	0.000	0.000	0.009
	Sub–Total (a):	0.226	0.198	0.051	0.000	0.000	0.000	0.051
	Total (IV):	0.226	0.198	0.051	0.000	0.000	0.000	0.051
	Gross Total (I to IV) :	9.197	8.102	0.891	1.112	0.188	0.270	2.462

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–63

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Gojra Khewra Phase–I	0.038	0.030	0.000	0.000	0.0002	0.000	0.000
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.027	0.058	0.022	0.000	0.106
3. Hadali Sub Unit	0.164	0.137	0.031	0.000	0.023	0.000	0.054
4. AWL TSMB Link	0.052	0.052	0.016	0.005	0.005	0.000	0.025
Sub–Total (a):	1.714	1.486	0.074	0.062	0.049	0.000	0.185
(b) Tile Drainage Projects							
SCARP V							
5. Drainage IV 1/	0.130	0.109	0.008	0.000	0.000	0.000	0.008
Sub–Total (b):	0.130	0.109	0.008	0.000	0.000	0.000	0.008
Total (I):	1.844	1.595	0.082	0.062	0.049	0.000	0.193
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.240	0.000	0.003	0.243
2. South Rohri (FGW)	0.541	0.375	0.020	0.000	0.000	0.000	0.020
Sub–Total (a):	0.982	0.775	0.020	0.240	0.000	0.003	0.263
(b) Surface Drainage Project							
3. North Dadu	0.514	0 445	0.000	0.148	0.000	0.363	0.511
4. LBOD Stage I	1.426	1.275	0.110	0.000	0.730	0.000	0.840
5. Kotri Surface Drainage	3.216	2.671	0.029	0.129	0.935	1.691	2.785
Sub–Total (b):	5.156	4.391	0.139	0.277	1.666	2.054	4.136
Total (II):	6.138	5.166	0.159	0.517	1.666	2.058	4.399
III. Balochistan							
(a) Surface Drainage Projects							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Total (III):	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Total (II+III):	6.228	5.253	0.159	0.517	1.666	2.081	4.422
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.043	0.000	0.000	0.000	0.043
2. Chashma Command Area Development	0.157	0.151	0.028	0.000	0.000	0.000	0.028
Sub–Total (a):	0.377	0.331	0.071	0.000	0.000	0.000	0.071
Total (IV):	0.377	0.331	0.071	0.000	0.000	0.000	0.071
Gross Total (I to IV):	8.449	7.179	0.311	0.579	1.715	2.081	4.686

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–64

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
Sub–Total (I):	15.885	12.552	0.327	0.641	0.250	0.086	1.304
II. Sind/Balochistan							
Gudu R. Bank	1.855		0.000	0.147	0.000	1.073	1.220
Gudu L. Bank	0.519		0.000	0.039	0.000	0.336	0.375
Sukkur R. Bank	1.373		0.045	0.012	1.002	0.000	1.059
Sukkur L. Bank	3.105		0.387	0.000	0.266	0.000	0.653
Sub–Total (II):	6.852	7.583	0.432	0.198	1.268	1.409	3.307
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.035	0.000	0.000	0.000	0.035
Gross Total (I TO III) :	23.519	20.645	0.794	0.839	1.518	1.495	4.646

Source: SCARP Monitoring, WAPDA

TABLE–65

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1988)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.236	0.000	0.012	0.248
Kandhkot (Pilot)	0.013	0.012	0.000	0.010	0.000	0.000	0.010
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.246	0.000	0.099	0.345
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.023	0.023
North Dadu	0.497	0.430	0.000	0.148	0.000	0.363	0.511
Kotri Surface Drainage	1.962	1.690	0.000	0.124	0.000	1.623	1.747
Sub–Total (II):	2.548	2.207	0.000	0.272	0.000	2.009	2.281
III. Remaining Area	1.762	1.700	0.000	0.000	0.000	0.309	0.309
Gross Total (I TO III) :	4.697	4.235	0.000	0.518	0.000	2.416	2.935

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–66

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	6.891	6.122	0.136	0.407	0.028	0.120	0.691
NWFP	0.226	0.198	0.034	0.000	0.000	0.000	0.034
Sindh/Balochistan	2.080	1.782	0.272	0.023	0.128	0.050	0.473
i. Non–Rice	1.693	1.454	0.272	0.000	0.128	0.000	0.400
ii. Rice	0.387	0.328	0.000	0.023	0.000	0.050	0.073
Total:	9.197	8.102	0.442	0.431	0.156	0.170	1.198
i. Non–Rice	8.810	7.774	0.442	0.408	0.156	0.120	1.125
ii. Rice	0.387	0.328	0.000	0.023	0.000	0.050	0.073
II. On–Going SCARPs							
Punjab	1.844	1.595	0.076	0.076	0.069	0.002	0.223
NWFP	0.377	0.331	0.042	0.000	0.000	0.000	0.042
Sindh/Balochistan	6.228	5.253	0.303	0.078	0.898	0.791	2.070
i. Non–Rice	3.680	3.046	0.303	0.013	0.898	0.000	1.214
ii. Rice	2.548	2.207	0.000	0.065	0.000	0.791	0.856
Total:	8.449	7.179	0.421	0.154	0.967	0.794	2.335
i. Non–Rice	5.901	4.972	0.421	0.089	0.967	0.003	1.479
ii. Rice	2.548	2.207	0.000	0.065	0.000	0.791	0.856
III. Out Side SCARPs							
Punjab	15.885	12.552	0.104	0.293	0.185	0.108	0.690
NWFP	0.782	0.510	0.034	0.000	0.000	0.000	0.034
Sindh/Balochistan	6.852	6.221	0.257	0.092	1.119	0.189	1.657
i. Non–Rice	5.090	4.521	0.257	0.004	1.119	0.106	1.486
ii. Rice	1.762	1.700	0.000	0.088	0.000	0.083	0.171
Total:	23.519	19.283	0.395	0.385	1.304	0.297	2.381
i. Non–Rice	21.757	17.583	0.395	0.297	1.304	0.213	2.210
ii. Rice	1.762	1.700	0.000	0.088	0.000	0.083	0.171
Gross Total (I TO III) :	41.165	34.564	1.257	0.969	2.427	1.261	5.914

Source: SCARP Monitoring, WAPDA

TABLE–67

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1989)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
	(a) Tubewell Drainage Project							
	1. SCARP I	1.217	1.141	0.016	0.001	0.000	0.000	0.017
	SCARP II							
	2. Non Saline	1.649	1.490	0.069	0.013	0.002	0.000	0.084
	3. Saline	0.690	0.550	0.001	0.036	0.007	0.000	0.044
	4. Shahpur	0.112	0.090	0.001	0.015	0.000	0.000	0.016
	SCARP III							
	5. Non Saline	1.138	0.950	0.000	0.272	0.000	0.023	0.295
	6. Saline	0.142	0.098	0.000	0.042	0.000	0.063	0.105
	SCARP IV							
	7. Muridke Mangtanwala	0.558	0.544	0.001	0.000	0.000	0.000	0.001
	SCARP V							
	8. Shorkot Kamalia (P)	0.169	0.154	0.007	0.000	0.000	0.000	0.007
	9. Sataina (P)	0.118	0.098	0.016	0.000	0.000	0.000	0.016
	10. Khairwala Unit	0.333	0.301	0.000	0.000	0.000	0.000	0.000
	SCARP VI							
	11. Allahabad	0.216	0.202	0.015	0.021	0.000	0.001	0.037
	SCARP VII							
	12. Pandoki (CBDC) Unit	0.196	0.174	0.000	0.000	0.000	0.000	0.000
	SCARP VIII							
	13. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
	14. Fordwah Sadiqia–II	0.111	0.100	0.000	0.001	0.012	0.034	0.047
	15. AWL TP Link	0.020	0.020	0.011	0.000	0.000	0.000	0.011
	16.7. AWL RQB Link	0.024	0.022	0.000	0.005	0.000	0.000	0.005
	Sub–Total (a):	6.771	6.007	0.136	0.407	0.022	0.120	0.685
	(b) Surface Drainage Projects							
	17. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
	18. AWL CJ Link	0.023	0.020	0.000	0.000	0.006	0.000	0.006
	Sub–Total (b):	0.120	0.115	0.000	0.000	0.006	0.000	0.006
	Total (I):	6.891	6.122	0.136	0.407	0.028	0.120	0.691
II.	**Sindh**							
	(a) Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.141	0.000	0.009	0.000	0.150
	2. North Rohri	0.793	0.686	0.045	0.000	0.000	0.000	0.045
	3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
	Sub–Total (a):	1.247	1.078	0.186	0.000	0.009	0.000	0.195
	(b) Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.077	0.023	0.103	0.000	0.203
	Sub–Total (b):	0.701	0.594	0.077	0.023	0.103	0.000	0.203
	(c) Tile Drainage Project							
	5. East Khairpur	0.045	0.036	0.009	0.000	0.016	0.000	0.025
	Sub–Total (c):	0.045	0.036	0.009	0.000	0.016	0.000	0.025
	Total (II):	1.993	1.708	0.272	0.023	0.128	0.000	0.423
III.	**Balochistan**							
	(a) Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.050	0.050
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.050	0.050
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.050	0.050
	Total (II+III):	2.080	1.782	0.272	0.023	0.128	0.050	0.473
IV.	**N.W.F.P.**							
	(a) Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.028	0.000	0.000	0.000	0.028
	2. Bannu	0.090	0.080	0.006	0.000	0.000	0.000	0.006
	Sub–Total (a):	0.226	0.198	0.034	0.000	0.000	0.000	0.034
	Total (IV):	0.226	0.198	0.034	0.000	0.000	0.000	0.034
	Gross Total (I to IV) :	9.197	8.102	0.442	0.431	0.156	0.170	1.198

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–68

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
	GA	CCA	P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Gojra Khewra Phase–I	0.038	0.030	0.001	0.000	0.000	0.000	0.001
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.044	0.074	0.067	0.002	0.187
SCARP VII							
3. Hadali Sub Unit	0.164	0.137	0.013	0.000	0.002	0.000	0.015
4. AWL TSMB Link	0.052	0.052	0.013	0.002	0.000	0.000	0.015
Sub–Total (a):	1.714	1.486	0.071	0.076	0.069	0.002	0.218
(b) Tile Drainage Projects							
SCARP V							
5. Drainage IV 1/	0.130	0.109	0.005	0.000	0.000	0.000	0.005
Sub–Total (b):	0.130	0.109	0.0050	0.000	0.000	0.000	0.0050
Total (I):	1.844	1.595	0.076	0.076	0.069	0.002	0.223
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.013	0.000	0.000	0.013
2. South Rohri (FGW)	0.541	0.375	0.005	0.000	0.000	0.000	0.005
Sub–Total (a):	0.982	0.775	0.005	0.013	0.000	0.000	0.018
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.002	0.005	0.000	0.039	0.046
4. LBOD Stage I	1.426	1.275	0.292	0.000	0.658	0.000	0.950
5. Kotri Surface Drainage	3.216	2.671	0.004	0.061	0.240	0.733	1.037
Sub–Total (b):	5.156	4.391	0.298	0.065	0.898	0.772	2.033
Total (II):	6.138	5.166	0.303	0.078	0.898	0.772	2.051
III. Balochistan							
(a) Surface Drainage Project							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.019	0.019
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.019	0.019
Total (III):	0.090	0.087	0.000	0.000	0.000	0.019	0.019
Total (II+III):	6.228	5.253	0.303	0.078	0.898	0.791	2.070
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.021	0.000	0.000	0.000	0.021
2. Chashma Command Area Development	0.157	0.151	0.021	0.000	0.000	0.000	0.021
Sub–Total (a):	0.377	0.331	0.042	0.000	0.000	0.000	0.042
Total (IV):	0.377	0.331	0.042	0.000	0.000	0.000	0.042
Gross Total (I to IV):	8.449	7.179	0.421	0.154	0.967	0.794	2.335

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–69

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	15.885	12.552	0.104	0.293	0.185	0.108	0.690
II. Sindh/Balochistan							
Gudu R. Bank	1.855		0.000	0.092	0.000	0.096	0.188
Gudu L. Bank	0.519		0.000	0.000	0.000	0.093	0.093
Sukkur R. Bank	1.373		0.141	0.000	0.697	0.000	0.838
Sukkur L. Bank	3.105		0.116	0.000	0.422	0.000	0.538
Sub–Total (II):	6.852	6.221	0.257	0.092	1.119	0.189	1.657
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.034	0.000	0.000	0.000	0.034
Gross Total (I TO III) :	23.519	19.283	0.395	0.385	1.304	0.297	2.381

Source: SCARP Monitoring, WAPDA

TABLE–70

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.023	0.000	0.000	0.023
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.050	0.050
Sub–Total (I):	0.387	0.328	0.000	0.023	0.000	0.050	0.073
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.019	0.019
North Dadu	0.497	0.430	0.000	0.005	0.000	0.039	0.044
Kotri Surface Drainage	1.962	1.690	0.000	0.060	0.000	0.733	0.793
Sub–Total (II):	2.548	2.207	0.000	0.065	0.000	0.791	0.856
III. Remaining Area	1.762	1.700	0.000	0.088	0.000	0.083	0.171
Gross Total (I TO III) :	4.697	4.235	0.000	0.176	0.000	0.924	1.100

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–71

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (October 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.891	6.122	0.145	0.793	0.043	0.155	1.137
NWFP	0.226	0.198	0.044	0.000	0.000	0.000	0.044
Sindh/Balochistan	2.080	1.782	0.648	0.212	0.269	0.107	1.235
i. Non–Rice	1.693	1.454	0.648	0.005	0.269	0.001	1.162
ii. Rice	0.387	0.328	0.000	0.207	0.000	0.106	0.073
Total:	9.197	8.102	0.837	1.005	0.313	0.262	2.416
i. Non–Rice	8.810	7.774	0.837	0.798	0.313	0.156	2.103
ii. Rice	0.387	0.328	0.000	0.207	0.000	0.106	0.313
II. On–Going SCARPs							
Punjab	1.844	1.595	0.081	0.062	0.079	0.002	0.224
NWFP	0.377	0.331	0.089	0.000	0.000	0.000	0.089
Sindh/Balochistan	6.228	5.253	0.417	0.589	1.406	2.061	4.474
i. Non–Rice	3.680	3.046	0.417	0.243	1.406	0.080	2.146
ii. Rice	2.548	2.207	0.000	0.346	0.000	1.982	2.328
Total:	8.449	7.179	0.587	0.650	1.486	2.064	4.787
i. Non–Rice	5.901	4.972	0.587	0.304	1.486	0.082	2.459
ii. Rice	2.548	2.207	0.000	0.346	0.000	1.982	2.328
III. Out Side SCARPs							
Punjab	15.885	12.552	0.326	0.650	0.252	0.065	1.293
NWFP	0.782	0.510	0.034	0.000	0.000	0.000	0.034
Sindh/Balochistan	6.852	6.221	0.591	0.998	1.343	0.705	3.637
i. Non–Rice	5.090	4.521	0.591	0.102	1.343	0.315	2.352
ii. Rice	1.762	1.700	0.000	0.896	0.000	0.389	1.285
Total:	23.519	19.283	0.951	1.648	1.595	0.770	4.964
i. Non–Rice	21.757	17.583	0.951	0.752	1.595	0.380	3.679
ii. Rice	1.762	1.700	0.000	0.896	0.000	0.389	1.285
Gross Total (I TO III) :	41.165	34.564	2.375	3.303	3.394	3.095	12.167

Source: SCARP Monitoring, WAPDA

TABLE–72

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1989)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	1. SCARP I SCARP II	1.217	1.141	0.032	0.008	0.000	0.000	0.040
	2. Non Saline	1.649	1.490	0.053	0.028	0.002	0.000	0.083
	3. Saline	0.690	0.550	0.010	0.089	0.015	0.000	0.114
	4. Shahpur SCARP III	0.112	0.090	0.003	0.056	0.000	0.000	0.059
	5. Non Saline	1.138	0.950	0.000	0.450	0.000	0.019	0.469
	6. Saline SCARP IV	0.142	0.098	0.000	0.060	0.000	0.061	0.121
	7. Muridke Mangtanwala SCARP V	0.558	0.544	0.001	0.005	0.000	0.000	0.006
	8. Shorkot Kamalia (P)	0.169	0.154	0.002	0.000	0.000	0.000	0.002
	9. Sataina (P)	0.118	0.098	0.015	0.000	0.000	0.000	0.015
	10. Khairwala Unit SCARP VI	0.333	0.301	0.000	0.000	0.000	0.000	0.000
	11. Allahabad SCARP VII	0.216	0.202	0.014	0.027	0.000	0.001	0.042
	12. Pandoki (CBDC) Unit SCARP VIII	0.196	0.174	0.006	0.000	0.000	0.000	0.006
	13. Minchinabad (P)	0.078	0.073	0.000	0.035	0.000	0.000	0.035
	14. Fordwah Sadiqia–II	0.111	0.100	0.000	0.022	0.015	0.074	0.111
	15. AWL TP Link	0.020	0.020	0.010	0.000	0.000	0.000	0.010
	16. AWL RQB Link	0.024	0.022	0.000	0.013	0.000	0.000	0.013
	Sub–Total (a):	6.771	6.007	0.145	0.793	0.032	0.155	1.126
(b)	Surface Drainage Projects							
	17. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
	18. AWL CJ Link	0.023	0.020	0.000	0.000	0.011	0.000	0.011
	Sub–Total (b):	0.120	0.115	0.000	0.000	0.011	0.000	0.011
	Total (I):	6.891	6.122	0.145	0.793	0.043	0.155	1.137
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.203	0.000	0.053	0.000	0.256
	2. North Rohri	0.793	0.686	0.163	0.000	0.000	0.000	0.163
	3. Kandhkot (P)	0.013	0.012	0.000	0.010	0.000	0.000	0.010
	Sub–Total (a):	1.247	1.078	0.366	0.010	0.053	0.000	0.429
(b)	Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.273	0.202	0.183	0.020	0.677
	Sub–Total (b):	0.701	0.594	0.273	0.202	0.183	0.020	0.677
(c)	Tile Drainage Project							
	5. East Khairpur	0.045	0.036	0.009	0.000	0.033	0.000	0.042
	Sub–Total (c):	0.045	0.036	0.009	0.000	0.033	0.000	0.042
	Total (II):	1.993	1.708	0.648	0.212	0.269	0.020	1.148
III.	**Balochistan**							
(a)	Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (II+III):	2.080	1.782	0.648	0.212	0.269	0.107	1.235
IV.	**N.W.F.P.**							
(a)	Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.037	0.000	0.000	0.000	0.037
	2. Bannu	0.090	0.080	0.007	0.000	0.000	0.000	0.007
	Sub–Total (a):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
	Total (IV):	0.226	0.198	0.044	0.000	0.000	0.000	0.044
	Gross Total (I to IV) :	9.197	8.102	0.837	1.005	0.313	0.262	2.416

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–73

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Gojra Khewra Phase–I	0.038	0.030	0.001	0.000	0.000	0.000	0.001
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.012	0.057	0.040	0.002	0.111
SCARP VII							
3. Hadali Sub Unit	0.164	0.137	0.037	0.000	0.037	0.000	0.074
4. AWL TSMB Link	0.052	0.052	0.025	0.005	0.002	0.000	0.032
Sub–Total (a):	1.714	1.486	0.075	0.062	0.079	0.002	0.218
(b) Tile Drainage Projects							
SCARP V							
5. Drainage IV 1/	0.130	0.109	0.006	0.000	0.000	0.000	0.006
Sub–Total (b):	0.130	0.109	0.0060	0.000	0.000	0.000	0.0060
Total (I):	1.844	1.595	0.081	0.062	0.079	0.002	0.224
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.237	0.000	0.010	0.247
2. South Rohri (FGW)	0.541	0.375	0.029	0.000	0.000	0.000	0.029
Sub–Total (a):	0.982	0.775	0.029	0.237	0.000	0.010	0.276
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.000	0.211	0.017	0.285	0.514
4. LBOD Stage I	1.426	1.275	0.342	0.000	0.541	0.000	0.883
5. Kotri Surface Drainage	3.216	2.671	0.046	0.140	0.848	1.722	2.757
Sub–Total (b):	5.156	4.391	0.388	0.352	1.406	2.008	4.154
Total (II):	6.138	5.166	0.417	0.589	1.406	2.017	4.430
III. Balochistan							
(a) Surface Drainage Project							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.044	0.044
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.044	0.044
Total (III):	0.090	0.087	0.000	0.000	0.000	0.044	0.044
Total (II+III):	6.228	5.253	0.417	0.589	1.406	2.061	4.474
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.027	0.000	0.000	0.000	0.027
2. Chashma Command Area Development	0.157	0.151	0.062	0.000	0.000	0.000	0.062
Sub–Total (a):	0.377	0.331	0.089	0.000	0.000	0.000	0.089
Total (IV):	0.377	0.331	0.089	0.000	0.000	0.000	0.089
Gross Total (I to IV):	8.449	7.179	0.587	0.650	1.486	2.064	4.787

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–I, Sataina and Paharang Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–74

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	15.885	12.552	0.326	0.650	0.252	0.065	1.293
II. Sindh/Balochistan							
Gudu R. Bank	1.855		0.000	0.922	0.000	0.330	1.252
Gudu L. Bank	0.519		0.000	0.041	0.000	0.366	0.407
Sukkur R. Bank	1.373		0.335	0.035	0.773	0.009	1.152
Sukkur L. Bank	3.105		0.256	0.000	0.570	0.000	0.826
Sub–Total (II):	6.852	6.221	0.591	0.998	1.343	0.705	3.637
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.034	0.000	0.000	0.000	0.034
Gross Total (I TO III) :	23.519	19.283	0.951	1.648	1.595	0.770	4.964

Source: SCARP Monitoring, WAPDA

TABLE–75

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1989)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.227	0.000	0.021	0.248
Kandhkot (Pilot)	0.013	0.012	0.000	0.010	0.000	0.000	0.010
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.237	0.000	0.108	0.345
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.044	0.044
North Dadu	0.497	0.430	0.000	0.212	0.000	0.285	0.497
Kotri Surface Drainage	1.962	1.690	0.000	0.135	0.000	1.652	1.787
Sub–Total (II):	2.548	2.207	0.000	0.346	0.000	1.982	2.328
III. Remaining Area	1.762	1.700	0.000	0.896	0.000	0.389	1.285
Gross Total (I TO III) :	4.697	4.235	0.000	1.479	0.000	2.479	3.958

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–76

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed SCARPs							
Punjab	6.981	6.204	0.195	0.410	0.023	0.171	0.799
NWFP	0.226	0.198	0.035	0.000	0.000	0.000	0.035
Sindh/Balochistan	2.080	1.782	0.109	0.004	0.176	0.074	0.363
i. Non–Rice	1.693	1.454	0.109	0.000	0.176	0.000	0.285
ii. Rice	0.387	0.328	0.000	0.004	0.000	0.074	0.078
Total:	9.287	8.184	0.339	0.414	0.199	0.245	1.197
i. Non–Rice	8.900	7.856	0.339	0.410	0.199	0.171	1.119
ii. Rice	0.387	0.328	0.000	0.004	0.000	0.074	0.078
II. On–Going SCARPs							
Punjab	1.754	1.513	0.035	0.045	0.060	0.000	0.140
NWFP	0.377	0.331	0.045	0.000	0.000	0.000	0.045
Sindh/Balochistan	6.228	5.253	0.393	0.091	0.701	0.500	1.686
i. Non–Rice	3.680	3.046	0.393	0.007	0.701	0.000	1.102
ii. Rice	2.548	2.207	0.000	0.084	0.000	0.500	0.584
Total:	8.359	7.097	0.473	0.136	0.761	0.500	1.871
i. Non–Rice	5.811	4.890	0.473	0.052	0.761	0.000	1.287
ii. Rice	2.548	2.207	0.000	0.084	0.000	0.500	0.584
III. Out Side SCARPs							
Punjab	15.885	12.552	0.123	0.392	0.259	0.124	0.898
NWFP	0.782	0.510	0.037	0.000	0.000	0.000	0.037
Sindh/Balochistan	6.852	6.221	0.242	0.139	0.717	0.321	1.419
i. Non–Rice	5.090	4.521	0.242	0.031	0.717	0.049	1.039
ii. Rice	1.762	1.700	0.000	0.108	0.000	0.272	0.380
Total:	23.519	19.283	0.403	0.530	0.976	0.445	2.354
i. Non–Rice	21.757	17.583	0.403	0.423	0.976	0.173	1.974
ii. Rice	1.762	1.700	0.000	0.108	0.000	0.272	0.380
Gross Total (I TO III) :	41.165	34.564	1.215	1.081	1.935	1.190	5.421

Source: SCARP Monitoring, WAPDA

TABLE–77

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1990)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
(a)	Tubewell Drainage Project							
	1. SCARP I SCARP II	1.217	1.141	0.003	0.001	0.000	0.000	0.004
	2. Non Saline	1.649	1.490	0.063	0.014	0.002	0.000	0.079
	3. Saline	0.690	0.550	0.006	0.031	0.007	0.000	0.044
	4. Shahpur SCARP III	0.112	0.090	0.002	0.026	0.000	0.000	0.028
	5. Non Saline	1.138	0.950	0.000	0.275	0.000	0.019	0.294
	6. Saline	0.142	0.098	0.000	0.036	0.000	0.073	0.109
	SCARP IV							
	7. Muridke Mangtanwala SCARP V	0.558	0.544	0.000	0.000	0.000	0.000	0.000
	8. Shorkot Kamalia (P)	0.169	0.154	0.007	0.000	0.000	0.000	0.007
	9. Sataina (P)	0.118	0.098	0.003	0.000	0.000	0.000	0.003
	10. Khairwala Unit	0.333	0.301	0.000	0.000	0.000	0.000	0.000
	11. Gojra & Khewra Phase–I SCARP VI	0.038	0.030	0.001	0.000	0.000	0.000	0.001
	12. Allahabad SCARP VII	0.216	0.202	0.009	0.021	0.000	0.003	0.033
	13. Pandoki (CBDC) Unit SCARP VIII	0.196	0.174	0.000	0.000	0.000	0.000	0.000
	14. Minchinabad (P)	0.078	0.073	0.000	0.000	0.000	0.000	0.000
	15. Fordwah Sadiqia–II	0.111	0.100	0.000	0.005	0.009	0.076	0.090
	16. AWL TP Link	0.020	0.020	0.082	0.000	0.000	0.000	0.082
	17. AWL RQB Link	0.024	0.022	0.000	0.002	0.000	0.000	0.002
	18. AWL T.S.M.B Link	0.052	0.052	0.020	0.000	0.000	0.000	0.000
	Sub–Total (a):	6.861	6.089	0.195	0.410	0.018	0.171	0.775
(b)	Surface Drainage Projects							
	19. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
	20. AWL CJ Link	0.023	0.020	0.000	0.000	0.004	0.000	0.004
	Sub–Total (b):	0.120	0.115	0.000	0.000	0.004	0.000	0.004
	Total (I):	6.981	6.204	0.195	0.410	0.023	0.171	0.779
II.	**Sindh**							
(a)	Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.019	0.000	0.074	0.000	0.093
	2. North Rohri	0.793	0.686	0.067	0.000	0.000	0.000	0.067
	3. Kandhkot (P)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
	Sub–Total (a):	1.247	1.078	0.086	0.000	0.074	0.000	0.160
(b)	Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.021	0.004	0.074	0.000	0.099
	Sub–Total (b):	0.701	0.594	0.021	0.004	0.074	0.000	0.099
(c)	Tile Drainage Project							
	1. East Khairpur	0.045	0.036	0.003	0.000	0.027	0.000	0.030
	Sub–Total (c):	0.045	0.036	0.003	0.000	0.027	0.000	0.030
	Total (II):	1.993	1.708	0.109	0.004	0.176	0.000	0.289
III.	**Balochistan**							
(a)	Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.074	0.074
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.074	0.074
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.074	0.074
	Total (II+III):	2.080	1.782	0.109	0.004	0.176	0.074	0.363
IV.	**N.W.F.P.**							
(a)	Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.028	0.000	0.000	0.000	0.028
	2. Bannu	0.090	0.080	0.007	0.000	0.000	0.000	0.007
	Sub–Total (a):	0.226	0.198	0.035	0.000	0.000	0.000	0.035
	Total (IV):	0.226	0.198	0.035	0.000	0.000	0.000	0.035
	Gross Total (I to IV) :	11.280	9.892	0.448	0.418	0.375	0.245	1.466

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–78

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Punjab							
(a) Tubewell Drainage Project							
SCARP VI							
1. Panjnad Abbasia (Rem.)	1.460	1.267	0.016	0.045	0.057	0.000	0.119
SCARP VII							
2. Hadali Sub Unit	0.164	0.137	0.016	0.000	0.002	0.000	0.019
Sub–Total (a):	1.624	1.404	0.032	0.045	0.060	0.000	0.137
(b) Tile Drainage Projects							
SCARP V							
5. Drainage IV 1/	0.130	0.109	0.0025	0.000	0.000	0.000	0.0025
Sub–Total (b):	0.130	0.109	0.0025	0.000	0.000	0.000	0.0025
Total (I):	1.754	1.513	0.035	0.045	0.060	0.000	0.140
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.007	0.000	0.000	0.007
2. South Rohri (FGW)	0.541	0.375	0.016	0.000	0.000	0.000	0.016
Sub–Total (a):	0.982	0.775	0.016	0.007	0.000	0.000	0.023
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.001	0.005	0.000	0.032	0.038
4. LBOD Stage I	1.426	1.275	0.371	0.000	0.485	0.000	0.856
5. Kotri Surface Drainage	3.216	2.671	0.005	0.080	0.216	0.445	0.746
Sub–Total (b):	5.156	4.391	0.377	0.085	0.701	0.477	1.640
Total (II):	6.138	5.166	0.393	0.092	0.701	0.477	1.663
III. Balochistan							
(a) Surface Drainage Project							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Total (III):	0.090	0.087	0.000	0.000	0.000	0.023	0.023
Total (II+III):	6.228	5.253	0.393	0.092	0.701	0.500	1.686
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.017	0.000	0.000	0.000	0.017
2. Chashma Command Area Development	0.157	0.151	0.028	0.000	0.000	0.000	0.028
Sub–Total (a):	0.377	0.331	0.045	0.000	0.000	0.000	0.045
Total (IV):	0.377	0.331	0.045	0.000	0.000	0.000	0.045
Gross Total (I to IV):	8.359	7.097	0.473	0.137	0.761	0.500	1.871

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and .269 MA respectively, which is partially overlapped by SCARP–1, Sataina and Paharang drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–79

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (April/June 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	15.885	12.552	0.123	0.392	0.259	0.124	0.898
II. Sindh/Balochistan							
Gudu R. Bank	1.840		0.000	0.110	0.000	0.269	0.379
Gudu L. Bank	0.519		0.000	0.029	0.000	0.052	0.081
Sukkur R. Bank	1.373		0.055	0.000	0.610	0.000	0.665
Sukkur L. Bank	3.105		0.187	0.000	0.107	0.000	0.294
Sub–Total (II):	6.852	6.221	0.242	0.139	0.717	0.321	1.419
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.037	0.000	0.000	0.000	0.037
Gross Total (I TO III) :	23.519	19.283	0.403	0.530	0.976	0.445	2.354

Source: SCARP Monitoring, WAPDA

TABLE–80

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (April/June 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.034	0.000	0.000	0.034
Kandhkot (Pilot)	0.013	0.012	0.000	0.000	0.000	0.000	0.000
Hairdin	0.087	0.074	0.000	0.000	0.000	0.074	0.074
Sub–Total (I):	0.387	0.328	0.000	0.034	0.000	0.074	0.108
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.023	0.023
North Dadu	0.497	0.430	0.000	0.005	0.000	0.032	0.037
Kotri Surface Drainage	1.962	1.690	0.000	0.080	0.000	0.444	0.524
Sub–Total (II):	2.548	2.207	0.000	0.084	0.000	0.500	0.584
III. Remaining Area	1.762	1.700	0.000	0.108	0.000	0.272	0.380
Gross Total (I TO III) :	4.697	4.235	0.000	0.226	0.000	0.846	1.072

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

TABLE–81

Detail of Gross Area in Completed SCARPs, On–going SCARPs and in the Canal Commands Outside SCARPs Having Groundwater Table Ranging from 0–5 feet Depth
(October 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
	GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
			P	NP	P	NP	
I. Completed SCARPs							
Punjab	6.981	6.204	0.328	0.775	0.043	0.179	1.326
NWFP	0.226	0.198	0.046	0.000	0.000	0.000	0.046
Sindh/Balochistan	2.080	1.782	0.654	0.235	0.267	0.087	1.244
i. Non–Rice	1.693	1.454	0.654	–0.062	0.267	0.000	0.860
ii. Rice	0.387	0.328	0.000	0.297	0.000	0.087	0.384
Total:	9.287	8.184	1.029	1.010	0.311	0.266	2.616
i. Non–Rice	8.900	7.856	1.029	0.713	0.311	0.179	2.232
ii. Rice	0.387	0.328	0.000	0.297	0.000	0.087	0.384
II. On–Going SCARPs							
Punjab	3.745	3.290	0.121	0.089	0.182	0.003	0.395
NWFP	0.377	0.331	0.069	0.000	0.000	0.000	0.069
Sindh/Balochistan	6.228	5.253	0.423	0.579	1.474	1.953	4.430
i. Non–Rice	3.680	3.046	0.423	0.232	1.474	0.010	2.139
ii. Rice	2.548	2.207	0.000	0.347	0.000	1.944	2.291
Total:	10.350	8.874	0.613	0.668	1.656	1.956	4.893
i. Non–Rice	7.802	6.667	0.613	0.321	1.656	0.012	2.602
ii. Rice	2.548	2.207	0.000	0.347	0.000	1.944	2.291
III. Out Side SCARPs							
Punjab	13.894	10.775	0.057	0.437	0.054	0.143	0.691
NWFP	0.782	0.510	0.059	0.000	0.000	0.000	0.059
Sindh/Balochistan	6.852	6.221	0.657	0.988	1.435	0.666	3.746
i. Non–Rice	5.090	4.521	0.657	0.159	1.435	0.271	2.523
ii. Rice	1.762	1.700	0.000	0.828	0.000	0.395	1.223
Total:	21.528	17.506	0.773	1.425	1.489	0.809	4.496
i. Non–Rice	19.766	15.806	0.773	0.596	1.489	0.414	3.273
ii. Rice	1.762	1.700	0.000	0.828	0.000	0.395	1.223
Gross Total (I TO III) :	41.165	34.564	2.416	3.103	3.456	3.031	12.005

Source: SCARP Monitoring, WAPDA

TABLE–82

Detail of Gross Area of Completed SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1990)

(Million Acres)

	SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
		GA	CCA	Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5 feet
				P	NP	P	NP	
I.	**Punjab**							
	(a) Tubewell Drainage Project							
	1. SCARP I SCARP II	1.217	1.141	0.014	0.006	0.000	0.000	0.020
	2. Non Saline	1.649	1.490	0.093	0.037	0.009	0.000	0.139
	3. Saline	0.690	0.550	0.014	0.080	0.013	0.000	0.107
	4. Shahpur SCARP III	0.112	0.090	0.000	0.068	0.000	0.000	0.068
	5. Non Saline	1.138	0.950	0.000	0.459	0.000	0.022	0.481
	6. Saline SCARP IV	0.142	0.098	0.000	0.053	0.000	0.071	0.124
	7. Muridke Mangtanwala SCARP V	0.558	0.544	0.001	0.012	0.000	0.000	0.013
	8. Shorkot Kamalia (P)	0.169	0.154	0.057	0.000	0.000	0.000	0.057
	9. Satuana (P)	0.118	0.098	0.015	0.000	0.000	0.000	0.015
	10. Khairwala Unit	0.333	0.301	0.000	0.000	0.000	0.000	0.000
	11. Gojra & Khewra Phase–I SCARP VI	0.038	0.030	0.001	0.000	0.000	0.000	0.001
	12. Allahabad (P) SCARP VII	0.216	0.202	0.013	0.022	0.000	0.001	0.036
	13. Pandoki (CBDC) Unit SCARP VIII	0.196	0.174	0.012	0.000	0.000	0.000	0.012
	14. Minchinabad (P)	0.078	0.073	0.000	0.014	0.000	0.000	0.014
	15. Fordwah Sadiqia–II	0.111	0.100	0.000	0.012	0.014	0.085	0.111
	16. AWL TP Link	0.020	0.020	0.094	0.000	0.000	0.000	0.094
	17. AWL RQB Link	0.024	0.022	0.000	0.012	0.000	0.000	0.012
	18. AWL T.S.M.B Link	0.052	0.052	0.015	0.000	0.001	0.000	0.016
	Sub–Total (a):	6.861	6.089	0.328	0.775	0.037	0.179	1.319
	(b) Surface Drainage Projects							
	19. Paharang Drain (Out Side Scarp–I)	0.097	0.095	0.000	0.000	0.000	0.000	0.000
	20. AWL CJ Link	0.023	0.020	0.000	0.000	0.006	0.000	0.006
	Sub–Total (b):	0.120	0.115	0.000	0.000	0.006	0.000	0.006
	Total (I):	6.981	6.204	0.328	0.775	0.043	0.179	1.326
II.	**Sindh**							
	(a) Tubewell Drainage Project							
	1. Khairpur	0.441	0.380	0.210	0.000	0.055	0.000	0.265
	2. North Rohri	0.793	0.686	0.148	0.000	0.000	0.000	0.148
	3. Kandhkot (P)	0.013	0.012	0.000	0.010	0.000	0.000	0.010
	Sub–Total (a):	1.247	1.078	0.358	0.010	0.055	0.000	0.423
	(b) Surface Drainage Project							
	4. Larkana Shikarpur 1/ (4 Schemes)	0.701	0.594	0.289	0.225	0.184	0.000	0.698
	Sub–Total (b):	0.701	0.594	0.289	0.225	0.184	0.000	0.698
	(c) Tile Drainage Project							
	1. East Khairpur	0.045	0.036	0.007	0.000	0.029	0.000	0.036
	Sub–Total (c):	0.045	0.036	0.007	0.000	0.029	0.000	0.036
	Total (II):	1.993	1.708	0.654	0.235	0.267	0.000	1.157
III.	**Balochistan**							
	(a) Surface Drainage Projects							
	1. Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Sub–Total (a):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (III):	0.087	0.074	0.000	0.000	0.000	0.087	0.087
	Total (II+III):	2.080	1.782	0.654	0.235	0.267	0.087	1.244
IV.	**N.W.F.P.**							
	(a) Tubwell Drainage Project							
	1. Peshawar 2/ (4 Schemes)	0.136	0.118	0.037	0.000	0.000	0.000	0.037
	2. Bannu	0.090	0.080	0.009	0.000	0.000	0.000	0.009
	Sub–Total (a):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
	Total (IV):	0.226	0.198	0.046	0.000	0.000	0.000	0.046
	Gross Total (I to IV) :	9.287	8.184	1.029	1.010	0.311	0.266	2.616

Source: SCARP Monitoring, WAPDA

1/ Includes: (a) Sukkur (b) Larkana (c) Shikarpur Pilot Projects and (d) Sukkur Right Bank FGW Project.

2/ Includes (a) Pabbi I & II (b) Jue Sheikh (c) Kafur Dheri and (d) AWL along Peshawar City.

TABLE–83

Detail of Gross Area of On–going SCARPs in Various Provinces, Having Groundwater Table Ranging from 0–5 feet Depth (October 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
(a) Tubewell Drainage Project							
SCARP V							
1. Gojra Khewra Phase–II	0.399	0.320	0.000	0.000	0.000	0.000	0.000
SCARP VI							
2. Panjnad Abbasia (Rem.)	1.460	1.267	0.012	0.057	0.034	0.003	0.106
SCARP VII							
3. Hadali Sub Unit	0.164	0.137	0.010	0.000	0.004	0.000	0.014
4. Shorkot Kamalia Saline Zone	0.056	0.054	0.014	0.001	0.000	0.000	0.015
Sub–Total (a):	2.079	1.778	0.036	0.058	0.038	0.003	0.135
(b) Surface Drainage Project							
5. Sukh Beas below B.S Link Phase I	0.813	0.738	0.010	0.025	0.012	0.000	0.047
6. Fordwah Sadiqia Phase I	0.618	0.571	0.007	0.005	0.132	0.000	0.144
Sub–Total (b):	1.431	1.309	0.017	0.031	0.143	0.000	0.191
(c) Tile Drainage Projects							
SCARP V							
7. Drainage IV 1/	0.130	0.109	0.0044	0.000	0.000	0.000	0.0044
8. Khushab Sub Unit	0.105	0.094	0.064	0.000	0.000	0.000	0.0635
Sub–Total (c):	0.235	0.203	0.068	0.000	0.000	0.000	0.068
Total (I):	3.745	3.290	0.121	0.089	0.182	0.003	0.395
II. Sindh							
(a) Tubewell Drainage Project							
1. Ghotki (FGW)	0.441	0.400	0.000	0.232	0.000	0.010	0.242
South Rohri (FGW)	0.541	0.375	0.030	0.000	0.000	0.000	0.030
Sub–Total (a):	0.982	0.775	0.030	0.232	0.000	0.010	0.272
(b) Surface Drainage Project							
3. North Dadu	0.514	0.445	0.007	0.215	0.010	0.282	0.514
4. LBOD Stage I	1.426	1.275	0.337	0.000	0.533	0.000	0.870
5. Kotri Surface Drainage	3.216	2.671	0.050	0.132	0.931	1.616	2.728
Sub–Total (b):	5.156	4.391	0.393	0.347	1.474	1.898	4.112
Total (II):	6.138	5.166	0.423	0.579	1.474	1.907	4.384
III. Balochistan							
(a) Surface Drainage Project							
1. Hairdin S. Drainage II	0.090	0.087	0.000	0.000	0.000	0.046	0.046
Sub–Total (a):	0.090	0.087	0.000	0.000	0.000	0.046	0.046
Total (III):	0.090	0.087	0.000	0.000	0.000	0.046	0.046
Total (II+III):	6.228	5.253	0.42314	0.57902	1.47412	1.95347	4.42976
IV. N.W.F.P.							
(a) Tile Drainage Project							
1. Mardan	0.220	0.180	0.020	0.000	0.000	0.000	0.020
2. Chashma Command Area Development	0.157	0.151	0.049	0.000	0.000	0.000	0.049
Sub–Total (a):	0.377	0.331	0.069	0.000	0.000	0.000	0.069
Total (IV):	0.377	0.331	0.069	0.000	0.000	0.000	0.069
Gross Total (I to IV):	10.350	8.787	0.613	0.668	1.656	1.910	4.847

Source: SCARP Monitoring, WAPDA

1/ GA and CCA of Drainage–IV project is 0.354 MA and 0.269 MA respectively, which is partially overlapped by SCARP–1, Satiana and Paharange Drainage Projects. The remaining GA and CCA is 0.130 MA and 0.109 MA respectively.

TABLE–84

Detail of Gross Area of Canal Commands Outside SCAPRs, Having Groundwater Table Ranging from 0–5 feet Depth (October 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Punjab							
Sub–Total (I):	13.894	10.775	0.057	0.437	0.054	0.143	0.691
II. Sindh/Balochistan							
Gudu R. Bank	1.855		0.000	0.903	0.000	0.288	1.191
Gudu L. Bank	0.519		0.000	0.049	0.000	0.370	0.419
Sukkur R. Bank	1.373		0.333	0.036	0.798	0.008	1.174
Sukkur L. Bank	3.105		0.325	0.000	0.637	0.000	0.962
Sub–Total (II):	6.852	6.221	0.657	0.988	1.435	0.666	3.746
III. N.W.F.P.							
Sub–Total (III):	0.782	0.510	0.059	0.000	0.000	0.000	0.059
Gross Total (I TO III) :	21.528	17.506	0.773	1.425	1.489	0.809	4.496

Source: SCARP Monitoring, WAPDA

TABLE–85

SCARP–wise Detail of Gross Area in Rice Canal Area* Having Depth to Watertable Ranging from 0–5 feet Depth (October 1990)

(Million Acres)

SCARPS	Total Area		G.A. with Depth to Watertable Less than 5 ft.				
			Non–Saline < 3000 PPM		Saline > 3000 PPM		Total < 5
	GA	CCA	P	NP	P	NP	feet
I. Completed							
Larkana Shikarpur	0.287	0.242	0.000	0.287	0.000	0.000	0.287
Kandhkot (Pilot)	0.013	0.012	0.000	0.010	0.000	0.000	0.010
Hairdin	0.087	0.074	0.000	0.000	0.000	0.087	0.087
Sub–Total (I):	0.387	0.328	0.000	0.297	0.000	0.087	0.384
II. On–Going							
Hairdin Phase–II	0.089	0.087	0.000	0.000	0.000	0.046	0.046
North Dadu	0.497	0.430	0.000	0.215	0.000	0.282	0.497
Kotri Surface Drainage	1.962	1.690	0.000	0.132	0.000	1.616	1.748
Sub–Total (II):	2.548	2.207	0.000	0.347	0.000	1.944	2.291
III. Remaining Area	1.762	1.700	0.000	0.828	0.000	0.395	1.223
Gross Total (I TO III) :	4.697	4.235	0.000	1.473	0.000	2.425	3.898

Source: SCARP Monitoring, WAPDA

* Rice area mean commanded area served by Pat, Desert, Begari, Rice, Fuleli and Pinyari canals.

A. TOTAL WESTERN AND EASTERN RIVERS INFLOWS AT RIM STATION

B. SCARPS-WISE DEPTH TO WATERTABLE

Table–1
SCARP–I
(Pindi Bhattian, Chichokimallian, Chuharkana, Jaranwala, Shahkot, Shadman, Zafarwal, Hafiqabad, Khangah Dogran, Sangla Hill Beranwala, Harse Sheikh Units)
Areas in Different Depth to Water Table Ranges

Gross Area = 1.217 MA CCA = 1.141 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1961 (Pre–Project)	0.165	13.5	0.745	61.2	0.307	25.3	8.20
2	June 1962	0.000	0.0	0.000	0.0	0.000	0.0	9.40
3	June 1963	0.000	0.0	0.000	0.0	0.000	0.0	11.47
4	June 1964	0.000	0.0	0.000	0.0	0.000	0.0	14.25
5	June 1965	0.000	0.0	0.291	23.9	0.926	76.1	12.62
6	June 1966	0.000	0.0	0.088	7.2	1.129	92.8	15.06
7	June 1967	0.000	0.0	0.185	15.2	1.032	84.8	14.36
8	June 1968	0.000	0.0	0.185	15.2	1.032	84.8	14.36
9	June 1969	0.000	0.0	0.161	13.2	1.056	86.8	14.65
10	June 1970	0.000	0.0	0.152	12.5	1.065	87.5	14.83
11	June 1971	0.000	0.0	0.099	8.1	1.118	91.9	16.66
12	June 1972	0.000	0.0	0.101	8.3	1.116	91.7	16.54
13	June 1973	0.005	0.4	0.112	9.2	1.100	90.4	16.27
14	June 1974	0.001	0.1	0.196	16.1	1.120	83.8	14.97
15	June 1975	0.012	1.0	0.189	15.6	1.016	83.4	15.32
16	October 1975	0.043	3.5	0.270	22.2	0.094	74.3	0.00
17	June 1976	0.031	2.5	0.251	20.6	0.935	76.9	14.07
18	October 1976	0.168	13.8	0.542	44.5	0.507	41.7	
19	June 1977	0.027	2.2	0.370	30.4	0.820	67.4	12.50
20	October 1977	0.063	5.2	0.452	37.1	0.702	57.7	
21	June 1978	0.015	1.2	0.317	26.0	0.885	72.8	13.00
22	October 1978	0.162	13.3	0.452	37.1	0.603	49.6	
23	June 1979	0.039	3.2	0.456	37.5	0.722	59.3	11.60
24	October 1979	0.094	7.7	0.426	35.0	0.697	57.3	
25	June 1980	0.014	1.2	0.323	26.5	0.880	72.3	12.36
26	October 1980	0.168	13.8	0.547	44.9	0.502	41.3	9.81
27	June 1981	0.049	4.0	0.549	45.1	0.619	50.9	10.85
28	October 1981	0.115	9.4	0.535	44.0	0.567	46.6	10.24
29	June 1982	0.109	9.0	0.501	41.2	0.607	49.8	10.50
30	October 1982	0.080	6.6	0.472	38.8	0.665	54.6	11.21
31	June 1983	0.064	5.3	0.471	38.7	0.682	56.0	11.09
32	October 1983	0.125	10.2	0.492	40.4	0.601	49.4	10.60
33	June 1984	0.010	0.8	0.309	25.4	0.898	73.8	12.60
34	October 1984	0.124	10.2	0.497	40.8	0.596	40.0	10.60
35	June 1985	0.001	0.1	0.262	21.5	0.954	78.4	13.10
36	October 1985	0.044	3.6	0.484	39.8	0.689	72.8	12.70
37	June 1986	0.032	2.6	0.299	24.5	0.886	72.8	12.70
38	October 1986	0.047	3.8	0.371	30.5	0.799	65.7	12.10
39	June 1987	0.027	2.2	0.301	24.7	0.889	73.1	12.70
40	October 1987	0.012	1.0	0.297	24.4	0.908	74.6	13.10
41	June 1988	0.005	0.4	0.126	10.4	1.086	89.2	14.50
42	October 1988	0.047	0.4	0.350	29.3	0.814	66.9	
43	June 1989	0.017	1.4	0.215	17.7	0.985	80.9	
44	October 1989	0.040	3.3	0.294	24.2	0.883	72.5	
45	June 1990	0.003	0.3	0.225	18.5	0.989	81.25	

Source: SCARP Monitoring, WAPDA.

Table–2

SCARP–II (NON–SALINE)

(Phalia, Sohawa, Busal, Mona, Lalian, Khadir, Lower Hujjan, Kot Momin, Shah Jewna & Ara Units)

Areas in Different Depth to Water Table Ranges

Gross Area = 1.649 MA CCA = 1.490 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (FL)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1964 (Pre–Project)	0.181	11.0	0.066	40.0	0.808	49.0	
2	June 1975	0.013	0.8	0.505	30.6	1.131	68.6	
3	June 1976	0.124	7.5	0.739	44.8	0.786	47.7	
4	June 1977	0.153	9.3	0.676	41.0	0.820	49.7	
5	June 1978	0.146	8.9	0.638	38.7	0.865	52.4	10.80
6	October 1978	0.384	23.3	0.657	39.8	0.608	36.9	9.00
7	June 1979	0.245	14.9	0.746	45.2	0.658	39.9	9.40
8	October 1979	0.369	22.4	0.638	38.7	0.642	38.9	9.30
9	June 1980	0.108	6.6	0.680	41.2	0.861	52.2	10.80
10	October 1980	0.369	22.4	0.542	32.9	0.738	44.7	8.60
11	June 1981	0.113	6.9	0.750	45.4	0.786	47.7	10.60
12	October 1981	0.355	21.5	0.552	33.5	0.742	45.0	9.20
13	June 1982	0.269	16.3	0.689	41.8	0.691	41.9	9.70
14	October 1982	0.276	16.7	0.698	42.4	0.675	40.9	9.70
15	June 1983	0.247	15.0	0.768	46.6	0.634	38.4	9.50
16	October 1983	0.484	29.4	0.620	37.6	0.545	33.0	8.60
17	June 1984	0.096	5.8	0.771	46.8	0.782	47.4	10.50
18	October 1984	0.387	23.5	0.734	44.3	0.528	32.0	8.80
19	June 1985	0.078	4.7	0.689	41.8	0.882	53.5	11.30
20	October 1985	0.269	16.3	0.638	38.7	0.724	45.0	10.20
21	June 1986	0.166	10.1	0.663	40.2	0.820	49.7	10.80
22	October 1986	0.209	12.7	0.630	38.2	0.810	49.1	10.70
23	June 1987	0.117	7.1	0.602	36.5	0.930	56.4	11.30
24	October 1987	0.072	4.4	0.556	33.7	1.021	61.9	12.00
25	June 1988	0.020	1.2	0.326	19.8	1.303	79.0	13.30
26	October 1988	0.219	13.3	0.546	33.1	0.884	53.6	
27	June 1989	0.084	5.1	0.518	31.4	1.047	63.5	
28	October 1989	0.083	5.0	0.576	35.0	0.990	60.0	
29	June 1990	0.079	4.8	0.479	29.0	1.091	66.2	

Source: SCARP Monitoring, WAPDA.

Table–3

SCARP–II (SALINE)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.690 MA — CCA = 0.550 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1980 (Pre–Project)	0.144	20.8	0.362	52.5	0.184	26.7	8.60
2	October 1980	0.287	41.6	0.231	33.5	0.172	24.9	7.40
3	June 1981	0.128	18.5	0.347	50.3	0.215	31.2	9.00
4	October 1981	0.292	42.3	0.229	33.2	0.169	24.5	7.50
5	June 1982	0.185	26.8	0.314	45.5	0.191	27.7	8.30
6	October 1982	0.275	39.8	0.255	37.0	0.160	23.2	7.30
7	June 1983	0.247	35.8	0.284	41.2	0.159	23.0	7.50
8	October 1983	0.427	61.9	0.148	21.4	0.115	16.7	5.90
9	June 1984	0.082	11.9	0.425	61.6	0.183	26.5	8.80
10	October 1984	0.331	48.0	0.183	26.5	0.176	25.5	7.20
11	June 1985	0.096	14.0	0.390	56.6	0.204	29.5	9.00
12	October 1985	0.250	36.2	0.240	34.8	0.200	29.0	8.00
13	June 1986	0.116	16.8	0.346	50.2	0.228	33.0	9.10
14	October 1986	0.210	30.4	0.207	30.0	0.273	39.6	8.40
15	June 1987	0.166	24.0	0.264	38.3	0.260	37.7	9.10
16	October 1987	0.126	18.3	0.287	41.6	0.277	40.1	9.70
17	June 1988	0.037	5.4	0.311	45.1	0.342	49.5	10.50
18	October 1988	0.207	30.0	0.274	39.7	0 209	30.3	
19	June 1989	0.044	6.4	0.326	47.2	0.320	46.4	
20	October 1989	0.114	16.5	0.326	47.2	0.250	36.3	
21	June 1990	0.044	6.4	0.370	53.6	0.276	40.0	

Source: SCARP Monitoring, WAPDA.

Table–4

SHAHPUR SCARP
Areas in Different Depth to Water Table Ranges

Gross Area = 0.112 MA — CCA = 0.092 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	August 1974 (Pre· Project)	0.062	55.4	0.048	42.8	0.002	1.8	5.10
2	June 1980	0.019	17.0	0.083	74.1	0.010	8.9	7.20
3	October 1980	0.056	50.0	0.034	48.2	0.002	1.8	5.30
4	June 1981	0.022	19.6	0.085	75.9	0.005	4.5	6.80
5	October 1981	0.053	47.3	0.057	50.9	0.002	1.8	5.50
6	June 1982	0.012	10.7	0.100	89.3	0.000	0.0	7.00
7	October 1982	0.027	24.1	0.085	75.9	0.000	0.0	6.40
8	June 1983	0.010	8.9	0.100	89.3	0.002	1.8	7.20
9	October 1983	0.089	79.4	0.022	19.6	0.001	1.0	4.00
10	June 1984	0.002	1.8	0.096	85.7	0.014	12.5	7.90
11	October 1984	0.069	61.5	0.042	37.5	0.001	1.0	4.80
12	June 1985	0.002	1.8	0.096	85.7	0.014	12.5	7.90
13	October 1985	0.027	24.1	0.074	66.1	0.011	9.8	6.80
14	June 1986	0.020	17.9	0.084	75.0	0.008	7.1	7.00
15	October 1986	0.059	52.8	0.049	43.7	0.004	3.5	5.10
16	June 1987	0.022	19.6	0.086	76.8	0.004	3.6	6.70
17	October 1987	0.059	52.7	0.042	37.5	0.011	9.8	5.50
18	June 1988	0.022	19.6	0.044	39.3	0.046	41.1	10.00
19	October 1988	0.075	67.0	0.023	20.5	0.014	12.5	
20	June 1989	0.016	14.3	0.075	67.0	0.021	18.7	
21	October 1989	0.059	52.7	0.041	36.6	0.012	10.7	
22	June 1990	0.028	25.0	0.062	55.4	0.022	19.6	

Source: SCARP Monitoring, WAPDA.

Table–5

SCARP–III (NON–SALINE)
(Alipur, Kot Adu and Rangpur)
Areas in Different Depth to Water Table Ranges

Gross Area = 1.138 MA CCA = 0.952 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (FL)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1969 (Pre–Project)	0.469	41.2	0.484	42.5	0.135	16.3	6.30
2	June 1974	0.114	10.0	0.357	31.4	0.667	58.6	10.60
3	June 1975	0.071	6.2	0.219	19.3	0.848	74.5	11.90
4	October 1975	0.223	19.6	0.599	52.6	0.316	27.8	
5	June 1976	0.124	10.9	0.470	41.3	0.544	47.8	9.50
6	October 1976	0.370	32.5	0.589	51.8	0.179	15.7	
7	June 1977	0.136	11.9	0.596	52.4	0.406	35.7	9.10
8	October 1977	0.487	42.8	0.435	38.2	0.216	19.0	
9	June 1978	0.218	19.2	0.641	56.3	0.279	24.5	8.30
10	October 1978	0.490	34.1	0.455	40.0	0.193	16.9	
11	June 1979	0.317	27.8	0.578	50.8	0.243	21.4	7.50
12	October 1979	0.394	34.6	0.542	47.6	0.202	17.8	
13	June 1980	0.261	22.9	0.616	54.1	0.261	23.0	7.90
14	October 1980	0.371	32.6	0.542	47.6	0.225	19.8	7.50
15	June 1981	0.294	25.8	0.602	52.9	0.242	21.3	7.80
16	October 1981	0.355	31.2	0.552	48.5	0.231	20.3	7.50
17	June 1982	0.146	12.8	0.680	59.7	0.312	27.5	8.90
18	October 1982	0.288	25.3	0.572	50.3	0.278	24.4	8.20
19	June 1983	0.166	14.6	0.650	57.1	0.322	29.3	8.80
20	October 1983	0.358	31.5	0.519	45.6	0.261	22.9	7.70
21	June 1984	0.101	8.9	0.598	52.5	0.439	38.6	9.60
22	October 1984	0.309	27.1	0.581	51.1	0.248	21.8	7.80
23	June 1985	0.017	1.5	0.346	30.4	0.775	68.1	11.80
24	October 1985	0.222	19.5	0.556	48.9	0.360	31.6	8.90
25	June 1986	0.109	9.6	0.479	42.1	0.550	48.3	10.50
26	October 1986	0.393	34.5	0.484	42.6	0.216	22.9	7.60
27	June 1987	0.262	23.0	0.613	53.8	0.263	23.2	8.00
28	October 1987	0.440	38.7	0.437	38.4	0.261	22.9	7.40
29	June 1988	0.170	15.0	0.576	50.6	0.392	34.4	9.10
30	October 1988	0.462	40.6	0.420	37.0	0.256	22.4	
31	June 1989	0.295	25.9	0.474	41.7	0.369	32.4	
32	October 1989	0.469	41.2	0.405	35.6	0.264	23.2	
33	June 1990	0.294	25.8	0.516	45.3	0.328	28.8	

Source: SCARP Monitoring, WAPDA.

Table–6

SCARP–III (SALINE)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.142 MA CCA = 0.098 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1975	0.063	44.6	0.061	43.3	0.018	12.1	6.10
2	October 1975	0.088	62.2	0.048	34.0	0.006	3.8	4.90
3	June 1976	0.094	66.2	0.048	33.8	0.000	0.0	4.50
4	October 1976	0.115	81.0	0.027	19.0	0.000	0.0	3.90
5	June 1977	0.083	58.5	0.059	41.5	0.000	0.0	4.90
6	October 1977	0.111	78.2	0.031	21.8	0.000	0.0	4.00
7	June 1978	0.091	64.1	0.051	35.9	0.000	0.0	4.60
8	October 1978	0.115	81.0	0.027	19.0	0.000	0.0	3.90
9	June 1979	0.090	63.4	0.052	36.6	0.000	0.0	4.70
10	October 1979	0.115	81.0	0.027	19.0	0.000	0.0	3.90
11	June 1980	0.074	52.1	0.068	47.9	0.000	0.0	5.20
12	October 1980	0.110	77.5	0.032	22.5	0.000	0.0	4.00
13	June 1981	0.072	50.7	0.070	49.3	0.000	0.0	5.20
14	October 1981	0.103	72.6	0.039	27.5	0.000	0.0	4.20
15	June 1982	0.085	59.9	0.057	40.1	0.000	0.0	4.80
16	October 1982	0.107	75.4	0.035	24.6	0.000	0.0	4.10
17	June 1983	0.095	66.9	0.047	33.1	0.000	0.0	4.50
18	October 1983	0.096	67.6	0.044	31.0	0.002	1.4	4.40
19	June 1984	0.072	50.7	0.070	49.3	0.000	0.0	5.20
20	October 1984	0.089	62.7	0.052	36.6	0.001	0.1	4.60
21	June 1985	0.059	41.8	0.072	50.5	0.011	7.7	8.80
22	October 1985	0.096	67.6	0.044	31.0	0.002	1.4	4.40
23	June 1986	0.096	67.6	0.044	31.0	0.002	1.4	4.40
24	October 1986	0.101	71.1	0.040	28.2	0.001	0.7	4.20
25	June 1987	0.101	71.1	0.041	28.9	0.000	0.0	4.30
26	October 1987	0.119	83.8	0.023	16.2	0.000	0.0	3.70
27	June 1988	0.102	71.8	0.040	28.2	0.000	0.0	4.30
28	October 1988	0.111	78.2	0.029	20.4	0.002	1.4	
29	June 1989	0.105	73.9	0.035	24.6	0.002	1.4	
30	October 1989	0.121	85.2	0.020	14.1	0.001	0.7	
31	June 1990	0.108	76.1	0.032	22.5	0.002	1.4	

Source: SCARP Monitoring, WAPDA.

Table–7

SCARP – IV
Areas in Different Depth to Water Table Ranges

Gross Area = 0.558 MA CCA = 0.544 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1969 (Pre–Project)	0.001	0.2	0.121	21.7	0.436	78.1	11.70
2	June 1972	0.000	0.0	0.097	17.4	0.461	82.6	14.40
3	June 1973	0.004	0.8	0.054	9.7	0.500	89.5	15.70
4	June 1974	0.000	0.0	0.040	7.2	0.518	92.8	15.60
5	June 1975	0.006	1.1	0.018	3.2	0.534	95.7	16.30
6	October 1975	0.009	1.6	0.062	11.1	0.487	87.3	
7	June 1976	0.001	0.2	0.037	6.6	0.520	93.2	15.50
8	October 1976	0.015	2.7	0.182	32.6	0.361	64.7	
9	June 1977	0.002	0.3	0.050	9.0	0.506	90.7	14.80
10	October 1977	0.013	2.3	0.149	26.7	0.396	71.0	
11	June 1978	0.000	0.0	0.039	6.9	0.519	93.1	15.10
12	October 1978	0.017	3.1	0.140	25.0	0.401	71.9	13.00
13	June 1979	0.001	0.2	0.066	11.8	0.491	88.0	14.20
14	October 1979	0.004	0.7	0.012	16.4	0.462	82.9	14.30
15	June 1980	0.004	0.7	0.029	5.2	0.525	94.1	16.00
16	October 1980	0.011	2.0	0.126	22.6	0.421	75.4	13.30
17	June 1981	0.001	0.2	0.054	9.7	0.503	90.1	14.90
18	October 1981	0.002	0.3	0.083	14.9	0.473	84.8	14.40
19	June 1982	0.001	0.2	0.070	12.5	0.487	87.3	14.60
20	October 1982	0.001	0.2	0.035	6.3	0.522	93.5	15.00
21	June 1983	0.002	0.3	0.082	14.7	0.474	85.0	14.40
22	October 1983	0.010	1.8	0.138	24.8	0.410	73.4	13.00
23	June 1984	0.000	0.0	0.025	4.5	0.533	95.5	15.90
24	October 1984	0.017	3.1	0.131	23.5	0.410	73.4	13.00
25	June 1985	0.000	0.0	0.030	5.4	0.528	94.6	16.10
26	October 1985	0.020	3.6	0.089	15.9	0.449	80.5	14.20
27	June 1986	0.001	0.2	0.049	8.8	0.408	91.0	15.90
28	October 1986	0.032	5.7	0.109	19.5	0.417	74.8	14.30
29	June 1987	0.002	0.3	0.049	8.8	0.507	90.9	15.70
30	October 1987	0.002	0.3	0.087	15.6	0.469	84.1	15.20
31	June 1988	0.000	0.0	0.017	3.0	0.541	97.0	15.80
32	October 1988	0.015	2.7	0.127	22.8	0.416	74.5	
33	June 1989	0.001	0.2	0.057	10.2	0.500	89.6	
34	October 1989	0.006	1.1	0.103	18.4	0.449	80.5	
35	June 1990	0.000	0.0	0.044	7.9	0.514	92.1	

Source: SCARP Monitoring, WAPDA.

Table–8

SCARP – V
SHORKOT KAMALIA (PILOT)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.169 MA CCA = 0.154 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1976 (Pre–Project)	0.016	9.5	0.017	10	0.136	80.5	
2	June 1977	0.013	7.7	0.020	11.8	0.136	80.5	
3	June 1978	0.016	9.5	0.011	6.5	0.142	84.0	
4	October 1978	0.015	8.9	0.013	7.7	0.141	83.4	16.00
5	June 1979	0.005	3.0	0.011	6.5	0.153	879.5	16.60
6	October 1979	0.006	3.5	0.014	8.3	0.149	88.2	16.90
7	June 1980	0.006	3.5	0.014	8.3	0.145	88.2	16.10
8	October 1980	0.008	4.7	0.016	9.5	0.135	85.8	15.10
9	June 1981	0.007	4.1	0.013	7.7	0.149	88.2	15.80
10	October 1981	0.008	4.7	0.016	9.5	0.145	85.8	16.40
11	June 1982	0.005	3.0	0.020	11.8	0.144	85.2	16.10
12	October 1982	0.002	1.2	0.009	5.3	0.158	93.5	16.60
13	June 1983	0.002	1.2	0.020	11.8	0.147	87.0	16.20
14	October 1983	0.002	1.2	0.015	8.9	0.152	89.9	14.90
15	June 1984	0.002	1.2	0.012	7.1	0.155	91.7	16.00
16	October 1984	0.002	1.2	0.005	3.0	0.162	95.8	16.50
17	June 1985	0.001	0.7	0.009	5.3	0.159	94.0	16.90
18	October 1985	0.001	0.6	0.007	4.4	0.161	95.0	17.20
19	June 1986	0.002	1.2	0.009	5.3	0.158	93.5	17.20
20	October 1986	0.002	1.2	0.005	3.0	0.162	95.8	17.50
21	June 1987	0.005	3.0	0.010	5.9	0.154	91.1	17.00
22	October 1987	0.005	3.0	0.005	3.0	0.159	94.0	18.80
23	June 1988	0.001	0.6	0.011	6.5	0.157	92.9	16.20
24	October 1988	0.006	3.6	0.014	8.3	0.149	88.1	
25	June 1989	0.005	3.0	0.013	7.7	0.151	89.3	
26	October 1989	0.003	1.8	0.010	5.9	0.156	92.3	
27	June 1990	0.002	1.2	0.013	7.7	0.154	91.1	

Source: SCARP Monitoring, WAPDA.

Table–9

SCARP – V
SATIANA (PILOT)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.118 MA CCA = 0.098 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (FL)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1976 (Pre–Project)	0.017	14.4	0.096	81.4	0.005	4.2	
2	June 1977	0.033	18.0	0.085	72.0	0.000	0.0	6.20
3	June 1978	0.013	11.0	0.098	83.1	0.007	5.9	7.30
4	October 1978	0.102	86.4	0.016	13.6	0.000	0.0	3.60
5	June 1979	0.026	22.1	0.089	75.4	0.003	2.5	6.60
6	October 1979	0.073	61.9	0.045	38.1	0.000	0.0	4.70
7	June 1980	0.007	5.9	0.104	88.2	0.007	5.9	7.50
8	October 1980	0.054	54.8	0.063	53.4	0.001	0.8	5.50
9	June 1981	0.031	26.3	0.084	71.2	0.003	2.5	6.40
10	October 1981	0.093	78.8	0.025	21.2	0.000	0.0	4.00
11	June 1982	0.098	83.1	0.019	16.1	0.001	0.8	3.80
12	October 1982	0.077	65.3	0.038	32.2	0.003	2.5	4.70
13	June 1983	0.010	8.5	0.106	89.8	0.002	1.7	7.20
14	October 1983	0.106	89.8	0.012	10.2	0.000	0.0	3.50
15	June 1984	0.002	1.7	0.106	89.8	0.010	8.5	7.80
16	October 1984	0.091	77.1	0.026	22.1	0.001	0.8	4.10
17	June 1985	0.002	1.7	0.107	90.7	0.009	7.6	7.80
18	October 1985	0.015	12.7	0.094	79.7	0.009	7.6	7.40
19	June 1986	0.005	4.3	0.104	88.1	0.009	7.6	7.70
20	October 1986	0.017	14.4	0.091	77.1	0.010	8.5	7.30
21	June 1987	0.032	27.1	0.086	72.9	0.000	0.0	6.30
22	October 1987	0.005	4.3	0.106	89.8	0.007	5.9	7.60
23	June 1988	0.009	7.6	0.090	76.3	0.019	16.1	8.00
24	October 1988	0.020	16.9	0.096	81.4	0.002	1.7	
25	June 1989	0.016	13.6	0.098	83.1	0.004	3.3	
26	October 1989	0.015	12.7	0.102	86.3	0.001	0.0	
27	June 1990	0.003	2.5	0.106	89.8	0.009	7.6	

Source: SCARP Monitoring, WAPDA.

Table–10

SCARP – V
(KHAIRWALA UNIT)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.333 MA CCA = 0.301 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (FL)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1979 (Pre–Project)	0.010	5.7	0.096	28.8	0.218	65.5	11.80
2	June 1982	0.060	18.0	0.165	49.5	0.108	32.5	8.30
3	October 1982	0.020	6.0	0.160	48.0	0.153	46.0	9.50
4	June 1983	0.006	1.8	0.202	60.7	0.125	37.5	9.70
5	October 1983	0.044	13.2	0.148	44.4	0.141	42.2	9.60
6	June 1984	0.015	4.5	0.122	36.6	0.196	58.9	11.40
7	October 1984	0.025	7.5	0.161	48.3	0.147	44.2	10.10
8	June 1985	0.005	1.5	0.103	30.9	0.225	67.6	12.50
9	October 1985	0.003	0.9	0.105	31.5	0.225	67.6	12.20
10	June 1986	0.005	1.5	0.099	29.7	0.229	68.8	12.50
11	October 1986	0.003	0.9	0.121	36.3	0.209	62.8	12.20
12	June 1987	0.002	0.6	0.119	35.7	0.212	63.7	12.00
13	October 1987	0.001	0.3	0.057	17.1	0.275	82.6	13.90
14	June 1988	0.000	0.0	0.028	8.4	0.305	91.6	13.80
15	October 1988	0.000	0.0	0.034	102.0	0.299	89.8	
16	June 1989	0.000	0.0	0.032	9.6	0.301	9.4	
17	October 1989	– Data Not Available –						
18	June 1990	0.000	0.0	0.035	10.5	0.298	89.5	

Source: SCARP Monitoring, WAPDA.

Table–11

SCARP – V
(DRAINAGE IV)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.354 MA CCA = 0.296 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1978 (Pre–Project)	0.099	27.8	0.253	71.6	0.002	0.6	6.30
2	June 1985	0.013	3.7	0.234	61.1	0.107	30.2	8.60
3	October 1985	0.034	9.6	0.259	73.2	0.061	17.2	8.00
4	June 1986	0.035	9.9	0.262	74.0	0.057	16.1	8.10
5	October 1987	0.060	16.9	0.231	65.3	0.063	17.8	7.80
6	June 1987	0.071	20.1	0.249	70.3	0.034	9.6	7.20
7	October 1987	0.014	4.0	0.239	67.5	0.101	28.5	8.90
8	June 1988	0.009	2.5	0.159	44.9	0.186	52.6	10.40
9	October 1988							
10	June 1989							
11	October 1989							
12	June 1990							

Source: SCARP Monitoring, WAPDA.

Table–12

SCARP – VI
ALLAHABAD (PILOT)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.216 MA CCA = 0.202 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1972 (Pre–Project)	0.062	27.8	0.154	71.2	0.002	1.0	6.30
2	June 1979	0.099	45.8	0.109	50.5	0.008	3.7	5.60
3	October 1979							
4	June 1980	0.041	19.0	0.175	81.0	0.000	0.0	6.60
5	October 1980	0.103	47.7	0.112	51.9	0.001	0.4	5.40
6	June 1981	0.044	20.4	0.125	57.8	0.047	21.8	7.70
7	October 1981	0.069	32.0	0.142	65.1	0.005	2.3	6.20
8	June 1982	0.051	23.6	0.119	55.1	0.046	21.3	7.50
9	October 1982	0.061	28.2	0.126	58.4	0.029	13.4	6.90
10	June 1983	0.052	24.1	0.102	47.2	0.062	28.7	7.90
11	October 1983	0.044	20.4	0.119	55.1	0.053	24.5	7.80
12	June 1984	0.049	22.7	0.096	44.4	0.071	32.9	8.10
13	October 1984	0.042	19.4	0.104	48.2	0.070	32.4	8.20
14	June 1985	0.030	13.7	0.133	61.8	0.053	24.5	8.10
15	October 1985	0.027	12.5	0.129	59.7	0.060	27.8	8.20
16	June 1986	0.012	5.5	0.138	63.9	0.066	30.6	8.70
17	October 1986	0.015	6.9	0.143	66.2	0.058	26.9	8.50
18	June 1987	0.059	27.4	0.096	44.4	0.061	28.2	7.60
19	October 1987	0.059	27.3	0.106	49.1	0.051	23.6	7.40
20	June 1988	0.042	19.4	0.114	52.8	0.060	27.8	8.00
21	October 1988	0.042	19.4	0.010	45.7	0.164	34.9	
22	June 1989	0.030	13.9	0.124	57.4	0.062	28.7	
23	October 1989	0.042	19.4	0.111	51.4	0.063	29.2	
24	June 1990	0.033	15.3	0.091	42.1	0.092	42.6	

Source: SCARP Monitoring, WAPDA.

Table–13

SCARP – VI
(PUNJNAD ABBASIA UNIT–II TO V)
Areas in Different Depth to Water Table Ranges

Gross Area = 1.460 MA CCA = 1.267 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1974 (Pre–Project)	0.204	14	0.721	49.4	0.535	36.6	8.70
2	June 1981	0.185	12.7	0.999	68.4	0.276	18.9	7.90
3	October 1981	0.333	22.8	0.981	67.2	0.146	10.0	7.00
4	June 1982	0.236	16.2	1.093	74.9	0.131	8.9	7.20
5	October 1982	0.415	28.4	0.932	63.8	0.113	7.8	6.60
6	June 1983	0.291	19.9	1.026	70.3	0.143	9.8	7.10
7	October 1983	0.514	35.2	0.843	57.7	0.103	7.1	6.30
8	June 1984	0.311	21.3	1.050	71.9	0.099	6.8	6.90
9	October 1984	0.467	32.0	0.932	63.8	0.062	4.2	6.20
10	June 1985	0.200	13.7	1.064	72.9	0.196	13.4	7.60
11	October 1985	0.306	21.0	1.021	69.9	0.133	9.1	7.10
12	June 1986	0.222	15.2	1.040	71.2	0.198	13.6	7.10
13	October 1986	0.262	18.0	1.085	74.3	0.113	7.7	7.10
14	June 1987	0.316	21.6	0.917	62.8	0.227	15.6	7.40
15	October 1987	0.279	19.1	1.061	69.6	0.165	11.3	7.20
16	June 1988	0.136	9.3	0.929	63.6	0.395	27.1	8.60
17	October 1988	0.106	7.3	1.015	69.5	0.339	23.2	
18	June 1989	0.187	12.8	0.834	57.1	0.439	30.1	
19	October 1989	0.111	7.6	0.906	62.1	0.443	30.3	
20	June 1990	0.119	8.2	0.774	53.0	0.569	38.8	

Source: SCARP Monitoring, WAPDA.

Table–14

SCARP – VII
CBDC (PONDOKI)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.196 MA CCA = 0.174 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June (Pre–Project)	0.118	60	0.035	18	0.043	22.0	
2	June 1982	0.052	26.5	0.088	44.9	0.056	28.6	7.70
3	October 1982	0.097	49.5	0.062	31.6	0.037	18.9	6.20
4	June 1983	0.062	31.6	0.062	31.6	0.072	36.8	9.20
5	October 1983	0.098	50.0	0.051	26.0	0.047	24.0	7.80
6	June 1984	0.069	35.2	0.068	34.7	0.059	30.1	8.90
7	October 1984	0.102	52.0	0.041	20.9	0.053	27.1	8.20
8	June 1985	0.007	3.6	0.110	56.1	0.079	40.3	9.80
9	October 1985	0.117	59.7	0.030	15.3	0.049	25.0	7.30
10	June 1986	0.012	6.1	0.104	53.1	0.080	40.8	10.50
11	October 1986	0.067	34.2	0.082	41.8	0.047	24.0	8.40
12	June 1987	0.007	3.6	0.109	55.6	0.080	40.8	10.70
13	October 1987	0.048	24.5	0.083	42.3	0.065	33.2	9.20
14	June 1988	0.000	0.0	0.063	32.1	0.133	67.9	12.40
15	October 1988	0.062	31.6	0.077	39.3	0.057	29.1	
16	June 1989	0.000	0.0	0.111	56.6	0.085	43.4	
17	October 1989	0.006	3.1	0.090	45.9	0.100	51.0	
18	June 1990	0.000	0.0	0.100	51.0	0.096	49.0	

Source: SCARP Monitoring, WAPDA.

Table–15

SCARP – VIII
MINCHINABAD (PILOT)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.078 MA CCA = 0.073 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1979 (Pre–Project)	0.000	0.0	0.068	87.2	0.01	12.8	8.10
2	October 1979							
3	June 1980	0.000	0.0	0.059	75.6	0.019	24.4	8.70
4	October 1980	0.016	20.5	0.053	68.0	0.009	11.5	7.20
5	June 1981	0.000	0.0	0.058	74.4	0.020	25.6	8.80
6	October 1981	0.015	19.2	0.044	61.6	0.015	19.2	7.60
7	June 1982	0.000	0.0	0.034	43.6	0.044	56.4	10.30
8	October 1982	0.015	19.2	0.049	62.8	0.014	18.0	7.50
9	June 1983	0.000	0.0	0.032	41.0	0.047	59.0	10.60
10	October 1983	0.049	62.8	0.017	21.8	0.012	15.4	5.40
11	June 1984	0.000	0.0	0.035	44.9	0.043	55.1	10.30
12	October 1984	0.037	47.4	0.030	38.5	0.011	14.1	6.20
13	June 1985	0.000	0.0	0.027	34.6	0.061	65.4	11.30
14	October 1985	0.015	19.2	0.054	69.3	0.009	11.5	7.40
15	June 1986	0.000	0.0	0.047	60.3	0.031	39.7	10.00
16	October 1986	0.040	51.3	0.032	41.0	0.006	7.7	6.00
17	June 1987	0.000	0.0	0.057	73.1	0.021	26.9	8.80
18	October 1987	0.022	28.2	0.010	12.8	0.046	59.0	11.50
19	June 1988	0.000	0.0	0.027	34.6	0.051	65.4	12.30
20	October 1988	0.021	27.0	0.016	20.5	0.041	52.5	
21	June 1989	0.000	0.0	0.040	51.3	0.038	48.7	
22	October 1989	0.035	44.9	0.025	32.1	0.018	23.0	
23	June 1990	0.000	0.0	0.032	41.0	0.046	59.0	

Source: SCARP Monitoring, WAPDA.

Table–16

SCARP – VIII
(FORDWAH SADIQIA II)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.111 MA CCA = 1.100 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1980 (Pre–Project)							
2	June 1982	0.095	85.6	0.016	14.4	0.000	0.000	3.60
3	October 1982	0.106	95.5	0.005	4.5	0.000	0.000	3.20
4	June 1983	0.098	88.3	0.013	11.7	0.000	0.000	3.50
5	October 1983	0.106	95.5	0.005	4.5	0.000	0.000	3.20
6	June 1984	0.079	71.2	0.032	28.8	0.000	0.000	4.30
7	October 1984	0.109	98.2	0.002	1.8	0.000	0.000	3.10
8	June 1985	0.065	58.6	0.046	41.4	0.000	0.000	4.90
9	October 1985	0.094	84.7	0.017	15.3	0.000	0.000	3.70
10	June 1986	0.070	63.1	0.040	36.0	0.001	0.900	4.70
11	October 1986	0.104	93.7	0.007	6.3	0.000	0.000	3.30
12	June 1987	0.093	83.8	0.017	15.3	0.001	0.900	3.80
13	October 1987	0.091	82.0	0.020	18.0	0.000	0.000	3.80
14	June 1988	0.057	51.4	0.052	46.8	0.002	1.800	5.30
15	October 1988	0.096	86.5	0.012	10.8	0.003	2.700	
16	June 1989	0.046	41.4	0.065	58.6	0.000	0.000	
17	October 1989	0.111	100.0	0.000	0.0	0.000	0.000	
18	June 1990	0.090	81.1	0.021	18.9	0.000	0.000	

Source: SCARP Monitoring, WAPDA.

Table–17

SCARP KHAIRPUR
Areas in Different Depth to Water Table Ranges

Gross Area = 0.441 MA | CCA = 0.380 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1960 (Pre–Project)	0.131	29.7	0.310	70.3	0.000	0.0	6.2
2	April 1966	0.164	37.3	0.277	62.7	0.000	0.0	5.8
3	April 1972	N.A	N.A	N.A	N.A	N.A	N.A	10.2
4	April 1975	N.A	N.A	N.A	N.A	N.A	N.A	7.9
5	April 1976	0.197	44.7	0.203	46.0	0.041	9.3	5.9
6	April 1977	0.110	25.0	0.273	62.0	0.058	13.3	6.9
7	April 1978	0.083	18.8	0.332	75.3	0.026	5.9	6.9
8	October 1978	0.129	29.3	0.262	59.4	0.050	11.3	6.6
9	April 1979	0.081	18.4	0.331	75.1	0.029	6.5	6.9
10	October 1979	0.088	20.0	0.300	68.0	0.053	12.0	7.2
11	April 1980	0.069	15.6	0.324	73.3	0.048	11.1	7.4
12	October 1980	0.100	22.7	0.274	62.1	0.967	15.2	7.4
13	April 1981	0.134	30.4	0.280	63.5	0.027	6.1	6.5
14	October 1981	0.126	28.6	0.286	64.9	0.029	6.5	6.6
15	April 1982	0.248	56.2	0.184	41.7	0.009	2.1	5.1
16	October 1982	0.148	33.6	0.245	55.6	0.048	10.6	6.7
17	April 1983	0.170	38.5	0.260	59.0	0.011	2.5	5.9
18	October 1983	0.136	30.8	0.263	59.6	0.042	9.6	6.7
19	April 1984	0.208	47.1	0.227	51.5	0.006	1.4	5.5
20	October 1984	0.183	41.5	0.242	55.0	0.016	3.5	5.8
21	April 1985	0.188	42.6	0.243	55.1	0.010	2.3	5.7
22	October 1985	0.193	44.0	0.245	55.0	0.003	1.0	5.6
23	April 1986	0.171	38.8	0.262	59.2	0.008	2.0	5.9
24	October 1986	0.220	49.9	0.215	48.8	0.006	1.3	5.6
25	April 1987	0.195	44.2	0.242	54.9	0.004	0.9	5.6
26	October 1987	0.175	39.6	0.245	55.6	0.021	4.7	6.0
27	April 1988	0.094	21.3	0.339	76.9	0.008	1.8	6.7
28	October 1988	0.152	34.5	0.278	63.0	0.011	2.5	
29	April 1989	0.150	34.0	0.290	65.8	0.001	0.2	
30	October 1988	0.256	58.1	0.184	41.7	0.001	0.2	
31	April 1990	0.093	21.1	0.341	77.3	0.007	1.6	

Source: SCARP Monitoring, WAPDA.

Table–18

SCARP NORTH ROHRI
Areas in Different Depth to Water Table Ranges

Gross Area = 0.793 MA | CCA = 0.686 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1966 (Pre–Project)	0.087	11.0	0.174	22.0	0.532	67.0	11.0
2	April 1976	0.016	2.0	0.349	44.0	0.428	54.0	10.9
3	April 1977	0.000	0.0	0.436	55.0	0.357	45.0	10.5
4	April 1978	0.005	0.6	0.237	29.9	0.551	69.5	13.4
5	October 1978	0.014	1.8	0.334	42.1	0.445	56.1	11.7
6	April 1979	0.006	0.8	0.301	37.9	0.486	61.3	12.2
7	October 1979	0.006	0.8	0.200	25.2	0.587	74.0	13.4
8	April 1980	0.002	0.3	0.278	35.1	0.513	64.6	12.8
9	October 1980	0.009	1.1	0.216	27.2	0.568	71.7	13.4
10	April 1981	0.007	0.9	0.347	43.8	0.439	55.3	12.4
11	October 1981	0.023	2.9	0.243	30.6	0.527	66.5	13.1
12	April 1982	0.025	3.2	0.351	44.3	0.417	52.5	12.4
13	October 1982	0.037	4.7	0.290	36.6	0.466	58.7	12.7
14	April 1983	0.025	3.2	0.359	45.3	0.409	51.5	12.7
15	October 1983	0.040	5.0	0.351	44.3	0.402	50.7	12.1
16	April 1984	0.049	6.2	0.348	43.9	0.396	49.9	12.2
17	October 1984	0.049	6.2	0.309	38.8	0.435	55.0	12.4
18	April 1985	0.035	4.4	0.326	41.1	0.432	54.5	12.6
19	October 1985	0.044	5.1	0.297	37.0	0.452	57.9	12.6
20	April 1986	0.020	2.5	0.238	30.0	0.535	67.5	13.5
21	October 1986	0.059	7.5	0.285	35.9	0.449	56.6	12.4
22	April 1987	0.031	3.9	0.333	42.0	0.429	54.1	12.9
23	October 1987	0.057	7.2	0.203	25.6	0.533	67.2	13.7
24	April 1988	0.049	6.2	0.330	41.6	0.414	52.2	11.5
25	October 1988	0.043	5.4	0.317	40.0	0.433	54.6	
26	April 1989	0.093	11.7	0.348	43.9	0.352	44.4	
27	October 1988	0.163	20.6	0.315	39.7	0.315	39.7	
28	April 1990	0.078	9.8	0.348	43.9	0.367	46.3	

Source: SCARP Monitoring, WAPDA.

Table--19

SUKKUR RIGHT BANK (FGW)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.151 MA | CCA = 0.130 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1979	0.020	13.3	0.124	82.1	0.007	4.6	5.6
2	October 1979	0.151	100.0	0.000	0.0	0.000	0.0	3.0
3	April 1980	0.012	7.9	0.128	84.8	0.011	7.3	7.5
4	October 1980	0.148	98.0	0.003	2.0	0.000	0.0	3.1
5	April 1981	0.018	11.9	0.132	87.4	0.001	0.7	7.0
6	October 1981	0.150	99.3	0.001	0.7	0.000	0.0	3.0
7	April 1982	0.016	10.6	0.125	82.8	0.010	6.6	7.4
8	October 1982	0.151	100.0	0.000	0.0	0.000	0.0	3.0
9	April 1983	0.015	10.0	0.132	87.4	0.004	2.6	7.2
10	October 1983	0.148	98.0	0.003	2.0	0.000	0.0	3.1
11	April 1984	0.010	6.6	0.141	93.4	0.000	0.0	7.2
12	October 1984	0.148	98.0	0.003	2.0	0.000	0.0	3.1
13	April 1985	0.007	4.6	0.138	91.4	0.006	4.0	7.4
14	October 1985	0.151	100.0	0.000	0.0	0.000	0.0	3.0
15	April 1986	0.010	6.6	0.137	90.8	0.004	2.6	7.3
16	October 1986	0.151	100.0	0.000	0.0	0.000	0.0	3.0
17	April 1987	0.002	1.3	0.142	94.0	0.007	4.7	7.7
18	October 1987	0.139	92.1	0.012	7.9	0.000	0.0	3.4
19	April 1988	0.000	0.0	0.146	96.7	0.005	3.3	7.7
20	October 1988	0.148	98.0	0.003	2.0	0.000	0.0	
21	April 1989	0.003	2.0	0.148	98.0	0.000	0.0	
22	October 1988	0.151	100.0	0.000	0.0	0.000	0.0	
23	April 1990	0.005	3.3	0.144	95.4	0.002	1.3	

Source: SCARP Monitoring, WAPDA.

Table–20

LARKANA SHIKARPUR SURFACE DRAINAGE PROJECT
Areas in Different Depth to Water Table Ranges

Gross Area = 0.701 MA — CCA = 0.594 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April (Pre–Project)							
2	April 1977	0.301	43.0	0.322	46.0	0.078	11.0	
3	October 1977							
4	April 1978	0.312	44.5	0.333	47.5	0.056	8.0	5.7
5	October 1978	0.693	98.9	0.008	1.1	0.000	0.0	3.1
6	April 1979	0.291	41.5	0.378	53.9	0.032	4.6	5.7
7	October 1979	0.690	98.4	0.011	1.6	0.000	0.0	3.1
8	April 1980	0.291	41.5	0.382	54.5	0.028	4.0	5.8
9	October 1980	0.691	98.5	0.010	1.5	0.000	0.0	3.1
10	April 1981	0.297	42.4	0.372	53.0	0.032	4.6	5.8
11	October 1981	0.691	98.5	0.010	1.5	0.000	0.0	3.1
12	April 1982	0.271	38.7	0.401	57.2	0.029	4.1	6.0
13	October 1982	0.692	98.7	0.005	0.7	0.004	0.6	3.1
14	April 1983	0.290	41.4	0.368	52.5	0.043	6.1	6.0
15	October 1983	0.682	97.3	0.012	1.7	0.007	1.0	3.1
16	April 1984	0.220	31.4	0.432	61.6	0.049	7.0	6.5
17	October 1984	0.687	98.0	0.007	1.0	0.007	1.0	3.1
18	April 1985	0.279	39.8	0.376	53.6	0.046	6.6	6.1
19	October 1985	0.689	98.3	0.005	0.7	0.007	1.0	3.1
20	April 1986	0.270	38.5	0.387	55.2	0.044	6.3	6.1
21	October 1986	0.687	98.0	0.005	0.7	0.009	1.0	3.1
22	April 1987	0.115	16.4	0.516	73.6	0.070	10.0	7.3
23	October 1987	0.672	95.9	0.025	3.5	0.004	0.6	3.2
24	April 1988	0.082	11.7	0.581	82.9	0.038	5.4	7.3
25	October 1988	0.681	97.1	0.012	1.7	0.008	1.2	
26	April 1989	0.201	28.7	0.440	62.8	0.060	8.5	
27	October 1988	0.682	97.3	0.011	1.6	0.008	1.1	
28	April 1990	0.181	25.8	0.485	69.2	0.035	5.0	

Source: SCARP Monitoring, WAPDA.

Table–21

SOUTH ROHRI (FGW)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.541 MA CCA = 0.375 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1975 (Pre–Project)	0.002	0.4	0.056	10.4	0.483	89.2	N.A
2	April 1976	0.008	1.5	0.073	13.5	0.460	85.0	N.A
3	April 1977	0.002	0.4	0.096	17.7	0.443	81.9	N.A
4	April 1978	0.009	1.7	0.096	17.7	0.436	80.6	15.7
5	October 1978	0.021	3.9	0.117	21.6	0.403	74.5	13.2
6	April 1979	0.018	3.3	0.139	25.7	0.384	71.0	14.6
7	October 1979	0.014	2.6	0.086	15.9	0.441	81.5	15.4
8	April 1980	0.010	1.8	0.088	16.3	0.443	81.9	15.6
9	October 1980	0.011	2.0	0.054	10.0	0.476	88.0	16.4
10	April 1981	0.003	0.6	0.092	17.0	0.446	82.4	15.7
11	October 1981	0.000	0.0	0.077	14.2	4.464	85.8	17.3
12	April 1982	0.001	0.2	0.100	18.5	0.440	81.3	15.8
13	October 1982	0.007	1.3	0.069	12.8	0.465	85.9	16.3
14	April 1983	0.008	1.5	0.096	17.7	0.437	80.0	15.8
15	October 1983	0.015	2.9	0.114	21.0	0.412	76.1	14.7
16	April 1984	0.010	1.8	0.163	30.1	0.368	68.1	14.4
17	October 1984	0.027	5.0	0.136	25.1	0.378	69.9	13.9
18	April 1985	0.010	1.8	0.146	27.0	0.385	71.2	14.3
19	October 1985	0.030	5.5	0.151	28.0	0.360	66.5	13.5
20	April 1986	0.016	3.0	0.139	25.7	0.386	71.3	14.1
21	October 1986	0.022	4.0	0.129	24.0	0.390	72.0	13.6
22	April 1987	0.036	6.7	0.17	31.4	0.335	61.9	12.9
23	October 1987	0.009	1.6	0.128	23.7	0.404	74.7	14.3
24	April 1988	0.004	0.7	0.125	23.1	0.412	76.2	13.8
25	October 1988	0.020	3.7	0.153	28.3	0.368	68.0	
26	April 1989	0.005	0.9	0.183	33.8	0.353	65.3	
27	October 1989	0.029	5.4	0.181	33.5	0.331	61.1	
28	April 1990	0.016	3.0	0.217	40.1	0.308	56.9	

Source: SCARP Monitoring, WAPDA.

Table–22

GHOTKI (FGW)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.541 MA | CCA = 0.375 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April (Pre–Project)							
2	April 1976	0.001	0.2	0.298	67.6	0.142	32.2	
3	April 1977	0.031	7.0	0.388	88.0	0.022	5.0	
4	April 1978	0.080	18.2	0.345	78.2	0.016	3.6	6.8
5	October 1978	0.267	60.6	0.169	38.3	0.005	1.1	4.5
6	April 1979	0.117	26.5	0.296	67.1	0.128	6.4	6.5
7	October 1979	0.355	80.5	0.082	18.6	0.004	0.9	3.9
8	April 1980	0.084	19.0	0.331	75.1	0.026	5.9	6.9
9	October 1980	0.284	66.7	0.139	31.5	0.008	1.8	4.6
10	April 1981	0.079	17.9	0.335	76.0	0.027	6.1	7.0
11	October 1981	0.326	73.9	0.106	24.0	0.009	2.1	4.3
12	April 1982	0.022	5.0	0.384	87.1	0.035	7.9	7.7
13	October 1982	0.343	77.8	0.090	20.4	0.008	1.8	4.1
14	April 1983	0.022	5.0	0.366	83.0	0.053	12.0	7.9
15	October 1983	0.400	90.7	0.037	8.4	0.004	0.9	3.5
16	April 1984	0.020	4.5	0.388	88.0	0.033	7.5	7.7
17	October 1984	0.297	67.3	0.139	31.3	0.006	1.4	4.5
18	April 1985	0.024	5.4	0.373	84.6	0.044	10.0	6.3
19	October 1985	0.311	70.5	0.121	27.5	0.009	2.0	4.5
20	April 1986	0.023	5.2	0.366	83.0	0.052	11.8	6.4
21	October 1986	0.400	90.7	0.041	9.3	0.000	0.0	3.4
22	April 1987	0.017	3.9	0.37	83.9	0.054	12.2	7.9
23	October 1987	0.267	60.5	0.173	39.3	0.001	0.2	4.8
24	April 1988	0.001	0.2	0.300	68.0	0.140	31.8	9.1
25	October 1988	0.243	44.9	0.199	36.8	0.099	18.3	
26	April 1989	0.013	2.4	0.380	70.2	0.148	27 4	
27	October 1989	0.247	45.7	0.195	36.0	0.099	18.3	
28	April 1990	0.007	1.3	0.380	70.2	0.154	28.5	

Source: SCARP Monitoring, WAPDA.

Table–23

LBOD STAGE–I
Areas in Different Depth to Water Table Ranges

Gross Area = 1.427 MA | CCA = 1.276 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1983 (Pre–Project)	0.614	43.0	0.641	44.9	0.172	12.1	
2	April 1987	0.752	52.7	0.618	43.3	0.057	4.0	5.3
3	October 1987	0.390	27.3	0.905	63.4	0.132	9.3	6.7
4	April 1988	0.693	48.6	0.641	44.9	0.093	6.5	5.6
5	October 1988	0.839	58.8	0.478	33.5	0.110	7.7	
6	April 1989	0.950	66.6	0.418	29.3	0.059	4.1	
7	October 1989	0.883	61.9	0.483	33.8	0.061	4.3	
8	April 1990	0.855	59.9	0.519	36.4	0.053	3.7	

Source: SCARP Monitoring, WAPDA.

Table–24

HAIRDIN
Areas in Different Depth to Water Table Ranges

Gross Area = 0.087 MA — CCA = 0.074 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April (Pre–Project)	0.202	22.7	0.067	77.3	0.000	0.000	6.5
2	April 1980	0.065	74.7	0.022	25.3	0.000	0.000	4.1
3	October 1980	0.087	100.0	0.000	0.0	0.000	0.000	3.0
4	April 1981	0.046	52.9	0.041	47.1	0.000	0.000	5.1
5	October 1981	0.087	100.0	0.000	0.0	0.000	0.000	3.0
6	April 1982	0.036	41.4	0.051	58.6	0.000	0.000	5.7
7	October 1982	0.087	100.0	0.000	0.0	0.000	0.000	3.0
8	April 1983	0.035	40.2	0.052	59.8	0.000	0.000	5.7
9	October 1983	0.087	100.0	0.000	0.0	0.000	0.000	3.0
10	April 1984	0.072	82.8	0.015	17.2	0.000	0.000	3.8
11	October 1984	0.087	100.0	0.000	0.0	0.000	0.000	3.0
12	April 1985	0.074	85.1	0.013	14.9	0.000	0.000	3.7
13	October 1985	0.087	100.0	0.000	0.0	0.000	0.000	3.0
14	April 1986	0.084	96.6	0.003	3.4	0.000	0.000	3.2
15	October 1987	0.087	100.0	0.000	0.0	0.000	0.000	3.0
16	April 1987	0.072	82.2	0.015	17.2	0.000	0.000	3.8
17	October 1987	0.087	100.0	0.000	0.0	0.000	0.000	3.0
18	April 1988	0.041	47.1	0.046	52.9	0.000	0.000	5.4
19	October 1988	0.087	100.0	0.000	0.0	0.000	0.000	
20	April 1989	0.049	56.3	0.038	43.7	0.000	0.000	
21	October 1989	0.087	100.0	0.000	0.0	0.000	0.000	
22	April 1990	0.074	85.1	0.013	14.9	0.000	0.000	

Source: SCARP Monitoring, WAPDA.

Table–25

HAIRDIN – II
Areas in Different Depth to Water Table Ranges

Gross Area = 0.089 MA — CCA = 0.087 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	April 1982 (Pre–Project)	0.038	42.7	0.045	50.6	6.700	5.9	
2	April 1987	0.032	36.0	0.023	25.8	0.034	38.2	9.2
3	October 1987	0.033	37.1	0.027	30.3	0.029	32.6	8.6
4	April 1988	0.016	18.0	0.034	38.2	0.039	43.8	9.4
5	October 1988	0.023	25.8	0.027	30.3	0.039	43.9	
6	April 1989	0.019	21.3	0.058	65.2	0.012	13.5	
7	October 1989	0.044	49.4	0.035	39.3	0.010	11.2	
8	April 1990	0.023	25.8	0.029	32.6	0.037	41.6	

Source: SCARP Monitoring, WAPDA.

Table–26

SCARP PESHAWAR

(Pabbi I & II, Jue Sheikh, Kafur Dheri, AWL along Peshawar City and Warsak Units)
Areas in Different Depth to Water Table Ranges

Gross Area = 0.227 MA — CCA = 0.185 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1976 (Pre–Project)	0.064	28.2	0.096	42.3	0.067	29.5	
2	June 1982	0.027	11.9	0.062	27.3	0.138	60.8	14.50
3	June 1983	0.038	16.8	0.060	26.4	0.129	56.8	14.00
4	October 1983	0.047	20.7	0.056	24.7	0.124	54.6	13.50
5	June 1984	0.037	16.3	0.063	27.8	0.127	55.9	13.90
6	October 1984	0.047	20.7	0.057	25.1	0.123	54.2	13.50
7	June 1985	0.029	12.8	0.069	30.4	0.129	56.8	14.10
8	October 1985	0.036	15.9	0.065	28.6	0.126	55.5	13.70
9	June 1986	0.027	11.9	0.072	31.7	0.128	56.4	14.00
10	October 1986	0.045	19.8	0.106	46.7	0.076	33.5	15.20
11	June 1987	0.032	14.1	0.068	30.0	0.127	55.9	12.00
12	October 1987	0.037	16.3	0.131	57.7	0.059	26.0	16.10
13	June 1988	0.039	17.2	0.056	24.7	0.132	58.1	12.10
14	October 1988	0.041	18.1	0.062	27.3	0.124	45.2	
15	June 1989	0.028	12.3	0.065	28.6	0.134	59.1	
16	October 1989	0.037	16.3	0.029	12.7	0.161	71.0	
17	June 1990	0.028	12.3	0.067	29.5	0.132	58.2	

Source: SCARP Monitoring, WAPDA.

Table–27

BANNU
Areas in Different Depth to Water Table Ranges

Gross Area = 0.090 MA — CCA = 0.080 MA

Sr. No.	Period	Areas in Different Depth to Water Table Ranges						Average Depth to Watertable (Ft.)
		Below 5'		5' – 10'		Above 10'		
		Area (MA)	%	Area (MA)	%	Area (MA)	%	
1	June 1980 (Pre–Project)	0.003	3.3	0.005	5.5	0.082	91.2	
2	June 1981	0.005	5.5	0.011	12.3	0.074	82.2	16.80
3	October 1981	0.004	4.4	0.010	11.1	0.076	84.5	17.90
4	June 1982	0.011	12.2	0.015	16.7	0.064	71.1	15.30
5	October 1982	0.003	3.4	0.008	8.9	0.079	87.7	18.40
6	June 1983	0.010	11.1	0.015	16.7	0.065	72.2	15.20
7	October 1983	0.007	7.8	0.015	16.7	0.068	75.5	15.90
8	June 1984	0.007	7.8	0.015	16.7	0.068	75.5	15.30
9	October 1984	0.007	7.8	0.015	16.7	0.068	75.5	15.90
10	June 1985	0.005	5.5	0.017	19.0	0.068	75.5	15.40
11	October 1985	0.005	5.5	0.017	19.0	0.068	75.5	16.10
12	June 1986	0.005	5.5	0.016	17.8	0.069	76.7	15.60
13	October 1986	0.006	6.7	0.014	15.5	0.070	77.8	16.20
14	June 1987	0.003	3.4	0.019	21.1	0.068	75.5	15.60
15	October 1987	0.006	6.7	0.014	15.5	0.070	77.8	16.60
16	June 1988	0.007	7.8	0.010	11.1	0.073	81.1	14.30
17	October 1988	0.007	7.8	0.013	14.4	0.070	77.8	
18	June 1989	0.007	7.8	0.014	15.6	0.069	76.6	
19	October 1989	0.008	8.9	0.011	12.2	0.071	78.9	
20	June 1990	0.006	6.7	0.012	13.3	0.072	80.0	

Source: SCARP Monitoring, WAPDA.

C. CANAL COMMAND-WISE DEPTH TO WATERTABLE

TABLE – 1
Canal Commandwise Depth to Watertable (1959)

Sr. No.	Canal Commands	Gross Area (M.Ac)	JUNE Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'
	Punjab	24.620	2.050	7.642	14.928
1	U. Dipalpur	0.451	0.057	0.207	0.187
2	Ravi Syphon	0.822	0.045	0.134	0.643
3	Raya Branch	0.453	0.054	0.281	0.118
4	U. Chenab	1.160	0.342	0.632	0.136
5	M.R. Link	0.175	0.023	0.125	0.027
6	Sadiqia	1.315	0.025	0.042	1.248
7	Fordwah	0.529	0.030	0.297	0.202
8	Pakpattan	1.066	Nil	0.030	1.036
9	L. Dipalpur	0.660	0.063	0.309	0.288
10	L.B.D.C.	1.972	0.026	0.164	1.782
11	Jhang	2.126	0.073	0.468	1.585
12	Gugera	1.846	0.159	0.715	0.972
13	U. Jhelum	0.711	0.068	0.333	0.31
14	L. Jhelum	1.740	0.339	1.020	0.381
15	Bahawal	0.856	0.019	0.052	0.785
16	Mailsi	1.076	Nil	0.048	1.028
17	Sidhnai	0.862	Nil	0.120	0.742
18	Havali	0.234	Nil	0.029	0.205
19	Rangpur	0.435	0.051	0.303	0.081
20	Panjnad	1.491	0.153	1.020	0.318
21	Abbasia	0.131	Nil	0.039	0.092
22	Thal	2.655	0.060	0.265	2.33
23	Muzaffargarh	0.845	0.099	0.572	0.174
24	D.G.Khan	1.009	0.364	0.387	0.258

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 2
Canal Commandwise Depth to Watertable (1964)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	Sindh & Balochistan	14.547	1.800	5.775	6.972	5.708	4.235	4.604
1	Pat & Desert	0.449	0.003	0.019	0.427	0.009	0.032	0.408
2	Begari	1.082	0.177	0.672	0.233	0.792	0.287	0.003
3	Ghotki	0.960	0.002	0.008	0.950	0.002	0.069	0.889
4	N. West	1.263	0.335	0.780	0.148	0.691	0.403	0.169
5	Rice	0.595	0.104	0.393	0.098	0.526	0.069	0.000
6	Dadu	0.631	0.088	0.356	0.187	0.142	0.308	0.181
7	Khairpur	0.870	0.204	0.349	0.317	0.234	0.358	0.278
8	Rohri	2.952	0.089	0.523	2.340	0.132	0.951	1.869
9	Nara	2.514	0.441	1.253	0.820	1.654	0.578	0.282
10	Kalri	0.687	0.077	0.320	0.290	0.426	0.133	0.128
11	Lined Channel	0.582	0.059	0.270	0.253	0.441	0.135	0.006
12	Fuleli & Pinyari	1.962	0.221	0.832	0.909	0.659	0.912	0.391

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 3
Canal Commandwise Depth to Watertable (1975)

Sr. No.	Canal Commands	Gross Area (M.Ac)	JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	Punjab	24.620	0.990	5.621	18.009	2.812	6.704	15.104
1	U. Dipalpur	0.451	Nil	0.141	0.310	0.020	0.182	0.249
2	Ravi Syphon	0.822	0.007	0.452	0.363	0.084	0.201	0.537
3	Raya Branch	0.453	Nil	0.052	0.401	0.042	0.119	0.292
4	U. Chenab	1.160	0.026	0.283	0.851	0.255	0.406	0.499
5	M.R. Link	0.175	Nil	0.003	0.172	0.014	0.107	0.054
6	Sadiqia	1.315	0.076	0.184	1.055	0.124	0.133	1.058
7	Fordwah	0.529	0.139	0.252	0.138	0.181	0.258	0.090
8	Pakpattan	1.066	Nil	Nil	1.066	0.031	0.088	0.947
9	L. Dipalpur	0.660	Nil	0.051	0.609	0.019	0.054	0.587
10	L.B.D.C.	1.972	Nil	0.072	1.900	0.009	0.135	1.828
11	Jhang	2.126	0.015	0.509	1.602	0.102	0.619	1.405
12	Gugera	1.846	0.021	0.214	1.611	0.153	0.284	1.409
13	U. Jhelum	0.711	0.017	0.284	0.410	0.189	0.321	0.201
14	L. Jhelum	1.740	0.077	0.543	1.120	0.414	0.569	0.757
15	Bahawal	0.856	Nil	0.011	0.845	0.006	0.253	0.597
16	Mailsi	1.076	Nil	0.060	1.016	Nil	0.037	1.039
17	Sidhnai	0.862	Nil	Nil	0.862	Nil	Nil	0.862
18	Havali	0.234	0.060	0.074	0.100	0.069	0.056	0.109
19	Rangpur	0.435	0.014	0.041	0.380	0.034	0.227	0.174
20	Panjnad	1.491	0.145	0.738	0.608	0.211	0.948	0.332
21	Abbasia	0.131	0.087	0.044	0.000	0.022	0.109	0.000
22	Thal	2.655	0.175	1.056	1.424	0.377	0.887	1.391
23	Muzaffargarh	0.845	0.127	0.316	0.402	0.303	0.413	0.129
24	D.G.Khan	1.009	0.004	0.241	0.764	0.153	0.298	0.558

Source: Depth to water tableMaps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E). Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 4
Canal Commandwise Depth to Watertable (1976)

Sr. No.	Canal Commands	Gross Area (M.Ac)	JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	Punjab	24.620	1.840	6.243	16.5	4.958	7.069	12.593
1	U. Dipalpur	0.451				0.063	0.167	0.221
2	Ravi Syphon	0.822				0.212	0.111	0.499
3	Raya Branch	0.453				0.065	0.110	0.278
4	U. Chenab	1.160				0.476	0.438	0.246
5	M.R. Link	0.175				0.033	0.115	0.027
6	Sadiqia	1.315				0.188	0.214	0.913
7	Fordwah	0.529				0.262	0.241	0.026
8	Pakpattan	1.066	Doab wise detail given			Nil	Nil	1.066
9	L. Dipalpur	0.660	in Table 5			0.032	0.065	0.563
10	L.B.D.C.	1.972				0.031	0.347	1.594
11	Jhang	2.126				0.326	0.586	1.214
12	Gugera	1.846				0.377	0.347	1.122
13	U. Jhelum	0.711				0.362	0.188	0.161
14	L. Jhelum	1.740				0.647	0.593	0.500
15	Bahawal	0.856				Nil	0.518	0.338
16	Mailsi	1.076				0.052	0.164	0.860
17	Sidhnai	0.862				0.017	0.037	0.808
18	Havali	0.234				0.120	0.114	0.000
19	Rangpur	0.435				0.058	0.258	0.119
20	Panjnad	1.491				0.484	0.777	0.230
21	Abbasia	0.131				Nil	0.117	0.014
22	Thal	2.655				0.457	0.904	1.294
23	Muzaffargarh	0.845				0.376	0.419	0.050
24	D.G.Khan	1.009				0.320	0.239	0.450
	Sindh	3.219	0.544	1.646	1.029	2.041	0.542	0.636
	N.W.F.P.	1.191	0.110	0.139	0.942	0.129	0.212	0.850

Areas observed in Sind and NWFP are only 3.219 M.Ac and 1.191 M.Ac respectively against total areas of 14.221 M.Ac and 1.333 M.Ac of each province.

Source: Depth to Watertable Maps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E). Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 5
Doab–wise Depth to Watertable (1976)

Sr. No.	Canal Commands	Gross Area (M.Ac)	JUNE Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'
	Punjab	24.620	2.022	6.060	16.538
1	Rechna	5.994	0.319	1.436	4.239
2	Chaj	2.451	0.358	1.088	1.005
3	Bari	6.909	0.065	0.622	6.222
4	Bahawalpur	4.322	0.614	1.202	2.506
5	Thal	3.935	0.500	1.440	1.995
6	D.G.Khan	1.009	0.166	0.272	0.571

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 6
Canal Commandwise Depth to Watertable (1977)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	Total for Pakistan not worked out.			11.781	11.261	17.458
	Punjab	24.620	2.371	7.001	15.2	4.091	7.312	13.217
1	U. Dipalpur	0.451				0.043	0.205	0.203
2	Ravi Syphon	0.822				0.160	0.263	0.399
3	Raya Branch	0.453				0.060	0.121	0.272
4	U. Chenab	1.160				0.356	0.380	0.424
5	M.R. Link	0.175				0.039	0.104	0.032
6	Sadiqia	1.315				0.140	0.339	0.836
7	Fordwah	0.529				0.241	0.248	0.040
8	Pakpattan	1.066	Doab wise detail given			0.033	0.030	1.003
9	L. Dipalpur	0.660	in Table 7.			0.037	0.117	0.506
10	L.B.D.C.	1.972				0.022	0.284	1.666
11	Jhang	2.126				0.171	0.716	1.239
12	Gugera	1.846				0.094	0.376	1.376
13	U. Jhelum	0.711				0.210	0.364	0.137
14	L. Jhelum	1.740				0.426	0.495	0.819
15	Bahawal	0.856				0.090	0.657	0.109
16	Mailsi	1.076				Nil	0.077	0.999
17	Sidhnai	0.862				Nil	0.046	0.816
18	Havali	0.234				0.058	0.079	0.097
19	Rangpur	0.435				0.085	0.132	0.218
20	Panjnad	1.491				0.513	0.763	0.215
21	Abbasia	0.131				0.032	0.099	0.000
22	Thal	2.655				0.369	0.901	1.385
23	Muzaffargarh	0.845				0.540	0.295	0.010
24	D.G.Khan	1.009				0.372	0.221	0.416
	Sindh & Balochistan	14.547	0.709	2.147	0.947	7.555	3.736	3.256
25	Pat*	0.087				0.087	Nil	0.000
26	Desert	0.362				0.362	Nil	0.000
27	Begari	1.082				1.077	0.005	0.000
28	Ghotki	0.960				0.513	0.269	0.178
29	N. West	1.263				1.232	0.031	0.000
30	Rice	0.595				0.581	0.014	0.000
31	Dadu	0.631	Total area observed in Sind/			0.262	0.260	0.109
32	Khairpur East	0.572	Baluchistan is only 3.803 M.Ac			0.159	0.286	0.127
33	Khairpur West	0.298	against Total Gross Area of			0.068	0.115	0.115
34	Rohri	2.952	14.547 M.Ac.			0.204	1.016	1.732
35	Nara	2.514				0.354	1.224	0.936
36	Kalri	0.687				0.662	0.025	0.000
37	Lined Channel	0.582				0.541	0.041	0.000
38	Fuleli	1.010				0.930	0.075	0.005
39	Pinyari	0.952				0.523	0.375	0.054
	N.W.F.P.	1.333	0.079	0.298	0.814	0.135	0.213	0.985
40	U. Swat & Pehur	0.440				0.020	0.068	0.352
41	L. Swat	0.210				0.064	0.067	0.079
42	Warsak	0.124	Total area observed in NWFP is			0.003	0.006	0.115
43	Kabul	0.103	only 1.191 M.Ac against Total			0.019	0.036	0.048
44	Paharpur	0.109	Gross Area of 1.333 M.Ac.			0.027	0.018	0.064
45	Bannu	0.347				0.002	0.018	0.327

* Gross Area is 0.80 M.Ac but observed area is 0.087 M.Ac.

Source: Depth to water tableMaps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E). Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 7
Doab–wise Depth to Watertable (1977)

Sr. No.	Canal Commands	Gross Area (M.Ac)	JUNE Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'
	Punjab	24.620	2.597	6.775	15.248
1	Rechna	5.994	0.208	1.749	4.037
2	Chaj	2.451	0.341	1.008	1.102
3	Bari	6.909	0.121	0.783	6.005
4	Bahawalpur	4.322	1.067	1.530	1.725
5	Thal	3.935	0.601	1.422	1.912
6	D.G.Khan	1.009	0.259	0.283	0.467

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 8
Canal Commandwise Depth to Watertable (1978)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	4.837	16.014	19.649	13.506	11.367	15.627
	Punjab	24.620	2.093	7.613	14.914	5.227	7.066	12.327
1	U. Dipalpur	0.451	0.007	0.078	0.366	0.046	0.295	0.110
2	Ravi Syphon	0.822	0.040	0.188	0.594	0.196	0.229	0.397
3	Raya Branch	0.453	0.008	0.110	0.335	0.104	0.141	0.208
4	U. Chenab	1.160	0.045	0.363	0.752	0.424	0.397	0.339
5	M.R. Link	0.175	0.004	0.028	0.143	0.044	0.115	0.016
6	Sadiqia	1.315	0.091	0.290	0.934	0.193	0.218	0.904
7	Fordwah	0.529	0.166	0.251	0.112	0.393	0.115	0.021
8	Pakpattan	1.066	Nil	Nil	1.066	Nil	0.020	1.046
9	L. Dipalpur	0.660	0.022	0.134	0.504	0.043	0.068	0.549
10	L.B.D.C.	1.972	0.005	0.195	1.772	0.037	0.283	1.652
11	Jhang	2.126	0.033	0.841	1.252	0.251	0.721	1.154
12	Gugera	1.846	0.057	0.329	1.460	0.405	0.338	1.103
13	U. Jhelum	0.711	0.100	0.393	0.218	0.341	0.275	0.095
14	L. Jhelum	1.740	0.192	0.681	0.867	0.415	0.620	0.705
15	Bahawal	0.856	0.002	0.538	0.316	0.040	0.626	0.190
16	Mailsi	1.076	Nil	0.073	1.003	0.010	0.086	0.980
17	Sidhnai	0.862	Nil	0.020	0.842	0.009	0.037	0.816
18	Havali	0.234	0.094	0.128	0.012	0.034	0.176	0.024
19	Rangpur	0.435	Nil	0.148	0.287	0.092	0.160	0.183
20	Panjnad	1.491	0.320	0.904	0.267	0.525	0.811	0.155
21	Abbasia	0.131	Nil	0.131	0.000	0.038	0.082	0.011
22	Thal	2.655	0.325	0.978	1.352	0.636	0.740	1.279
23	Muzaffargarh	0.845	0.301	0.529	0.015	0.560	0.271	0.014
24	D.G.Khan	1.009	0.281	0.283	0.445	0.391	0.242	0.376
	Sindh & Balochistan	14.547	2.67	8.08	3.797	8.113	3.946	2.488
25	Pat*	0.087	0.017	0.07	0.000	0.087	Nil	0.000
26	Desert	0.362	0.03	0.29	0.042	0.362	Nil	0.000
27	Begari	1.082	0.056	0.973	0.053	1.076	0.006	0.000
28	Ghotki	0.960	0.1	0.718	0.142	0.452	0.344	0.164
29	N. West	1.263	1.077	0.186	0.000	1.256	0.007	0.000
30	Rice	0.595	0.054	0.517	0.024	0.595	Nil	0.000
31	Dadu	0.631	0.138	0.367	0.126	0.422	0.133	0.076
32	Khairpur East	0.572	0.125	0.306	0.141	0.182	0.256	0.134
33	Khairpur West	0.298	0.021	0.232	0.045	0.03	0.221	0.047
34	Rohri	2.952	0.247	0.924	1.781	0.426	1.248	1.278
35	Nara	2.514	0.51	1.234	0.770	0.81	1.03	0.674
36	Kalri	0.687	0.07	0.582	0.035	0.62	0.048	0.019
37	Lined Channel	0.582	0.169	0.329	0.084	0.532	0.05	0.000
38	Fuleli	1.010	0.042	0.71	0.258	0.623	0.301	0.086
39	Pinyari	0.952	0.014	0.642	0.296	0.64	0.302	0.010
	N.W.F.P.	1.333	0.074	0.321	0.938	0.166	0.355	0.812
40	U. Swat & Pehur	0.440	0.006	0.113	0.321	0.04	0.13	0.270
41	L. Swat	0.210	0.019	0.095	0.096	0.049	0.089	0.072
42	Warsak	0.124	0.003	0.008	0.113	0.007	0.023	0.094
43	Kabul	0.103	0.008	0.074	0.021	0.052	0.039	0.012
44	Paharpur	0.109	0.036	0.025	0.048	0.014	0.05	0.045
45	Bannu	0.347	0.002	0.006	0.339	0.004	0.024	0.319

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: Depth to water tableMaps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E). Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 9
Canal Commandwise Depth to Watertable (1979)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	6.156	15.933	18.411	12.215	11.635	16.650
	Punjab	24.620	3.011	7.634	13.975	4.352	7.509	12.759
1	U. Dipalpur	0.451	0.006	0.088	0.357	0.018	0.158	0.275
2	Ravi Syphon	0.822	0.044	0.219	0.559	0.150	0.220	0.452
3	Raya Branch	0.453	0.003	0.124	0.326	0.080	0.106	0.267
4	U. Chenab	1.160	0.054	0.453	0.653	0.157	0.485	0.518
5	M.R. Link	0.175	0.002	0.033	0.140	0.011	0.097	0.067
6	Sadiqia	1.315	0.195	0.301	0.819	0.178	0.352	0.785
7	Fordwah	0.529	0.195	0.276	0.058	0.307	0.178	0.044
8	Pakpattan	1.066	Nil	Nil	1.066	Nil	0.017	1.049
9	L. Dipalpur	0.660	0.039	0.146	0.475	0.048	0.123	0.489
10	L.B.D.C.	1.972	0.027	0.185	1.760	0.011	0.260	1.701
11	Jhang	2.126	0.137	0.949	1.040	0.250	0.832	1.044
12	Gugera	1.846	0.093	0.403	1.350	0.312	0.293	1.241
13	U. Jhelum	0.711	0.125	0.477	0.109	0.248	0.351	0.112
14	L. Jhelum	1.740	0.430	0.580	0.730	0.516	0.531	0.693
15	Bahawal	0.856	0.010	0.407	0.439	0.094	0.449	0.313
16	Mailsi	1.076	0.003	0.070	1.003	Nil	0.065	1.011
17	Sidhnai	0.862	0.005	0.015	0.842	Nil	0.065	0.797
18	Havali	0.234	0.025	0.171	0.038	0.040	0.144	0.050
19	Rangpur	0.435	0.063	0.168	0.204	0.121	0.111	0.203
20	Panjnad	1.491	0.286	0.857	0.348	0.485	0.883	0.123
21	Abbasia	0.131	0.013	0.118	0.000	0.017	0.114	0.000
22	Thal	2.655	0.540	0.901	1.214	0.506	1.041	1.108
23	Muzaffargarh	0.845	0.381	0.447	0.017	0.410	0.411	0.024
24	D.G.Khan	1.009	0.335	0.246	0.428	0.393	0.223	0.393
	Sindh & Balochistan	14.547	3.068	7.953	3.526	7.759	3.741	3.047
25	Pat*	0.087	0.025	0.060	0.002	0.087	Nil	0.000
26	Desert	0.362	0.104	0.251	0.007	0.362	Nil	0.000
27	Begari	1.082	0.115	0.954	0.013	1.071	0.011	0.000
28	Ghotki	0.960	0.229	0.555	0.176	0.634	0.194	0.132
29	N. West	1.263	0.816	0.418	0.029	1.252	0.011	0.000
30	Rice	0.595	0.109	0.472	0.014	0.591	0.004	0.000
31	Dadu	0.631	0.184	0.325	0.122	0.282	0.18	0.169
32	Khairpur East	0.572	0.135	0.317	0.120	0.094	0.303	0.175
33	Khairpur West	0.298	0.028	0.193	0.077	0.031	0.202	0.065
34	Rohri	2.952	0.377	1.116	1.459	0.221	1.071	1.660
35	Nara	2.514	0.67	1.065	0.779	0.377	1.344	0.793
36	Kalri	0.687	0.021	0.644	0.022	0.66	0.027	0.000
37	Lined Channel	0.582	0.223	0.311	0.048	0.447	0.082	0.053
38	Fuleli	1.010	0.009	0.701	0.300	0.879	0.131	0.000
39	Pinyari	0.952	0.023	0.571	0.358	0.771	0.181	0.000
	N.W.F.P.	1.333	0.077	0.346	0.910	0.104	0.385	0.844
40	U. Swat & Pehur	0.440	0.012	0.141	0.287	0.01	0.12	0.310
41	L. Swat	0.210	0.026	0.109	0.075	0.044	0.117	0.049
42	Warsak	0.124	0.004	0.006	0.114	0.01	0.017	0.097
43	Kabul	0.103	0.025	0.054	0.024	0.029	0.055	0.019
44	Paharpur	0.109	0.007	0.026	0.076	0.009	0.045	0.055
45	Bannu	0.347	0.003	0.010	0.334	0.002	0.031	0.314

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: Depth to water tableMaps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E). Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 10
Canal Commandwise Depth to Watertable (1980)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.223	16.027	19.25	12.21	11.37	16.92
	Punjab	24.620	2.200	7.320	15.10	4.266	7.631	12.72
1	U. Dipalpur	0.451	0.008	0.057	0.39	0.011	0.184	0.26
2	Ravi Syphon	0.822	0.020	0.217	0.59	0.215	0.189	0.42
3	Raya Branch	0.453	0.001	0.046	0.41	0.088	0.125	0.24
4	U. Chenab	1.160	0.019	0.380	0.76	0.347	0.380	0.43
5	M.R. Link	0.175	0.003	0.016	0.16	0.010	0.067	0.10
6	Sadiqia	1.315	0.123	0.428	0.76	0.131	0.376	0.81
7	Fordwah	0.529	0.176	0.230	0.12	0.298	0.196	0.03
8	Pakpattan	1.066	Nil	Nil	1.07	Nil	Nil	1.07
9	L. Dipalpur	0.660	0.037	0.140	0.48	0.045	0.124	0.49
10	L.B.D.C.	1.972	0.023	0.197	1.75	0.011	0.290	1.67
11	Jhang	2.126	0.076	0.702	1.35	0.278	0.855	0.99
12	Gugera	1.846	0.039	0.261	1.55	0.140	0.520	1.19
13	U. Jhelum	0.711	0.089	0.411	0.21	0.216	0.392	0.10
14	L. Jhelum	1.740	0.182	0.714	0.84	0.496	0.536	0.71
15	Bahawal	0.856	Nil	0.457	0.40	0.033	0.410	0.41
16	Mailsi	1.076	Nil	0.034	1.04	Nil	0.015	1.06
17	Sidhnai	0.862	Nil	Nil	0.86	Nil	0.052	0.81
18	Havali	0.234	0.018	0.149	0.07	0.043	0.138	0.05
19	Rangpur	0.435	0.029	0.166	0.24	0.050	0.172	0.21
20	Panjnad	1.491	0.202	1.067	0.22	0.505	0.920	0.07
21	Abbasia	0.131	0.004	0.071	0.06	0.010	0.121	0.00
22	Thal	2.655	0.489	0.842	1.32	0.454	0.987	1.21
23	Muzaffargarh	0.845	0.319	0.502	0.02	0.490	0.348	0.01
24	D.G.Khan	1.009	0.343	0.233	0.43	0.395	0.234	0.38
	Sindh & Balochistan	14.547	2.964	8.368	3.21	7.811	3.363	3.37
25	Pat*	0.087	0.021	0.066	–0.00	0.087	Nil	0.00
26	Desert	0.362	0.087	0.275	–0.00	0.362	Nil	0.00
27	Begari	1.082	0.107	0.954	0.02	1.079	0.003	0.00
28	Ghotki	0.960	0.209	0.58	0.17	0.64	0.19	0.13
29	N. West	1.263	0.819	0.44	0.00	1.257	0.006	0.00
30	Rice	0.595	0.03	0.553	0.01	0.592	0.003	0.00
31	Dadu	0.631	0.128	0.342	0.16	0.266	0.172	0.19
32	Khairpur East	0.572	0.11	0.313	0.15	0.089	0.358	0.13
33	Khairpur West	0.298	0.022	0.221	0.05	0.055	0.175	0.07
34	Rohri	2.952	0.232	1.145	1.58	0.09	1.004	1.86
35	Nara	2.514	0.509	1.159	0.85	0.249	1.276	0.99
36	Kalri	0.687	0.488	0.199	0.00	0.66	0.027	0.00
37	Lined Channel	0.582	0.173	0.375	0.03	0.536	0.036	0.01
38	Fuleli	1.010	0.019	0.956	0.04	0.976	0.034	0.00
39	Pinyari	0.952	0.01	0.79	0.15	0.873	0.079	0.00
	N.W.F.P.	1.333	0.059	0.339	0.93	0.136	0.373	0.82
40	U. Swat & Pehur	0.440	0.006	0.088	0.35	0.018	0.154	0.27
41	L. Swat	0.210	0.019	0.139	0.05	0.052	0.108	0.05
42	Warsak	0.124	0.002	0.003	0.12	0.005	0.007	0.11
43	Kabul	0.103	0.021	0.059	0.02	0.039	0.058	0.01
44	Paharpur	0.109	0.006	0.034	0.07	0.014	0.029	0.07
45	Bannu	0.347	0.005	0.016	0.33	0.008	0.017	0.32

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: Depth to water tableMaps supplied by SCARP Monitoring (S.M), WAPDA, (Ex. CMO/M&E).
Plannimetering partially done by S.M and partially by Drainage Section of Planning Division, WAPDA.

TABLE – 11
Canal Commandwise Depth to Watertable (1981)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.151	17.168	18.181	11.771	11.730	16.993
	Punjab	24.620	2.365	7.954	14.301	3.680	8.072	12.868
1	U. Dipalpur	0.451	0.015	0.070	0.366	0.016	0.180	0.255
2	Ravi Syphon	0.822	0.038	0.200	0.584	0.147	0.226	0.449
3	Raya Branch	0.453	0.014	0.147	0.292	0.056	0.138	0.259
4	U. Chenab	1.160	0.059	0.513	0.588	0.144	0.480	0.536
5	M.R. Link	0.175	0.002	0.036	0.137	0.009	0.062	0.104
6	Sadiqia	1.315	0.153	0.447	0.715	0.168	0.418	0.729
7	Fordwah	0.529	0.180	0.223	0.126	0.250	0.175	0.104
8	Pakpattan	1.066	Nil	0.007	1.059	Nil	0.023	1.043
9	L. Dipalpur	0.660	0.036	0.138	0.486	0.050	0.126	0.484
10	L.B.D.C.	1.972	0.009	0.191	1.772	0.013	0.313	1.646
11	Jhang	2.126	0.141	0.947	1.038	0.265	0.957	0.904
12	Gugera	1.846	0.062	0.425	1.359	0.233	0.516	1.097
13	U. Jhelum	0.711	0.073	0.439	0.199	0.213	0.371	0.127
14	L. Jhelum	1.740	0.190	0.743	0.807	0.487	0.574	0.679
15	Bahawal	0.856	0.002	0.409	0.445	0.048	0.348	0.460
16	Mailsi	1.076	Nil	0.013	1.063	Nil	0.020	1.056
17	Sidhnai	0.862	Nil	Nil	0.862	Nil	0.059	0.803
18	Havali	0.234	0.009	0.177	0.048	0.042	0.151	0.041
19	Rangpur	0.435	0.031	0.176	0.228	0.059	0.160	0.216
20	Panjnad	1.491	0.225	0.989	0.277	0.368	0.979	0.144
21	Abbasia	0.131	0.004	0.081	0.046	0.034	0.090	0.007
22	Thal	2.655	0.445	0.844	1.366	0.294	1.044	1.317
23	Muzaffargarh	0.845	0.335	0.496	0.014	0.399	0.431	0.015
24	D.G.Khan	1.009	0.342	0.243	0.424	0.385	0.231	0.393
	Sindh & Balochistan	14.547	2.679	8.863	3.005	7.966	3.292	3.289
25	Pat*	0.087	0.012	0.075	0.000	0.087	Nil	0.000
26	Desert	0.362	0.047	0.315	0.000	0.362	Nil	0.000
27	Begari	1.082	0.094	0.969	0.019	1.081	0.001	0.000
28	Ghotki	0.960	0.274	0.518	0.168	0.617	0.215	0.128
29	N. West	1.263	0.83	0.429	0.004	1.262	0.001	0.000
30	Rice	0.595	0.02	0.566	0.009	0.595	Nil	0.000
31	Dadu	0.631	0.129	0.335	0.167	0.297	0.149	0.185
32	Khairpur East	0.572	0.149	0.3	0.123	0.098	0.329	0.145
33	Khairpur West	0.298	0.051	0.196	0.051	0.088	0.169	0.041
34	Rohri	2.952	0.285	1.247	1.420	0.18	1.052	1.720
35	Nara	2.514	0.479	1.197	0.838	0.294	1.152	1.068
36	Kalri	0.687	0.089	0.598	0.000	0.632	0.055	0.000
37	Lined Channel	0.582	0.166	0.375	0.041	0.532	0.048	0.002
38	Fuleli	1.010	0.025	0.953	0.032	0.975	0.035	0.000
39	Pinyari	0.952	0.029	0.79	0.133	0.866	0.086	0.000
	N.W.F.P.	1.333	0.107	0.351	0.875	0.131	0.366	0.836
40	U. Swat & Pehur	0.440	0.016	0.144	0.280	0.014	0.159	0.267
41	L. Swat	0.210	0.044	0.091	0.075	0.059	0.093	0.058
42	Warsak	0.124	0.005	0.005	0.114	0.006	0.005	0.113
43	Kabul	0.103	0.029	0.052	0.022	0.038	0.046	0.019
44	Paharpur	0.109	0.006	0.043	0.060	0.007	0.044	0.058
45	Bannu	0.347	0.007	0.016	0.324	0.007	0.019	0.321

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 12
Canal Commandwise Depth to Watertable (1982)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.478	17.525	17.50	11.161	11.844	17.50
	Punjab	24.620	2.800	8.139	13.68	3.345	7.834	13.44
1	U. Dipalpur	0.451	0.015	0.088	0.35	0.013	0.202	0.24
2	Ravi Syphon	0.822	0.090	0.220	0.51	0.127	0.212	0.48
3	Raya Branch	0.453	0.015	0.128	0.31	0.027	0.109	0.32
4	U. Chenab	1.160	0.091	0.441	0.63	0.099	0.346	0.72
5	M.R. Link	0.175	0.002	0.076	0.10	0.009	0.055	0.11
6	Sadiqia	1.315	0.182	0.488	0.65	0.159	0.469	0.69
7	Fordwah	0.529	0.177	0.212	0.14	0.253	0.182	0.09
8	Pakpattan	1.066	Nil	0.004	1.06	Nil	0.028	1.04
9	L. Dipalpur	0.660	0.041	0.141	0.48	0.050	0.116	0.49
10	L.B.D.C.	1.972	0.005	0.240	1.73	0.013	0.299	1.66
11	Jhang	2.126	0.288	0.896	0.94	0.202	0.949	0.98
12	Gugera	1.846	0.209	0.424	1.21	0.155	0.325	1.37
13	U. Jhelum	0.711	0.150	0.388	0.17	0.168	0.397	0.15
14	L. Jhelum	1.740	0.317	0.714	0.71	0.410	0.641	0.69
15	Bahawal	0.856	Nil	0.398	0.46	0.036	0.324	0.50
16	Mailsi	1.076	Nil	0.017	1.06	Nil	0.038	1.04
17	Sidhnai	0.862	Nil	0.002	0.86	Nil	0.059	0.80
18	Havali	0.234	0.017	0.162	0.06	0.042	0.151	0.04
19	Rangpur	0.435	0.017	0.187	0.23	0.040	0.175	0.22
20	Panjnad	1.491	0.255	1.063	0.17	0.435	0.915	0.14
21	Abbasia	0.131	0.032	0.095	0.00	0.041	0.090	0.00
22	Thal	2.655	0.326	0.968	1.36	0.303	1.120	1.23
23	Muzaffargarh	0.845	0.214	0.549	0.08	0.355	0.432	0.06
24	D.G.Khan	1.009	0.357	0.238	0.41	0.408	0.200	0.40
	Sindh & Balochistan	14.547	2.587	9.013	2.95	7.691	3.669	3.19
25	Pat*	0.087	0.016	0.071	0.00	0.087	Nil	0.00
26	Desert	0.362	0.063	0.299	0.00	0.362	Nil	0.00
27	Begari	1.082	0.114	0.953	0.02	1.081	0.001	0.00
28	Ghotki	0.960	0.072	0.748	0.14	0.676	0.173	0.11
29	N. West	1.263	0.808	0.442	0.01	1.259	0.004	0.00
30	Rice	0.595	0.06	0.526	0.01	0.593	0.002	0.00
31	Dadu	0.631	0.136	0.328	0.17	0.284	0.164	0.18
32	Khairpur East	0.572	0.185	0.263	0.12	0.102	0.315	0.16
33	Khairpur West	0.298	0.132	0.135	0.03	0.092	0.153	0.05
34	Rohri	2.952	0.218	1.242	1.49	0.147	1.143	1.66
35	Nara	2.514	0.413	1.204	0.90	0.41	1.094	1.01
36	Kalri	0.687	0.079	0.607	0.00	0.588	0.099	0.00
37	Lined Channel	0.582	0.213	0.361	0.01	0.479	0.096	0.01
38	Fuleli	1.010	0.018	0.99	0.00	0.796	0.212	0.00
39	Pinyari	0.952	0.06	0.844	0.05	0.735	0.213	0.00
	N.W.F.P.	1.333	0.091	0.373	0.87	0.125	0.341	0.87
40	U. Swat & Pehur	0.440	0.008	0.142	0.29	0.012	0.149	0.28
41	L. Swat	0.210	0.036	0.11	0.06	0.059	0.086	0.07
42	Warsak	0.124	0.003	0.007	0.11	0.006	0.007	0.11
43	Kabul	0.103	0.024	0.055	0.02	0.037	0.048	0.02
44	Paharpur	0.109	0.004	0.04	0.07	0.007	0.037	0.07
45	Bannu	0.347	0.016	0.019	0.31	0.004	0.014	0.33

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 13
Canal Commandwise Depth to Watertable (1983)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.480	16.966	18.054	13.595	11.160	15.745
	Punjab	24.620	2.617	7.880	14.123	4.742	7.670	12.208
1	U. Dipalpur	0.451	0.012	0.069	0.370	0.024	0.198	0.229
2	Ravi Syphon	0.822	0.122	0.168	0.532	0.156	0.230	0.436
3	Raya Branch	0.453	0.022	0.121	0.310	0.077	0.114	0.262
4	U. Chenab	1.160	0.102	0.356	0.703	0.175	0.479	0.506
5	M.R. Link	0.175	0.007	0.084	0.084	0.099	0.062	0.014
6	Sadiqia	1.315	0.267	0.398	0.650	0.375	0.259	0.681
7	Fordwah	0.529	0.186	0.167	0.176	0.309	0.153	0.067
8	Pakpattan	1.066	Nil	0.003	1.063	Nil	0.052	1.014
9	L. Dipalpur	0.660	0.032	0.153	0.475	0.067	0.160	0.433
10	L.B.D.C.	1.972	0.007	0.185	1.780	0.062	0.415	1.495
11	Jhang	2.126	0.170	0.993	0.963	0.378	0.909	0.839
12	Gugera	1.846	0.035	0.457	1.354	0.222	0.427	1.197
13	U. Jhelum	0.711	0.119	0.467	0.125	0.255	0.361	0.095
14	L. Jhelum	1.740	0.385	0.685	0.670	0.746	0.430	0.564
15	Bahawal	0.856	Nil	0.363	0.493	0.035	0.343	0.478
16	Mailsi	1.076	Nil	0.022	1.054	Nil	0.025	1.051
17	Sidhnai	0.862	Nil	0.005	0.857	Nil	0.062	0.800
18	Havali	0.234	0.032	0.173	0.029	0.084	0.129	0.021
19	Rangpur	0.435	0.022	0.173	0.240	0.101	0.124	0.210
20	Panjnad	1.491	0.311	0.981	0.199	0.524	0.811	0.156
21	Abbasia	0.131	0.032	0.092	0.007	0.035	0.096	0.000
22	Thal	2.655	0.314	0.981	1.360	0.240	1.205	1.210
23	Muzaffargarh	0.845	0.237	0.524	0.084	0.353	0.440	0.052
24	D.G.Khan	1.009	0.203	0.260	0.546	0.425	0.186	0.398
	Sindh & Balochistan	14.547	2.735	8.716	3.096	8.681	3.174	2.692
25	Pat*	0.087	0.009	0.078	0.000	0.087	Nil	0.000
26	Desert	0.362	0.039	0.322	0.001	0.362	Nil	0.000
27	Begari	1.082	0.068	0.988	0.026	1.08	0.002	0.000
28	Ghotki	0.960	0.042	0.763	0.155	0.774	0.109	0.077
29	N. West	1.263	0.823	0.439	0.001	1.263	Nil	0.000
30	Rice	0.595	0.045	0.539	0.011	0.59	0.005	0.000
31	Dadu	0.631	0.146	0.292	0.193	0.357	0.146	0.128
32	Khairpur East	0.572	0.161	0.283	0.128	0.144	0.314	0.114
33	Khairpur West	0.298	0.091	0.182	0.025	0.103	0.153	0.042
34	Rohri	2.952	0.298	1.198	1.456	0.268	1.245	1.439
35	Nara	2.514	0.608	0.993	0.913	0.713	0.909	0.892
36	Kalri	0.687	0.139	0.532	0.016	0.657	0.03	0.000
37	Lined Channel	0.582	0.218	0.354	0.010	0.533	0.049	0.000
38	Fuleli	1.010	0.026	0.944	0.040	0.914	0.096	0.000
39	Pinyari	0.952	0.022	0.809	0.121	0.836	0.116	0.000
	N.W.F.P.	1.333	0.128	0.370	0.835	0.172	0.316	0.845
40	U. Swat & Pehur	0.440	0.013	0.141	0.286	0.016	0.13	0.294
41	L. Swat	0.210	0.052	0.101	0.057	0.077	0.077	0.056
42	Warsak	0.124	0.005	0.007	0.112	0.005	0.008	0.111
43	Kabul	0.103	0.033	0.053	0.017	0.042	0.047	0.014
44	Paharpur	0.109	0.011	0.034	0.064	0.022	0.025	0.062
45	Bannu	0.347	0.014	0.034	0.299	0.01	0.029	0.308

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 14
Canal Commandwise Depth to Watertable (1984)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.062	16.909	18.529	13.023	11.980	15.497
	Punjab	24.620	1.945	7.692	14.983	4.255	8.108	12.257
1	U. Dipalpur	0.451	0.007	0.086	0.358	0.040	0.180	0.231
2	Ravi Syphon	0.822	0.124	0.170	0.528	0.150	0.242	0.430
3	Raya Branch	0.453	0.002	0.054	0.397	0.101	0.099	0.253
4	U. Chenab	1.160	0.025	0.198	0.937	0.198	0.487	0.475
5	M.R. Link	0.175	Nil	0.044	0.131	0.077	0.079	0.019
6	Sadiqia	1.315	0.213	0.462	0.640	0.314	0.306	0.695
7	Fordwah	0.529	0.143	0.255	0.131	0.262	0.183	0.084
8	Pakpattan	1.066	Nil	0.027	1.039	Nil	0.067	0.999
9	L. Dipalpur	0.660	0.035	0.106	0.519	0.057	0.121	0.482
10	L.B.D.C.	1.972	0.015	0.242	1.715	0.035	0.385	1.552
11	Jhang	2.126	0.062	0.754	1.310	0.262	0.932	0.932
12	Gugera	1.846	0.035	0.388	1.423	0.242	0.430	1.174
13	U. Jhelum	0.711	0.054	0.423	0.234	0.200	0.413	0.098
14	L. Jhelum	1.740	0.126	0.870	0.744	0.588	0.546	0.606
15	Bahawal	0.856	0.002	0.371	0.483	0.032	0.365	0.459
16	Mailsi	1.076	Nil	0.007	1.069	Nil	0.035	1.041
17	Sidhnai	0.862	Nil	Nil	0.862	Nil	0.069	0.793
18	Havali	0.234	0.037	0.160	0.037	0.054	0.143	0.037
19	Rangpur	0.435	0.020	0.138	0.277	0.062	0.168	0.206
20	Panjnad	1.491	0.353	0.967	0.171	0.482	0.881	0.128
21	Abbasia	0.131	0.007	0.124	0.000	0.027	0.104	0.000
22	Thal	2.655	0.334	0.983	1.338	0.341	1.203	1.111
23	Muzaffargarh	0.845	0.153	0.529	0.163	0.336	0.465	0.044
24	D.G.Khan	1.009	0.198	0.334	0.477	0.395	0.205	0.409
	Sindh & Balochistan	14.547	2.99	8.924	2.633	8.593	3.552	2.402
25	Pat*	0.087	0.022	0.065	0.000	0.087	Nil	0.000
26	Desert	0.362	0.077	0.282	0.003	0.362	Nil	0.000
27	Begari	1.082	0.074	0.990	0.018	1.08	0.002	0.000
28	Ghotki	0.960	0.072	0.756	0.132	0.6	0.269	0.091
29	N. West	1.263	0.662	0.601	0.000	1.263	Nil	0.000
30	Rice	0.595	0.062	0.506	0.027	0.595	Nil	0.000
31	Dadu	0.631	0.158	0.326	0.147	0.336	0.164	0.131
32	Khairpur East	0.572	0.148	0.275	0.149	0.175	0.293	0.104
33	Khairpur West	0.298	0.104	0.166	0.028	0.11	0.147	0.041
34	Rohri	2.952	0.351	1.300	1.301	0.306	1.282	1.364
35	Nara	2.514	0.675	1.055	0.784	0.717	1.132	0.665
36	Kalri	0.687	0.049	0.633	0.005	0.657	0.03	0.000
37	Lined Channel	0.582	0.336	0.244	0.002	0.514	0.067	0.001
38	Fuleli	1.010	0.143	0.867	0.000	0.963	0.047	0.000
39	Pinyari	0.952	0.057	0.858	0.037	0.828	0.119	0.005
	N.W.F.P.	1.333	0.127	0.293	0.913	0.175	0.32	0.838
40	U. Swat & Pehur	0.440	0.005	0.108	0.327	0.019	0.133	0.288
41	L. Swat	0.210	0.049	0.077	0.084	0.077	0.077	0.056
42	Warsak	0.124	0.004	0.007	0.113	0.005	0.009	0.110
43	Kabul	0.103	0.037	0.048	0.018	0.042	0.046	0.015
44	Paharpur	0.109	0.022	0.024	0.063	0.022	0.025	0.062
45	Bannu	0.347	0.01	0.029	0.308	0.01	0.03	0.307

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 15
Canal Commandwise Depth to Watertable (1985)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	4.454	16.040	20.006	11.779	11.891	16.830
	Punjab	24.620	1.217	6.990	16.413	2.982	8.083	13.555
1	U. Dipalpur	0.451	0.012	0.057	0.382	0.035	0.135	0.281
2	Ravi Syphon	0.822	0.020	0.235	0.567	0.185	0.235	0.402
3	Raya Branch	0.453	Nil	0.025	0.428	0.047	0.124	0.282
4	U. Chenab	1.160	0.012	0.205	0.943	0.158	0.425	0.577
5	M.R. Link	0.175	Nil	0.039	0.136	0.042	0.094	0.039
6	Sadiqia	1.315	0.183	0.474	0.658	0.191	0.460	0.664
7	Fordwah	0.529	0.142	0.178	0.209	0.208	0.188	0.133
8	Pakpattan	1.066	Nil	0.020	1.046	0.001	0.013	1.052
9	L. Dipalpur	0.660	0.027	0.116	0.517	0.037	0.094	0.529
10	L.B.D.C.	1.972	0.005	0.256	1.711	0.020	0.398	1.554
11	Jhang	2.126	0.043	0.598	1.485	0.095	0.857	1.174
12	Gugera	1.846	0.032	0.300	1.514	0.043	0.395	1.408
13	U. Jhelum	0.711	0.042	0.399	0.270	0.178	0.390	0.143
14	L. Jhelum	1.740	0.135	0.777	0.830	0.368	0.561	0.811
15	Bahawal	0.856	0.002	0.318	0.536	0.011	0.333	0.512
16	Mailsi	1.076	Nil	Nil	1.076	Nil	0.005	1.071
17	Sidhnai	0.862	Nil	Nil	0.862	Nil	0.064	0.798
18	Havali	0.234	0.014	0.148	0.072	0.017	0.146	0.071
19	Rangpur	0.435	0.005	0.084	0.346	0.057	0.139	0.239
20	Panjnad	1.491	0.227	1.048	0.216	0.319	0.991	0.181
21	Abbasia	0.131	0.002	0.104	0.025	0.002	0.121	0.008
22	Thal	2.655	0.212	0.761	1.682	0.353	1.176	1.126
23	Muzaffargarh	0.845	0.072	0.334	0.439	0.262	0.462	0.121
24	D.G.Khan	1.009	0.030	0.514	0.465	0.353	0.277	0.379
	Sindh & Balochistan	14.547	3.141	8.727	2.679	8.658	3.47	2.419
25	Pat*	0.087	0.027	0.06	0.000	0.087	Nil	0.000
26	Desert	0.362	0.091	0.269	0.002	0.362	Nil	0.000
27	Begari	1.082	0.114	0.948	0.020	1.082	Nil	0.000
28	Ghotki	0.960	0.193	0.648	0.119	0.696	0.188	0.076
29	N. West	1.263	0.801	0.46	0.002	1.26	0.003	0.000
30	Rice	0.595	0.034	0.526	0.035	0.595	Nil	0.000
31	Dadu	0.631	0.161	0.304	0.166	0.316	0.166	0.149
32	Khairpur East	0.572	0.146	0.252	0.174	0.213	0.262	0.097
33	Khairpur West	0.298	0.106	0.166	0.026	0.106	0.168	0.024
34	Rohri	2.952	0.331	1.275	1.346	0.351	1.265	1.336
35	Nara	2.514	0.808	1.075	0.631	0.603	1.174	0.737
36	Kalri	0.687	0.002	0.672	0.013	0.63	0.057	0.000
37	Lined Channel	0.582	0.297	0.283	0.002	0.523	0.059	0.000
38	Fuleli	1.010	0.027	0.973	0.010	0.991	0.019	0.000
39	Pinyari	0.952	0.003	0.816	0.133	0.843	0.109	0.000
	N.W.F.P.	1.333	0.096	0.323	0.914	0.139	0.338	0.856
40	U. Swat & Pehur	0.440	0.005	0.101	0.334	0.007	0.143	0.290
41	L. Swat	0.219	0.047	0.084	0.079	0.069	0.072	0.069
42	Warsak	0.124	Nil	0.01	0.114	0.003	0.01	0.111
43	Kabul	0.103	0.031	0.054	0.018	0.032	0.054	0.017
44	Paharpur	0.109	0.008	0.042	0.059	0.018	0.025	0.066
45	Bannu	0.347	0.005	0.032	0.310	0.01	0.034	0.303

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 16
Canal Commandwise Depth to Watertable (1986)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	40.500	5.191	15.171	20.138	12.131	11.291	17.078
	Punjab	24.620	1.804	6.739	16.077	3.194	7.623	13.803
1	U. Dipalpur	0.451	0.008	0.062	0.38	0.027	0.101	0.323
2	Ravi Syphon	0.822	0.017	0.259	0.55	0.116	0.224	0.482
3	Raya Branch	0.453	0.001	0.064	0.39	0.104	0.124	0.225
4	U. Chenab	1.160	0.037	0.264	0.86	0.123	0.390	0.647
5	M.R. Link	0.175	0.002	0.076	0.10	0.054	0.091	0.030
6	Sadiqia	1.315	0.183	0.467	0.67	0.354	0.319	0.642
7	Fordwah	0.529	0.165	0.198	0.17	0.245	0.153	0.124
8	Pakpattan	1.066	–	0.018	1.05	–	0.013	1.053
9	L. Dipalpur	0.660	0.025	0.121	0.51	0.030	0.099	0.53i
10	L.B.D.C.	1.972	–	0.272	1.70	0.017	0.450	1.505
11	Jhang	2.126	0.101	0.070	1.95	0.124	0.806	1.196
12	Gugera	1.846	0.036	0.326	1.48	0.045	0.348	1.453
13	U. Jhelum	0.711	0.008	0.390	0.31	0.141	0.378	0.192
14	L. Jhelum	1.740	0.200	0.700	0.84	0.337	0.568	0.835
15	Bahawal	0.856	–	0.355	0.50	0.022	0.338	0.496
16	Mailsi	1.076	–	–	1.08	–	–	1.076
17	Sidhnai	0.862	–	–	0.86	–	–	0.862
18	Havali	0.234	0.017	0.158	0.06	0.017	0.145	0.072
19	Rangpur	0.435	0.025	0.117	0.29	0.096	0.126	0.213
20	Panjnad	1.491	0.220	1.039	0.23	0.252	1.077	0.162
21	Abbasia	0.131	0.008	0.101	0.02	0.015	0.111	0.005
22	Thal	2.655	0.299	0.926	1.43	0.296	1.104	1.255
23	Muzaffargarh	0.845	0.180	0.407	0.26	0.398	0.398	0.049
24	D.G.Khan	1.009	0.272	0.349	0.39	0.381	0.260	0.368
	Sindh & Balochistan	14.547	3.289	8.129	3.129	8.787	3.394	2.366
25	Pat*	0.087	0.02	0.067	0.00	0.087	–	0.000
26	Desert	0.362	0.128	0.234	0.00	0.362	–	0.000
27	Begari	1.082	0.139	0.93	0.01	1.082	–	0.000
28	Ghotki	0.960	0.225	0.639	0.10	0.782	0.166	0.012
29	N. West	1.263	0.74	0.5	0.02	1.245	0.018	0.000
30	Rice	0.595	0.038	0.532	0.03	0.595	–	0.000
31	Dadu	0.631	0.137	0.326	0.17	0.366	0.111	0.154
32	Khairpur East	0.572	0.204	0.278	0.09	0.237	0.25	0.085
33	Khairpur West	0.298	0.085	0.17	0.04	0.123	0.146	0.029
34	Rohri	2.952	0.406	1.268	1.28	0.393	1.23	1.329
35	Nara	2.514	1.037	0.843	0.63	0.531	1.231	0.752
36	Kalri	0.687	0.014	0.067	0.61	0.583	0.104	0.000
37	Lined Channel	0.582	0.101	0.481	0.00	0.521	0.056	0.005
38	Fuleli	1.010	0.01	0.972	0.03	0.983	0.027	0.000
39	Pinyari	0.952	0.005	0.822	0.13	0.897	0.055	0.000
	N.W.F.P.	1.333	0.098	0.303	0.932	0.15	0.274	0.909
40	U. Swat & Pehur	0.440	0.005	0.101	0.33	0.005	0.091	0.344
41	L. Swat	0.210	0.044	0.034	0.13	0.062	0.081	0.067
42	Warsak	0.124	0.008	0.008	0.11	0.006	0.007	0.111
43	Kabul	0.103	0.03	0.059	0.01	0.036	0.05	0.017
44	Paharpur	0.109	0.006	0.07	0.03	0.028	0.019	0.062
45	Bannu	0.347	0.005	0.031	0.31	0.013	0.026	0.308

* Gross Area is 0.808 M.Ac but observed area is 0.087 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 17
Canal Commandwise Depth to Watertable (1987)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	41.113	5.383	16.979	18.750	10.526	11.040	19.545
	Punjab	24.620	2.375	7.349	14.895	2.328	6.958	15.333
1	U. Dipalpur	0.451	0.007	0.042	0.402	0.010	0.040	0.401
2	Ravi Syphon	0.822	0.022	0.242	0.558	0.064	0.306	0.452
3	Raya Branch	0.453	0.002	0.035	0.416	0.022	0.109	0.322
4	U. Chenab	1.160	0.042	0.259	0.859	0.084	0.331	0.745
5	M.R. Link	0.175	0.005	0.079	0.091	0.025	0.104	0.046
6	Sadiqia	1.315	0.314	0.376	0.625	0.215	0.363	0.737
7	Fordwah	0.529	0.151	0.240	0.138	0.196	0.143	0.190
8	Pakpattan	1.066	–	0.013	1.053	–	0.001	1.065
9	L. Dipalpur	0.660	0.037	0.086	0.537	0.022	0.096	0.542
10	L.B.D.C.	1.972	–	0.215	1.757	0.012	0.341	1.619
11	Jhang	2.126	0.104	0.653	1.369	0.047	0.519	1.560
12	Gugera	1.846	0.072	0.353	1.421	0.047	0.321	1.478
13	U. Jhelum	0.711	0.071	0.336	0.304	0.052	0.375	0.284
14	L. Jhelum	1.740	0.234	0.617	0.889	0.206	0.509	1.025
15	Bahawal	0.856	0.002	0.383	0.471	0.022	0.331	0.503
16	Mailsi	1.076	–	–	1.076	–	–	1.076
17	Sidhnai	0.862	–	–	0.861	–	0.037	0.824
18	Havali	0.234	0.023	0.141	0.070	0.017	0.146	0.071
19	Rangpur	0.435	0.047	0.183	0.205	0.082	0.132	0.221
20	Panjnad	1.491	0.306	0.969	0.216	0.031	0.975	0.485
21	Abbasia	0.131	0.020	0.111	0.000	0.005	0.111	0.015
22	Thal	2.655	0.259	1.245	1.151	0.326	1.067	1.262
23	Muzaffargarh	0.845	0.316	0.469	0.060	0.477	0.329	0.039
24	D.G.Khan	1.009	0.341	0.302	0.366	0.366	0.272	0.371
	Sindh & Balochistan	15.159	2.904	9.295	2.960	8.067	3.781	3.311
25	Pat*	0.699	0.159	0.234	0.306	0.217	0.121	0.361
26	Desert	0.362	0.077	0.274	0.011	0.334	0.016	0.012
27	Begari	1.082	0.070	0.975	0.037	1.068	0.014	0.000
28	Ghotki	0.960	0.119	0.757	0.084	0.641	0.306	0.013
29	N. West	1.263	0.548	0.646	0.069	1.139	0.070	0.054
30	Rice	0.595	0.041	0.497	0.057	0.590	0.005	0.000
31	Dadu	0.631	0.128	0.349	0.154	0.315	0.146	0.170
32	Khairpur East	0.572	0.185	0.274	0.113	0.213	0.242	0.117
33	Khairpur West	0.298	0.100	0.175	0.023	0.105	0.158	0.034
34	Rohri	2.952	0.496	1.254	1.202	0.386	0.963	1.603
35	Nara	2.514	0.699	1.038	0.776	0.404	1.179	0.931
36	Kalri	0.687	0.009	0.658	0.019	0.588	0.098	0.000
37	Lined Channel	0.582	0.191	0.391	0.000	0.378	0.194	0.010
38	Fuleli	1.010	0.072	0.937	0.001	0.893	0.114	0.004
39	Pinyari	0.952	0.010	0.836	0.106	0.797	0.153	0.002
	N.W.F.P.	1.333	0.103	0.334	0.895	0.130	0.302	0.901
40	U. Swat & Pehur	0.440	0.004	0.102	0.334	0.004	0.089	0.346
41	L. Swat	0.210	0.037	0.104	0.069	0.041	0.104	0.065
42	Warsak	0.124	0.005	0.009	0.109	0.004	0.009	0.110
43	Kabul	0.103	0.034	0.055	0.015	0.034	0.052	0.017
44	Paharpur	0.109	0.019	0.032	0.059	0.037	0.019	0.053
45	Bannu	0.347	0.004	0.032	0.310	0.010	0.028	0.309

* Gross Area is 0.808 M.Ac but observed area is 0.699 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 18
Canal Commandwise Depth to Watertable (1988)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	41.113	3.706	15.502	21.905	11.782	9.508	19.822
	Punjab	24.620	1.346	5.622	17.652	2.908	6.830	14.883
1	U. Dipalpur	0.451	0.005	0.022	0.424	0.011	0.065	0.375
2	Ravi Syphon	0.822	0.000	0.104	0.718	0.068	0.309	0.446
3	Raya Branch	0.453	0.000	0.035	0.418	0.074	0.146	0.233
4	U. Chenab	1.160	0.016	0.079	1.065	0.126	0.388	0.646
5	M.R. Link	0.175	0.000	0.041	0.134	0.057	0.086	0.032
6	Sadiqia	1.315	0.168	0.412	0.735	0.180	0.410	0.725
7	Fordwah	0.529	0.091	0.227	0.210	0.222	0.165	0.141
8	Pakpattan	1.066	0.000	0.005	1.061	0.000	0.001	1.065
9	L. Dipalpur	0.660	0.005	0.052	0.603	0.050	0.043	0.567
10	L.B.D.C.	1.972	0.000	0.178	1.794	0.019	0.433	1.520
11	Jhang	2.126	0.016	0.294	1.816	0.084	0.566	1.476
12	Gugera	1.846	0.027	0.220	1.599	0.077	0.373	1.396
13	U. Jhelum	0.711	0.015	0.188	0.508	0.163	0.311	0.237
14	L. Jhelum	1.740	0.064	0.494	1.182	0.339	0.533	0.869
15	Bahawal	0.856	0.005	0.146	0.705	0.007	0.096	0.752
16	Mailsi	1.076	0.000	0.000	1.076	0.000	0.000	1.076
17	Sidhnai	0.862	0.000	0.000	0.862	0.000	0.000	0.862
18	Havali	0.234	0.005	0.085	0.144	0.031	0.100	0.103
19	Rangpur	0.435	0.030	0.138	0.267	0.111	0.128	0.195
20	Panjnad	1.491	0.163	0.902	0.426	0.141	0.892	0.459
21	Abbasia	0.131	0.010	0.106	0.015	0.004	0.100	0.027
22	Thal	2.655	0.227	1.119	1.309	0.326	1.121	1.208
23	Muzaffargarh	0.845	0.242	0.477	0.126	0.462	0.321	0.062
24	D.G.Khan	1.009	0.257	0.299	0.453	0.356	0.242	0.411
	Sindh & Balochistan	15.159	2.219	9.601	3.341	8.718	2.416	4.025
25	Pat*	0.699	0.043	0.276	0.380	0.123	0.106	0.470
26	Desert	0.362	0.067	0.282	0.013	0.335	0.019	0.008
27	Begari	1.082	0.014	1.063	0.006	1.066	0.016	0.000
28	Ghotki	0.960	0.028	0.720	0.212	0.618	0.337	0.005
29	N. West	1.263	0.479	0.600	0.183	1.082	0.050	0.131
30	Rice	0.595	0.025	0.519	0.051	0.592	0.003	0.000
31	Dadu	0.631	0.092	0.349	0.190	0.393	0.118	0.120
32	Khairpur East	0.572	0.158	0.339	0.075	0.206	0.242	0.125
33	Khairpur West	0.298	0.033	0.236	0.030	0.084	0.190	0.024
34	Rohri	2.952	0.322	1.221	1.409	0.383	0.324	2.245
35	Nara	2.514	0.823	0.942	0.749	1.053	0.566	0.895
36	Kalri	0.687	0.016	0.671	0.000	0.591	0.096	0.000
37	Lined Channel	0.582	0.082	0.501	0.000	0.447	0.135	0.000
38	Fuleli	1.010	0.026	0.977	0.007	0.980	0.030	0.000
39	Pinyari	0.952	0.012	0.904	0.037	0.767	0.184	0.001
	N.W.F.P.	1.333	0.141	0.280	0.912	0.157	0.263	0.914
40	U. Swat & Pehur	0.440	0.008	0.063	0.369	0.010	0.096	0.334
41	L. Swat	0.210	0.058	0.091	0.061	0.047	0.042	0.122
42	Warsak	0.124	0.005	0.007	0.112	0.003	0.012	0.109
43	Kabul	0.103	0.035	0.049	0.019	0.036	0.053	0.014
44	C.R.B.C Stage I	0.109	0.022	0.048	0.038	0.043	0.036	0.030
45	Bannu	0.347	0.013	0.021	0.313	0.018	0.023	0.306

* Gross Area is 0.808 M.Ac but observed area is 0.699 M.Ac.

Source: SCARP Monitoring (S.M), WAPDA.

TABLE – 19
Canal Commandwise Depth to Watertable (1989)

Sr. No.	Canal Commands	Gross Area (M.Ac)	APRIL/JUNE Area under Various Depth to Watertable Ranges (Million Acres)			OCTOBER Area under Various Depth to Watertable Ranges (Million Acres)		
			< 5'	5'–10'	>10'	< 5'	5'–10'	>10'
	PAKISTAN	41.160	5.915	14.189	21.056	12.147	10.383	18.630
	Punjab	24.620	1.607	5.697	17.316	2.649	6.646	15.325
1	U. Dipalpur	0.451	0.002	0.033	0.415	0.001	0.077	0.373
2	Ravi Syphon	0.822	0.012	0.058	0.752	0.001	0.259	0.562
3	Raya Branch	0.453	0.000	0.000	0.453	0.035	0.165	0.253
4	U. Chenab	1.160	0.005	0.084	1.071	0.047	0.385	0.728
5	M.R. Link	0.175	0.000	0.054	0.121	0.042	0.101	0.032
6	Sadiqia	1.315	0.183	0.314	0.819	0.240	0.398	0.678
7	Fordwah	0.529	0.109	0.232	0.188	0.259	0.143	0.126
8	Pakpattan	1.066	0.000	0.005	1.061	0.000	0.003	1.063
9	L. Dipalpur	0.660	0.012	0.033	0.614	0.031	0.059	0.571
10	L.B.D.C.	1.972	0.004	0.190	1.778	0.009	0.362	1.602
11	Jhang	2.126	0.035	0.412	1.679	0.086	0.536	1.504
12	Gugera	1.846	0.044	0.232	1.569	0.057	0.324	1.466
13	U. Jhelum	0.711	0.074	0.315	0.322	0.059	0.358	0.294
14	L. Jhelum	1.740	0.071	0.604	1.065	0.197	0.585	0.958
15	Bahawal	0.856	0.000	0.131	0.725	0.015	0.156	0.686
16	Mailsi	1.076	0.000	0.000	1.076	0.000	0.000	1.076
17	Sidhnai	0.862	0.000	0.000	0.862	0.000	0.000	0.862
18	Havali	0.234	0.000	0.117	0.117	0.027	0.128	0.078
19	Rangpur	0.435	0.027	0.142	0.266	0.074	0.148	0.213
20	Panjnad	1.491	0.175	0.857	0.459	0.106	0.914	0.471
21	Abbasia	0.131	0.042	0.069	0.020	0.044	0.069	0.017
22	Thal	2.655	0.133	1.171	1.351	0.306	1.060	1.289
23	Muzaffargarh	0.845	0.373	0.368	0.104	0.516	0.277	0.052
24	D.G.Khan	1.009	0.305	0.274	0.430	0.496	0.141	0.372
	Sindh & Balochistan	15.159	4.199	8.026	2.935	9.347	3.418	2.394
25	Pat*	0.699	0.013	0.283	0.403	0.147	0.149	0.403
26	Desert	0.362	0.091	0.270	0.001	0.348	0.011	0.002
27	Begari	1.082	0.107	0.940	0.035	1.074	0.007	0.000
28	Ghotki	0.960	0.106	0.780	0.074	0.656	0.290	0.014
29	N. West	1.263	0.820	0.438	0.005	1.134	0.083	0.046
30	Rice	0.595	0.046	0.514	0.036	0.595	0.000	0.000
31	Dadu	0.631	0.197	0.338	0.096	0.454	0.088	0.089
32	Khairpur East	0.572	0.178	0.281	0.113	0.298	0.190	0.085
33	Khairpur West	0.298	0.064	0.201	0.033	0.156	0.130	0.012
34	Rohri	2.952	0.418	1.227	1.307	0.625	1.246	1.081
35	Nara	2.514	1.123	0.675	0.715	1.104	0.754	0.656
36	Kalri	0.687	0.099	0.582	0.006	0.546	0.140	0.000
37	Lined Channel	0.582	0.145	0.394	0.043	0.423	0.159	0.000
38	Fuleli	1.010	0.503	0.506	0.001	0.948	0.062	0.000
39	Pinyari	0.952	0.290	0.596	0.066	0.839	0.108	0.005
	N.W.F.P.	1.381	0.109	0.467	0.805	0.151	0.319	0.911
40	U. Swat & Pehur	0.440	0.019	0.214	0.207	0.026	0.086	0.328
41	L. Swat	0.210	0.033	0.106	0.071	0.031	0.115	0.064
42	Warsak	0.124	0.001	0.008	0.115	0.002	0.012	0.110
43	Kabul	0.103	0.026	0.058	0.019	0.035	0.044	0.024
44	C.R.B.C Stage I	0.157	0.021	0.062	0.074	0.045	0.036	0.076
45	Bannu	0.347	0.010	0.019	0.318	0.012	0.025	0.310

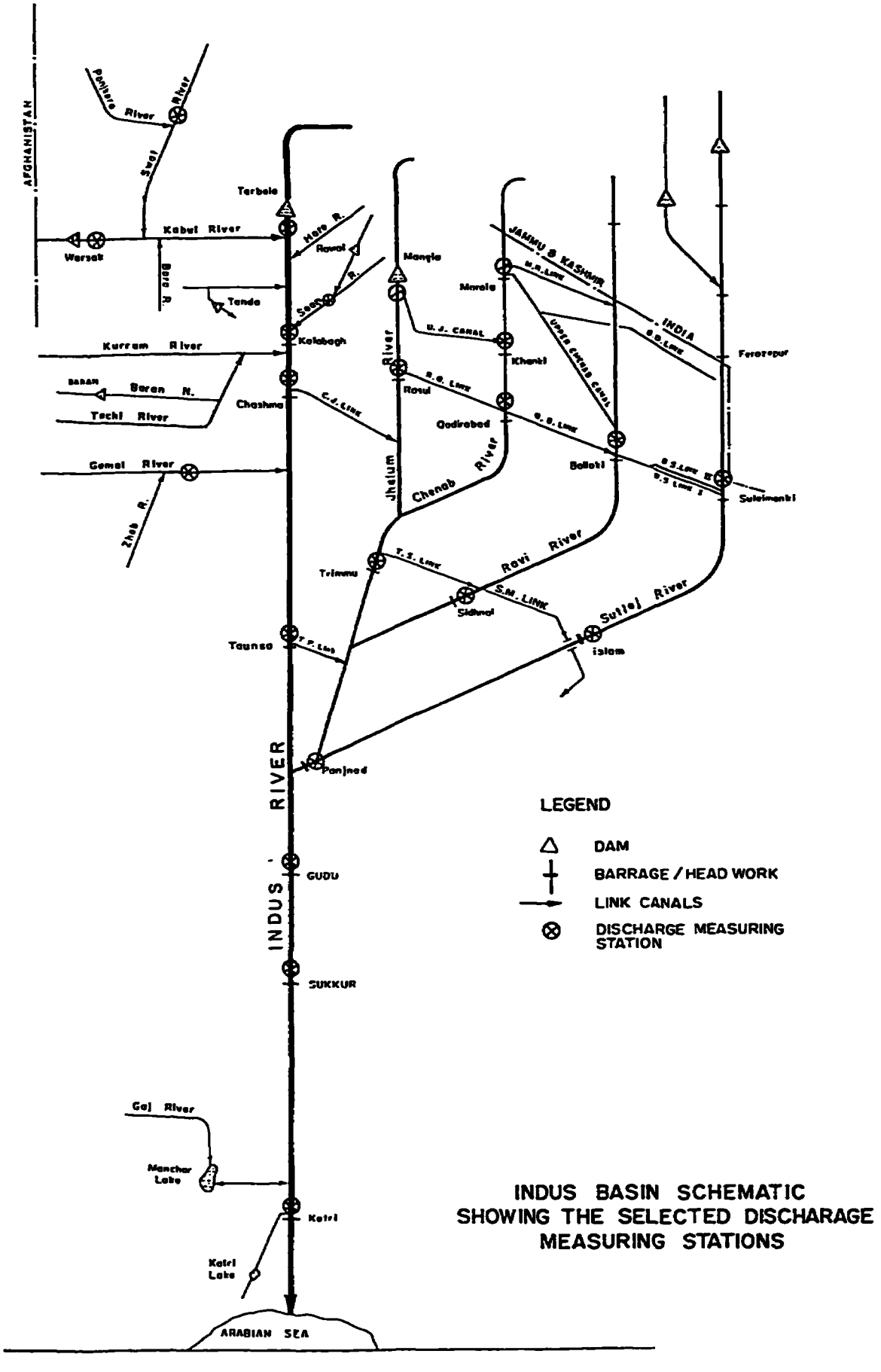
* Gross Area is 0.808 M.Ac but observed area is 0.699 M.Ac.
Source: SCARP Monitoring (S.M), WAPDA.

ANNEX - II

A. TOTAL WESTERN AND EASTERN RIVERS INFLOWS AT RIM STATION.

B. TEN DAILY FLOWS UPSTREAM AND DOWNSTREAM OF BARRAGES.

C. MONTHLY FLOW OF TRIBUTARY RIVERS AT RIM STATIONS.



INDUS BASIN SCHEMATIC
SHOWING THE SELECTED DISCHARAGE
MEASURING STATIONS

Total Western and Eastern Rivers Inflows at Rim–Station

(All values in MAF)

Years	Kharif			Rabi			Annual		
	Western Rivers	Eastern Rivers	Total	Western Rivers	Eastern Rivers	Total	Kharif	Rabi	Total
1977–78	104.36	10.36	114.72	23.1	2.14	25.24	114.72	25.24	139.96
1978–79	137.45	16.48	153.93	26.02	3.49	29.51	153.93	29.51	183.44
1979–80	108.84	4.19	113.03	23.14	0.92	24.06	113.03	24.06	137.09
1980–81	109.81	6.77	116.58	26.59	2.08	28.67	116.58	28.67	145.25
1981–82	117.68	5.28	122.96	22.93	1.95	24.88	122.96	24.88	147.84
1982–83	97.11	4.16	101.27	25.27	1.8	27.07	101.27	27.07	128.34
1983–84	128.29	5.97	134.26	21.67	1.54	23.21	134.26	23.21	157.47
1984–85	115.99	3.75	119.74	18.93	0.96	19.89	119.74	19.89	139.63
1985–86	91.66	4.12	95.78	26.02	1.79	27.81	95.78	27.81	123.59
1986–87	116.38	4.55	120.93	30.27	1.79	32.06	120.93	32.06	152.99
1987–88	111.77	1.88	113.65	29.28	1.12	30.4	113.65	30.4	144.05
1988–89	136.56	10.26	146.82	24.84	6.32	31.16	146.82	31.16	177.98
1989–90	102.01	2.2	104.21	29.31	0.81	30.12	104.21	30.12	134.33
1990–91	130.97	7.94	138.91	35.14	1.97	37.11	138.91	37.11	176.02
Average	114.92	6.28	121.20	25.89	2.05	27.94	121.20	27.94	149.14

Source: Planning Division, WAPDA.

River Indus Flow Upstream Tarbela (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	30.2	22.1	27.2	29.5	30.2	23.8	26.5	28.7	20.9	25.4	34.5	26.7	28.0	29.7
	2	26.9	37.5	36.0	32.7	42.0	28.0	35.9	21.9	21.0	35.4	31.2	44.4	23.8	32.2
	3	30.6	30.5	48.0	40.0	59.7	38.6	37.6	29.7	26.1	46.4	48.5	62.2	28.6	37.9
May	1	31.0	48.1	57.7	58.9	90.7	55.9	41.2	38.0	35.8	43.9	49.1	89.4	45.2	63.8
	2	37.2	71.1	42.6	65.5	106.2	52.2	73.3	53.1	41.0	70.1	45.7	94.0	54.2	174.4
	3	72.3	105.6	51.5	73.6	139.3	67.6	89.0	83.5	70.9	61.5	66.8	118.3	90.4	159.5
Jun	1	104.3	154.4	59.0	133.0	99.9	77.8	89.0	190.8	104.2	69.8	132.3	135.4	190.0	149.9
	2	93.4	162.3	136.3	164.8	87.6	151.0	110.9	220.7	123.5	117.7	127.7	131.2	143.0	154.8
	3	218.9	272.8	249.6	228.4	191.5	95.8	157.4	214.2	128.9	243.9	124.4	300.7	162.6	303.3
Jul	1	305.9	380.7	249.0	214.7	224.8	123.2	217.2	227.0	165.8	258.3	217.3	323.4	156.2	320.8
	2	309.3	275.0	297.9	254.5	246.1	120.0	129.7	177.8	235.3	257.8	193.8	341.5	239.6	255.3
	3	232.7	294.7	236.7	225.7	275.9	221.3	239.2	233.0	245.2	284.0	249.4	333.3	252.6	223.8
Aug	1	255.9	304.8	244.0	263.3	236.3	296.2	342.0	262.6	273.9	324.8	190.6	317.9	243.3	303.2
	2	216.6	280.0	224.6	166.9	256.9	290.2	238.5	300.3	219.4	231.8	212.9	276.0	175.7	246.4
	3	180.6	223.0	165.8	131.5	150.7	152.8	268.4	278.4	194.4	171.6	248.1	201.4	176.4	200.7
Sep	1	142.2	125.7	111.5	121.7	99.4	89.1	230.9	201.3	147.7	118.5	198.1	123.7	90.5	171.8
	2	85.4	111.9	94.4	97.6	66.8	83.8	142.2	83.3	82.0	83.5	113.7	122.6	89.1	127.9
	3	71.3	72.2	55.7	66.6	65.0	57.1	68.4	56.6	74.8	64.0	83.1	76.6	83.1	104.4
Oct	1	51.9	56.2	38.4	47.5	46.6	32.4	57.5	38.6	49.9	51.8	55.0	50.2	51.9	59.1
	2	40.4	39.7	34.9	38.7	35.0	30.3	44.7	32.7	35.2	43.0	61.7	37.3	43.0	44.1
	3	34.5	32.9	30.0	31.6	28.4	29.1	31.5	29.3	27.5	29.2	38.3	32.0	31.4	32.6
Nov	1	26.9	30.2	27.2	27.5	25.9	24.1	27.7	24.0	23.6	25.0	33.3	28.9	27.7	29.3
	2	20.1	25.3	22.7	23.5	22.3	24.1	25.1	21.6	21.0	27.1	28.8	26.9	24.3	25.3
	3	18.8	22.8	21.3	21.9	21.3	20.4	25.3	20.3	19.6	24.7	24.9	24.2	22.5	23.5
Dec	1	19.5	20.6	19.8	20.4	20.0	19.9	19.4	18.2	19.2	21.3	24.0	22.1	22.4	20.1
	2	18.7	19.5	18.6	19.0	18.3	19.4	19.8	17.7	18.1	25.1	22.5	20.1	20.6	21.3
	3	18.5	17.9	17.5	18.2	17.2	16.7	17.8	16.1	20.3	19.0	20.3	21.3	19.2	27.0
Jan	1	16.6	17.1	17.3	17.5	16.8	16.2	16.9	15.9	16.1	17.1	20.1	20.7	18.1	22.0
	2	16.8	18.0	16.1	16.0	15.6	15.5	15.8	15.3	15.1	16.5	18.8	17.5	16.8	18.5
	3	16.4	16.5	16.6	16.8	15.9	16.8	15.2	16.5	15.1	16.6	18.6	17.3	18.8	25.0
Feb	1	16.6	15.7	16.1	16.6	15.7	15.2	15.3	15.7	14.2	15.1	17.5	16.6	20.5	23.1
	2	17.3	17.5	16.9	17.8	15.8	15.2	16.2	14.7	20.2	14.0	15.8	14.6	26.5	38.2
	3	16.9	18.7	17.8	17.0	15.5	16.5	15.6	18.1	18.1	18.7	19.0	14.9	24.1	25.6
Mar	1	19.0	22.5	20.2	18.9	17.1	20.3	14.8	16.6	16.7	21.5	20.0	16.4	18.4	30.8
	2	28.6	19.7	19.1	16.0	17.9	24.2	20.9	17.4	28.1	24.8	32.7	16.8	26.8	36.0
	3	21.5	19.3	24.4	29.1	25.5	27.3	21.4	19.0	26.3	33.9	27.0	22.7	37.0	37.3
Mean Flow		79.5	94.5	77.3	77.3	79.4	66.9	82.2	85.2	72.6	82.0	79.6	98.3	76.2	100.0

Source: Planning Division, WAPDA.

B. TEN DAILY FLOWS UPSTREAM AND DOWNSTREAM OF BARRAGES

River Indus Flow Downstream Tarbela (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	34.0	28.7	55.1	23.2	14.8	23.9	19.7	29.7	24.7	24.5	55.2	25.6	36.0	16.0
	2	36.3	40.4	69.6	23.8	15.7	28.5	19.4	39.7	21.7	29.1	58.7	34.8	46.1	16.5
	3	50.5	62.3	68.6	37.0	79.4	38.8	19.1	54.7	38.2	24.6	74.6	44.5	54.2	20.3
May	1	33.2	78.5	76.1	50.1	109.3	55.0	21.0	52.3	41.4	45.6	56.0	79.9	54.9	28.6
	2	38.0	86.6	83.1	65.5	142.9	38.5	64.6	64.5	41.4	73.9	44.6	84.1	65.7	41.9
	3	72.5	115.0	51.9	97.1	146.2	80.3	55.7	79.1	70.4	102.6	61.6	114.7	87.7	139.2
Jun	1	103.5	154.8	58.4	120.0	129.2	114.3	126.5	73.3	100.9	97.4	102.1	140.7	65.6	159.9
	2	93.6	151.7	127.0	151.6	88.3	98.0	114.8	135.9	122.5	109.8	35.7	155.2	90.3	154.7
	3	133.4	152.4	155.9	159.0	153.0	136.0	139.7	153.5	128.0	97.3	125.4	134.0	138.4	297.2
Jul	1	124.7	154.5	143.5	97.8	149.2	126.1	149.9	156.7	145.9	122.4	139.0	233.4	151.0	317.3
	2	195.2	256.8	122.6	107.5	127.9	129.7	123.2	158.7	107.3	204.5	125.5	336.3	124.7	239.1
	3	224.2	297.9	218.8	217.7	124.8	69.5	74.4	195.0	67.1	259.5	167.0	303.7	205.1	196.6
Aug	1	236.6	304.1	230.2	242.5	196.0	53.9	291.7	233.3	207.7	295.9	161.7	297.0	215.0	278.1
	2	181.0	279.6	180.3	110.5	227.0	252.8	208.3	276.3	180.3	204.4	169.2	239.7	146.2	216.9
	3	169.6	216.0	154.3	130.5	151.1	142.0	269.4	284.1	192.7	165.8	242.0	201.4	176.4	200.7
Sep	1	126.7	110.6	118.1	128.3	115.8	87.9	230.2	193.2	147.6	118.6	198.7	123.7	92.8	171.8
	2	88.5	113.8	123.3	126.8	124.5	121.8	145.9	82.6	98.0	121.6	113.1	122.6	110.1	127.9
	3	87.2	65.3	121.8	125.0	105.1	151.2	64.3	62.8	142.7	122.8	128.8	76.8	113.6	104.4
Oct	1	101.1	86.6	84.3	102.1	98.4	109.0	69.8	81.7	109.3	103.5	139.5	50.8	109.7	86.5
	2	95.3	81.5	60.4	66.2	53.7	65.4	55.3	54.0	55.1	51.6	71.3	39.7	76.3	70.8
	3	44.5	53.5	55.6	43.1	47.9	48.0	39.4	48.9	52.3	45.5	55.2	37.5	46.2	55.1
Nov	1	38.3	42.4	40.4	31.7	36.0	41.3	37.5	57.7	45.2	48.5	55.0	55.0	57.4	36.2
	2	54.3	28.4	46.6	43.1	36.6	34.2	45.2	50.4	44.6	49.6	49.5	53.8	58.0	52.5
	3	47.6	29.7	40.7	45.4	42.9	33.7	64.7	49.4	64.3	50.2	53.8	62.2	62.2	60.3
Dec	1	19.9	28.1	44.0	42.3	40.9	43.1	60.0	49.7	45.1	31.3	52.2	48.8	55.4	61.0
	2	22.6	27.8	41.9	44.1	38.1	31.7	48.7	36.8	24.8	31.7	28.9	41.4	51.3	48.9
	3	20.3	15.5	33.4	25.2	24.6	20.0	31.0	19.9	18.0	18.3	17.2	24.4	23.8	33.0
Jan	1	21.1	16.7	22.7	22.0	23.6	23.5	27.0	18.7	15.5	8.6	31.4	21.9	17.5	15.6
	2	19.2	16.3	22.6	40.7	23.3	22.5	21.8	34.9	17.2	21.3	27.2	26.4	13.6	24.0
	3	28.4	19.2	22.4	23.9	23.1	18.1	25.9	35.0	28.3	27.2	26.5	33.1	18.0	31.1
Feb	1	47.1	55.0	22.2	22.7	34.9	17.6	30.9	37.2	33.4	59.9	46.9	33.7	27.4	30.4
	2	49.7	49.2	35.5	30.8	47.2	38.0	48.2	61.4	39.8	51.4	63.3	50.4	43.9	39.2
	3	53.0	35.0	41.1	42.2	44.5	42.6	50.6	52.1	27.4	43.8	33.5	42.6	34.5	33.8
Mar	1	50.6	27.6	47.4	42.1	34.9	24.9	63.6	51.8	30.8	24.9	38.9	53.7	17.3	52.3
	2	28.3	22.4	39.2	42.3	18.3	36.2	63.9	33.4	29.0	20.1	16.0	55.1	28.8	58.7
	3	16.9	71.9	35.9	15.1	29.0	22.1	31.6	25.8	18.9	29.3	22.4	49.4	14.5	57.4
Mean Flow		77.4	93.8	80.4	76.1	80.5	67.2	82.0	86.8	71.6	81.6	80.2	98.0	75.8	99.3

Source: Planning Division, WAPDA.

River Indus Flow Upstream Kalabagh (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	65.7	36.8	132.4	55.0	48.2	45.0	41.6	54.9	37.2	43.4	93.0	57.3	64.6	39.9
	2	57.5	73.4	108.8	56.4	58.0	52.4	70.2	44.9	36.6	62.0	98.1	70.4	73.3	48.1
	3	68.0	81.4	125.2	88.6	160.0	63.0	78.3	72.9	54.5	72.6	124.0	82.9	72.6	51.3
May	1	43.3	113.3	151.3	94.4	217.7	65.3	85.4	74.7	57.5	94.2	128.9	127.5	97.3	52.7
	2	63.8	140.8	155.3	133.0	242.9	68.6	128.1	89.7	58.2	185.9	126.4	173.5	121.3	65.8
	3	80.9	166.6	81.3	140.5	233.1	119.0	147.0	104.1	101.3	159.7	120.0	174.5	147.5	205.0
Jun	1	135.7	212.9	86.6	214.5	231.5	139.8	183.2	161.6	146.7	175.9	176.4	198.8	167.2	197.3
	2	135.0	234.6	210.5	259.4	156.5	148.4	203.1	206.4	174.6	168.7	109.1	197.5	154.7	197.9
	3	178.0	217.0	240.0	295.3	230.4	191.0	203.8	221.1	164.2	226.4	181.8	200.7	216.4	401.4
Jul	1	206.8	317.0	234.9	218.2	235.8	175.7	234.8	216.3	190.0	218.2	240.7	307.0	228.3	410.6
	2	313.7	347.2	226.0	206.9	216.1	178.0	129.6	214.8	194.7	285.7	197.6	444.5	202.7	325.7
	3	301.3	370.2	324.5	313.1	217.5	136.0	157.0	260.1	138.5	321.3	236.6	366.9	252.8	250.3
Aug	1	311.3	404.6	313.3	344.1	261.0	108.6	350.8	289.0	272.6	383.9	199.1	362.9	359.0	341.5
	2	239.6	377.7	233.8	204.7	278.6	361.2	288.8	357.3	166.0	252.3	209.4	304.5	197.7	327.0
	3	208.0	303.7	212.8	182.5	175.4	200.1	365.8	418.8	171.0	246.9	285.4	240.3	220.2	238.0
Sep	1	147.1	146.6	145.8	210.3	131.4	118.7	347.0	297.0	174.7	166.1	251.6	135.9	147.3	225.4
	2	95.4	118.7	166.0	150.7	128.8	121.0	186.5	148.7	116.9	134.8	163.7	121.0	117.7	185.5
	3	93.5	83.8	177.5	125.8	123.2	151.1	81.8	106.6	151.7	136.2	141.1	95.3	127.6	156.0
Oct	1	104.6	47.3	106.2	121.9	110.1	105.0	73.4	110.7	117.6	115.4	141.7	77.3	102.5	121.4
	2	106.8	83.5	70.0	82.7	56.4	57.1	43.2	51.4	80.3	67.0	106.9	57.4	87.5	78.5
	3	60.1	62.1	61.8	50.1	57.0	43.0	30.7	50.2	58.2	55.9	36.3	36.4	54.0	68.4
Nov	1	53.0	58.0	50.9	46.9	51.6	36.2	31.6	38.7	47.5	43.7	40.7	45.9	57.7	52.6
	2	91.5	47.6	43.8	49.2	48.6	41.8	46.3	33.4	38.5	55.2	47.0	35.3	59.3	61.0
	3	60.5	28.7	45.7	52.6	57.4	38.1	42.6	35.6	51.0	50.3	38.9	50.5	61.8	55.3
Dec	1	18.0	25.3	54.9	50.3	38.7	38.2	40.1	31.6	46.2	46.6	38.2	54.2	62.2	56.4
	2	19.3	31.6	49.7	52.0	37.4	35.8	38.4	34.8	41.5	43.0	35.5	57.2	56.9	52.9
	3	19.9	25.4	44.0	36.5	37.9	33.9	32.1	26.1	34.9	26.3	19.3	44.1	23.2	55.7
Jan	1	21.7	34.3	39.6	31.2	33.7	30.8	26.0	21.7	32.5	26.2	30.1	31.3	21.2	36.5
	2	23.8	40.4	30.0	55.6	32.5	32.0	26.0	32.4	29.0	28.1	36.4	41.8	20.2	43.4
	3	31.0	25.2	34.9	55.4	38.7	29.4	30.6	26.5	43.3	36.8	28.2	49.9	23.8	49.3
Feb	1	47.4	62.1	36.4	42.8	46.2	32.1	33.0	39.8	40.4	56.1	41.8	47.8	30.8	39.8
	2	53.4	50.9	44.4	44.4	65.3	34.7	43.9	54.6	45.7	75.2	77.7	54.2	36.6	42.1
	3	54.4	74.0	59.9	46.9	63.9	40.1	38.3	71.4	44.3	59.3	52.8	49.3	36.0	57.4
Mar	1	58.1	54.2	66.7	49.0	58.5	36.0	59.3	59.1	34.6	63.2	45.8	60.5	31.7	60.5
	2	70.1	39.9	47.8	52.8	47.0	41.9	68.3	42.4	51.9	59.4	60.8	62.3	37.3	60.5
	3	37.6	65.4	61.4	47.8	51.0	37.4	50.8	30.0	40.2	71.4	53.3	71.7	56.1	99.0
Mean Flow		102.1	127.8	118.7	118.4	118.8	88.5	112.2	114.7	91.2	119.8	111.5	127.5	106.3	133.6

Source: Planning Division, WAPDA.

River Indus Flow Downstream Kalabagh (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	59.7	30.2	125.6	49.1	43.3	39.2	34.3	49.2	32.7	36.5	86.8	50.3	58.9	33.6
	2	51.4	66.3	102.0	50.1	52.2	46.4	64.4	38.4	31.3	54.9	91.2	62.9	65.9	40.8
	3	62.0	74.2	118.4	82.6	153.9	56.6	74.1	65.4	49.2	65.4	116.7	75.4	65.1	44.0
May	1	36.6	106.0	144.8	87.8	211.5	58.6	79.6	67.0	52.0	86.7	121.5	120.0	90.3	45.2
	2	56.7	133.4	148.6	126.8	236.4	62.1	120.9	82.1	52.7	170.2	119.4	166.1	113.9	58.2
	3	74.1	159.2	74.9	133.8	226.4	112.4	140.6	96.9	95.8	152.0	113.0	167.0	140.0	197.5
Jun	1	129.7	205.5	80.1	207.7	224.3	133.1	176.4	154.8	141.4	168.6	169.1	191.3	159.7	189.8
	2	129.0	227.3	204.0	252.6	148.8	141.6	196.2	199.5	169.3	161.1	101.6	190.0	147.2	190.5
	3	171.7	209.7	234.9	288.8	222.6	184.1	196.9	214.0	158.6	218.7	174.3	193.2	208.8	393.9
Jul	1	200.2	309.9	228.5	212.9	228.4	168.7	227.2	210.5	183.9	210.6	233.5	299.5	220.8	403.4
	2	307.2	343.0	219.6	201.2	208.4	171.0	122.2	207.9	188.0	278.7	190.2	437.8	195.4	318.1
	3	295.2	366.7	318.5	306.2	211.9	129.2	149.3	253.1	131.8	313.6	229.2	360.8	245.3	243.0
Aug	1	305.3	401.2	307.0	337.4	255.4	100.8	343.1	282.2	265.6	376.1	191.7	356.9	352.6	334.0
	2	233.5	373.4	227.9	198.0	271.9	353.3	280.6	350.4	158.9	244.8	201.8	297.7	190.9	319.5
	3	201.8	298.4	206.9	175.7	168.2	192.2	358.9	411.9	163.6	239.1	277.8	233.5	212.8	230.4
Sep	1	140.7	140.5	139.3	203.5	123.7	111.5	340.2	289.8	167.2	158.3	244.1	129.3	140.2	217.9
	2	88.9	112.2	159.9	143.9	121.0	113.8	179.6	141.8	109.3	126.9	156.2	113.8	110.2	178.0
	3	87.6	77.0	171.2	119.0	115.4	144.0	75.0	99.6	144.1	128.2	133.6	87.8	120.1	148.4
Oct	1	98.2	42.2	99.7	115.1	102.8	97.8	66.5	103.9	110.8	107.2	134.2	69.7	95.0	113.9
	2	100.3	79.9	64.2	76.0	49.1	49.9	36.1	44.6	73.5	62.6	101.5	49.8	79.9	70.9
	3	53.6	56.2	56.0	43.3	49.7	36.1	23.6	43.3	51.4	49.3	26.8	28.9	46.4	61.1
Nov	1	47.0	52.0	45.0	40.1	45.0	29.3	24.6	31.3	40.7	42.0	33.2	38.2	50.1	45.1
	2	85.5	41.6	38.4	42.3	41.9	34.9	39.4	26.1	31.5	40.5	39.5	27.9	51.8	53.4
	3	54.5	22.7	48.9	46.1	50.8	31.3	34.8	28.2	44.1	43.6	31.7	43.3	54.3	47.7
Dec	1	12.8	19.3	49.1	44.2	32.1	30.8	33.2	24.8	39.3	40.0	31.1	46.8	54.7	48.9
	2	14.5	25.6	44.0	45.8	31.1	28.8	31.6	28.1	36.6	37.2	28.2	49.8	49.8	45.4
	3	14.5	20.8	38.2	30.4	32.5	27.9	26.5	19.4	29.9	25.4	11.8	38.4	17.6	48.4
Jan	1	19.9	34.3	36.3	28.1	32.1	29.4	26.0	14.8	27.1	20.3	24.7	25.7	15.5	30.5
	2	23.8	40.4	29.4	55.5	32.5	32.0	24.8	31.2	29.0	23.9	36.4	41.6	19.9	43.1
	3	29.4	21.8	30.0	55.4	37.2	28.4	24.8	26.5	42.8	36.8	28.2	49.9	23.8	49.3
Feb	1	42.8	56.5	30.5	39.1	41.2	26.7	26.7	35.9	35.8	53.6	36.4	43.8	27.3	38.8
	2	48.4	44.1	38.4	38.6	59.7	28.1	37.7	48.2	39.0	60.2	70.2	46.6	29.8	36.3
	3	49.3	73.3	53.9	40.4	57.9	33.4	32.2	65.2	39.7	51.8	45.3	41.7	29.0	49.9
Mar	1	51.9	54.2	61.2	42.4	53.1	29.3	53.3	52.1	27.2	55.7	38.3	52.9	25.6	53.0
	2	64.2	36.3	41.8	46.2	42.7	35.0	62.3	36.4	46.5	52.8	55.9	54.7	30.8	53.0
	3	32.0	58.8	55.4	42.0	46.1	30.9	44.9	25.0	34.5	66.4	46.7	65.0	50.4	91.5
Mean Flow		96.5	122.6	113.1	112.4	112.8	82.2	105.8	108.3	85.4	112.8	104.8	120.8	99.7	126.8

Source: Planning Division, WAPDA.

River Indus Flow Upstream Chashma Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	54.1	31.2	92.8	60.2	57.1	36.0	46.5	58.7	44.7	39.5	90.9	48.1	66.8	48.8
	2	57.6	74.0	101.4	48.6	58.0	43.9	89.0	47.2	39.4	63.2	96.7	65.6	62.6	53.6
	3	61.9	100.1	137.5	78.3	125.5	70.5	81.0	79.9	47.0	74.1	121.4	87.4	65.4	58.3
May	1	45.3	132.3	135.6	99.1	194.0	92.7	84.2	75.4	52.6	80.4	112.6	127.9	88.7	70.7
	2	52.6	142.5	119.2	118.3	230.1	71.5	127.3	99.1	60.9	123.2	92.9	152.7	86.3	128.3
	3	90.5	183.0	89.8	139.1	239.2	117.6	146.3	109.4	87.2	138.0	112.9	172.6	126.1	206.6
Jun	1	147.1	221.7	77.7	194.8	217.9	160.5	182.7	163.6	129.6	142.0	184.8	205.5	147.6	206.9
	2	140.9	251.9	210.1	242.5	140.6	152.5	184.7	205.5	161.0	165.7	115.3	216.4	147.6	214.9
	3	204.3	226.6	258.0	273.4	224.3	178.5	207.4	238.5	163.0	197.2	163.5	217.6	188.3	374.5
Jul	1	220.5	289.4	250.1	198.1	260.6	175.6	248.8	243.1	183.9	203.1	234.0	313.0	213.5	440.4
	2	292.2	393.1	219.9	184.1	205.3	178.0	166.1	219.4	206.0	279.1	184.3	435.5	202.4	335.1
	3	318.1	413.3	313.6	285.0	239.4	134.2	156.6	273.2	138.9	345.0	251.8	420.3	226.1	247.3
Aug	1	311.8	452.1	313.2	328.9	260.4	106.1	358.3	305.4	276.5	411.6	204.1	407.2	336.2	346.5
	2	276.0	392.0	253.2	200.4	312.3	346.2	311.7	373.7	228.4	277.3	216.3	343.5	187.3	329.5
	3	220.8	344.6	238.7	166.2	191.9	224.2	364.0	391.0	252.6	242.4	283.9	283.7	229.6	258.4
Sep	1	167.6	168.8	134.1	168.6	139.9	107.6	326.1	300.7	190.6	151.5	256.2	157.0	125.8	249.5
	2	88.0	151.2	159.6	145.0	131.8	110.1	213.2	130.6	110.0	137.1	166.4	154.5	116.6	165.7
	3	93.3	99.6	134.7	145.7	122.2	182.4	96.6	87.9	152.1	139.7	127.3	140.8	129.0	150.0
Oct	1	99.1	107.9	102.6	107.3	127.0	124.6	75.4	94.2	148.2	126.6	156.5	76.2	118.1	100.7
	2	98.4	101.6	68.5	102.0	66.5	85.4	74.7	66.6	71.9	73.7	116.0	55.4	99.2	101.0
	3	53.2	72.9	66.9	50.8	58.8	53.5	49.1	53.6	60.4	55.3	65.1	46.6	53.9	75.2
Nov	1	49.1	62.6	54.8	40.4	43.5	46.5	42.0	65.1	52.1	53.4	65.8	64.3	64.5	55.1
	2	65.8	39.5	51.6	47.0	39.9	59.2	48.7	61.6	51.4	62.3	64.6	64.8	65.9	60.8
	3	57.4	36.3	49.3	53.5	52.3	33.9	69.4	64.4	71.3	63.1	55.1	73.4	70.3	64.0
Dec	1	22.2	36.1	45.8	48.4	39.4	39.7	69.0	59.1	54.7	44.7	71.9	56.5	63.3	70.2
	2	23.8	34.9	50.5	50.2	51.2	42.7	63.8	50.3	40.2	58.5	38.1	55.4	64.6	60.1
	3	25.7	24.8	44.0	35.1	32.5	32.2	45.9	33.3	38.7	38.6	22.8	42.0	39.4	47.3
Jan	1	25.9	32.8	36.7	36.4	34.9	40.5	43.5	27.1	28.9	18.8	36.0	34.9	28.1	34.1
	2	27.3	37.3	32.9	55.0	38.8	40.6	39.7	52.2	33.7	36.2	48.8	47.5	30.4	34.7
	3	27.3	29.2	31.2	37.8	35.8	38.1	33.8	56.8	44.0	43.5	43.7	50.1	31.7	53.9
Feb	1	55.7	63.0	32.3	37.4	39.4	28.3	36.6	47.7	40.4	65.0	49.8	46.8	40.3	53.4
	2	48.8	61.3	35.5	33.1	53.6	36.1	53.2	64.1	56.6	64.6	78.3	54.0	52.5	79.6
	3	53.8	80.3	47.3	48.2	48.5	58.3	55.5	47.7	44.0	62.0	42.9	51.9	50.0	50.3
Mar	1	59.4	57.2	78.9	58.5	60.6	40.0	69.7	63.3	32.7	51.0	46.2	58.6	32.1	77.8
	2	76.5	35.0	45.5	48.8	38.5	44.3	83.4	37.7	61.7	52.0	62.0	60.4	51.2	89.6
	3	29.1	80.6	61.6	76.3	50.0	50.7	55.6	31.5	42.8	74.1	51.4	71.7	65.2	117.5
Mean Flow		103.9	140.6	116.0	112.3	118.4	94.5	122.2	121.6	97.2	118.2	114.7	137.8	104.6	142.0

* Mean Inflow = Outflow + Paharpur at Head + CJ at Head + Chasma Storage

Source: Planning Division, WAPDA.

River Indus Flow Downstream Chashma Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	48.9	47.2	81.0	54.0	68.3	41.5	42.1	56.7	38.8	48.3	97.6	56.0	56.9	47.3
	2	60.5	64.7	103.9	55.2	50.7	49.8	94.5	61.2	43.9	51.6	91.7	61.1	68.9	55.2
	3	57.8	98.6	116.2	75.8	139.7	70.5	73.9	76.8	51.8	72.1	117.0	65.8	73.6	61.0
May	1	66.7	115.3	165.6	89.3	197.8	95.0	92.9	81.4	56.2	94.9	114.4	137.4	87.7	66.9
	2	55.3	150.6	151.0	118.3	230.1	87.6	138.1	91.5	60.9	119.3	91.1	146.5	89.1	134.7
	3	90.9	193.3	91.6	140.1	239.2	113.2	146.1	111.3	87.2	139.9	117.6	169.1	124.8	209.7
Jun	1	142.8	213.1	78.9	177.9	217.9	151.5	187.6	160.0	129.1	143.8	182.9	200.9	145.8	206.9
	2	140.9	270.5	205.8	242.5	140.8	160.0	184.7	211.1	161.5	163.0	116.8	215.4	144.5	215.9
	3	204.9	226.6	258.0	281.9	227.9	170.8	208.1	240.3	162.9	189.8	177.7	226.4	189.8	376.8
Jul	1	219.2	287.9	243.4	208.1	256.8	177.9	248.8	243.1	183.2	202.6	225.0	314.7	212.7	440.4
	2	290.8	399.9	223.7	190.8	205.3	175.1	166.1	219.4	194.7	293.7	194.2	435.5	203.8	335.1
	3	319.4	407.0	313.6	286.7	246.0	126.4	156.6	273.2	142.3	344.8	238.7	417.4	226.4	247.3
Aug	1	310.9	447.4	313.2	331.4	261.9	117.7	353.0	305.4	282.2	408.2	220.0	407.0	335.5	346.5
	2	249.8	401.8	248.2	204.0	299.9	343.1	311.7	373.2	221.0	282.7	210.3	341.4	174.4	329.5
	3	218.8	346.1	230.5	136.6	173.2	209.9	364.0	389.4	244.5	226.3	288.0	280.6	242.4	255.4
Sep	1	167.4	134.2	126.4	176.3	161.7	113.2	326.1	297.5	185.1	146.3	236.7	147.9	120.7	238.3
	2	87.8	151.2	156.2	156.5	141.9	135.4	193.7	112.8	131.2	151.0	166.9	154.5	121.5	162.1
	3	93.3	99.6	143.8	139.2	121.0	151.7	99.6	104.5	152.7	141.0	149.4	108.8	129.3	151.1
Oct	1	99.1	107.9	119.0	118.8	103.1	135.9	94.4	94.9	140.2	122.9	152.3	75.9	119.4	112.7
	2	101.4	101.5	70.9	80.5	78.6	91.7	71.8	66.6	73.8	76.4	107.2	58.5	89.1	93.8
	3	66.6	78.3	65.4	60.2	54.1	58.4	51.1	56.1	60.4	56.1	70.5	55.8	61.2	75.0
Nov	1	52.5	59.2	53.9	48.7	46.5	47.0	43.3	63.7	51.6	54.1	65.6	59.9	62.1	59.7
	2	54.2	51.6	52.6	51.8	45.0	44.8	50.3	62.0	51.8	63.4	60.0	68.0	68.7	59.1
	3	49.1	47.2	49.4	52.4	45.7	50.9	67.3	62.8	77.0	59.5	62.4	72.9	70.7	68.9
Dec	1	31.3	39.3	50.4	52.0	49.6	55.5	70.5	61.3	50.7	49.1	71.3	60.0	63.0	75.4
	2	29.7	34.6	49.1	48.1	51.4	48.7	57.3	50.3	43.4	46.6	45.0	52.7	61.3	61.0
	3	19.8	24.6	35.4	33.8	29.9	30.3	44.3	35.1	31.9	43.0	19.8	45.9	43.6	49.8
Jan	1	31.8	23.4	34.9	33.2	37.4	29.2	49.9	23.8	26.7	25.3	37.0	27.5	23.9	28.7
	2	31.6	33.5	33.7	55.0	37.9	38.7	36.3	41.7	32.5	34.7	39.2	43.8	28.6	33.5
	3	38.4	43.3	32.0	42.9	37.5	37.7	34.7	54.0	41.4	45.0	45.2	48.6	35.3	54.4
Feb	1	52.9	51.4	33.0	31.8	42.0	36.7	37.2	57.0	52.0	66.5	56.5	53.1	43.9	61.1
	2	53.9	57.4	42.5	37.0	53.8	44.9	51.3	67.1	52.3	64.8	71.9	57.3	50.5	71.7
	3	53.0	61.6	50.8	45.1	52.2	49.8	60.7	57.4	40.8	52.2	49.5	52.6	43.0	63.3
Mar	1	51.0	66.6	62.4	55.1	46.1	40.7	70.1	54.2	42.5	45.3	42.5	57.2	34.9	70.8
	2	50.4	57.2	56.9	57.9	32.7	48.6	73.8	36.5	47.4	53.4	52.8	63.2	44.7	85.8
	3	39.9	79.7	60.5	50.2	40.3	35.7	51.7	34.5	44.3	72.6	47.4	68.3	61.2	117.5
Mean Flow		103.7	140.9	116.8	111.6	118.4	94.9	122.3	121.9	96.9	118.1	114.8	136.3	104.2	142.3

Source: Planning Division, WAPDA.

River Indus Flow Upstream Taunsa (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	46.4	44.2	88.3	49.0	63.4	39.3	34.5	39.2	30.3	43.8	87.0	48.8	55.0	42.0
	2	55.2	49.0	88.8	45.4	39.3	38.0	66.3	41.3	40.3	43.7	96.5	52.1	58.6	48.9
	3	54.3	73.8	106.7	66.9	103.7	57.8	63.6	49.2	40.8	57.4	95.6	75.2	53.6	52.3
May	1	59.3	90.2	126.6	80.9	167.0	79.1	79.6	66.0	48.5	80.0	112.0	117.4	70.9	60.0
	2	60.3	133.4	139.1	95.6	218.6	98.7	105.5	85.8	56.7	109.7	94.3	132.0	76.9	98.5
	3	72.2	166.8	94.9	124.5	229.0	69.3	142.8	91.2	76.7	139.7	103.9	142.3	110.4	188.3
Jun	1	137.8	189.2	76.2	157.4	225.8	135.3	166.5	146.1	115.7	143.7	158.6	179.6	142.0	190.6
	2	142.6	271.0	184.9	207.5	148.9	149.0	170.6	176.6	167.2	150.1	134.1	193.2	131.0	204.4
	3	169.6	206.1	237.7	225.4	185.7	150.5	202.6	241.1	164.3	172.5	139.9	213.6	166.5	305.6
Jul	1	207.7	271.6	236.5	232.4	265.8	169.2	265.1	251.5	176.4	195.9	219.8	263.5	204.4	450.5
	2	217.4	423.0	206.7	193.7	204.6	174.3	157.2	218.4	197.3	267.5	190.9	412.0	196.9	356.3
	3	336.0	404.7	273.7	257.1	248.9	133.1	163.3	264.8	141.8	337.2	217.5	471.9	197.0	235.5
Aug	1	287.1	422.5	296.6	312.3	260.2	105.8	296.3	314.7	266.6	394.8	232.6	417.6	366.6	310.7
	2	247.7	455.8	263.5	249.7	326.1	297.6	381.1	380.5	234.9	317.1	203.7	336.8	174.5	338.3
	3	200.2	365.4	236.2	125.0	172.0	235.0	359.1	428.6	242.5	247.7	256.0	284.4	232.5	241.2
Sep	1	148.0	150.9	108.4	162.0	162.8	109.9	385.0	334.7	182.0	143.5	250.3	151.9	124.3	225.5
	2	101.1	128.8	132.7	138.2	134.6	115.3	202.7	120.3	117.6	136.9	161.0	133.6	105.7	137.1
	3	92.1	84.7	131.3	129.6	118.2	142.5	114.5	99.9	125.6	128.7	122.7	103.0	104.9	130.6
Oct	1	90.1	87.2	107.2	109.0	101.3	123.1	84.2	83.7	123.4	113.6	140.9	75.3	101.2	90.4
	2	102.6	82.0	69.1	88.5	68.0	97.0	67.8	63.3	81.7	78.1	100.4	54.4	83.2	78.9
	3	64.1	71.0	61.9	64.6	46.2	61.2	51.2	50.4	60.6	56.0	67.3	49.7	51.1	57.7
Nov	1	50.4	52.1	52.3	48.2	38.2	46.1	41.5	55.7	46.2	48.6	53.9	55.5	39.9	48.5
	2	52.6	45.2	48.0	47.5	37.1	39.6	43.4	49.2	43.4	55.6	53.7	53.5	44.2	42.8
	3	48.1	42.4	44.2	47.1	38.1	40.9	48.3	48.7	48.6	45.3	43.2	53.9	52.9	51.1
Dec	1	38.1	39.4	43.0	47.6	37.1	42.6	50.8	52.6	44.7	41.9	52.4	41.1	49.8	54.0
	2	34.3	33.7	46.2	47.6	35.2	44.0	41.5	42.4	38.7	41.0	40.9	43.5	45.8	51.4
	3	26.6	27.3	39.3	32.4	23.0	23.5	28.7	23.7	33.0	41.6	15.3	42.1	38.2	40.6
Jan	1	28.3	8.5	14.5	13.5	14.5	18.8	25.0	5.2	22.0	25.3	20.9	13.1	21.2	21.1
	2	34.9	18.2	29.3	34.3	30.4	35.3	23.9	13.4	28.6	29.1	35.5	38.2	24.5	19.0
	3	37.2	33.6	31.6	44.2	34.8	34.5	14.0	38.7	34.7	37.0	42.3	44.4	29.5	36.7
Feb	1	48.4	41.5	30.4	36.2	34.4	35.0	17.3	46.3	42.8	55.2	44.5	49.2	37.5	35.8
	2	45.6	48.4	35.8	35.4	41.4	35.2	22.7	50.5	45.0	62.0	55.1	50.1	46.6	43.0
	3	44.4	51.5	41.6	41.0	44.7	39.4	37.4	52.3	37.0	49.2	48.2	48.1	39.6	39.9
Mar	1	46.9	57.4	48.8	45.8	42.5	34.6	43.0	53.2	33.4	40.3	38.8	47.4	30.8	47.4
	2	44.2	66.7	47.0	48.9	34.5	40.0	45.9	37.6	39.4	45.3	43.7	51.3	38.5	63.5
	3	47.1	64.7	48.5	49.2	38.0	33.1	43.2	30.6	42.6	63.2	40.8	59.0	51.1	99.0
Mean Flow		97.7	133.4	107.4	103.7	111.5	86.2	113.5	115.2	90.9	112.2	106.0	127.7	94.4	126.0

Source: Planning Division, WAPDA.

* Above = Below + Muzaffargarh Canal at Head + D.G.Khan at Head + T–P at Head.

River Indus Flow Downstream Taunsa (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	36.2	37.3	77.9	42.5	60.5	35.7	28.1	39.2	34.3	37.5	82.0	41.9	47.7	38.6
	2	44.7	36.2	82.5	44.9	34.3	34.0	58.5	39.8	36.7	38.2	89.3	51.8	53.8	44.9
	3	44.4	60.8	95.0	51.8	95.5	49.5	59.5	42.0	36.6	51.9	84.8	70.4	40.5	41.7
May	1	44.7	75.7	108.3	61.0	157.9	68.4	69.0	58.1	41.9	65.0	100.8	102.6	55.1	47.8
	2	49.5	115.5	121.2	70.6	205.0	86.0	91.4	67.1	47.5	91.5	81.6	110.4	62.6	83.9
	3	60.8	148.3	80.1	99.4	213.8	83.3	129.1	70.1	37.8	120.2	92.0	122.2	99.4	173.0
Jun	1	120.0	170.4	62.7	131.5	210.0	118.9	152.8	121.8	101.5	127.8	143.8	161.1	125.5	173.8
	2	122.0	254.1	167.7	181.1	129.0	129.6	155.2	156.9	146.2	131.6	118.3	174.7	113.0	185.7
	3	144.9	186.1	214.0	205.3	162.6	132.9	187.1	219.5	143.9	154.7	124.5	190.2	147.0	287.2
Jul	1	186.4	254.3	218.7	212.8	246.7	148.5	250.5	230.9	150.4	176.7	203.6	244.7	184.7	434.4
	2	199.5	421.5	190.1	173.4	181.4	150.6	142.5	195.6	173.7	248.8	174.3	397.8	180.6	339.3
	3	319.5	401.9	256.3	237.7	232.2	110.6	151.0	245.4	128.6	320.4	201.3	464.0	180.4	218.2
Aug	1	271.4	416.4	280.7	296.2	246.7	87.3	285.7	264.9	251.6	378.9	216.5	405.2	359.5	293.4
	2	232.3	444.8	253.4	235.9	312.4	284.0	371.4	364.9	219.7	307.8	188.6	321.2	160.3	323.7
	3	183.5	349.8	221.4	106.6	153.9	214.8	352.2	421.7	227.2	233.9	241.0	268.8	217.0	224.2
Sep	1	129.2	135.2	87.6	139.1	140.9	90.1	379.5	324.5	158.6	126.9	234.4	136.8	105.9	208.1
	2	86.2	113.3	109.0	113.6	114.2	92.9	197.6	99.8	96.8	116.9	141.0	109.5	84.3	120.0
	3	79.0	70.0	108.9	107.9	94.0	117.1	102.3	83.1	101.6	105.7	100.4	85.5	85.1	113.8
Oct	1	71.4	73.1	86.5	83.5	78.4	98.9	64.6	65.4	98.3	90.0	119.2	62.3	82.3	75.3
	2	83.4	64.1	54.0	68.4	49.1	80.0	47.7	55.3	65.9	64.1	84.3	41.7	68.4	60.5
	3	49.2	55.0	51.0	51.1	37.7	55.1	40.8	49.9	52.3	51.5	65.8	43.9	45.2	46.9
Nov	1	42.2	43.1	45.7	38.5	34.8	40.8	37.4	43.5	42.5	44.5	50.9	51.4	36.5	45.4
	2	45.4	37.3	40.6	37.8	32.7	34.6	38.6	40.6	40.4	39.8	43.8	49.4	40.6	39.8
	3	45.8	37.7	38.6	39.8	33.0	36.1	37.8	44.5	39.2	37.2	39.4	43.3	40.7	40.4
Dec	1	37.2	37.5	36.1	40.3	32.9	38.0	35.4	42.0	37.9	37.4	43.6	36.1	42.0	43.5
	2	31.2	31.8	37.3	41.2	30.4	39.6	26.1	34.1	37.6	37.6	38.4	40.1	42.3	42.3
	3	26.6	27.3	33.8	30.1	23.0	14.2	11.9	23.1	32.0	34.7	15.3	35.1	35.7	35.6
Jan	1	26.7	7.8	14.5	13.5	14.3	12.7	8.1	5.2	22.0	19.8	20.4	13.1	21.2	19.5
	2	30.3	13.3	29.3	33.5	27.4	35.3	7.4	13.4	27.5	29.1	33.9	37.3	23.3	15.1
	3	34.4	27.6	29.9	39.8	31.7	32.7	5.0	26.5	28.7	36.5	39.8	32.8	22.3	26.6
Feb	1	45.7	36.0	25.8	31.7	31.8	30.8	4.9	43.0	40.6	47.8	41.7	38.5	34.7	35.8
	2	42.8	45.4	31.0	31.2	36.8	32.3	7.5	46.6	41.6	49.5	48.5	43.2	40.1	43.0
	3	41.5	51.5	35.7	34.3	40.1	36.0	18.7	48.2	35.1	45.9	42.9	45.1	37.7	39.9
Mar	1	43.1	56.5	39.4	37.6	39.0	30.9	24.3	46.4	31.0	37.4	37.3	45.5	24.9	40.6
	2	40.7	65.7	37.1	41.3	31.7	33.8	32.2	36.9	35.9	35.1	37.6	48.4	25.2	50.1
	3	41.8	59.1	39.4	40.0	35.0	28.2	40.7	30.0	36.0	54.3	33.1	53.4	46.2	83.4
Mean Flow		87.04	123.9	95.58	90.13	100.8	76.22	101.4	103.8	79.97	100.7	95.94	117.2	83.65	114.8

Source: Planning Division, WAPDA.

Indus River Flow Upstream Guddu Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	31.2	42.0	84.8	43.7	73.1	56.2	48.2	40.6	25.3	37.1	93.9	34.0	48.7	54.4
	2	36.8	37.2	87.2	42.7	55.5	56.9	86.9	40.5	31.6	43.2	105.1	41.9	40.9	44.5
	3	47.1	54.3	103.0	46.7	72.3	52.0	193.4	39.5	31.0	46.5	76.9	49.3	39.9	38.8
May	1	41.8	76.8	109.7	56.3	181.3	75.6	136.3	50.7	34.2	63.8	105.6	80.7	42.6	53.2
	2	49.0	119.2	141.8	68.7	203.4	102.4	98.0	57.5	39.6	76.2	109.4	108.4	51.0	75.8
	3	44.2	155.5	107.7	87.6	267.3	78.9	148.5	61.9	43.3	109.5	107.1	113.9	80.5	181.5
Jun	1	98.0	199.7	82.7	110.5	258.6	114.2	167.6	101.0	80.1	123.9	158.7	153.3	136.0	196.4
	2	115.1	279.4	141.8	169.2	156.9	138.6	185.0	137.9	122.4	113.7	247.3	172.2	149.9	195.0
	3	115.7	215.1	204.1	193.3	136.3	155.6	190.3	204.2	124.6	151.5	182.3	174.5	160.2	214.1
Jul	1	211.6	330.5	284.7	256.3	241.9	161.0	267.9	227.1	130.4	245.4	232.0	245.2	206.8	522.9
	2	241.8	705.8	241.4	236.2	214.9	162.0	237.1	206.2	171.4	340.1	229.9	400.1	209.6	491.4
	3	432.7	800.6	320.9	317.5	349.3	147.8	209.6	232.7	211.3	479.9	216.9	972.6	218.8	262.7
Aug	1	385.8	752.0	349.5	403.7	549.7	146.6	329.3	315.2	306.8	689.9	232.3	876.4	616.3	310.2
	2	395.4	1066.4	349.8	490.9	456.6	352.9	642.5	388.9	347.1	900.1	196.7	622.8	286.6	429.9
	3	259.5	653.4	277.0	149.4	256.9	346.4	492.6	524.2	290.0	397.2	218.0	423.5	245.2	319.2
Sep	1	189.0	309.4	102.5	167.9	156.9	138.8	693.1	545.9	203.0	177.6	296.7	235.3	175.4	277.8
	2	176.0	178.4	102.7	122.3	128.0	92.2	453.4	282.4	119.7	137.8	172.5	130.9	93.9	278.2
	3	132.2	129.9	118.4	120.4	100.0	105.4	227.9	143.9	98.2	111.1	100.2	124.7	87.0	196.8
Oct	1	92.6	90.4	94.7	89.1	84.0	102.7	144.5	80.0	106.0	94.5	121.0	336.7	83.7	106.9
	2	85.9	70.9	52.0	75.4	66.3	89.8	87.4	59.1	83.4	70.5	92.5	205.6	72.0	71.2
	3	67.3	61.9	49.2	53.7	44.9	65.4	53.7	51.4	54.3	53.9	66.0	80.9	55.9	52.3
Nov	1	42.1	45.0	44.9	43.7	42.3	43.8	46.0	47.8	44.2	47.7	58.0	60.4	40.6	50.0
	2	39.2	41.6	42.3	35.8	38.3	36.0	44.5	43.1	39.7	42.8	45.7	57.2	38.6	41.9
	3	43.3	42.3	40.6	37.5	37.0	35.6	43.5	41.0	38.4	36.9	38.8	51.1	37.2	42.8
Dec	1	44.7	40.5	36.7	39.7	36.7	38.4	43.7	40.0	38.5	37.6	43.1	40.5	36.5	42.8
	2	33.6	34.5	37.8	40.5	35.5	38.7	47.1	36.2	37.0	38.0	40.2	39.3	35.6	41.2
	3	30.3	33.0	35.5	38.2	35.4	33.8	43.5	35.5	34.7	41.1	27.1	38.0	35.4	39.2
Jan	1	21.6	23.0	29.7	26.8	30.8	24.3	35.1	26.0	29.7	28.9	19.5	57.9	31.0	63.3
	2	32.3	23.4	37.2	42.1	32.6	33.7	39.1	34.9	27.8	24.3	39.3	73.8	36.2	49.0
	3	36.5	36.9	38.1	54.5	39.4	43.9	57.4	32.3	30.2	39.7	37.5	69.5	32.0	50.3
Feb	1	36.1	37.2	30.7	63.0	32.5	40.2	54.2	34.1	33.1	40.8	37.6	48.5	35.7	47.2
	2	44.2	41.9	28.6	47.1	35.7	37.4	35.0	37.6	41.5	55.7	41.9	41.1	33.8	44.6
	3	42.3	55.9	31.7	36.5	40.3	34.6	35.5	39.9	38.5	48.6	42.8	45.1	39.4	54.6
Mar	1	38.3	87.3	37.4	36.3	40.8	34.4	40.0	38.5	30.7	46.2	39.8	41.1	41.2	55.1
	2	43.5	121.5	37.0	37.9	43.4	34.0	46.2	33.5	30.1	37.8	35.4	38.4	31.9	61.6
	3	54.5	70.9	37.6	51.2	38.0	43.0	53.4	25.5	38.1	47.1	41.6	48.3	70.2	73.9
Mean Flow		106.4	196.2	109.8	109.2	128.1	91.5	159.9	120.5	88.5	141.0	109.7	175.9	102.1	142.5

Source: Planning Division, WAPDA.

River Indus Flow Downstream Guddu Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	31.2	42.0	85.1	43.6	74.3	56.2	47.4	41.3	25.2	37.1	95.4	35.5	47.2	54.9
	2	36.8	36.4	87.2	43.0	55.5	57.0	86.9	40.5	31.8	43.2	105.1	41.9	42.6	44.5
	3	44.4	51.8	101.6	45.5	72.3	49.0	193.4	39.5	31.0	46.5	76.9	49.3	39.9	38.2
May	1	37.3	71.6	105.9	50.4	180.2	76.7	135.5	50.4	34.2	63.0	104.0	73.6	42.5	51.9
	2	43.9	111.3	136.2	60.9	199.1	98.5	94.4	53.9	37.0	70.5	102.1	97.4	49.5	66.5
	3	37.5	139.8	96.7	76.3	253.1	70.0	140.2	54.3	41.6	93.6	90.1	91.5	71.2	162.3
Jun	1	81.5	177.7	62.0	91.2	234.5	93.8	145.7	81.5	70.6	99.7	127.0	120.5	106.6	164.4
	2	90.4	251.6	116.3	140.0	126.7	109.5	153.7	105.2	96.2	87.2	208.7	131.7	113.1	155.7
	3	84.4	179.6	169.4	157.3	106.5	109.5	153.1	166.7	92.4	116.6	143.3	134.5	121.0	173.7
Jul	1	174.2	295.3	243.8	218.7	204.0	122.7	229.4	190.2	106.3	206.5	192.8	209.0	170.3	486.8
	2	210.2	695.1	201.6	196.1	176.9	125.1	199.4	170.0	140.2	300.6	192.7	366.3	171.1	452.6
	3	405.4	790.4	284.4	284.1	327.5	114.8	178.5	202.7	179.7	450.5	180.7	944.9	204.4	230.8
Aug	1	352.7	733.6	327.1	373.9	519.3	121.6	301.0	285.9	276.4	671.2	197.2	859.4	589.3	285.9
	2	370.4	1044.9	321.1	461.8	428.0	328.9	618.2	364.4	314.6	892.9	163.0	615.6	254.1	397.7
	3	235.5	633.4	247.5	120.1	233.0	321.0	471.6	502.7	257.8	370.2	188.6	397.4	215.7	287.7
Sep	1	166.4	285.6	79.0	139.0	128.0	117.3	664.5	524.5	172.2	153.0	269.5	208.1	148.6	248.4
	2	152.9	157.8	74.8	95.1	101.1	72.5	427.6	259.4	97.2	115.4	151.6	108.4	71.7	252.0
	3	106.0	106.1	92.3	94.1	75.3	93.6	201.9	129.0	78.1	87.5	74.5	104.2	62.1	176.1
Oct	1	72.6	69.2	72.0	67.5	64.8	84.0	131.0	60.4	88.9	77.8	102.5	326.2	69.0	95.1
	2	72.8	54.4	49.2	60.9	55.9	80.0	77.5	47.5	71.6	61.7	82.5	197.0	63.1	61.7
	3	59.3	49.4	40.7	44.3	38.2	53.5	47.5	41.3	48.6	48.4	60.0	73.6	48.5	45.7
Nov	1	37.1	37.6	36.6	38.8	35.8	40.5	41.5	39.1	40.3	43.2	52.2	53.8	35.1	43.1
	2	35.8	34.8	34.7	31.8	33.9	33.4	40.2	36.4	36.2	38.4	39.8	51.9	33.9	35.7
	3	40.0	36.0	35.3	31.9	33.1	33.6	38.3	36.1	34.9	33.0	34.0	45.0	33.4	42.8
Dec	1	42.1	36.1	32.7	35.2	33.3	35.0	39.3	36.0	36.0	35.5	37.9	36.1	33.3	37.8
	2	32.4	30.6	35.1	36.3	32.9	35.3	43.8	33.6	35.8	35.7	36.6	35.8	33.1	37.3
	3	30.1	31.5	32.1	35.6	32.9	27.9	40.4	32.9	33.5	38.2	25.8	34.9	33.1	36.1
Jan	1	14.4	17.6	22.1	21.4	24.8	19.2	29.1	19.6	22.8	20.2	8.4	50.5	25.6	59.0
	2	20.7	10.2	23.1	26.1	20.1	31.1	28.8	18.4	17.9	13.4	24.2	50.6	16.2	29.2
	3	32.5	28.6	27.6	45.1	32.3	41.2	51.9	25.7	26.3	33.7	35.2	58.1	25.5	35.6
Feb	1	35.2	34.1	30.7	60.0	31.5	38.2	49.0	33.9	32.5	39.1	36.4	41.9	31.8	41.8
	2	40.3	36.1	28.1	45.3	35.4	34.3	33.6	36.5	39.0	54.5	37.5	35.2	31.2	41.2
	3	37.0	52.3	29.7	34.1	38.3	31.1	33.4	38.9	36.4	46.6	37.4	40.8	35.4	48.9
Mar	1	34.5	82.1	32.9	33.5	38.1	31.6	36.7	36.7	29.3	41.7	37.8	37.4	36.1	49.9
	2	38.7	113.7	32.4	33.6	39.9	31.6	40.9	32.0	28.4	35.0	33.4	34.1	30.0	56.3
	3	54.3	67.0	35.2	51.0	38.6	38.9	48.6	24.3	35.2	43.4	39.3	45.5	66.1	69.9
Mean Flow		94.2	184.0	96.2	95.1	115.4	79.4	147.1	108.1	77.1	129.0	95.1	162.2	88.9	127.7

Source: Planning Division, WAPDA.

River Indus Flow Upstream Sukkur Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	29.7	41.8	85.7	43.1	67.0	51.8	48.6	47.0	23.3	34.8	85.1	33.0	46.9	55.0
	2	33.0	37.0	83.4	42.4	62.8	55.7	70.0	40.7	29.8	41.2	101.4	40.3	42.0	43.4
	3	40.3	45.5	97.8	44.7	61.9	52.3	174.2	39.6	29.9	44.4	73.9	44.8	39.2	40.1
May	1	34.6	66.3	97.9	47.5	166.0	64.9	113.0	45.0	31.1	58.3	101.3	66.7	38.9	46.1
	2	43.3	101.6	135.3	55.1	196.5	96.9	79.7	48.2	34.3	64.9	94.1	90.7	49.0	63.5
	3	35.7	138.3	102.0	71.2	233.2	67.3	148.8	52.7	39.4	88.6	88.9	87.0	64.1	146.2
Jun	1	70.9	176.0	66.6	88.9	233.3	84.4	136.6	67.4	59.5	99.7	108.7	110.6	101.8	166.8
	2	91.7	241.9	102.3	125.0	140.8	100.2	148.4	97.8	86.5	86.4	201.5	124.5	110.9	153.4
	3	78.8	175.2	163.5	154.0	91.8	118.6	152.8	161.6	89.4	103.5	138.3	124.1	116.7	149.6
Jul	1	162.8	278.9	249.0	213.6	200.0	121.5	185.2	182.4	96.8	197.1	181.5	178.7	159.8	443.0
	2	195.3	663.6	185.1	189.4	180.9	126.6	214.3	176.1	121.5	278.0	189.0	323.5	166.1	498.7
	3	403.8	752.3	279.2	267.6	294.1	112.7	175.9	193.8	169.8	422.8	171.5	848.0	197.0	242.5
Aug	1	356.2	780.8	317.5	346.9	501.6	122.0	253.3	261.5	249.8	610.8	188.9	967.5	505.0	251.0
	2	403.3	957.3	352.7	502.6	389.1	285.5	668.3	334.2	324.1	1031.6	160.3	715.9	411.3	390.5
	3	240.6	689.4	258.0	137.3	264.0	333.5	519.3	487.0	258.4	422.8	173.2	444.2	204.2	323.0
Sep	1	174.8	320.5	85.4	136.1	129.6	130.0	596.4	561.1	182.4	170.3	280.8	242.4	170.1	249.0
	2	152.3	159.1	74.4	97.8	103.3	65.0	473.9	326.7	102.6	115.6	159.1	116.3	71.8	254.6
	3	103.2	119.5	93.2	99.8	75.5	83.8	225.8	141.6	70.7	89.8	72.2	95.0	59.1	198.0
Oct	1	76.3	70.2	75.8	77.0	63.3	84.0	133.6	67.7	88.6	78.6	99.2	293.6	67.1	104.3
	2	71.8	62.3	54.1	70.8	55.4	77.9	75.5	51.4	71.0	65.0	85.5	237.7	61.4	61.3
	3	64.1	55.8	45.1	50.6	42.8	61.3	48.5	44.3	53.7	50.3	61.6	88.0	48.6	47.4
Nov	1	38.2	42.5	38.8	43.4	40.2	47.8	41.8	40.7	39.5	44.8	53.5	54.9	34.0	42.9
	2	35.7	36.4	36.2	38.2	37.4	39.1	44.5	37.6	35.6	40.5	41.8	51.5	33.0	36.0
	3	41.0	38.1	36.8	35.2	35.8	31.8	39.8	36.9	33.8	33.2	33.2	45.1	32.5	35.5
Dec	1	42.0	38.8	34.0	37.3	36.3	32.3	42.5	36.9	35.6	35.5	37.0	38.8	32.9	37.5
	2	33.8	35.3	37.2	39.3	35.4	33.8	48.0	34.8	35.7	35.4	36.3	35.3	32.1	36.7
	3	33.1	35.6	36.8	38.9	32.9	33.4	42.6	33.3	37.8	39.3	28.4	34.9	33.8	36.1
Jan	1	15.1	26.1	28.2	28.6	29.0	25.8	32.5	21.9	25.1	26.1	11.4	46.9	28.7	53.7
	2	17.5	13.6	23.0	22.9	22.3	28.6	26.8	16.1	15.7	9.0	16.7	47.0	13.6	33.5
	3	30.1	27.2	28.0	42.4	33.5	39.9	52.6	26.4	24.4	30.0	28.9	56.9	23.8	33.4
Feb	1	36.1	36.1	31.8	56.8	33.2	37.8	47.8	32.3	29.8	37.6	34.5	42.5	30.1	40.6
	2	40.6	38.0	29.6	46.3	33.7	39.5	36.2	36.1	37.4	47.6	36.2	34.5	30.7	36.3
	3	38.0	49.8	28.3	38.2	40.1	31.9	31.5	37.1	36.4	46.3	36.1	38.8	34.4	48.9
Mar	1	36.2	76.7	31.5	33.5	39.1	31.0	36.0	36.3	28.7	42.3	37.5	37.3	33.3	46.0
	2	39.6	120.5	32.3	34.2	40.0	31.4	42.7	29.6	26.2	35.0	32.4	33.5	33.6	56.2
	3	55.1	65.9	34.3	43.8	35.6	34.9	48.2	23.2	33.1	36.9	38.5	42.4	56.2	64.3
Mean Flow		94.3	183.7	97.0	95.6	113.3	78.2	146.0	108.5	74.7	130.4	92.2	164.2	89.3	126.8

Source: Planning Division, WAPDA.

River Indus Flow Downstream Sukkur Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	1.3	13.3	56.5	13.6	35.6	21.2	16.2	15.3	0.5	6.4	56.1	3.1	17.3	31.0
	2	5.5	8.4	53.0	12.0	31.4	26.2	44.3	14.7	3.1	14.2	75.0	12.5	13.8	19.6
	3	15.3	18.9	69.8	17.5	33.5	25.1	148.4	13.4	5.3	17.6	47.6	16.7	10.6	14.7
May	1	5.9	35.5	67.4	17.3	135.9	40.3	84.3	14.8	8.5	29.5	72.9	35.1	9.8	18.0
	2	11.2	66.3	100.9	20.3	161.6	63.4	46.7	14.0	10.2	32.5	60.8	53.6	18.0	30.0
	3	2.7	99.8	65.0	32.8	194.2	30.0	111.1	15.9	10.9	53.8	52.4	45.1	26.9	108.9
Jun	1	30.9	133.0	25.1	44.9	189.5	43.0	93.6	27.3	22.7	55.3	63.7	61.1	58.4	123.0
	2	45.7	194.2	55.9	73.6	89.8	50.1	98.8	49.4	41.5	39.9	147.8	69.4	62.3	102.9
	3	26.4	124.8	109.3	98.0	41.7	60.8	99.4	108.6	39.3	52.4	83.1	67.2	61.6	94.8
Jul	1	11.5	230.0	192.0	158.1	145.8	66.3	129.5	126.9	45.4	140.3	125.8	120.6	104.9	388.9
	2	144.1	642.1	127.1	132.8	127.2	70.8	161.8	120.4	73.2	220.6	133.0	269.4	111.3	443.6
	3	352.4	724.9	233.4	212.1	249.8	57.0	127.4	140.1	117.3	366.7	114.5	801.5	167.7	188.1
Aug	1	307.7	744.8	273.2	292.4	454.2	70.1	204.0	211.2	200.3	558.7	132.1	928.5	466.3	205.5
	2	358.9	920.0	304.8	447.1	340.2	235.8	621.2	289.0	270.3	987.8	104.5	677.5	366.3	352.1
	3	190.7	651.4	208.5	85.9	218.4	292.1	478.0	439.3	203.2	369.2	118.2	396.1	160.2	275.1
Sep	1	132.7	276.2	39.3	80.9	79.0	82.4	554.3	512.0	135.6	119.2	227.9	198.4	125.4	201.6
	2	101.9	117.0	25.1	44.4	53.5	16.0	426.6	278.1	48.9	63.2	109.8	63.7	24.1	204.9
	3	52.0	72.7	45.8	51.8	27.3	34.4	183.0	97.5	21.3	38.5	20.6	47.4	12.2	156.7
Oct	1	29.7	27.3	31.4	30.5	18.8	38.2	86.8	23.3	42.1	31.7	54.1	253.3	24.2	57.5
	2	33.0	25.9	14.5	32.0	15.3	36.6	36.4	13.6	28.9	24.9	46.4	201.6	25.2	23.9
	3	29.7	23.0	12.8	16.2	8.5	24.8	13.4	10.6	18.3	13.7	26.5	53.2	17.6	14.2
Nov	1	3.7	10.6	8.3	13.1	6.9	13.0	8.0	8.3	6.5	9.1	19.5	20.5	4.0	9.6
	2	2.8	4.6	4.9	5.9	4.4	5.5	10.7	5.5	2.6	4.4	8.4	16.5	2.3	2.4
	3	8.3	7.2	5.1	2.2	2.5	0.2	5.4	5.1	1.7	0.9	1.6	9.6	1.5	2.2
Dec	1	8.7	7.6	2.1	4.2	3.0	0.0	7.9	4.9	2.6	2.5	4.5	3.2	1.6	3.5
	2	0.5	1.9	4.9	5.8	2.3	4.6	14.1	3.0	3.0	2.5	4.2	1.2	1.4	3.5
	3	0.0	2.1	5.9	6.2	0.0	33.4	8.4	1.0	0.5	6.2	1.0	0.7	3.3	2.3
Jan	1	12.8	7.7	12.7	11.7	10.9	20.2	14.5	8.8	11.1	12.7	11.4	32.7	14.5	42.4
	2	11.8	13.6	23.0	22.9	22.3	1.4	26.8	16.1	15.7	9.0	8.7	47.0	13.6	33.5
	3	1.2	9.2	10.0	26.3	18.8	5.1	38.2	3.7	7.7	12.1	1.6	38.6	8.1	22.0
Feb	1	1.6	1.5	0.3	24.7	2.0	4.6	14.3	0.2	0.1	7.2	0.6	10.5	0.2	10.2
	2	6.1	5.9	0.0	12.2	0.8	9.3	1.8	0.6	3.3	14.0	3.7	0.9	1.3	3.3
	3	4.1	21.2	0.0	6.5	6.1	2.2	0.3	1.7	3.0	13.2	4.3	5.0	6.5	15.8
Mar	1	2.9	47.9	1.8	1.5	9.6	0.0	1.2	2.8	0.5	9.2	5.7	3.1	4.6	12.8
	2	8.0	91.0	2.4	5.3	8.8	0.0	8.8	0.5	0.7	3.2	1.5	0.2	4.8	23.2
	3	27.9	35.7	4.8	15.5	6.2	2.5	14.9	0.9	7.1	7.7	8.4	12.5	31.9	38.6
Mean Flow		55.3	150.5	61.0	57.7	76.6	41.3	109.5	72.2	39.2	93.1	54.4	127.1	55.1	91.1

Source: Planning Division, WAPDA.

River Indus Flow Upstream Kotri Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	11.6	20.1	49.5	6.1	16.1	7.7	3.9	10.3	0.8	4.9	20.3	5.8	12.3	33.5
	2	10.5	5.5	48.2	11.4	37.1	14.9	13.6	11.8	0.7	5.6	60.6	4.0	10.5	20.8
	3	16.3	6.2	55.4	9.6	17.7	19.6	95.5	13.8	1.7	9.2	51.0	10.2	11.4	17.9
May	1	19.7	14.1	54.8	11.9	70.7	17.2	61.2	12.1	2.4	10.6	43.5	12.0	8.3	10.2
	2	10.4	18.8	79.3	11.0	134.1	44.3	55.4	14.0	5.5	18.3	51.5	25.5	11.4	16.8
	3	12.7	70.2	78.4	18.6	155.5	40.3	70.5	9.9	6.9	24.1	48.6	28.0	12.3	46.8
Jun	1	9.7	105.5	40.7	27.9	159.1	20.9	78.0	8.5	9.9	29.9	32.9	31.4	25.1	96.8
	2	44.2	128.1	19.0	37.6	107.0	27.5	109.3	24.5	16.4	25.9	91.7	38.8	49.9	88.1
	3	34.9	126.7	82.7	81.7	24.7	35.8	78.4	54.2	24.4	21.7	74.5	39.4	35.4	69.6
Jul	1	58.7	146.1	160.5	125.5	86.5	39.5	93.2	85.4	18.0	72.9	57.4	51.9	51.8	177.0
	2	122.2	293.6	120.8	98.2	117.0	39.8	148.8	101.2	31.0	104.5	100.8	118.1	74.9	271.0
	3	237.0	426.8	174.7	135.4	132.2	39.2	99.2	107.0	88.9	154.0	89.6	221.9	107.1	248.7
Aug	1	276.6	570.7	242.8	196.3	226.8	34.6	125.4	158.9	96.5	224.3	113.3	423.0	193.7	158.1
	2	329.0	620.8	290.6	251.8	282.5	91.9	207.9	205.6	176.5	314.7	99.5	651.3	240.5	253.2
	3	204.4	676.2	175.2	143.8	251.4	217.5	335.3	256.3	143.9	406.8	101.2	528.4	122.3	255.2
Sep	1	158.7	478.0	84.7	52.5	97.7	115.8	331.1	339.1	120.0	168.2	139.2	241.4	157.5	177.7
	2	107.6	149.6	19.8	66.5	42.6	17.0	448.6	281.8	62.4	66.3	109.5	56.8	36.0	167.5
	3	52.1	101.5	31.4	38.4	31.2	15.3	187.6	70.6	14.6	41.7	37.8	45.6	18.6	141.2
Oct	1	51.8	47.9	33.1	26.3	17.7	25.7	35.0	32.4	26.6	22.3	24.4	140.5	18.0	74.2
	2	22.3	46.0	19.2	20.6	15.1	16.9	20.9	15.2	23.5	17.8	19.9	188.4	21.8	43.5
	3	24.6	34.1	10.9	14.8	10.6	14.0	12.4	13.8	13.8	10.3	17.1	37.5	16.6	15.1
Nov	1	17.5	25.6	9.3	8.6	9.3	14.0	12.3	12.1	8.7	7.4	13.5	29.9	10.7	16.5
	2	9.5	15.9	8.6	7.1	8.1	4.6	10.9	10.7	3.7	6.0	10.7	28.0	8.4	14.5
	3	8.8	9.0	7.0	3.9	4.8	2.0	10.4	8.4	1.6	2.6	6.4	27.2	8.0	11.2
Dec	1	7.6	8.7	6.8	2.6	3.4	1.3	7.1	8.8	1.5	1.6	3.3	23.9	7.1	11.5
	2	6.6	7.6	4.9	4.5	3.1	1.2	9.1	8.9	1.8	1.5	4.6	10.0	6.0	11.3
	3	3.1	6.0	3.7	6.3	3.9	14.4	12.4	5.8	2.0	1.9	1.0	5.6	5.9	10.1
Jan	1	3.9	4.1	2.8	4.7	3.4	17.7	9.1	2.7	0.4	4.2	4.0	10.3	5.2	18.6
	2	11.6	7.6	16.9	17.6	14.0	8.4	23.2	15.9	15.1	12.2	8.8	32.6	21.5	42.0
	3	7.7	12.2	15.6	27.3	16.1	2.3	35.4	16.0	12.6	4.6	6.3	33.8	15.1	28.1
Feb	1	3.5	8.0	9.0	27.8	11.7	6.3	26.8	6.5	6.6	7.4	2.5	25.8	10.9	25.7
	2	3.2	6.7	2.6	20.0	2.2	6.2	10.5	3.7	2.5	2.3	1.7	10.1	5.0	16.4
	3	4.5	8.3	1.8	14.2	1.3	7.7	2.9	3.0	1.4	11.3	3.6	5.1	5.7	14.7
Mar	1	4.0	37.0	1.2	6.4	3.5	3.3	2.3	3.3	1.4	8.3	4.0	7.2	9.7	14.7
	2	3.8	85.9	2.0	3.4	6.6	1.4	2.0	1.7	0.9	5.0	4.5	5.1	7.2	18.2
	3	14.1	46.0	2.8	6.0	4.9	1.0	4.4	1.0	1.2	1.9	3.3	4.8	4.6	21.6
Mean Flow		53.5	121.5	54.6	43.0	59.2	27.4	77.5	53.7	26.3	50.9	40.6	87.8	38.0	73.8

Source: Planning Division, WAPDA.

River Indus Flow Downstream Kotri Barrage (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	6.6	13.6	46.4	0.0	9.7	0.8	0.0	0.0	0.0	0.0	11.1	0.0	0.0	21.1
	2	7.4	0.4	44.8	0.0	33.0	6.6	2.3	0.0	0.0	0.0	47.9	0.0	0.0	11.9
	3	12.1	0.0	49.7	0.0	14.1	13.3	84.2	0.9	0.0	0.0	38.9	0.0	0.0	8.1
May	1	10.8	4.4	43.4	0.0	59.6	8.5	47.3	0.2	0.0	0.0	32.3	0.2	0.0	0.9
	2	0.2	6.3	64.8	0.0	117.7	28.6	38.4	0.0	0.0	0.0	35.1	2.9	0.0	0.2
	3	0.0	52.2	56.5	0.0	133.0	22.8	48.2	0.0	0.0	1.2	26.9	2.5	0.0	25.8
Jun	1	0.0	81.9	12.7	0.0	131.1	2.0	49.5	0.0	0.0	4.2	7.9	2.3	1.3	71.3
	2	16.0	102.7	0.0	5.1	74.1	0.6	74.7	0.0	0.0	1.5	61.6	4.1	21.3	58.8
	3	5.1	97.9	49.8	56.7	2.8	3.2	41.5	21.4	0.0	0.2	40.2	3.4	3.8	37.8
Jul	1	45.5	116.2	125.6	98.5	52.8	4.0	62.9	8.7	0.0	40.2	25.0	19.9	17.9	146.2
	2	106.3	277.0	87.1	67.2	93.0	9.8	118.5	8.9	5.7	68.0	64.3	86.3	41.2	239.7
	3	211.6	410.5	142.9	105.1	111.1	6.3	68.6	8.9	59.7	121.1	53.7	196.1	96.4	216.2
Aug	1	256.2	555.5	223.5	166.4	200.8	4.3	102.4	138.6	74.3	198.6	78.6	406.3	184.5	138.3
	2	307.8	605.9	279.1	223.8	264.9	66.0	185.4	194.8	148.0	295.4	64.7	640.6	216.5	234.6
	3	177.5	667.5	153.3	121.5	239.0	189.3	327.0	234.6	115.0	376.4	67.7	509.7	99.2	232.4
Sep	1	143.8	462.9	57.8	22.0	75.1	85.2	314.2	316.3	98.3	137.8	108.7	220.0	133.2	165.5
	2	85.9	131.2	0.0	39.5	19.6	2.3	429.9	259.9	32.6	37.2	82.4	34.6	14.3	145.5
	3	30.4	83.4	8.6	16.2	10.8	0.0	170.0	48.5	0.3	15.6	17.9	27.0	0.3	122.1
Oct	1	33.1	31.9	12.2	9.2	3.7	2.1	20.6	13.5	0.3	0.9	0.5	112.9	0.5	58.3
	2	7.9	34.8	2.7	2.1	2.9	2.0	10.1	0.1	0.0	0.4	0.8	175.9	0.0	30.1
	3	12.0	25.2	0.0	1.0	0.0	2.1	2.6	0.8	0.0	0.0	0.5	23.3	0.0	3.8
Nov	1	6.3	17.9	0.0	0.0	0.0	2.0	2.1	0.7	0.0	0.0	0.3	16.7	0.0	3.8
	2	0.0	8.3	0.0	0.0	0.0	0.2	0.7	0.1	0.0	0.0	0.0	13.0	0.0	1.8
	3	0.0	1.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	15.8	0.0	0.0
Dec	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11.9	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	1.6	0.0	9.1
	3	1.4	2.9	0.9	2.9	2.4	2.2	11.7	2.2	1.4	2.5	1.6	4.2	4.4	5.9
Jan	1	2.3	4.1	2.8	4.7	3.4	1.6	9.1		0.4	4.2	0.1	9.2	5.2	18.6
	2	0.0	0.1	0.7	9.0	0.1	0.0	8.7	11.5	0.0	0.8	0.0	15.9	8.3	29.1
	3	0.0	0.0	1.9	8.2	0.1	0.0	20.9	14.3	0.0	0.0	0.0	18.1	0.0	12.6
Feb	1	0.0	0.0	0.0	13.8	0.3	0.0	14.9	0.7	0.0	0.0	0.0	14.2	0.0	13.3
	2	0.0	0.0	0.0	11.4	0.0	0.0	0.4	0.0	0.0	0.0	0.0	1.1	0.0	4.9
	3	0.0	0.9	0.0	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.7
Mar	1	0.0	27.8	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.5
	2	0.0	77.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.4
	3	5.2	41.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11.9
Mean Flow		41.4	109.5	40.8	27.5	46.0	12.9	63.0	36.7	14.9	36.3	24.1	71.9	23.6	58.1

Source: Planning Division, WAPDA.

River Jhelum Flow Upstream Mangla (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	31.4	28.6	40.0	40.4	53.4	39.0	47.8	50.4	24.6	39.9	54.4	38.6	40.5	51.0
	2	30.6	53.3	46.7	47.8	65.7	51.9	72.5	38.0	30.2	54.5	44.9	54.9	33.8	54.2
	3	36.5	54.0	48.8	56.2	81.2	59.5	60.2	44.9	30.3	76.9	63.9	61.2	36.8	65.4
May	1	42.8	64.5	49.1	62.1	79.6	64.1	63.6	43.5	36.6	75.1	68.4	61.4	51.7	78.7
	2	38.4	64.7	51.8	57.9	72.1	51.7	82.5	49.6	47.0	78.6	64.9	54.3	61.6	98.3
	3	46.2	69.9	50.8	56.4	69.4	51.2	79.3	56.7	46.0	56.5	94.9	51.8	73.7	82.1
Jun	1	47.2	68.1	49.6	59.8	58.3	51.9	72.6	67.5	40.8	48.1	104.9	49.5	90.0	62.2
	2	39.8	58.3	56.6	64.8	45.7	56.7	64.0	59.6	37.0	61.0	93.2	40.9	65.5	52.6
	3	58.7	63.0	67.5	77.7	52.7	42.2	65.3	48.9	31.0	86.8	72.6	59.5	50.7	73.3
Jul	1	59.6	80.3	60.5	64.3	54.2	45.3	77.7	37.9	30.6	81.6	72.9	60.5	46.0	71.2
	2	72.5	59.8	58.1	57.9	55.7	38.6	56.5	30.3	39.8	69.8	62.1	105.9	55.0	46.0
	3	47.1	65.0	41.3	47.6	63.3	48.3	73.9	33.5	47.6	61.8	61.3	94.7	69.4	40.2
Aug	1	45.5	72.4	40.7	52.9	51.3	56.8	77.5	34.4	55.8	89.6	47.8	87.6	78.5	48.8
	2	33.3	62.9	28.0	34.3	37.6	49.9	55.2	41.9	37.4	53.8	40.5	54.3	42.5	41.6
	3	30.5	41.9	26.7	25.5	25.6	27.2	61.8	46.6	21.8	35.4	42.3	35.3	37.0	35.6
Sep	1	24.5	25.0	18.3	21.2	17.1	15.1	60.5	47.7	14.6	23.2	30.8	23.6	22.1	33.2
	2	19.9	22.1	19.0	33.8	14.6	12.5	32.7	27.0	14.1	16.6	21.7	19.9	16.7	24.1
	3	16.3	23.3	16.7	20.3	12.6	14.3	20.2	17.5	11.9	16.9	16.4	23.3	18.4	19.2
Oct	1	13.2	17.1	14.2	15.1	12.0	11.0	15.7	12.2	11.2	14.2	13.9	26.6	12.0	13.3
	2	9.6	13.0	11.1	11.8	10.4	10.5	14.7	8.9	10.8	14.6	36.8	15.2	17.8	12.6
	3	13.1	10.1	8.3	8.9	8.1	11.1	11.2	7.1	8.5	11.4	26.4	10.2	11.2	9.2
Nov	1	12.7	13.0	7.5	10.2	7.2	11.2	9.8	7.9	7.5	9.3	19.0	9.2	10.7	8.2
	2	9.8	12.9	7.3	7.8	8.1	14.9	9.4	6.6	5.7	17.6	14.0	7.6	9.6	6.6
	3	8.2	11.7	9.4	8.4	7.1	11.1	9.1	7.7	5.5	16.1	10.5	6.6	8.4	5.1
Dec	1	8.0	9.2	9.0	8.0	6.0	11.2	7.7	6.4	5.8	14.6	10.0	5.7	7.9	6.5
	2	7.5	8.8	5.6	5.9	4.2	11.7	6.4	6.6	9.4	35.9	7.1	6.8	7.7	8.1
	3	9.6	6.4	5.7	6.1	4.9	7.6	5.4	4.9	20.9	18.1	7.7	9.5	11.1	29.0
Jan	1	7.2	5.4	11.0	7.5	5.1	8.4	4.1	7.0	12.2	15.2	7.1	12.9	9.3	21.0
	2	8.2	6.8	6.6	4.7	4.2	7.9	4.4	6.8	9.5	11.7	6.4	9.4	8.2	13.9
	3	10.5	7.5	7.2	11.3	5.8	12.9	5.2	9.1	8.7	11.9	7.2	7.6	12.5	16.0
Feb	1	10.5	8.0	11.9	16.5	6.6	10.7	4.7	7.9	11.0	13.7	9.4	7.1	18.5	18.7
	2	10.8	12.8	15.8	21.3	8.3	10.3	5.4	6.4	21.2	16.2	10.4	8.0	22.5	42.1
	3	12.0	16.9	23.3	21.4	12.0	13.3	8.5	6.1	19.6	29.3	14.5	7.9	18.8	29.7
Mar	1	20.9	32.5	35.8	34.4	22.3	28.0	9.1	11.3	15.7	31.2	18.7	8.8	20.1	43.5
	2	48.9	26.3	30.0	31.3	28.1	39.3	17.8	9.8	53.2	33.7	58.9	16.7	40.4	44.6
	3	31.7	26.6	38.9	53.5	48.1	49.6	24.3	16.7	40.5	46.4	40.6	33.4	83.0	54.6
Mean Flow		27.0	33.9	28.6	32.4	31.1	29.4	36.0	25.7	24.3	38.5	38.2	33.0	33.9	37.8

Source: Planning Division, WAPDA.

* Total Outflow – Mean Spillway Discharge + Mean Power Station+Mean Jari Outlet.

River Jhelum Flow Downstream Mangla (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	19.2	11.3	7.5	11.8	9.2	23.0	26.8	12.3	13.6	27.5	4.0	23.6	9.4	22.7
	2	19.8	12.1	7.7	18.2	9.0	17.9	26.5	18.7	11.4	27.1	6.1	16.5	14.4	22.3
	3	21.0	30.0	9.5	17.1	75.6	16.7	26.5	21.0	18.3	24.4	14.5	16.6	16.6	33.5
May	1	25.8	52.1	44.0	40.2	51.8	16.3	25.5	23.4	21.1	36.9	29.5	22.2	18.6	64.4
	2	30.3	53.5	52.3	36.6	83.6	26.7	37.1	38.2	23.4	41.1	64.5	34.5	33.8	64.4
	3	37.3	50.1	53.3	45.1	60.9	18.6	72.6	36.8	23.0	41.1	59.0	36.7	49.4	59.7
Jun	1	32.6	51.0	62.8	25.9	47.2	34.1	49.3	22.4	18.1	41.1	104.2	32.6	49.4	57.5
	2	31.5	47.5	34.1	20.1	32.7	37.2	42.2	18.0	17.6	41.1	120.6	25.8	47.8	36.6
	3	13.5	44.9	56.3	48.0	33.1	23.1	43.1	11.9	13.7	77.6	51.1	42.9	26.0	58.7
Jul	1	6.2	72.4	46.3	53.1	18.6	36.8	70.3	8.3	13.0	59.9	50.0	46.4	23.9	52.0
	2	2.3	50.4	43.5	48.6	47.3	26.5	42.8	5.1	5.1	46.5	36.1	90.1	21.8	27.9
	3	4.3	41.4	12.3	29.4	48.2	38.3	60.8	4.6	3.7	43.2	43.6	81.8	50.0	25.7
Aug	1	11.4	36.4	9.0	36.3	38.0	38.3	63.8	10.9	3.4	74.2	31.0	65.2	64.5	19.3
	2	19.9	44.9	10.6	20.6	24.1	34.6	40.7	33.7	4.7	35.9	33.3	-37.4	29.9	29.3
	3	19.3	28.7	5.4	10.8	12.9	18.8	54.3	37.9	4.3	26.0	33.4	24.7	27.4	28.1
Sep	1	14.1	9.9	8.0	13.2	12.3	13.2	52.9	40.1	5.1	14.2	21.8	17.6	17.0	30.6
	2	11.4	13.6	20.1	24.8	19.4	15.7	22.6	18.7	14.8	15.4	18.1	13.1	17.0	15.7
	3	7.7	14.1	20.7	21.8	25.9	19.9	11.4	15.5	18.9	21.2	20.0	13.8	18.1	12.0
Oct	1	18.2	12.3	21.8	23.2	22.2	20.4	19.0	19.5	21.4	21.3	21.6	16.0	22.4	15.5
	2	21.2	20.6	20.0	16.7	20.6	19.5	13.2	18.4	6.5	15.4	20.4	8.9	19.7	21.6
	3	18.4	20.0	17.6	15.2	16.8	19.0	13.0	13.3	11.3	11.6	15.2	8.8	17.8	23.7
Nov	1	13.5	16.3	18.1	15.4	17.2	18.8	12.9	18.1	13.8	21.4	19.7	8.8	16.7	16.0
	2	15.2	15.7	16.9	17.3	16.3	18.0	17.8	20.6	16.3	21.1	20.8	13.6	18.1	17.7
	3	15.5	15.3	18.1	17.3	16.9	17.4	22.0	19.2	20.6	20.1	22.3	15.5	19.2	18.3
Dec	1	15.7	15.3	16.8	17.3	17.3	16.6	22.9	20.3	19.2	17.4	20.9	16.4	18.8	18.8
	2	12.4	15.3	17.1	17.4	15.8	11.8	18.1	17.4	18.4	20.9	19.8	18.0	18.7	18.8
	3	5.7	5.0	16.4	13.9	6.7	14.0	14.4	16.4	13.2	20.0	21.3	20.7	11.7	10.4
Jan	1	12.3	9.6	3.8	3.7	6.0	1.7	16.5	1.6	8.6	19.3	17.9	15.9	5.3	13.3
	2	21.9	19.3	6.5	6.3	14.1	15.1	23.0	8.0	8.9	27.3	10.0	14.5	9.3	17.8
	3	18.2	20.6	17.2	16.7	12.8	22.9	15.9	16.1	15.3	23.1	6.5	15.0	15.1	17.0
Feb	1	18.1	20.3	17.4	13.3	6.2	17.3	11.3	19.4	16.8	20.2	10.8	17.9	25.5	20.0
	2	17.8	19.5	14.2	8.7	7.6	16.7	21.2	23.4	21.6	27.1	29.3	30.9	29.9	25.9
	3	18.2	7.2	13.5	8.2	7.8	15.0	12.3	18.4	20.5	22.9	32.9	28.3	33.4	36.6
Mar	1	16.0	7.3	19.1	10.6	7.5	25.3	16.2	8.7	20.1	31.0	24.5	19.8	34.0	21.1
	2	7.8	8.0	17.5	26.9	6.4	24.1	8.2	9.9	20.7	35.0	23.4	17.6	31.2	18.5
	3	5.9	7.9	15.4	16.3	14.7	30.2	10.7	9.2	27.4	21.5	21.8	17.3	21.3	19.9
Mean Flow		16.65	25.55	21.96	21.83	24.51	21.65	29.38	18.20	14.82	30.30	30.55	26.26	25.08	28.09

Source: Planning Division, WAPDA.

* Total Reservoir Outflow – UJC at Skya.

River Jhelum Flow Upstream Rasul (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	18.7	12.7	9.3	13.3	9.4	19.4	28.7	12.4	11.4	29.4	8.4	23.9	10.0	22.0
	2	19.8	14.9	9.5	18.8	9.9	19.9	32.5	17.4	9.8	29.3	6.1	16.9	14.7	22.9
	3	21.8	30.4	10.8	18.4	75.4	20.4	28.1	22.7	17.5	26.2	15.2	16.9	18.9	31.5
May	1	27.4	50.3	43.8	39.8	54.3	20.5	30.0	20.9	19.4	38.3	27.6	22.4	20.0	68.2
	2	29.2	53.8	54.3	39.6	86.3	27.2	40.2	37.4	21.6	42.2	66.2	33.9	32.4	62.1
	3	39.6	48.8	54.5	45.7	64.5	21.0	75.0	39.6	21.7	43.0	62.8	38.7	45.6	59.8
Jun	1	38.4	53.4	63.1	29.0	49.6	35.7	52.1	63.5	20.0	41.3	107.5	28.5	46.6	57.5
	2	31.5	48.4	40.3	21.6	35.9	40.0	39.0	39.8	17.1	42.3	126.4	29.4	48.4	39.0
	3	19.2	46.5	54.0	48.4	37.6	27.6	39.5	24.8	14.5	79.4	52.0	42.0	25.3	58.8
Jul	1	11.9	79.3	50.0	60.0	21.9	39.0	73.9	13.8	14.8	63.4	52.2	47.4	26.3	59.0
	2	9.2	56.2	46.3	52.8	54.0	29.5	46.2	11.5	11.0	54.2	39.2	102.8	24.5	29.7
	3	11.6	55.1	19.8	37.0	56.2	39.0	69.7	11.7	10.2	50.2	44.6	89.4	52.5	28.9
Aug	1	18.2	45.8	16.6	40.4	46.2	39.5	76.5	4.6	13.6	85.2	32.1	75.8	74.2	22.7
	2	26.1	55.9	15.6	25.2	32.6	44.4	49.7	42.9	8.2	44.6	32.4	46.4	34.4	34.4
	3	23.3	33.0	12.0	13.6	15.6	25.9	61.2	45.5	6.1	31.0	41.3	27.2	36.9	31.9
Sep	1	18.7	11.6	7.5	14.6	13.9	15.5	65.8	44.2	4.6	15.7	23.7	20.9	20.2	38.2
	2	15.0	18.7	22.4	28.6	20.5	16.4	23.5	24.2	15.7	15.9	19.4	15.1	21.8	18.8
	3	11.0	14.0	22.8	22.4	26.5	20.0	15.0	18.2	20.1	23.2	21.9	18.0	21.8	15.8
Oct	1	20.0	15.8	23.0	23.7	22.5	21.1	19.8	21.5	22.8	22.6	23.8	21.1	23.3	16.1
	2	23.1	21.5	21.1	19.6	21.6	20.6	16.6	21.1	7.7	18.7	22.7	9.7	22.4	23.6
	3	21.6	21.5	19.0	17.6	19.2	19.7	14.5	16.3	12.6	11.7	16.2	11.0	21.4	22.6
Nov	1	15.1	18.3	20.0	17.9	19.5	19.3	13.5	17.1	15.5	21.7	20.3	9.6	19.7	19.0
	2	16.8	16.8	19.3	19.5	17.7	19.0	18.1	22.1	18.3	23.5	21.9	13.7	20.5	20.6
	3	16.6	16.4	20.0	19.7	19.3	17.8	21.3	22.5	23.0	24.5	22.8	16.0	23.5	21.4
Dec	1	17.3	17.1	18.9	19.6	19.4	17.8	22.9	21.0	21.4	18.5	21.7	15.9	21.4	21.5
	2	14.7	17.0	19.5	19.9	17.2	15.3	20.3	20.5	20.5	18.9	21.1	16.6	21.6	21.6
	3	7.4	8.1	18.4	17.9	9.0	17.2	16.7	17.6	14.0	22.2	22.2	23.1	14.9	22.2
Jan	1	13.6	8.7	5.2	9.1	8.0	3.2	20.5	7.1	12.5	21.7	19.7	20.7	9.3	23.0
	2	22.5	20.8	7.0	9.1	11.7	10.0	24.9	10.1	11.2	31.3	11.8	17.0	12.2	23.0
	3	21.3	22.2	17.6	16.5	13.4	20.1	18.4	16.2	14.3	22.6	8.8	16.5	15.4	18.1
Feb	1	20.1	20.9	18.6	12.6	5.6	16.8	13.0	19.1	17.3	20.8	13.5	19.5	25.6	21.4
	2	19.3	19.3	16.3	6.9	6.7	16.2	18.8	19.7	20.5	25.2	27.0	31.8	29.5	26.2
	3	19.0	8.4	13.6	9.6	6.4	14.8	13.5	16.2	19.9	24.4	35.1	32.8	32.3	39.1
Mar	1	16.9	10.4	20.4	12.2	8.3	21.5	15.9	8.1	19.8	30.3	27.0	23.3	29.7	23.8
	2	11.1	10.8	17.9	28.9	8.1	20.3	9.9	9.1	22.3	33.2	23.4	18.8	29.3	19.5
	3	7.9	10.9	16.1	18.0	15.0	25.4	10.4	8.3	31.1	22.7	22.8	15.8	20.5	22.7
Mean Flow		19.3	28.2	24.0	24.1	26.6	22.7	32.1	21.9	16.2	32.5	32.2	28.6	26.9	30.7

Source: Planning Division, WAPDA.

* LJC Feeder + RQ Head + Rasul Below. Average of 8 Days.

River Jhelum Flow Downstream Rasul (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	4.2	0.0	0.0	0.0	3.7	7.6	9.5	0.0	0.0	14.8	8.4	4.9	2.8	3.3
	2	0.0	0.0	0.0	4.1	3.4	7.0	22.4	1.1	0.0	11.6	2.1	0.0	0.0	2.6
	3	0.4	13.7	0.0	10.9	62.9	7.6	24.4	2.7	0.0	8.2	0.0	0.0	0.0	11.2
May	1	9.8	50.7	24.3	19.2	37.9	3.6	18.1	0.0	0.0	20.2	11.4	5.8	6.8	47.8
	2	12.1	33.4	34.6	18.2	77.8	7.3	23.5	15.7	0.0	23.9	54.3	19.1	14.2	41.4
	3	21.8	28.4	33.1	23.8	58.1	1.2	56.7	17.8	0.0	22.6	46.0	20.6	29.7	38.9
Jun	1	25.9	33.0	41.5	7.1	39.0	15.1	33.7	33.0	0.0	19.2	90.8	15.6	37.5	36.5
	2	16.3	31.9	22.4	0.0	14.5	19.0	20.6	23.3	0.0	22.3	113.2	7.6	41.8	18.2
	3	5.0	39.3	41.4	32.9	16.9	6.7	21.0	13.0	0.0	61.2	36.2	23.3	7.0	37.9
Jul	1	7.9	75.4	46.1	56.0	11.9	18.0	55.4	7.2	0.0	49.0	37.0	29.8	7.1	42.6
	2	5.1	53.1	42.4	48.8	41.4	8.8	34.1	6.4	4.5	41.1	34.9	95.2	4.9	16.0
	3	7.7	52.3	15.9	32.9	49.2	25.1	61.7	10.0	7.0	43.1	41.4	87.4	36.8	15.1
Aug	1	14.3	43.8	12.5	36.3	43.7	36.0	73.2	2.3	10.5	81.9	28.4	72.5	74.2	9.1
	2	22.2	52.6	11.3	21.2	29.8	40.7	47.3	40.2	4.1	41.6	28.2	43.1	34.2	21.0
	3	19.4	29.3	8.1	0.4	8.4	17.7	60.0	43.1	0.8	24.5	37.1	23.1	33.8	18.2
Sep	1	14.8	7.9	0.8	0.1	0.1	0.0	65.2	42.4	0.0	1.8	17.9	5.4	4.4	25.0
	2	11.1	4.2	1.1	7.6	0.1	0.0	22.1	8.9	0.0	0.0	2.9	0.0	0.0	2.4
	3	4.8	0.0	0.1	0.1	4.0	0.0	0.9	0.0	0.0	0.5	0.0	5.5	0.0	0.0
Oct	1	2.5	0.0	0.1	0.1	0.0	0.1	1.5	0.0	0.0	0.0	0.0	18.1	0.0	0.0
	2	0.2	0.4	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	6.7	0.0	1.1
	3	0.8	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.2
Nov	1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.6	0.0	2.2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	2.6	0.0	0.0	1.6	0.0
Dec	1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0
	2	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.5	0.0	0.0	0.6	0.0
	3	7.4	7.3	0.0	0.1	9.1	8.7	1.4	0.0	0.0	5.9	1.9	16.6	6.8	13.8
Jan	1	11.0	5.0	5.2	9.1	6.7	3.2	24.9	7.1	12.5	7.7	2.6	20.7	9.3	23.0
	2	0.5	0.0	7.0	9.1	0.0	1.5	18.4	3.2	5.0		11.1	17.0	3.5	23.0
	3	0.0	0.0	0.3	2.6	0.0	0.1	2.7	0.0	0.0	3.3	8.5	2.8	0.0	6.7
Feb	1	0.0	0.0	0.2	6.8	0.0	0.0	0.0	0.0	0.0	1.0	2.5	5.5	5.1	3.5
	2	0.0	0.0	0.1	2.7	0.0	0.0	0.0	0.0	0.6	5.0	6.5	14.5	9.7	8.8
	3	0.0	6.8	0.1	0.1	0.0	0.1	0.0	0.0	0.0	5.0	11.9	10.5	12.2	19.9
Mar	1	0.0	0.0	0.0	1.3	0.0	3.8	0.0	0.0	0.0	11.7	6.2	1.5	9.9	3.8
	2	0.0	0.0	0.0	14.7	0.0	0.3	0.0	0.0	7.6	14.7	4.2	0.0	9.2	0.0
	3	0.0	0.0	0.0	7.8	5.4	5.8	0.0	0.0	21.5	7.6	4.0	1.5	11.0	2.9
Mean Flow		6.3	15.8	9.7	10.4	14.6	6.8	19.4	7.7	2.1	15.9	18.0	16.0	11.5	13.7

Source: Planning Division, WAPDA.

River Chenab Flow Upstream Marala (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	16.1	19.1	32.3	17.8	33.2	30.6	30.7	20.9	11.1	18.7	23.6	21.0	19.4	25.7
	2	15.7	34.8	34.4	18.6	37.3	34.8	76.2	16.8	13.5	25.9	22.5	37.2	17.8	29.4
	3	17.3	30.3	37.5	22.3	39.9	45.6	44.5	20.0	14.4	42.8	32.1	33.2	18.5	32.5
May	1	23.0	44.7	35.1	31.4	56.1	50.1	48.4	24.8	21.5	35.9	38.6	36.4	29.6	43.4
	2	22.7	51.8	32.7	31.2	54.2	51.6	53.6	30.1	24.0	48.6	40.7	47.2	34.7	73.4
	3	32.9	56.9	31.6	32.2	61.8	47.1	52.0	48.2	43.3	29.0	47.5	42.9	45.4	66.0
Jun	1	41.6	64.6	31.7	52.1	48.9	52.7	50.9	76.6	49.3	31.9	68.2	50.7	78.4	55.5
	2	28.4	61.9	57.2	66.4	39.1	77.6	47.0	75.4	57.5	61.1	59.3	51.5	59.3	51.2
	3	79.5	99.1	92.4	95.1	79.2	62.3	67.4	77.7	57.6	94.0	55.4	91.7	54.9	100.9
Jul	1	107.4	142.7	81.0	88.6	73.8	65.4	86.6	82.0	68.5	83.8	76.1	90.7	55.6	110.9
	2	141.0	109.9	110.6	116.0	97.4	68.3	51.8	62.5	122.0	94.1	75.3	148.3	87.9	78.2
	3	92.7	118.4	83.9	94.0	140.6	107.6	105.0	88.3	101.5	127.0	93.2	146.7	140.9	72.3
Aug	1	138.9	144.6	101.6	123.1	117.5	172.8	135.6	97.7	115.9	132.7	75.1	130.0	86.2	104.5
	2	88.8	115.5	80.5	68.3	112.6	106.0	103.7	97.8	79.9	97.3	76.0	97.2	66.2	82.2
	3	62.5	99.8	61.3	54.7	66.9	62.2	119.2	96.0	70.2	66.9	83.3	77.7	72.8	70.5
Sep	1	73.6	48.2	49.0	44.0	42.3	40.4	118.6	73.3	42.7	51.3	67.2	50.8	38.1	76.4
	2	52.2	49.5	36.1	38.0	27.3	34.6	66.7	38.2	35.7	28.2	41.4	40.8	34.5	49.5
	3	36.7	37.2	17.7	20.5	27.1	22.0	35.4	23.3	26.5	27.1	30.9	163.5	29.3	37.2
Oct	1	22.6	22.7	14.0	18.5	16.1	12.1	24.7	12.1	27.8	21.8	17.1	43.6	17.6	23.4
	2	15.2	16.5	12.6	12.9	12.0	10.1	18.3	10.6	19.4	17.6	19.3	25.3	13.6	15.9
	3	14.8	13.9	9.8	9.6	10.4	8.8	12.7	9.6	11.1	13.3	15.4	17.5	11.0	13.1
Nov	1	12.2	13.5	8.6	8.1	10.2	8.6	10.6	8.0	8.7	10.2	11.2	14.9	9.2	10.2
	2	8.2	11.8	7.7	6.8	8.6	9.4	9.4	7.1	7.4	13.3	9.6	11.8	7.7	8.6
	3	6.7	8.3	8.7	9.0	7.2	8.0	7.7	7.2	6.5	15.3	9.4	10.0	7.2	8.0
Dec	1	6.2	7.0	7.1	6.9	6.8	8.8	7.1	6.2	5.9	12.8	7.0	8.7	6.6	7.1
	2	5.4	6.1	6.3	6.1	6.6	8.1	6.6	6.9	8.0	24.5	6.8	8.2	6.3	9.5
	3	10.4	6.3	6.8	9.6	7.1	7.3	6.3	5.3	25.7	14.4	7.5	14.3	8.1	33.6
Jan	1	6.9	5.9	9.6	11.4	9.7	8.2	5.8	8.6	10.8	11.4	6.7	15.6	7.2	21.2
	2	5.7	7.5	7.6	11.8	5.9	7.5	5.9	6.9	8.6	10.0	8.0	12.6	6.9	13.8
	3	7.6	10.3	6.8	25.2	6.2	12.3	5.9	7.6	7.8	9.2	7.7	9.9	15.0	13.0
Feb	1	7.9	6.7	11.5	29.7	6.5	9.3	5.4	7.5	6.9	8.1	6.7	9.3	11.8	12.5
	2	10.2	6.4	14.4	21.5	8.5	10.9	6.4	6.8	13.5	9.3	6.2	9.8	13.3	32.0
	3	9.9	28.8	11.8	15.3	10.6	15.6	9.0	7.0	15.5	26.5	10.6	8.3	23.4	20.0
Mar	1	28.3	49.7	22.3	21.8	32.1	27.4	8.8	7.1	11.2	15.0	12.7	9.7	19.2	31.2
	2	48.9	25.1	15.5	17.2	24.2	36.8	14.9	6.5	27.1	15.8	49.9	9.5	23.5	21.2
	3	20.4	24.3	16.0	40.4	40.8	35.4	14.1	8.6	21.6	28.2	25.6	20.3	72.4	31.5
Mean Flow		36.62	44.43	33.43	36.01	38.46	38.23	40.91	33.03	33.23	38.13	34.55	44.91	34.70	41.26

Source: Planning Division, WAPDA.

River Chenab Flow Downstream Marala (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	5.1	10.0	23.1	7.9	23.6	25.3	22.2	10.4	1.1	7.8	23.6	10.3	11.5	15.2
	2	3.6	23.9	24.5	18.6	27.0	28.7	69.2	5.7	2.6	15.1	11.1	20.2	9.4	18.9
	3	2.5	15.1	25.8	11.6	28.5	39.0	37.3	8.2	3.0	27.2	14.4	18.3	7.5	22.0
May	1	7.4	28.0	18.0	16.1	34.2	44.2	40.1	9.0	6.5	13.4	25.7	24.2	15.1	27.6
	2	7.8	32.8	13.6	6.2	28.8	47.2	42.4	6.4	3.4	24.0	32.5	25.4	16.7	49.2
	3	11.6	37.3	10.9	6.3	33.6	39.6	32.1	22.5	17.3	5.5	33.6	16.6	16.5	41.5
Jun	1	16.5	40.7	10.2	23.7	19.9	38.0	32.1	49.8	24.5	6.3	46.7	23.2	57.1	29.8
	2	3.2	36.2	31.8	37.1	9.5	55.6	25.1	48.4	30.4	32.6	33.1	23.5	35.0	24.1
	3	51.9	73.0	67.4	67.7	48.9	35.4	44.5	50.2	29.5	66.8	33.8	63.2	27.8	72.9
Jul	1	81.0	120.9	54.5	58.8	43.3	35.3	62.5	54.6	40.0	57.1	49.5	62.7	27.6	82.4
	2	128.1	85.5	84.1	88.1	69.0	37.5	30.3	35.6	100.2	67.6	48.4	126.9	60.5	49.7
	3	69.0	94.1	57.4	65.0	118.8	80.1	82.5	61.3	86.8	106.1	66.1	125.8	116.7	43.8
Aug	1	115.3	122.5	77.3	64.9	92.4	148.5	114.3	70.5	96.2	121.4	47.8	112.8	78.2	76.0
	2	62.3	94.6	54.5	40.5	83.1	77.1	77.6	72.2	54.5	78.6	48.2	73.4	44.5	54.0
	3	35.9	67.4	35.2	24.8	36.2	33.4	95.5	72.3	42.9	43.1	55.0	49.2	45.4	42.8
Sep	1	46.1	22.8	22.0	14.3	10.9	12.4	93.5	48.6	15.7	25.7	38.8	21.3	10.5	53.0
	2	24.2	24.2	8.7	11.2	3.8	8.0	41.2	13.0	8.5	5.8	12.9	11.1	6.7	24.7
	3	12.7	13.5	2.3	3.2	4.0	6.1	9.7	4.9	4.4	3.4	5.0	148.4	6.7	12.7
Oct	1	8.7	7.1	1.5	2.3	4.0	4.4	6.3	1.6	7.9	2.0	2.3	41.5	35.0	5.0
	2	2.1	2.8	1.7	1.7	1.7	1.5	2.4	0.3	3.2	2.2	3.2	23.0	3.0	3.5
	3	1.2	2.4	0.4	0.0	0.6	0.0	2.2	0.0	0.6	1.0	2.9	8.6	3.0	3.0
Nov	1	2.6	1.2	0.1	0.3	1.5	0.0	2.0	0.0	2.0	1.0	3.0	5.6	3.0	3.0
	2	0.0	0.6	0.1	0.0	1.1	0.3	1.6	0.0	1.8	1.7	2.9	3.8	2.5	2.5
	3	0.0	0.0	0.7	1.6	1.0	0.0	1.0	0.0	0.0	1.9	2.0	3.0	2.0	2.0
Dec	1	0.0	0.0	0.0	0.0	1.0	1.3	0.3	0.0	0.3	0.0	1.4	3.0	2.0	2.0
	2	0.0	0.0	0.0	0.0	1.0	0.3	0.0	0.0	0.4	7.2	0.8	3.0	2.0	2.3
	3	10.4	6.3	0.4	2.0	7.1	4.3	0.0	0.0	15.5	0.0	0.0	10.6	5.2	30.8
Jan	1	5.5	4.3	9.6	11.4	7.8	8.2	0.0	8.6	10.8	0.0	0.0	15.6	7.2	21.2
	2	0.0	0.1	7.6	11.8	0.0	3.2	5.9	4.3	6.3	0.0	8.0	12.0	6.5	8.5
	3	1.6	2.4	1.6	23.5	0.0	5.4	5.9	0.0	0.0	0.0	7.0	7.6	11.2	3.0
Feb	1	0.1	0.0	4.2	24.2	0.0	1.2	1.2	0.0	0.0	0.0	2.9	3.2	4.3	3.3
	2	1.8	0.0	5.4	10.0	1.5	2.6	1.2	0.0	1.7	0.0	0.0	2.9	3.4	23.6
	3	0.6	24.3	2.6	5.7	6.6	6.8	1.7	0.0	4.4	13.7	2.3	3.0	14.1	9.2
Mar	1	18.5	43.9	12.1	9.4	28.6	19.0	0.0	0.0	2.1	5.0	1.1	3.0	10.1	20.7
	2	41.8	18.9	5.0	7.6	18.1	27.9	5.7	0.0	17.1	5.0	38.4	3.0	13.9	12.1
	3	13.6	16.6	4.5	33.1	32.6	27.2	3.6	0.0	11.5	18.9	15.2	12.3	63.9	21.7
Mean Flow		22.0	29.8	18.9	19.7	23.0	25.1	27.6	18.3	16.1	21.3	20.0	31.1	21.8	25.5

Source: Planning Division, WAPDA.

River Chenab Flow Upstream Khanki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	8.2	13.1	25.8	11.1	25.5	24.9	23.8	13.3	6.1	11.8	28.9	16.0	15.3	17.7
	2	6.1	24.4	26.8	21.4	28.0	28.4	57.1	9.1	8.3	19.8	17.3	26.2	14.3	21.4
	3	8.3	17.8	27.4	15.5	33.4	45.3	28.6	9.8	8.4	52.3	19.8	28.0	11.0	27.5
May	1	10.7	27.1	21.0	18.6	33.3	37.1	33.0	12.3	10.6	19.0	40.7	37.2	18.9	29.0
	2	11.5	32.5	15.8	11.6	27.0	40.6	41.2	10.5	7.8	30.4	49.9	31.8	21.7	57.4
	3	17.8	40.8	13.2	10.9	35.7	32.0	40.9	23.2	18.3	8.8	37.6	25.3	33.1	42.7
Jun	1	26.2	44.9	9.5	25.3	22.4	29.5	29.2	56.9	23.7	9.8	52.0	25.4	68.3	32.1
	2	9.1	40.7	27.9	37.4	12.4	48.0	26.6	60.2	25.8	36.2	43.3	24.3	42.5	29.9
	3	63.6	79.9	63.3	64.8	49.9	31.5	29.9	46.2	29.3	69.4	40.6	70.9	30.8	83.6
Jul	1	96.1	148.1	60.4	59.9	52.5	28.6	77.2	60.5	34.5	70.7	63.3	82.6	34.6	95.1
	2	200.4	112.8	97.5	80.3	84.9	20.8	26.1	47.9	100.3	62.3	59.8	178.5	77.6	62.5
	3	91.2	123.9	58.0	64.7	168.7	42.2	105.3	60.1	95.0	135.3	70.8	159.2	131.9	56.6
Aug	1	89.8	132.8	94.6	92.5	101.4	132.5	128.3	87.7	106.0	164.7	52.6	162.7	98.1	90.5
	2	68.2	111.7	58.2	42.1	85.1	103.1	82.3	90.0	74.9	99.7	49.8	88.1	46.1	62.1
	3	47.6	75.5	36.1	25.5	37.6	42.8	93.5	91.7	58.2	49.6	64.8	53.7	57.7	58.3
Sep	1	44.2	25.9	23.0	16.0	15.3	12.5	122.1	18.2	31.8	30.5	43.9	24.2	13.0	70.6
	2	29.1	29.7	10.8	19.1	8.3	10.6	52.8	16.8	12.8	9.0	19.7	17.5	10.5	43.9
	3	16.1	15.1	9.7	9.9	9.0	10.0	14.6	13.2	10.0	9.1	10.0	212.5	10.5	22.8
Oct	1	13.5	11.2	8.4	9.3	9.0	9.2	10.3	8.0	12.2	9.1	8.8	46.0	8.4	8.4
	2	10.7	7.9	7.8	9.0	7.6	7.6	9.7	6.9	12.2	9.0	9.1	27.9	7.6	8.4
	3	8.4	9.2	7.2	7.5	6.7	6.2	8.6	6.5	7.3	7.4	9.1	13.3	6.8	7.2
Nov	1	10.1	8.1	7.0	7.1	7.1	6.3	8.0	6.2	7.8	7.8	9.2	9.9	7.3	7.1
	2	7.6	7.8	7.2	6.4	6.7	6.4	7.4	6.2	7.7	7.4	9.1	9.0	6.8	7.3
	3	7.4	7.3	7.0	7.4	6.8	6.9	7.6	6.0	6.8	8.8	8.9	7.6	7.2	6.9
Dec	1	7.4	7.1	8.0	7.5	6.9	6.9	8.6	7.2	7.3	7.3	9.0	8.0	6.8	6.7
	2	7.3	7.3	7.9	7.3	6.7	6.9	7.1	6.0	7.2	10.1	8.0	9.0	7.1	7.4
	3	13.6	11.3	7.9	8.1	10.0	7.5	9.4	5.8	21.0	7.9	4.4	19.0	10.0	43.7
Jan	1	10.8	9.2	14.3	14.2	9.3	13.4	6.6	13.8	17.1	7.3	0.5	28.4	11.7	30.3
	2	0.9	0.8	12.4	13.3	0.5	5.2	3.2	7.2	12.4	1.6	8.1	9.3	5.7	9.4
	3	4.5	3.3	3.6	32.5	1.6	12.9	1.7	0.8	1.5	0.4	12.6	5.2	9.1	4.0
Feb	1	7.4	7.0	4.3	32.9	5.4	7.1	6.9	3.2	6.0	5.8	6.6	1.6	7.3	6.0
	2	8.4	8.1	8.8	13.0	5.6	7.1	6.5	6.4	8.5	6.3	5.2	5.2	10.2	25.9
	3	8.1	16.6	10.0	10.8	10.2	7.5	8.4	6.1	11.1	12.7	7.7	9.3	17.6	12.5
Mar	1	21.6	48.0	16.1	14.6	26.7	23.9	6.7	5.5	7.3	10.9	8.9	9.0	15.8	22.8
	2	39.2	22.4	9.5	13.7	22.7	24.2	12.3	5.1	22.7	11.4	45.4	7.8	15.0	17.8
	3	24.0	24.5	9.4	30.3	41.5	26.1	10.6	4.5	17.5	22.7	20.4	16.9	83.7	24.1
Mean Flow		29.3	36.5	23.2	24.2	28.4	25.3	32.0	23.3	23.8	29.0	26.6	41.8	27.2	32.2

Source: Planning Division, WAPDA.

River Chenab Flow Downstream Khanki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	2.5	3.3	16.3	2.3	16.6	16.6	14.7	3.4	0.3	4.0	18.4	6.5	7.3	10.0
	2	0.6	15.1	17.4	10.7	17.8	19.1	54.2	0.7	0.4	11.2	7.0	16.8	6.2	13.3
	3	2.0	7.8	17.6	10.3	23.3	35.7	24.8	0.2	0.3	23.4	9.3	18.0	2.9	19.5
May	1	2.6	16.8	11.4	8.7	23.3	27.6	24.3	2.6	1.7	9.7	30.2	27.8	10.8	20.4
	2	2.5	22.1	5.6	2.9	17.1	34.2	31.4	1.6	0.0	21.3	39.6	23.0	13.6	43.5
	3	8.2	30.4	4.7	2.5	25.5	22.5	31.6	13.2	8.0	0.4	27.1	15.5	24.5	33.6
Jun	1	15.8	34.3	3.3	16.7	12.6	19.2	18.9	46.3	14.6	0.9	41.4	16.5	58.9	22.9
	2	2.3	29.8	17.9	26.7	3.5	37.7	16.2	50.2	16.0	26.3	32.6	14.5	32.9	20.2
	3	54.4	69.2	53.2	55.2	39.7	21.1	19.2	35.6	19.3	58.7	29.9	60.5	21.0	73.8
Jul	1	88.1	140.5	50.1	49.9	44.4	18.0	67.1	50.1	23.9	60.1	52.6	71.9	25.4	85.9
	2	196.5	104.7	87.1	70.5	80.1	10.0	15.7	37.2	91.6	71.7	49.1	171.7	68.4	52.7
	3	80.9	120.7	48.3	54.2	168.7	33.0	97.0	49.5	85.2	127.0	60.1	153.8	123.8	46.8
Aug	1	81.4	128.8	85.0	82.3	96.8	125.2	121.3	77.2	98.8	158.2	41.9	154.6	93.9	80.8
	2	58.5	103.6	47.9	32.1	76.4	97.7	71.6	79.5	64.7	89.6	39.2	77.5	37.5	52.5
	3	36.5	65.2	25.5	15.3	28.1	33.4	85.0	81.9	48.0	39.0	54.1	43.1	49.1	48.6
Sep	1	33.9	15.5	12.0	6.1	5.5	2.3	115.6	8.2	21.1	19.9	33.3	13.9	4.0	61.1
	2	18.9	18.8	1.6	8.9	0.9	0.3	42.6	6.8	2.6	0.1	9.2	7.2	1.1	34.1
	3	6.4	5.7	0.8	0.1	0.5	0.0	4.9	5.2	0.0	0.0	0.6	205.8	1.6	13.1
Oct	1	3.7	1.0	0.8	0.4	0.3	0.0	0.0	0.2	3.4	0.0	0.0	38.4	0.2	1.5
	2	1.6	0.6	0.8	0.6	0.3	0.0	0.0	0.2	3.7	0.2	0.0	17.4	0.0	0.8
	3	0.6	0.5	0.8	0.4	0.5	0.0	0.0	0.2	0.9	0.0	0.0	2.8	0.0	0.0
Nov	1	1.3	0.5	0.6	0.4	0.4	0.0	0.0	0.2	0.0	0.0	0.0	0.3	0.0	0.0
	2	0.7	0.6	0.6	0.4	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0
	3	0.6	0.4	0.6	0.8	0.0	0.4	0.0	0.2	0.0	1.2	0.0	0.0	0.0	0.0
Dec	1	0.6	0.5	0.6	0.4	0.0	0.3	1.0	0.8	0.4	0.0	0.0	0.0	0.0	0.0
	2	0.6	0.6	0.6	0.4	0.0	0.3	0.0	0.1	0.4	10.0	0.0	0.0	0.0	0.4
	3	4.0	1.2	0.6	1.2	0.7	0.3	2.6	0.2	13.9	0.0	0.4	9.9	2.0	36.7
Jan	1	3.8	1.6	4.1	5.1	0.4	4.3	0.0	4.4	8.3	0.0	0.5	18.5	6.8	21.7
	2	0.9	0.8	1.7	3.5	0.5	1.8	2.2	2.3	5.5	0.4	4.9	7.6	4.1	7.[illegible]
	3	2.0	1.7	3.6	32.5	1.6	12.9	1.7	0.8	1.5	0.4	4.3	5.2	9.1	4.0
Feb	1	0.9	1.5	4.3	32.9	0.6	4.2	1.0	0.6	2.6	1.4	0.0	0.3	6.2	3.2
	2	0.7	2.2	3.7	9.3	0.2	0.8	0.0	0.5	2.3	0.0	0.0	0.0	4.7	20.1
	3	0.7	16.6	1.6	2.1	3.8	0.0	1.2	0.4	4.8	4.7	0.4	0.3	11.0	6.5
Mar	1	12.7	48.0	8.6	4.7	19.0	15.0	0.0	0.3	0.7	1.0	1.1	0.0	7.6	16.8
	2	30.6	18.8	1.8	3.4	16.1	14.8	6.0	0.3	15.5	1.8	38.3	0.0	6.7	10.0
	3	15.4	16.0	1.7	22.6	30.8	17.1	1.4	0.3	10.9	12.9	11.8	9.1	76.8	15.9
Mean Flow		21.5	29.0	15.1	16.0	21.0	17.4	24.3	15.6	15.9	21.0	17.7	33.6	19.9	24.4

Source: Planning Division, WAPDA.

River Chenab Flow Upstream Qadirabad (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	11.2	15.6	21.6	14.8	24.4	32.5	27.9	15.7	10.4	19.1	20.3	27.1	13.8	28.0
	2	17.9	29.3	25.1	23.2	22.7	33.0	78.2	15.1	9.7	27.3	11.9	34.9	18.4	32.4
	3	19.9	20.9	24.2	17.3	35.4	41.1	34.3	20.1	14.4	35.5	22.2	30.4	19.4	35.3
May	1	18.0	34.9	27.2	30.3	44.8	45.7	38.2	22.2	19.1	24.7	40.2	38.7	21.5	40.1
	2	16.4	47.0	23.1	23.3	27.9	62.2	47.2	21.8	18.1	32.6	49.5	31.3	27.5	66.2
	3	17.8	51.4	25.6	22.0	35.6	47.3	41.3	30.8	25.7	20.7	41.7	26.1	37.6	53.0
Jun	1	20.1	51.4	22.7	45.1	22.6	40.8	39.2	47.6	29.6	20.9	59.0	28.2	59.3	44.1
	2	12.9	42.0	35.2	54.0	26.4	58.2	36.1	48.3	32.4	40.8	45.5	31.7	32.5	36.6
	3	45.3	73.2	60.5	74.4	53.7	40.7	46.5	40.9	32.3	74.4	41.2	67.9	35.8	91.8
Jul	1	100.4	143.0	48.9	49.6	51.3	45.6	74.0	46.6	40.9	65.7	65.6	86.0	39.8	94.1
	2	209.1	97.5	83.9	69.7	89.6	42.6	33.0	25.7	103.7	73.8	51.4	191.2	76.6	63.3
	3	81.8	111.9	46.5	56.5	173.6	84.9	85.9	47.6	94.9	128.6	57.8	179.6	126.4	54.6
Aug	1	121.1	136.6	94.0	76.8	100.2	131.1	120.2	65.2	120.9	173.7	43.5	160.0	100.1	88.8
	2	61.5	104.2	50.3	35.6	84.8	107.5	73.0	74.1	58.0	90.8	40.8	85.3	40.4	64.2
	3	35.0	61.9	28.2	25.5	33.0	32.3	103.3	74.6	44.1	46.1	55.2	44.0	48.6	63.0
Sep	1	32.2	17.1	18.0	21.2	18.9	19.1	110.3	51.8	24.0	32.2	32.1	27.1	19.9	71.2
	2	20.5	29.6	21.3	27.3	17.7	16.8	45.2	23.7	18.4	16.4	23.3	26.2	20.9	49.6
	3	11.0	14.4	18.5	18.9	19.3	16.8	17.8	18.1	17.7	19.7	21.3	204.7	21.0	28.0
Oct	1	15.3	13.2	18.3	18.4	18.2	17.5	15.7	18.0	18.2	17.8	18.7	42.7	19.5	16.9
	2	18.6	18.5	17.5	16.1	18.4	17.6	14.8	16.8	9.5	17.5	18.6	16.9	18.7	21.9
	3	16.9	17.8	16.4	14.9	17.1	17.5	11.8	13.3	11.0	10.7	14.7	12.3	18.1	21.8
Nov	1	12.2	14.8	17.3	14.8	18.1	16.1	11.6	15.1	14.3	18.0	18.4	11.3	18.8	18.6
	2	13.1	14.4	16.4	17.3	15.8	16.8	14.8	18.8	15.6	22.8	19.5	15.6	18.4	19.1
	3	13.4	13.6	17.1	17.3	17.1	16.1	16.5	18.1	19.0	21.3	19.6	16.8	18.9	19.1
Dec	1	14.4	13.9	16.4	17.4	17.3	16.4	17.3	18.0	17.6	17.3	20.5	16.5	18.8	18.6
	2	12.6	14.1	16.8	18.2	16.9	12.1	16.6	18.1	17.9	30.6	19.6	17.1	18.6	19.2
	3	2.9	5.0	16.7	16.6	3.1	11.1	14.3	15.9	21.2	15.9	18.4	14.6	10.4	40.6
Jan	1	4.5	4.8	9.5	6.9	2.0	4.1	16.9	4.5	9.2	15.3	14.2	11.1	6.0	14.1
	2	15.4	14.9	2.8	4.7	12.6	7.4	5.5	5.0	7.0	15.8	7.7	7.9	10.9	8.0
	3	16.1	14.5	13.5	39.3	11.7	21.7	4.6	12.5	12.8	17.8	4.4	21.3	21.3	12.5
Feb	1	16.9	16.4	16.6	29.0	7.1	20.2	8.1	16.9	15.0	20.9	7.2	18.3	26.5	20.0
	2	17.2	17.2	14.3	11.0	6.5	16.6	14.4	17.7	20.3	17.8	17.6	17.5	26.6	36.2
	3	16.5	20.7	11.2	10.8	10.0	14.1	12.5	16.0	26.5	24.3	21.7	20.4	28.5	24.6
Mar	1	19.2	64.9	27.0	13.7	32.4	27.5	12.7	6.5	19.4	22.8	20.8	18.5	31.8	35.8
	2	44.6	32.6	17.8	13.9	28.1	28.0	10.2	9.0	28.5	22.2	56.9	17.1	29.0	28.2
	3	18.4	25.1	15.4	35.4	36.3	41.0	12.5	6.6	20.9	34.9	30.1	24.7	81.2	36.4
Mean Flow		31.7	39.4	26.6	27.8	32.5	33.9	35.6	26.0	28.3	36.3	29.8	45.6	32.8	39.3

Source: Planning Division, WAPDA.

* Above = Below + QB Link at Head

River Chenab Flow Downstream Qadirabad (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.0	3.3	7.6	0.0	14.5	27.5	13.1	1.6	0.0	1.3	20.3	8.7	1.1	13.7
	2	0.0	15.6	12.0	7.5	7.2	23.8	66.3	1.1	0.0	9.4	10.1	16.3	0.5	17.3
	3	1.6	8.9	10.9	5.9	18.3	25.9	29.3	2.8	0.0	17.6	8.4	11.8	0.3	17.3
May	1	1.8	29.0	16.6	12.9	27.6	35.4	22.7	4.7	2.2	6.8	28.7	20.1	4.5	22.1
	2	0.0	40.7	8.9	5.7	10.7	47.8	31.0	4.3	0.0	14.9	44.0	12.7	7.9	48.2
	3	1.6	41.6	10.8	3.4	18.4	31.5	25.1	13.1	7.8	2.0	25.1	9.0	18.5	34.8
Jun	1	3.3	40.6	6.7	28.2	7.4	24.9	23.0	30.5	13.0	2.5	41.6	20.1	40.7	25.5
	2	0.0	26.1	21.3	37.4	10.4	41.8	19.7	31.2	15.2	22.7	28.9	13.6	13.9	17.8
	3	28.1	59.1	48.2	58.7	37.7	24.2	30.1	24.2	14.5	56.9	23.5	49.3	22.3	72.8
Jul	1	90.9	140.9	39.0	33.0	38.1	29.2	57.7	29.7	24.4	48.4	48.7	67.7	21.7	75.1
	2	207.8	96.3	79.5	53.5	74.3	26.0	16.7	9.4	89.3	56.4	34.8	183.8	57.2	44.3
	3	78.7	111.6	45.1	39.9	165.8	68.7	72.0	32.7	85.1	114.3	39.5	174.7	107.8	35.6
Aug	1	111.1	136.6	90.1	62.8	87.2	115.9	108.8	48.8	114.9	115.9	24.9	151.4	91.8	69.8
	2	48.0	103.5	43.7	33.2	71.6	98.9	58.2	57.7	46.6	73.6	22.2	75.8	22.6	45.2
	3	20.8	51.3	13.5	9.9	23.6	17.4	88.0	58.6	29.1	28.9	36.6	30.4	29.8	44.0
Sep	1	19.6	11.7	3.6	6.5	3.6	2.5	98.0	35.8	8.2	14.9	13.5	8.5	2.0	52.2
	2	8.6	26.2	3.3	9.7	0.8	0.5	29.3	6.9	1.7	0.3	5.2	14.4	1.5	30.9
	3	1.7	11.1	0.5	1.9	0.8	0.3	3.1	0.9	0.0	1.4	2.3	200.1	1.9	14.7
Oct	1	0.0	3.8	0.0	0.0	0.0	0.7	1.0	0.8	0.3	0.0	0.0	39.5	0.3	3.5
	2	0.0	0.0	0.0	0.0	0.0	0.2	0.8	0.0	0.0	0.0	0.0	16.8	0.0	2.4
	3	0.0	0.2	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	1.2	0.0	2.0
Nov	1	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.8	0.0	4.2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3	0.4	2.7	0.0	0.0	0.0	0.0
Dec	1	0.0	0.0	0.0	0.0	0.1	0.9	0.5	0.0	0.0	0.3	1.1	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.4	0.0	12.7	0.4	0.0	0.0	0.0
	3	2.4	3.8	0.0	1.4	2.4	2.3	0.0	0.2	5.6	2.4	1.8	5.8	1.8	31.6
Jan	1	2.4	3.8	9.5	6.9	1.5	4.1	0.0	4.0	8.6	3.3	0.3	11.1	6.0	14.1
	2	0.1	0.0	2.8	4.7	0.0	1.4	4.5	2.0	3.1	1.4	6.8	7.9	3.7	3.6
	3	0.4	0.0	0.0	27.3	0.0	7.3	4.6	0.0	0.2	3.3	4.4	9.5	6.3	1.2
Feb	1	0.0	0.0	1.1	16.0	0.1	5.2	0.0	0.4	0.0	7.2	0.0	2.2	9.9	4.0
	2	0.0	0.0	1.3	0.4	0.0	1.0	0.0	0.0	2.3	1.5	0.4	1.3	9.3	20.1
	3	0.0	18.7	0.0	0.0	0.0	0.2	0.6	0.0	7.9	13.1	2.5	2.8	9.8	6.0
Mar	1	3.6	52.3	11.5	0.8	24.9	11.7	0.0	0.0	0.8	15.9	1.9	0.0	13.8	18.1
	2	31.5	30.5	1.3	0.0	15.6	11.8	1.6	0.0	10.1	9.8	40.3	0.4	12.6	17.6
	3	3.1	12.2	0.0	20.9	24.9	24.4	0.2	0.0	3.6	29.3	12.1	6.2	65.4	29.2
Mean Flow		18.5	30.0	13.6	13.6	19.1	19.8	22.4	11.2	13.7	19.3	14.7	32.6	16.2	23.2

Source: Planning Division, WAPDA.

River Chenab Flow Upstream Trimmu (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	18.5	9.5	8.2	11.6	22.9	35.2	25.0	12.5	6.4	22.7	28.1	13.3	6.3	20.0
	2	27.1	18.3	16.0	10.5	18.1	31.3	99.9	12.5	6.7	17.5	17.2	14.0	5.8	18.5
	3	18.9	24.4	14.6	19.7	58.4	23.9	68.2	18.3	6.6	26.0	17.5	11.5	9.3	13.8
May	1	23.2	34.2	26.6	23.4	70.7	40.3	46.8	15.9	6.4	24.0	21.4	21.4	12.1	57.4
	2	11.5	61.8	55.3	32.8	73.3	52.5	35.9	20.0	3.8	42.0	88.1	41.6	8.8	70.3
	3	20.0	57.0	47.7	26.8	75.2	26.7	82.8	22.7	3.6	31.8	65.0	42.1	40.2	81.3
Jun	1	26.4	58.1	51.5	35.3	62.2	21.2	62.9	33.5	9.6	24.2	111.6	52.1	78.5	69.4
	2	19.6	68.5	51.1	40.0	21.5	68.1	48.0	39.0	17.3	34.2	154.4	34.0	68.6	44.1
	3	35.6	64.5	71.4	73.0	37.2	42.3	52.6	32.6	16.2	102.5	60.7	58.5	30.5	81.3
Jul	1	106.6	176.8	101.6	93.0	55.6	42.7	128.0	45.5	17.7	106.0	68.8	104.0	49.1	121.5
	2	192.2	155.2	129.6	108.3	83.5	32.9	62.9	28.4	81.4	106.4	77.4	232.9	55.9	76.6
	3	128.3	189.3	102.1	85.0	159.3	73.7	113.7	37.0	121.5	144.5	73.0	363.5	91.4	50.8
Aug	1	134.2	184.8	99.2	100.0	134.8	114.7	185.6	58.6	121.9	266.1	58.7	180.6	212.7	63.8
	2	124.7	166.3	64.5	100.6	134.1	127.9	118.9	95.8	91.4	118.6	50.1	145.0	63.7	63.8
	3	88.2	118.4	43.1	28.3	47.5	68.5	151.7	112.0	41.7	88.2	64.7	57.3	73.2	50.7
Sep	1	142.0	48.8	25.4	32.8	27.5	20.9	169.4	107.6	22.2	30.1	43.2	20.7	25.3	72.5
	2	125.1	44.2	23.1	36.6	18.7	16.2	79.9	42.9	18.3	18.0	30.2	24.9	15.2	57.8
	3	113.4	38.0	22.0	24.4	19.2	18.2	21.6	19.3	19.1	17.7	19.1	133.0	20.5	24.8
Oct	1	31.4	33.5	18.6	21.2	18.7	20.1	15.2	20.7	20.1	17.9	18.5	67.5	19.0	15.5
	2	33.8	28.6	17.9	17.9	16.0	16.2	17.5	15.9	15.8	12.4	16.8	37.1	14.3	16.7
	3	30.3	30.3	14.7	12.2	11.8	10.0	11.8	9.3	8.4	6.2	7.0	13.7	7.3	15.7
Nov	1	8.2	25.1	11.4	10.3	9.6	9.1	8.5	5.3	9.0	5.0	5.9	3.8	7.4	8.4
	2	8.6	16.3	8.8	11.1	9.2	9.7	8.5	11.0	9.0	6.5	6.0	7.0	13.0	7.8
	3	11.0	11.4	9.0	11.4	8.2	10.1	14.1	14.1	19.4	15.0	7.0	12.9	10.0	7.7
Dec	1	13.1	8.6	9.6	11.2	9.1	15.3	17.6	9.9	10.1	7.0	17.5	15.5	7.2	16.0
	2	11.5	8.5	10.6	10.5	18.3	9.6	18.6	9.4	7.2	13.0	8.7	6.8	7.8	8.8
	3	14.8	15.3	13.7	12.9	21.3	16.9	18.9	18.2	12.2	7.7	17.8	27.0	14.4	14.5
Jan	1	21.1	45.2	25.2	24.7	26.1	19.5	20.7	29.8	17.5	6.7	15.8	58.6	14.9	68.0
	2	11.2	17.0	15.6	24.7	11.6	14.9	40.9	31.5	15.8	11.4	14.8	26.0	11.5	21.8
	3	7.0	6.0	17.1	28.4	8.4	11.4	54.1	14.6	8.2	6.5	16.0	15.6	3.8	16.4
Feb	1	9.8	11.5	11.9	33.3	3.8	12.8	32.2	8.9	5.9	6.6	8.0	6.7	7.2	25.1
	2	12.3	13.0	10.9	28.7	6.1	9.1	18.1	17.2	9.5	7.1	11.3	13.4	13.7	40.9
	3	17.5	23.7	11.4	13.9	7.8	12.1	17.7	10.3	7.0	13.5	15.1	17.7	11.0	36.4
Mar	1	18.2	41.2	15.8	12.1	18.5	13.9	12.1	7.7	3.1	19.0	10.6	8.7	25.0	38.7
	2	16.4	29.0	18.5	16.9	17.5	13.3	14.4	7.0	16.5	23.3	27.9	6.3	12.6	26.6
	3	13.8	19.1	12.7	32.5	23.6	28.0	7.9	6.8	28.9	36.1	14.4	9.0	61.9	26.5
Mean Flow		45.7	52.8	33.5	33.8	37.9	30.8	52.9	27.8	23.2	40.0	35.8	52.9	31.4	40.3

Source: Planning Division, WAPDA.

* Includes contribution from CJ Link which started operating on May 25, 1971.

River Chenab Flow Downstream Trimmu (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	2.3	1.8	3.6	4.4	17.9	28.5	18.2	5.4	1.7	15.1	28.1	2.2	0.0	13.9
	2	7.1	4.5	10.2	3.9	9.7	22.9	94.4	4.4	1.7	5.6	10.7	2.9	0.0	9.8
	3	1.0	7.8	4.4	5.8	44.2	8.2	63.4	3.1	1.6	14.5	2.4	0.0	0.0	3.2
May	1	4.2	18.1	14.2	8.5	53.9	24.4	36.4	2.6	1.6	12.2	4.6	5.7	0.0	41.9
	2	0.9	45.1	40.0	14.8	56.4	36.4	21.7	3.1	1.5	26.2	71.5	23.5	0.0	52.8
	3	6.2	40.2	31.1	8.6	58.3	10.1	67.8	5.0	1.5	14.6	48.1	23.3	22.8	66.0
Jun	1	10.5	40.8	34.4	16.9	45.1	4.2	47.2	15.7	1.6	7.3	94.1	33.3	61.0	53.5
	2	3.3	45.7	33.4	21.6	3.8	50.6	31.8	21.8	1.6	16.0	137.8	15.2	53.4	27.0
	3	11.4	47.4	53.8	54.9	19.3	24.8	35.4	15.3	2.1	85.7	42.7	39.4	16.4	63.0
Jul	1	73.4	159.9	84.5	76.3	37.8	25.2	112.1	28.3	1.6	93.6	50.9	85.9	34.2	107.3
	2	163.4	150.0	112.4	98.7	70.0	15.7	47.6	14.0	45.6	95.5	59.6	221.2	41.3	64.6
	3	103.5	187.6	85.2	80.2	155.6	57.9	102.7	22.1	114.5	135.4	54.9	363.4	80.9	36.9
Aug	1	105.2	184.8	83.2	97.7	132.6	100.4	182.4	48.5	115.8	266.1	41.6	178.8	209.0	50.7
	2	98.2	165.7	49.0	98.3	127.6	114.5	116.2	85.1	84.8	114.6	31.4	140.9	49.2	51.9
	3	60.6	114.7	25.5	17.2	31.3	52.3	147.8	106.1	30.8	80.5	47.8	47.3	58.2	38.8
Sep	1	41.7	40.4	7.6	19.2	9.9	4.6	167.9	104.4	6.5	15.5	27.8	5.4	11.5	60.6
	2	16.5	20.5	5.3	19.9	1.8	0.8	75.0	31.6	3.0	0.3	12.5	8.4	0.0	45.9
	3	7.9	13.1	4.1	7.0	1.8	0.9	11.0	4.8	2.6	0.0	0.0	120.6	2.5	12 1
Oct	1	2.4	8.1	2.1	3.8	1.7	2.5	3.1	3.7	2.6	0.0	0.0	65.9	13.2	3.3
	2	1.6	1.2	2.8	3.4	2.5	2.3	3.0	3.0	6.4	0.0	0.0	32.2	0.0	2.6
	3	0.9	1.1	2.6	4.5	3.1	2.3	2.6	2.8	1.8	0.1	0.0	6.4	0.0	1.4
Nov	1	0.0	1.2	2.5	3.5	2.8	2.4	2.4	0.0	3.0	0.0	0.0	0.0	0.0	0.0
	2	0.0	1.1	2.5	3.4	2.6	2.4	2.3	0.0	3.0	0.0	0.0	0.0	0.0	0.0
	3	0.0	1.0	2.5	3.4	2.2	2.4	2.2	0.0	4.5	0.0	0.0	0.0	0.0	0.0
Dec	1	1.6	1.5	2.5	3.3	2.1	2.3	1.9	2.7	2.2	0.0	0.0	0.0	0.0	0.0
	2	1.7	1.5	2.5	3.3	1.9	2.3	10.3	3.2	2.0	6.3	0.9	0.0	0.0	0.0
	3	1.8	1.5	6.7	6.2	13.5	9.5	13.5	12.8	6.2	0.0	11.3	19.3	7.5	8.3
Jan	1	4.3	13.7	16.5	17.6	13.2	8.9	14.9	28.7	6.6	0.0	9.8	50.2	9.0	59.5
	2	11.1	10.1	5.4	14.5	9.6	6.9	48.9	30.2	10.8	3.1	14.8	26.0	11.5	21.8
	3	4.3	5.5	17.1	27.4	8.4	11.4	52.7	8.6	8.2	0.9	16.3	15.3	3.4	15.6
Feb	1	1.7	3.7	11.9	33.3	1.7	11.3	25.5	3.2	3.0	0.0	2.3	2.3	1.5	20.9
	2	1.7	1.0	6.5	22.8	1.6	2.7	10.8	3.2	3.1	0.0	1.6	1.4	3.2	32.3
	3	1.8	22.7	4.2	6.8	1.4	3.7	12.4	3.0	3.1	2.6	0.0	5.1	3.6	26.4
Mar	1	3.1	38.7	4.8	3.9	6.9	3.9	7.0	3.0	3.1	9.0	0.0	0.0	18.0	31.0
	2	8.4	26.3	4.9	4.7	3.9	2.8	9.2	3.0	7.7	10.7	18.5	0.0	3.4	19.3
	3	4.6	12.2	3.9	23.3	16.5	22.3	3.0	3.0	20.0	22.6	4.4	0.0	53.6	17.3
Mean Flow		21.3	40.0	21.8	23.4	27.0	19.0	44.6	17.7	14.4	29.3	23.5	42.8	21.3	29.4

Source: Planning Division, WAPDA.

River Chenab Flow Upstream Panjnad (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	4.2	11.9	10.1	5.7	23.1	22.5	22.5	3.2	1.9	14.9	23.2	8.8	8.9	16.7
	2	10.9	6.5	14.8	3.4	15.0	24.1	71.0	5.1	2.1	10.6	21.0	7.2	5.5	13.4
	3	9.6	10.4	12.9	8.4	18.5	15.8	81.9	6.8	2.0	10.8	7.7	6.7	5.9	13.8
May	1	6.9	14.8	15.4	9.1	49.1	25.4	54.9	6.7	1.8	17.2	10.6	3.7	9.0	32.8
	2	8.6	43.1	34.5	14.9	36.1	25.8	24.6	7.4	3.1	16.1	54.5	9.6	9.6	37.2
	3	6.6	32.9	26.4	15.1	64.4	18.0	52.3	11.3	2.6	19.7	39.0	20.9	11.5	63.0
Jun	1	12.7	40.4	25.9	18.7	37.3	12.4	50.8	16.0	4.1	14.2	70.0	22.6	34.6	45.1
	2	12.8	42.6	26.6	18.8	18.1	23.7	32.7	21.3	5.8	10.6	136.9	17.5	45.9	33.1
	3	12.0	39.7	35.4	30.7	15.3	34.0	23.7	19.5	6.7	36.0	69.9	17.9	30.8	31.0
Jul	1	43.7	120.6	83.6	62.5	33.2	19.2	78.1	25.7	8.6	107.8	33.2	73.5	35.1	95.3
	2	115.5	188.9	78.0	85.6	43.9	24.1	69.2	26.0	16.2	117.2	62.9	97.6	31.1	87.8
	3	151.2	226.9	96.1	113.1	139.5	38.4	62.4	23.1	108.1	104.4	32.3	422.6	50.4	51.5
Aug	1	98.6	218.3	80.3	127.9	226.6	80.0	178.9	44.4	102.0	239.7	56.8	218.7	243.1	62.2
	2	122.5	276.3	56.8	153.2	147.8	119.8	162.9	69.2	136.2	169.7	25.3	197.3	62.9	78.7
	3	88.3	214.6	30.8	40.6	47.6	86.5	131.3	97.7	48.8	100.4	34.8	97.7	48.9	85.6
Sep	1	54.6	151.4	14.7	28.7	20.8	16.9	221.2	135.4	21.1	32.5	39.7	46.8	35.5	75.6
	2	52.4	51.9	13.2	24.7	15.0	10.5	128.2	88.7	10.8	12.0	21.0	17.1	15.0	135.6
	3	48.6	33.5	18.8	32.5	11.3	11.2	47.4	26.3	8.2	10.5	14.6	47.1	11.6	58.9
Oct	1	16.4	19.4	13.3	23.4	11.9	11.2	10.6	12.1	10.4	10.6	12.9	351.7	10.7	26.1
	2	12.9	10.3	11.8	15.8	10.3	10.4	10.1	8.4	14.1	9.5	9.9	140.6	8.2	13.3
	3	10.0	10.1	7.8	9.2	8.2	5.9	11.0	7.8	7.4	3.5	4.4	33.1	6.5	7.7
Nov	1	6.2	7.2	6.7	7.7	5.0	3.8	4.6	3.5	5.2	2.5	3.8	11.0	6.5	9.3
	2	4.9	5.4	5.3	6.7	4.2	3.6	4.5	3.3	3.9	2.8	3.3	8.0	4.5	6.4
	3	3.8	4.0	4.8	6.5	3.8	3.5	4.4	6.4	3.3	3.2	2.4	7.8	4.9	4.2
Dec	1	3.5	3.8	4.7	6.5	3.3	3.1	6.2	6.4	3.6	7.2	10.2	9.0	7.0	5.3
	2	3.5	3.7	6.4	5.1	3.1	3.7	16.6	3.5	2.8	6.9	6.4	6.3	6.7	8.6
	3	2.9	6.0	5.7	7.0	5.4	5.5	23.4	7.7	2.3	8.7	4.4	9.5	4.9	7.6
Jan	1	3.2	8.4	14.0	21.6	14.5	15.6	24.3	18.4	6.0	4.8	10.6	48.7	11.6	49.0
	2	12.5	13.0	11.8	21.1	12.4	5.8	38.6	34.9	8.2	3.9	5.6	56.6	13.0	36.6
	3	9.5	11.3	10.9	16.8	10.9	8.6	59.9	15.3	9.6	6.9	13.8	48.1	7.8	21.9
Feb	1	5.1	5.4	12.6	29.7	5.6	13.8	47.0	6.0	3.4	5.7	10.0	13.9	6.8	21.4
	2	3.6	5.2	6.7	19.5	4.1	7.5	25.0	5.4	3.5	3.8	3.9	6.7	6.9	22.5
	3	2.9	20.7	3.5	9.9	4.1	5.7	22.7	5.4	2.0	2.8	2.8	10.2	6.6	21.1
Mar	1	2.9	46.0	3.4	4.9	4.2	4.5	20.5	4.3	3.2	8.9	1.8	5.6	20.6	30.2
	2	7.2	57.3	8.6	4.8	17.3	8.0	20.1	3.3	2.3	10.8	12.9	4.4	12.8	28.3
	3	20.0	22.9	6.8	15.5	10.3	17.8	13.0	2.2	6.7	13.1	11.9	3.7	53.3	18.2
Mean Flow		27.5	55.1	22.8	28.5	30.6	20.7	51.6	21.9	16.3	32.2	24.6	58.6	24.9	37.6

Source: Planning Division, WAPDA.

* Upatream (Including TP Contribution) = Panjnad Canal at Head
+ Abbasia Canal at Head + Panjnad Below.

River Chenab Flow Downstream Panjnad (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.0	5.4	6.2	1.7	19.2	19.0	17.2	0.0	0.4	7.8	22.6	2.2	3.1	8.9
	2	4.3	0.5	8.4	3.4	8.4	19.2	64.9	0.0	0.5	4.6	17.5	1.0	0.3	6.4
	3	2.0	2.0	5.2	2.4	10.6	9.6	75.7	0.0	0.5	2.2	1.6	0.5	0.0	4.9
May	1	0.1	6.6	7.0	2.4	40.6	18.2	46.9	1.2	0.1	8.3	4.2	0.0	0.7	32.8
	2	1.5	32.6	25.2	5.6	27.0	16.9	15.1	1.0	0.0	6.6	45.3	2.7	0.8	29.1
	3	0.3	21.6	16.2	4.8	54.6	8.9	42.2	1.2	0.0	9.7	29.5	10.5	4.4	52.4
Jun	1	3.3	28.9	15.2	7.9	26.9	2.1	40.7	5.4	0.0	4.4	60.6	13.1	24.9	34.1
	2	2.6	31.5	15.5	7.9	7.9	14.8	22.7	10.4	0.0	1.0	127.3	6.9	35.5	22.1
	3	1.8	28.8	24.4	19.7	4.8	23.2	13.3	8.9	0.0	26.9	60.2	7.3	21.1	20.1
Jul	1	34.4	110.5	72.9	51.9	22.3	8.4	68.1	15.4	0.5	96.4	23.1	64.0	25.7	84.9
	2	104.5	183.4	67.7	74.9	33.0	13.3	59.0	15.6	7.1	107.3	52.4	87.3	21.9	77.1
	3	140.2	222.0	85.3	102.3	130.1	27.8	53.1	12.5	99.4	94.4	21.8	421.4	42.6	40.4
Aug	1	87.0	211.5	72.1	116.8	217.3	69.2	169.2	33.4	92.9	230.2	46.2	210.3	235.7	51.1
	2	111.0	265.5	48.0	142.1	137.2	109.2	152.8	58.1	126.9	168.1	14.6	187.6	53.1	67.6
	3	76.6	203.8	20.0	29.1	36.8	75.6	121.4	87.1	39.4	98.8	24.0	87.6	39.9	74.5
Sep	1	43.1	141.3	3.4	17.0	9.6	5.9	214.0	124.8	11.6	21.9	28.9	36.7	24.9	64.8
	2	41.0	41.6	2.8	13.2	4.4	0.2	121.6	78.3	1.9	1.4	9.8	6.9	4.1	124.5
	3	34.7	22.9	7.4	21.1	0.9	0.5	37.3	16.1	0.0	0.0	3.2	36.9	0.7	47.8
Oct	1	5.0	8.9	3.2	12.3	0.9	0.4	0.0	1.9	0.0	0.0	1.7	349.2	0.1	15.2
	2	1.7	0.8	2.9	7.2	0.4	0.5	0.0	0.0	4.9	0.0	0.2	132.8	0.5	4.2
	3	1.1	1.0	1.8	1.5	0.0	0.2	2.4	0.0	0.1	0.0	0.3	23.2	2.2	2.9
Nov	1	0.8	1.5	1.6	1.6	0.0	0.0	0.7	0.0	0.0	0.0	0.0	4.4	2.0	4.7
	2	1.1	0.9	0.9	1.1	0.6	0.0	0.9	0.0	0.0	0.0	0.0	3.5	0.2	2.0
	3	0.1	0.3	1.0	1.1	0.4	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0	0.0
Dec	1	0.0	0.0	1.0	0.9	0.0	0.1	2.4	0.0	0.0	0.0	1.7	0.9	0.0	0.0
	2	0.0	0.0	1.6	1.2	0.0	0.6	12.1	0.0	0.0	1.5	0.1	0.0	0.0	0.0
	3	2.3	5.3	1.4	4.5	4.8	2.4	20.2	4.6	0.0	5.4	2.8	5.8	1.4	0.0
Jan	1	2.7	8.4	14.0	15.9	14.5	12.8	21.0	18.0	6.9	1.2	10.6	42.1	8.4	44.6
	2	6.8	11.0	11.8	17.2	10.5	5.8	35.6	34.8	7.7	1.4	3.3	55.2	12.1	34.9
	3	1.6	7.7	6.9	16.8	7.8	8.6	59.9	12.6	6.4	7.3	4.8	48.1	7.8	21.9
Feb	1	1.0	2.6	5.2	29.7	2.1	11.8	45.4	1.6	1.5	0.5	1.3	9.9	4.1	16.7
	2	0.3	2.6	1.4	16.4	1.2	4.6	21.7	0.7	0.0	2.4	0.0	0.8	1.4	18.1
	3	0.2	20.4	0.3	3.7	0.9	0.5	19.4	0.6	0.0	0.0	0.0	3.2	2.0	12.6
Mar	1	0.0	43.8	0.5	0.6	0.7	1.3	17.5	0.4	0.0	1.7	0.0	0.0	15.2	24.6
	2	4.2	53.9	3.7	0.8	10.7	3.2	17.0	0.3	0.0	2.4	10.5	0.0	7.2	21.6
	3	15.6	19.1	2.2	9.3	6.0	13.9	9.3	0.3	0.0	7.2	5.5	0.0	45.3	9.5
Mean Flow		20.4	48.6	15.7	21.3	23.7	14.1	45.0	15.1	11.4	25.6	17.7	51.8	18.0	29.9

Source: Planning Division, WAPDA.

River Ravi Flow Upstream Balloki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	11.4	16.9	24.9	17.3	24.9	19.6	24.3	18.4	13.2	–	5.3	22.4	19.2	20.3
	2	18.1	21.1	23.9	14.9	25.3	17.8	48.2	15.9	14.4	–	7.5	24.6	20.4	19.8
	3	22.5	20.5	20.5	13.8	26.4	25.4	21.1	19.7	15.6	–	21.9	29.8	23.1	24.0
May	1	26.7	20.8	23.0	18.5	30.0	23.2	22.3	19.3	18.7	–	27.0	22.8	24.5	22.7
	2	23.7	17.2	22.6	23.1	29.0	22.4	25.4	25.5	22.5	–	20.6	24.1	27.3	34.6
	3	21.7	19.1	25.9	26.7	29.6	17.9	34.0	27.9	26.4	–	23.3	28.2	28.5	31.5
Jun	1	27.3	20.1	23.8	29.8	27.9	21.8	31.6	29.3	27.5	–	26.7	21.5	28.6	30.8
	2	23.1	26.4	26.2	29.8	28.5	27.1	29.2	28.6	26.8	–	31.9	28.6	27.9	28.7
	3	31.6	27.4	24.6	32.6	28.2	27.9	26.7	27.2	26.5	–	26.3	29.0	23.8	32.0
Jul	1	48.6	47.2	24.0	32.6	30.4	28.0	36.8	31.2	26.1	–	24.7	29.7	27.6	42.3
	2	76.0	54.4	27.7	84.9	41.1	26.5	26.9	35.4	33.1	–	27.2	45.4	33.1	41.2
	3	54.2	55.3	31.0	53.9	82.9	29.2	38.0	33.4	44.9	–	29.8	73.5	39.7	39.3
Aug	1	70.2	68.2	27.4	105.0	45.4	44.1	47.9	48.4	62.2	–	28.8	96.7	54.6	48.9
	2	63.8	66.5	25.4	59.1	54.4	55.1	43.6	41.0	49.9	–	31.8	51.4	28.4	59.5
	3	39.8	54.4	27.8	34.8	30.7	31.2	49.2	49.4	35.8	–	34.7	38.8	37.5	51.6
Sep	1	33.7	29.9	24.6	31.6	28.5	29.3	45.3	67.4	32.4	–	35.7	33.5	31.2	54.9
	2	44.3	25.9	29.3	31.4	23.7	27.4	42.2	35.5	25.9	–	31.0	27.9	29.4	61.6
	3	31.5	18.9	23.9	23.9	24.8	22.7	30.8	28.6	26.4	–	27.7	121.6	28.7	37.7
Oct	1	20.4	16.9	19.5	22.1	21.0	15.5	23.9	20.8	23.0	–	21.1	87.6	20.3	26.2
	2	21.3	20.7	17.9	20.0	17.5	15.9	22.9	18.4	37.7	–	20.2	16.7	18.1	23.7
	3	21.2	19.2	16.1	17.1	16.0	17.8	16.7	15.6	16.2	–	19.3	18.0	16.9	24.3
Nov	1	16.8	17.3	15.4	15.0	17.2	14.9	12.9	14.7	15.0	–	16.4	15.3	17.1	19.9
	2	14.3	16.9	14.6	15.4	15.0	16.0	13.9	16.9	14.8	–	18.1	16.4	15.1	17.0
	3	14.1	15.0	16.8	15.8	15.1	16.6	16.9	18.0	17.1	–	18.4	19.3	16.7	17.8
Dec	1	14.6	14.4	16.5	18.9	15.3	16.8	18.6	17.9	16.6	–	16.6	18.0	16.9	17.2
	2	13.1	14.0	15.1	16.9	15.3	14.2	17.5	19.0	18.8	–	16.9	17.7	16.4	18.0
	3	4.6	4.8	14.8	18.7	3.7	11.0	13.7	15.3	19.4	–	16.0	17.4	11.5	15.7
Jan	1	2.8	2.1	6.5	5.8	0.6	4.2	15.0	5.3	6.6	–	15.4	10.3	3.0	12.1
	2	17.3	16.4	3.0	3.7	9.9	6.2	5.6	4.0	4.5	–	5.4	5.7	4.3	4.1
	3	16.5	16.9	10.8	14.0	13.5	16.8	2.3	16.3	16.6	–	2.1	13.1	14.5	17.0
Feb	1	17.5	17.7	12.1	23.5	9.5	22.8	6.6	16.0	16.4	–	5.4	18.2	16.9	21.0
	2	18.2	17.0	16.3	23.4	8.1	17.8	10.7	14.7	21.0	–	12.3	16.3	21.4	24.1
	3	17.9	19.8	14.5	18.9	11.3	19.7	15.9	15.1	26.0	–	15.8	17.7	20.5	22.0
Mar	1	27.7	45.9	18.5	19.0	26.4	26.0	12.9	8.6	21.7	–	19.0	17.8	27.6	35.5
	2	31.4	27.9	18.5	22.0	24.6	22.5	12.0	11.5	20.7	–	35.7	16.6	20.1	23.4
	3	28.5	23.6	17.0	36.0	31.7	34.1	17.4	8.6	26.5	–	24.5	23.2	38.0	21.5
Mean Flow		27.4	26.0	20.0	27.5	24.5	22.4	24.4	23.3	24.1		21.1	31.0	23.6	28.9

* Above = Below + LBDC at Head + BS at Head.

Source: Planning Division, WAPDA.

River Ravi Flow Downstream Balloki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.2	1.3	5.3	0.1	8.2	0.9	5.2	0.2	0.0	–	3.5	4.3	3.8	1.3
	2	0.3	0.9	5.2	0.1	1.2	0.3	30.0	0.1	1.4	–	0.7	1.2	2.7	0.3
	3	0.4	0.3	1.8	0.1	0.9	2.3	4.4	0.1	0.5	–	0.6	4.2	0.3	0.0
May	1	1.5	0.4	3.1	0.2	4.4	3.2	1.6	0.1	0.6	–	2.9	0.0	3.7	0.0
	2	0.7	0.5	0.6	0.3	2.3	1.7	1.6	0.3	1.0	–	1.3	1.2	2.9	7.8
	3	0.3	0.6	0.8	1.1	3.1	0.7	9.9	1.8	1.2	–	0.3	0.6	0.8	3.1
Jun	1	1.7	0.8	0.6	3.4	2.1	1.0	6.5	2.6	2.2	–	1.3	0.0	0.9	2.4
	2	0.4	3.2	0.9	2.9	1.5	2.9	3.2	1.8	0.7	–	6.5	0.0	2.7	0.1
	3	6.6	4.4	0.5	5.4	0.8	2.1	0.5	0.4	0.5	–	0.3	0.0	3.0	2.6
Jul	1	22.3	22.8	0.5	5.4	3.8	1.5	12.4	4.2	0.4	–	0.1	0.5	1.1	13.9
	2	51.8	30.2	3.4	60.3	18.4	0.7	4.4	8.7	9.1	–	0.1	16.8	5.4	12.6
	3	30.1	39.6	6.7	27.2	65.5	2.8	12.2	6.9	23.6	–	2.2	46.2	12.1	12.1
Aug	1	45.4	54.6	4.8	90.0	26.3	17.2	23.2	21.6	39.4	–	0.7	69.4	31.2	21.5
	2	38.0	47.9	2.4	43.5	29.5	31.3	19.7	14.2	27.7	–	3.8	32.1	2.7	32.6
	3	14.0	34.3	2.1	9.8	3.5	5.1	25.9	22.2	13.1	–	6.2	19.9	8.2	24.7
Sep	1	7.9	7.8	0.7	5.6	1.7	2.0	25.4	43.1	6.4	–	7.0	4.9	2.4	28.0
	2	23.3	4.2	2.8	4.2	1.0	1.3	22.1	8.4	0.0	–	2.2	1.6	0.4	39.6
	3	13.0	1.0	1.1	1.6	0.7	1.3	7.7	3.2	0.4	–	0.1	101.0	0.8	14.7
Oct	1	2.1	1.2	0.5	1.2	0.5	1.0	1.3	0.6	0.8	–	0.0	6.2	0.0	2.8
	2	0.3	0.9	0.4	0.6	0.4	0.7	0.6	0.4	15.4	–	0.0	77.6	0.0	0.0
	3	0.2	0.3	0.5	0.5	0.3	0.8	0.4	0.2	0.1	–	0.8	3.0	0.0	0.0
Nov	1	0.1	0.2	0.4	0.7	0.2	0.8	0.3	0.2	0.1	–	0.0	8.6	0.0	0.0
	2	0.1	0.2	0.4	0.8	0.2	0.8	0.2	0.1	0.2	–	0.0	2.5	0.0	0.0
	3	0.1	0.0	0.5	0.7	0.2	0.6	0.1	0.5	0.2	–	0.0	0.1	0.0	0.0
Dec	1	0.1	0.0	0.3	0.8	0.1	0.4	0.5	0.4	3.0	–	0.0	0.0	0.4	0.0
	2	0.1	0.0	0.3	0.6	0.1	0.3	0.3	0.8	0.1	–	0.0	0.0	0.0	0.8
	3	2.1	2.0	0.3	0.9	1.1	1.4	0.4	0.4	2.2	–	0.0	0.0	1.5	5.6
Jan	1	2.3	2.1	6.5	5.8	0.6	4.2	0.6	3.0	3.7	–	0.0	4.8	3.0	12.1
	2	0.8	3.4	3.0	3.7	3.8	5.1	3.2	3.5	3.1	–	2.7	10.3	3.8	3.1
	3	0.2	0.9	1.4	2.6	0.9	3.6	2.3	3.0	1.6	–	2.1	5.7	5.1	7.9
Feb	1	0.2	1.5	0.5	4.9	0.2	6.1	3.7	0.3	0.2	–	2.4	8.9	0.1	4.2
	2	0.2	1.2	0.5	4.2	0.1	1.0	0.4	0.2	2.3	–	0.8	1.5	3.5	5.8
	3	0.2	9.9	0.2	0.9	0.1	4.1	1.7	0.2	1.9	–	0.0	0.1	2.2	3.5
Mar	1	7.0	36.5	0.1	1.3	8.4	6.7	0.2	0.2	0.2	–	0.0	0.0	9.7	16.4
	2	11.4	19.1	0.1	1.1	3.1	3.0	0.0	0.2	3.4	–	12.3	0.0	1.3	4.2
	3	10.7	5.5	0.1	14.6	9.3	14.5	0.0	0.2	9.0	–	3.8	4.6	20.0	2.5
Mean Flow		8.2	9.4	1.6	8.5	5.7	3.7	6.4	4.3	4.9		1.8	12.2	3.8	8.0

Source: Planning Division, WAPDA

River Ravi Flow Upstream Sidhnai (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	7.2	6.6	10.5	6.7	13.4	8.3	11.8	7.1	4.0	12.7	5.8	18.2	11.9	10.4
	2	8.2	6.7	10.5	5.9	10.4	7.5	27.1	7.6	4.7	12.9	5.9	11.4	6.9	9.1
	3	8.1	10.7	10.8	10.7	12.4	11.3	20.3	12.8	4.2	14.3	11.7	13.5	9.9	9.9
May	1	10.3	13.6	12.4	11.0	15.5	15.0	11.9	10.2	4.7	15.1	14.7	13.2	11.2	12.7
	2	10.4	14.3	13.6	13.8	16.9	14.9	13.6	12.4	2.7	15.4	15.8	14.2	9.7	16.7
	3	13.1	13.7	14.2	14.2	16.2	14.0	19.2	13.9	2.3	15.6	14.5	15.5	14.8	17.3
Jun	1	14.6	14.1	13.9	14.9	16.2	14.2	17.9	14.2	7.7	14.9	15.2	15.1	14.7	16.2
	2	14.7	15.1	14.2	15.2	15.0	14.8	16.4	14.3	12.8	15.1	20.5	14.5	14.4	15.6
	3	14.0	16.5	14.6	17.1	14.3	17.0	15.0	14.2	13.1	10.6	15.2	15.5	15.2	15.6
Jul	1	25.7	24.5	14.3	19.5	17.4	15.4	23.4	14.4	13.1	21.7	15.2	15.8	13.6	21.7
	2	37.2	27.7	16.3	25.3	22.8	14.8	20.0	18.1	14.1	24.2	14.8	17.8	15.4	20.9
	3	39.1	39.5	18.1	35.0	54.7	15.0	17.9	17.1	26.4	23.9	15.5	28.5	17.2	21.9
Aug	1	35.4	46.9	17.0	81.7	43.2	24.6	19.3	26.2	27.0	32.8	16.8	64.6	32.5	26.1
	2	42.0	45.0	14.9	43.7	41.6	41.3	24.9	21.2	32.2	27.5	16.6	57.1	13.6	41.0
	3	23.6	29.7	15.1	18.3	18.8	25.0	28.4	21.4	24.3	28.5	19.2	35.2	20.2	30.7
Sep	1	21.7	21.2	14.3	17.4	16.9	15.8	26.7	34.0	17.9	18.2	18.7	23.4	16.3	36.2
	2	29.5	18.5	15.2	17.0	15.5	13.0	26.3	26.1	15.0	15.1	17.6	18.3	14.3	48.9
	3	23.4	15.4	16.5	15.7	14.9	14.2	18.6	16.7	15.5	14.2	16.0	24.2	15.6	25.6
Oct	1	14.1	15.4	13.3	15.3	14.4	14.2	13.0	14.8	15.1	14.7	14.9	169.2	15.2	18.3
	2	14.5	14.6	12.7	13.7	12.0	12.4	13.8	11.5	18.6	10.8	13.8	16.6	11.7	14.1
	3	14.0	14.4	11.1	8.2	7.6	7.9	9.6	6.5	7.8	6.3	6.3	14.0	5.7	12.6
Nov	1	7.0	11.6	8.9	6.9	6.0	6.3	6.8	5.3	6.3	4.5	4.6	10.8	6.4	8.0
	2	4.9	7.9	6.4	7.0	5.4	6.2	5.8	9.1	5.6	6.7	5.5	9.0	11.3	7.4
	3	5.1	6.8	6.5	7.2	5.2	7.3	10.1	12.1	11.4	13.4	6.1	11.2	9.0	7.4
Dec	1	5.4	5.2	7.0	7.4	6.1	11.4	12.6	5.8	7.3	7.5	13.4	13.5	6.3	12.9
	2	5.2	4.4	6.3	6.1	12.4	6.5	8.5	6.1	4.7	8.2	7.1	10.0	6.0	9.3
	3	8.0	7.3	6.4	5.8	7.4	6.1	5.5	5.2	4.1	9.9	6.5	10.2	6.2	8.1
Jan	1	14.0	12.3	9.1	8.3	12.0	12.0	5.5	3.0	12.9	9.2	5.9	16.7	7.3	18.0
	2	3.1	6.8	10.9	11.7	5.7	7.0	3.5	3.2	9.3	14.9	3.2	6.7	2.6	4.8
	3	1.5	3.9	2.9	4.8	1.7	5.6	4.3	7.2	5.8	12.6	4.8	5.0	5.3	5.2
Feb	1	4.4	4.6	2.2	2.7	2.5	6.9	9.5	6.4	2.3	7.8	4.8	7.6	6.0	9.0
	2	4.7	6.3	3.8	9.2	4.2	5.8	9.0	12.0	7.2	6.6	11.0	11.8	10.6	13.9
	3	6.7	10.0	6.6	9.7	6.2	9.3	7.1	6.4	4.3	9.3	12.1	12.4	9.5	13.0
Mar	1	11.9	20.2	8.6	8.0	14.7	14.8	5.8	4.1	5.0	10.7	8.4	8.9	16.7	20.1
	2	10.3	29.1	12.4	11.0	15.3	12.8	5.5	3.5	10.9	11.4	15.5	6.0	11.5	14.3
	3	19.4	12.9	8.9	18.5	13.8	17.6	5.3	3.4	16.4	12.4	11.8	9.2	23.6	12.6
Mean Flow		14.8	15.9	11.1	15.1	14.7	13.0	13.9	11.9	11.0	14.2	11.8	21.2	12.2	16.8

Source: Planning Division, WAPDA.

* Above = Below + Sidhnai Canal + SM Link at Head.

River Ravi Flow Downstream Sidhnai (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.8	1.0	4.4	0.1	7.7	2.1	4.5	0.0	0.0	5.7	5.8	10.1	2.2	3.4
	2	1.8	0.0	2.8	2.4	3.9	1.7	22.2	0.4	0.0	7.0	1.4	2.8	0.0	2.0
	3	0.6	0.0	0.3	3.7	0.5	2.6	14.9	0.9	0.0	4.1	1.4	3.4	0.1	2.0
May	1	0.2	0.5	1.1	0.0	1.8	3.7	2.3	0.0	0.2	2.8	1.2	0.0	0.0	0.0
	2	0.2	0.3	0.4	0.0	3.0	1.8	1.1	0.0	0.1	1.0	1.6	0.0	0.0	1.8
	3	0.0	0.0	0.4	0.0	1.7	0.3	5.6	0.0	0.1	0.9	0.4	0.7	0.0	2.0
Jun	1	0.8	0.2	0.0	0.0	1.4	0.0	3.8	0.1	0.1	0.4	0.3	0.0	2.1	0.6
	2	0.9	0.8	0.0	0.1	0.2	0.0	4.0	0.2	0.1	0.3	6.6	0.0	3.1	0.2
	3	0.6	2.3	0.4	2.0	0.0	1.9	1.1	0.0	0.1	4.9	0.9	0.1	1.5	0.8
Jul	1	11.0	11.8	0.0	4.3	2.6	0.2	10.6	0.6	0.1	13.3	2.9	1.6	2.4	8.0
	2	23.3	22.4	1.9	12.0	9.2	0.0	8.0	4.0	1.1	12.5	0.2	5.5	3.1	6.2
	3	25.5	36.2	3.6	21.3	43.3	0.7	5.7	2.6	13.4	10.0	0.0	17.5	9.6	7.5
Aug	1	22.0	46.4	4.1	67.2	31.1	9.4	6.1	11.6	13.4	18.4	1.0	50.8	26.4	12.7
	2	28.0	35.2	2.4	29.3	27.4	26.4	11.4	6.6	17.3	18.5	0.8	41.4	0.0	26.4
	3	9.0	16.4	1.1	3.5	3.7	10.2	16.5	8.2	9.7	13.6	3.1	19.4	5.1	15.3
Sep	1	6.8	7.5	0.1	1.8	1.8	11.8	16.1	19.7	3.0	3.0	2.6	7.3	1.2	20.8
	2	14.9	4.5	1.0	1.5	0.5	0.0	13.5	11.4	0.7	0.0	1.5	3.6	0.0	33.5
	3	8.5	1.3	2.3	0.4	0.0	0.0	4.5	2.9	0.5	0.0	0.0	8.7	0.0	10.1
Oct	1	0.0	1.3	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.0	0.0	160.4	0.0	3.9
	2	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	8.5	0.0	0.0	2.8	0.0	0.0
	3	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.3	0.0	0.2
Nov	1	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.6	0.0	0.4
	2	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.0	0.0	0.0	0.1
	3	0.0	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0
Dec	1	0.0	0.4	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.2	2.2	0.0	0.6	0.0	0.0
	3	0.5	0.7	0.0	0.2	0.0	0.0	0.3	0.0	0.2	1.9	0.0	4.4	1.3	1.2
Jan	1	1.4	0.2	0.4	1.7	1.0	2.7	0.0	1.5	1.2	0.0	0.0	8.2	2.9	10.2
	2	1.8	1.8	0.3	1.7	2.4	2.9	2.9	2.8	2.9	3.7	2.9	6.7	2.6	4.8
	3	0.6	3.9	2.9	4.7	1.7	5.6	4.0	1.8	5.8	3.2	4.8	4.6	5.3	5.2
Feb	1	0.0	0.2	2.2	2.7	0.1	5.9	3.9	0.2	0.0	0.2	0.0	0.6	2.6	2.4
	2	0.0	0.2	0.3	4.0	0.0	0.0	0.0	0.5	1.0	0.0	0.0	0.1	2.2	2.5
	3	0.0	10.0	0.0	1.5	0.0	0.1	0.1	0.0	0.8	0.5	0.0	0.0	0.6	3.0
Mar	1	2.6	18.9	0.5	0.0	7.2	4.1	0.0	0.3	0.0	2.4	0.0	0.0	9.0	10.3
	2	2.2	24.1	1.3	0.0	6.5	3.8	0.0	0.3	1.6	0.0	8.3	0.0	0.3	4.3
	3	9.9	3.7	0.4	8.5	4.9	13.1	0.0	0.3	7.6	0.3	2.4	1.5	12.3	4.3
Mean Flow		4.8	7.1	1.0	4.9	4.5	3.1	4.6	2.1	2.5	3.6	1.4	10.1	2.7	5.7

Source: Planning Division, WAPDA.

River Sutlej Flow Upstream Sulcimanki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	4.3	9.9	11.5	0.1	8.4	11.0	10.7	9.6	6.1	0.6	4.6	12.6	9.0	10.7
	2	9.9	13.1	12.5	10.2	9.5	9.1	10.7	8.7	6.4	0.5	4.2	10.8	7.9	10.2
	3	12.2	15.6	12.7	5.2	12.2	10.8	14.0	10.1	7.6	9.1	13.2	11.8	14.5	12.8
May	1	13.9	18.7	15.1	8.4	12.7	13.5	13.7	10.6	9.1	10.7	17.7	9.4	15.6	11.1
	2	12.5	16.0	10.9	10.6	13.0	13.5	13.7	12.5	10.6	11.0	10.9	8.7	11.6	12.9
	3	10.4	17.5	13.8	12.5	13.1	13.7	13.2	13.4	12.2	11.9	12.2	11.9	12.9	14.0
Jun	1	14.8	12.6	14.6	13.3	12.7	10.3	14.5	13.0	12.7	12.2	11.5	11.4	13.7	13.6
	2	11.3	17.4	17.4	14.0	13.2	11.5	14.0	13.2	12.6	12.3	13.1	14.2	14.3	13.4
	3	12.2	16.1	16.0	14.3	13.8	11.6	14.2	13.3	12.9	11.6	12.3	13.8	14.7	13.3
Jul	1	16.0	21.0	15.6	14.0	14.0	12.6	14.9	13.1	12.8	14.5	12.2	13.4	13.9	27.9
	2	28.7	28.1	17.4	30.6	13.6	11.6	12.1	14.1	12.6	16.9	11.9	15.2	14.0	28.7
	3	22.5	32.9	24.6	28.2	18.0	12.2	14.2	12.9	25.9	12.2	12.1	29.5	14.1	22.9
Aug	1	24.9	47.4	17.9	35.8	18.4	15.7	14.6	13.9	16.5	20.1	12.9	42.6	20.2	25.2
	2	34.4	120.3	17.7	23.6	13.1	21.4	14.1	14.0	31.9	16.0	13.5	54.0	12.7	31.8
	3	35.8	124.0	17.6	13.8	14.7	15.3	12.9	14.5	21.4	21.3	13.2	41.1	16.1	25.6
Sep	1	27.8	62.0	13.2	14.6	13.7	14.0	16.7	19.7	12.3	12.9	13.1	20.2	15.6	54.1
	2	36.3	35.1	14.7	14.8	11.6	13.4	41.9	17.2	11.7	10.8	13.7	13.3	14.1	102.7
	3	37.4	20.4	13.0	12.7	12.4	10.9	17.0	13.2	12.1	11.2	13.5	81.9	13.9	34.0
Oct	1	10.1	13.5	10.3	10.9	10.8	7.0	13.7	10.3	11.0	12.4	8.8	123.8	9.7	18.7
	2	12.9	11.7	8.5	10.6	9.2	8.7	14.6	10.6	16.4	9.6	9.2	27.4	8.8	12.9
	3	12.7	11.0	8.3	9.7	8.5	8.8	12.5	7.1	8.1	7.6	9.8	12.0	9.0	15.2
Nov	1	9.8	9.9	8.7	7.7	9.1	8.3	8.5	7.3	8.2	7.2	7.0	10.9	7.5	9.6
	2	8.2	8.9	8.0	8.0	8.3	8.4	6.9	8.0	7.4	6.8	6.7	8.3	6.8	7.5
	3	7.6	8.4	9.4	8.3	8.9	9.3	9.7	7.8	9.7	10.4	8.3	10.4	8.6	10.6
Dec	1	8.4	7.8	9.8	10.4	9.4	9.5	10.5	7.9	9.2	12.0	8.3	11.3	8.6	9.8
	2	7.8	7.9	8.7	10.4	9.0	9.1	9.9	10.0	8.7	9.5	7.9	9.6	8.6	8.5
	3	3.6	3.8	8.4	9.9	3.6	5.8	7.9	7.8	9.7	12.6	7.4	12.7	7.1	6.9
Jan	1	3.4	2.5	1.9	0.6	0.3	1.0	7.9	2.0	4.7	8.9	6.7	6.6	4.3	4.2
	2	8.3	10.2	0.4	0.3	2.0	0.9	4.1	0.4	1.6	7.9	3.4	14.9	1.2	0.9
	3	8.1	9.5	4.6	4.5	6.4	5.7	2.1	4.9	6.2	7.1	0.8	7.8	3.4	4.4
Feb	1	9.5	9.8	9.0	9.9	6.7	11.6	3.0	7.3	7.6	8.0	0.0	9.4	7.7	8.5
	2	9.0	8.7	8.3	10.7	5.8	10.5	4.8	7.6	7.5	6.9	3.8	7.3	8.6	11.0
	3	9.6	14.3	8.9	9.3	4.8	9.2	9.8	7.9	11.3	6.2	7.1	10.2	10.0	11.5
Mar	1	12.5	14.1	10.2	9.0	9.3	10.5	9.2	3.9	11.3	8.1	8.0	10.1	11.0	11.6
	2	13.3	13.0	10.1	1.5	14.1	8.8	7.2	5.6	8.7	8.3	10.1	8.9	10.5	11.1
	3	13.8	12.1	9.4	9.8	12.4	9.3	9.5	4.1	9.0	8.7	11.6	9.3	10.5	10.8
Mean Flow		14.8	22.4	11.6	11.6	10.5	10.4	11.9	9.9	11.2	10.4	9.5	20.2	10.9	17.7

Source: Planning Division, WAPDA.

* Above = Below + Upper Pakpattan Canal + Eastern Sadiqia Canal + Fordwah Canal

River Sutlej Flow Downstream Suleimanki (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.0	2.9	4.1	0.2	0.4	3.2	2.8	1.4	0.0	0.0	4.1	3.0	1.4	0.8
	2	0.5	3.8	4.3	0.6	0.5	0.5	3.7	0.9	0.0	0.0	0.0	3.4	0.2	0.5
	3	2.0	4.7	2.2	0.3	0.5	1.6	7.7	1.0	0.0	0.0	4.6	2.5	3.2	1.8
May	1	2.4	7.4	3.1	0.2	0.4	3.7	3.0	1.2	0.4	0.4	7.9	0.0	4.8	1.0
	2	1.1	3.8	0.1	0.2	0.4	3.4	2.1	1.1	0.4	0.0	0.1	0.0	0.2	1.3
	3	0.6	5.2	0.7	0.2	0.3	2.3	2.2	1.0	0.4	0.0	0.8	0.0	0.0	1.2
Jun	1	1.1	1.1	1.3	0.2	0.3	0.2	2.2	0.6	0.4	0.0	0.0	0.0	0.9	0.8
	2	0.3	4.7	4.0	0.2	0.2	0.2	2.3	0.5	0.3	0.0	2.3	0.7	1.5	0.6
	3	1.0	3.5	2.8	0.2	0.2	0.2	2.0	0.9	0.3	0.0	1.1	0.0	3.1	0.0
Jul	1	3.7	9.3	2.0	0.1	0.2	0.2	4.8	1.0	0.3	5.6	0.7	0.0	1.2	15.3
	2	18.2	19.5	3.4	17.1	0.7	0.2	2.5	1.7	0.4	5.9	0.0	3.6	0.4	15.5
	3	11.3	29.4	10.0	15.3	9.6	0.2	1.7	0.8	16.8	0.5	0.0	18.9	0.1	9.4
Aug	1	12.7	41.9	4.4	24.2	9.4	1.8	2.3	0.7	4.8	8.1	0.0	30.6	6.9	11.6
	2	22.1	111.2	4.3	13.1	1.0	10.3	2.1	0.9	19.1	11.3	0.0	43.4	0.0	18.6
	3	22.8	112.4	4.4	0.3	1.0	2.5	4.3	2.3	8.2	10.8	0.0	28.2	1.7	11.5
Sep	1	14.6	49.3	0.8	0.3	0.3	0.3	8.9	8.6	0.1	0.0	0.0	6.6	2.3	40.7
	2	24.0	21.7	0.8	0.3	0.2	0.3	31.4	4.2	0.1	0.0	0.0	0.0	0.0	96.1
	3	24.7	7.5	0.9	0.3	0.2	0.3	4.4	0.8	0.1	0.0	0.0	73.0	0.0	19.5
Oct	1	0.7	2.6	0.7	0.2	0.2	0.3	1.5	0.9	0.3	0.6	0.0	123.8	0.0	4.8
	2	0.9	1.3	0.3	0.2	0.2	0.3	1.4	0.8	4.1	0.0	0.0	27.4	0.0	0.0
	3	1.0	1.4	0.3	0.2	0.2	0.3	1.2	0.7	0.3	0.0	0.0	9.6	0.0	3.2
Nov	1	0.9	0.4	0.3	0.1	0.2	0.3	0.7	0.3	0.1	0.0	0.0	4.5	0.0	0.1
	2	0.4	0.4	0.3	0.2	0.2	0.3	0.5	0.3	0.1	0.0	0.0	0.0	0.0	0.0
	3	0.4	0.4	0.3	0.2	0.0	0.3	0.4	0.3	0.1	0.1	0.0	0.0	0.0	0.1
Dec	1	0.3	0.0	0.3	0.1	0.0	0.2	1.0	0.4	0.1	2.4	0.2	0.6	0.0	0.0
	2	0.3	0.1	0.3	0.1	0.0	0.2	0.9	0.3	0.1	1.6	0.0	0.3	0.0	0.2
	3	1.3	1.6	0.2	0.1	1.1	0.2	0.9	0.4	0.1	4.3	0.0	5.1	0.9	1.4
Jan	1	3.4	2.5	1.8	0.3	0.3	1.0	0.8	0.4	2.1	0.1	0.0	6.6	4.3	4.2
	2	1.4	5.9	0.4	0.3	0.3	0.9	1.2	0.4	1.6	3.1	1.6	14.9	1.2	0.9
	3	0.5	0.7	0.2	0.4	0.2	0.8	2.1	0.4	0.1	2.4	0.8	4.9	0.1	0.4
Feb	1	0.4	0.7	0.2	1.0	0.2	2.7	1.2	0.5	0.1	0.6	0.0	0.0	0.0	0.2
	2	0.4	0.5	0.2	0.3	0.2	1.1	0.8	0.5	0.1	0.0	0.0	0.0	0.0	0.0
	3	1.1	14.2	0.4	0.3	0.2	0.8	1.0	0.4	0.1	0.7	0.0	0.0	0.3	0.2
Mar	1	4.1	11.3	0.2	0.3	0.2	1.3	0.9	0.4	0.8	0.7	0.0	0.0	1.3	0.0
	2	4.4	7.6	0.4	0.3	2.6	0.0	0.8	0.4	0.1	0.5	1.5	0.0	0.0	0.0
	3	5.5	2.8	0.3	0.7	3.9	1.1	0.8	0.4	0.3	1.5	1.3	0.1	0.0	0.1
Mean Flow		5.3	13.7	1.7	2.2	1.0	1.2	3.0	1.1	1.7	1.7	0.8	11.4	1.0	7.3

Source: Planning Division, WAPDA.

River Sutlej Flow Upstream Islam (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	3.3	4.2	0.5	2.2	3.7	2.1	1.0	0.3	0.9	1.3	1.7	2.0	0.1
	2	3.9	5.7	2.5	3.1	1.5	3.8	1.3	0.3	0.5	0.9	2.7	0.5	0.6
	3	3.9	2.0	1.4	2.4	3.2	9.2	1.6	0.3	0.5	3.6	3.2	1.6	0.9
May	1	8.1	5.0	0.9	1.6	5.5	4.0	1.3	0.3	0.5	6.6	0.8	5.4	1.5
	2	4.6	1.4	0.4	1.5	4.9	1.9	0.9	0.3	0.9	1.6	0.0	1.0	1.3
	3	6.4	2.1	0.8	1.2	1.9	1.7	0.9	0.7	0.0	1.9	1.0	1.3	1.4
Jun	1	3.0	2.6	0.9	0.5	0.5	1.6	0.6	0.6	0.6	0.8	0.8	1.2	1.4
	2	4.4	4.0	0.7	0.3	0.5	1.3	0.4	0.6	0.7	2.0	1.3	1.7	1.1
	3	3.2	2.6	0.9	0.8	0.7	1.3	0.8	0.7	0.6	1.3	0.8	4.6	1.3
Jul	1	8.6	2.0	0.9	0.4	0.7	2.8	0.8	0.6	3.4	0.8	0.8	0.6	10.7
	2	19.1	2.7	7.1	2.2	0.7	2.8	2.6	0.4	5.1	0.9	2.1	0.8	15.6
	3	30.5	8.2	27.4	8.1	0.6	1.4	1.5	1.2	1.3	0.6	10.6	0.5	10.4
Aug	1	40.7	5.6	16.9	9.0	1.0	1.8	1.1	5.3	3.6	0.5	20.0	5.2	10.9
	2	86.4	4.7	2.3	2.2	10.6	1.8	0.9	13.0	11.2	0.9	36.4	1.1	10.8
	3	115.2	3.6	1.1	1.6	5.6	4.2	1.1	7.9	8.3	0.9	28.8	1.7	14.6
Sep	1	74.2	1.4	1.9	1.2	1.8	14.2	16.7	1.3	0.9	0.9	11.0	2.3	30.3
	2	32.9	1.1	2.3	0.7	1.3	29.6	12.3	0.7	0.9	0.8	1.4	0.9	78.8
	3	11.8	1.2	1.8	0.6	1.8	5.9	2.6	0.8	0.4	0.9	10.8	1.0	22.6
Oct	1	5.7	0.5	0.7	0.4	0.5	1.5	1.2	0.6	0.9	0.8	182.7	0.9	6.1
	2	1.7	0.5	0.6	0.7	0.3	0.9	0.7	2.4	0.5	0.8	43.0	0.4	1.2
	3	3.4	0.6	0.1	0.4	0.4	1.3	0.2	0.8	0.7	0.1	10.3	0.5	3.6
Nov	1	2.1	0.6	0.1	0.0	0.3	2.8	0.4	0.4	0.7	0.0	4.1	0.7	1.6
	2	0.3	0.4	0.1	0.0	0.4	0.4	0.9	0.4	0.6	0.2	0.9	0.0	1.2
	3	0.4	0.2	0.4	0.1	0.2	1.1	1.4	0.2	1.9	0.0	3.5	0.0	1.4
Dec	1	0.6	1.0	0.7	0.4	0.8	2.3	0.9	0.4	4.1	0.6	1.5	0.2	1.2
	2	1.1	0.5	0.7	0.5	0.7	1.7	0.7	0.5	1.6	0.1	0.4	0.9	0.9
	3	1.8	0.2	1.1	0.4	0.3	0.8	1.3	1.1	4.4	0.3	4.2	0.6	2.1
Jan	1	1.7	1.5	4.6	0.7	0.6	0.7	0.8	1.2	1.5	0.0	5.6	1.2	2.5
	2	3.8	1.5	2.3	2.3	1.5	2.1	0.2	2.1	4.0	0.5	5.4	0.8	1.4
	3	2.3	0.6	3.7	1.1	1.5	4.6	0.6	1.2	1.9	0.3	5.1	0.8	1.8
Feb	1	2.6	0.5	2.9	0.2	2.4	5.8	0.8	0.8	1.0	0.0	0.3	0.6	0.7
	2	1.4	0.3	0.7	0.2	2.1	1.1	0.7	0.8	0.6	0.1	0.2	0.4	0.7
	3	5.8	0.6	0.6	0.2	1.4	0.2	0.7	0.4	0.0	0.5	0.5	0.8	0.2
Mar	1	10.1	0.4	0.5	0.1	2.1	0.2	0.5	1.2	2.2	0.6	0.7	0.6	0.6
	2	8.2	1.5	0.9	4.6	0.9	0.5	0.6	1.6	0.8	1.4	0.7	0.8	0.6
	3	2.7	0.8	3.6	5.6	1.8	0.5	0.6	0.7	2.4	2.7	1.2	0.6	0.7
Mean Flow		14.3	2.0	2.6	1.6	1.8	3.3	1.7	1.4	1.9	1.0	11.2	1.2	6.7

* Above = Below + Qaimpur Canal + Upper Bahawal Canal + Upper Mailsi Canal.

Source: Planning Division, WAPDA.

River Sutlej Flow Downstream Islam (1977–91)

10–Daily Discharges in 1000 Cusecs

Period		77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87	87–88	88–89	89–90	90–91
Apr	1	0.0	3.2	2.0	0.1	2.2	3.4	1.5	0.5	0.0	0.4	0.6	1.0	1.5	0.0
	2	0.0	3.3	5.7	2.2	2.5	1.2	3.3	0.8	0.0	0.2	0.2	2.1	0.0	0.1
	3	0.0	2.5	1.6	1.2	1.5	2.4	8.7	1.1	0.0	0.4	3.1	2.6	0.8	0.5
May	1	0.2	6.3	3.2	0.2	0.7	4.8	3.3	0.4	0.0	0.0	6.0	0.1	4.5	0.9
	2	0.0	3.1	0.2	0.2	0.6	4.3	1.3	0.3	0.0	0.0	0.9	0.0	0.4	0.4
	3	0.0	4.1	0.9	0.1	0.4	1.1	1.0	0.2	0.1	0.0	1.1	0.0	0.4	0.2
Jun	1	0.0	1.5	1.2	0.1	0.1	0.0	0.6	0.1	0.0	0.0	0.0	0.0	0.4	0.0
	2	0.0	3.0	2.4	0.1	0.1	0.0	0.2	0.1	0.0	0.0	1.2	0.5	0.5	0.1
	3	0.0	1.8	0.7	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.8	0.0	3.9	0.4
Jul	1	1.4	7.8	0.4	0.1	0.1	0.1	1.8	0.1	0.0	3.3	0.1	0.0	0.2	10.1
	2	11.6	18.6	1.2	6.1	1.5	0.1	1.6	1.8	0.0	4.5	0.1	1.6	0.0	14.7
	3	11.2	30.0	6.7	25.8	7.4	0.1	0.2	0.5	0.5	0.8	0.0	10.2	0.0	9.3
Aug	1	7.7	38.7	4.1	15.4	8.1	0.3	0.7	0.1	4.5	2.9	0.0	19.2	4.4	10.3
	2	16.7	83.6	2.4	0.5	1.4	9.3	0.8	0.1	12.0	10.9	0.0	35.3	0.4	9.7
	3	21.5	112.4	1.9	0.4	0.4	3.9	3.4	0.5	6.6	7.4	0.0	27.5	0.8	13.4
Sep	1	13.2	71.6	0.2	0.2	0.1	0.4	13.5	15.6	0.3	0.0	0.0	9.7	0.9	29.0
	2	23.2	29.9	0.1	0.6	0.1	0.2	28.6	11.1	0.0	0.0	0.0	0.0	0.0	78.0
	3	26.6	8.9	0.1	0.2	0.1	0.4	4.8	1.2	0.0	0.0	0.0	9.8	0.0	21.3
Oct	1	1.5	2.8	0.1	0.1	0.1	0.1	0.3	0.2	0.0	0.1	0.0	182.7	0.0	5.1
	2	0.2	0.0	0.1	0.1	0.1	0.1	0.1	0.2	1.6	0.0	0.0	42.9	0.0	0.0
	3	0.1	1.4	0.0	0.1	0.1	0.1	0.7	0.2	0.1	0.0	0.0	9.5	0.0	2.5
Nov	1	0.6	0.3	0.1	0.1	0.0	0.1	2.1	0.2	0.0	0.0	0.0	3.9	0.6	0.1
	2	0.5	0.2	0.1	0.1	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.1	0.0	0.0
	3	0.2	0.2	0.0	0.1	0.1	0.0	0.2	0.1	0.0	1.2	0.0	2.8	0.0	0.1
Dec	1	0.2	0.4	0.0	0.2	0.0	0.1	0.7	0.1	0.0	2.5	0.0	0.8	0.0	0.0
	2	0.2	0.6	0.1	0.1	0.1	0.1	0.7	0.1	0.0	1.0	0.0	0.0	0.0	0.0
	3	0.1	0.3	0.1	0.7	0.1	0.1	0.2	0.8	0.0	3.9	0.0	3.8	0.0	1.5
Jan	1	1.5	0.1	0.2	4.2	0.1	0.4	0.2	0.8	0.1	0.3	0.0	5.4	0.9	2.5
	2	2.2	2.2	0.1	0.4	2.0	0.3	1.5	0.2	2.1	3.8	0.0	5.4	0.8	1.4
	3	0.7	0.6	0.6	3.7	1.1	1.5	3.9	0.6	0.9	1.9	0.0	5.1	0.8	1.8
Feb	1	0.3	2.2	0.5	2.9	0.2	2.4	5.7	0.1	0.1	0.1	0.0	0.0	0.6	0.0
	2	0.2	1.4	0.0	0.4	0.1	1.8	1.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0
	3	0.1	5.8	0.1	0.2	0.1	0.7	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.0
Mar	1	1.8	9.4	0.0	0.2	0.1	1.4	0.2	0.0	0.1	1.5	0.6	0.0	0.4	0.0
	2	2.7	6.9	0.6	0.6	3.9	0.9	0.1	0.0	0.1	0.1	1.0	0.0	0.0	0.0
	3	4.4	0.6	0.2	3.0	4.5	1.7	0.1	0.0	0.1	1.8	2.0	0.6	0.0	0.0
Mean Flow		4.2	12.9	1.1	2.0	1.1	1.2	2.6	1.1	0.8	1.4	0.5	10.6	0.6	5.9

Source: Planning Division, WAPDA.

C. MONTHLY FLOW OF TRIBUTARY RIVERS AT RIM STATIONS

Flow of River Swat at Chakdara (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	5951	6841	7413	7682	10590	5093	6068	4623	4118	7564
May	8052	11290	8231	10899	14218	7369	9533	8833	6539	10330
Jun	13700	15935	13431	15717	11985	8169	11229	17734	9077	10674
Jul	14754	15958	15064	12705	14413	7938	10639	11029	11387	15242
Aug	7971	11338	9142	8231	8524	8866	11176	10004	7727	11322
Sep	3950	3917	4757	4051	3715	2605	5732	5362	3278	3311
KHARIF	54378.34	65279.61	58037.95	59285.11	63444.66	40041.43	54376.18	57585.72	42126.90	58444.09
Oct	2798	2147	2050	2473	2212	2131	2375	2033	2164	2098
Nov	1849	1984	1597	1950	1513	2555	1731	1815	1093	1496
Dec	1497	1415	1318	1562	1090	1903	1659	1562	1334	1610
Jan	1350	1220	1334	1383	1139	1545	1009	1464	1155	1301
Feb	1287	1409	1837	1687	1182	1652	1297	1356	1652	1495
Mar	4327	2700	4652	4262	2424	4034	2212	1903	374	3009
RABI	13107.71	10875.01	12787.65	13315.70	9560.13	13821.00	10283.34	10134.19	7771.19	11011.30
TOTAL	67486.05	76154.62	70825.60	72600.81	73004.78	53862.43	64659.51	67719.91	49898.09	69455.39

Flow of River Kabul above Warsak (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	22508	18911	26626	28744	28005	12758	26576	15398	9346	16490
May	21538	37512	36488	41693	55943	26369	47907	27703	14966	25149
Jun	38948	53253	55522	60514	51942	31400	47638	61842	25013	35334
Jul	43775	61344	64467	44979	56692	30095	49322	42474	38472	47891
Aug	32583	44800	45760	34552	33934	32681	48656	43531	30273	34926
Sep	14994	19197	17969	18776	13313	9531	19012	16339	13868	11817
KHARIF	174346	235016	246832	229259	239828	142835	239111	207287	131938	171607
Oct	8687	9956	8052	11956	7532	5807	7922	6670	7206	7093
Nov	5715	7598	11397	8590	6606	5984	6875	6018	5144	7363
Dec	4587	5938	9777	6458	6295	6637	6686	5775	5303	8410
Jan	4229	5059	8492	6393	6019	6881	6653	5840	5791	5840
Feb	3773	4886	6664	5999	5425	5199	7654	4556	5286	4852
Mar	5433	8980	8361	9077	5677	7093	7271	5287	6182	11452
RABI	32426	42416	52743	48474	37555	37602	43062	34145	34912	45009
TOTAL	206772	277433	299575	277733	277383	180436	282173	241432	166850	216616

Flow of River Gomal at Kot Murtaza (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	748	588	773	807	908	1479	4555	269	202	387
May	244	195	228	163	1041	163	911	130	195	602
Jun	1362	101	118	504	269	202	437	235	2168	1748
Jul	2115	7239	407	667	1741	293	1366	1692	2164	3367
Aug	1334	3302	1285	358	569	1545	2603	4457	260	130
Sep	740	857	168	605	353	118	689	773	134	134
KHARIF	6542	12283	2979	3104	4881	3800	10562	7557	5124	6369
Oct	114	195	130	163	130	846	195	195	130	146
Nov	622	151	101	403	134	471	202	185	168	168
Dec	390	163	114	179	146	342	228	195	163	163
Jan	781	260	228	374	163	244	228	228	244	277
Feb	730	417	270	452	209	417	198	435	3043	556
Mar	911	439	732	1383	1659	1659	325	1399	976	976
RABI	3548	1626	1575	2954	2442	3979	1376	2637	4724	2286
TOTAL	10091	13909	4553	6058	7322	7778	11938	10193	9848	8655

Flow of River Haro at Garaila (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	455	353	840	605	1765	874	2706	572	403	1042
May	569	244	472	277	699	830	1171	342	342	618
Jun	824	471	370	370	454	403	857	639	336	622
Jul	3449	3595	1464	2473	2115	651	1480	2229	3546	1383
Aug	2928	6800	1562	2489	1708	1350	2310	5189	1122	2505
Sep	1177	1395	538	756	521	622	2068	2084	958	689
KHARIF	9402	12858	5246	6969	7262	4730	10593	11054	6708	6859
Oct	699	732	342	439	374	407	667	618	927	618
Nov	555	672	336	353	370	437	555	588	504	891
Dec	407	455	472	325	342	455	634	521	732	765
Jan	374	521	586	423	358	992	521	667	797	504
Feb	330	748	558	626	365	643	504	470	1165	539
Mar	748	1806	1334	1692	1074	1041	667	407	1594	895
RABI	3114	4934	3627	3858	2882	3976	3548	3270	5720	4212
TOTAL	12516	17792	8873	10828	10145	8706	14141	14324	12428	11071

Flow of River Soan at Dhok Pathan (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	423	185	622	387	1059	1076	3799	319	387	504
May	277	65	293	81	455	1122	944	130	130	342
Jun	908	134	252	1025	185	202	656	471	101	672
Jul	9305	4783	3188	4067	4604	1920	4164	4718	2082	3319
Aug	6621	14331	3985	3953	2928	10053	5938	7011	3514	4018
Sep	1563	2605	1076	2017	840	941	4034	5749	2168	874
KHARIF	19096	22104	9417	11530	10072	15314	19534	18398	8382	9729
Oct	911	1415	423	651	277	651	618	521	878	1415
Nov	656	756	252	303	235	941	370	471	286	891
Dec	293	374	260	228	228	504	407	521	895	1155
Jan	374	846	716	618	342	992	325	569	423	455
Feb	330	1582	648	1078	435	852	828	348	991	678
Mar	699	1968	1399	2326	2294	1854	1155	277	2408	1236
RABI	3263	6942	3698	5204	3810	5795	3703	2705	5881	5831
TOTAL	22360	29047	13115	16734	13881	21109	23238	21103	14263	15560

Flow of River Kunhar at Gari Habib Ullah (1977–87)

Monthly Discharges in Cusecs

Period	77–78	78–79	79–80	80–81	81–82	82–83	83–84	84–85	85–86	86–87
Apr	2684	3698	3513	4421	2437	3261	2942	3715	2084	3379
May	4652	8215	4718	7206	4262	6344	6670	7125	4311	5742
Jun	8354	11212	8825	11430	9565	8455	9464	10287	5867	4186
Jul	7743	7841	9272	8134	8296	7402	8980	5319	4945	9142
Aug	3628	5108	4506	4343	4116	4913	7158	4213	3400	4164
Sep	2202	2370	2421	2521	1899	1933	3194	2488	1614	2538
KHARIF	29264	38444	33255	38056	30575	32308	38406	33148	22221	29152
Oct	1448	1301	1318	1399	1399	1204	1497	1253	1074	1366
Nov	1126	1160	1042	1009	975	1126	908	941	807	1076
Dec	944	911	813	813	830	992	813	765	765	1025
Jan	878	765	830	716	748	830	797	748	651	781
Feb	922	713	901	748	748	713	792	696	661	817
Mar	1399	1057	1513	1383	1301	1741	1399	992	1188	1236
RABI	6717	5907	6416	6067	6001	6606	6206	5395	5144	6302
TOTAL	35980	44351	39671	44123	36576	38913	44612	38543	27365	35453

ANNEX - III

WATER QUALITY DATA ON RIVERS AND DRAINS

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1.	Indus River at Partab Bridge	1963	–	140	224	220	232	–	200	180	180	–	185	130
		1965	192	179	154	205	–	134	134	134	122	192	186	198
		1967	–	158	158	–	136	84	–	–	–	–	–	–
		1969	–	–	–	–	–	–	–	–	–	–	–	–
		1981	–	–	–	–	–	–	–	104	–	–	–	–
		1982	–	–	–	–	–	76	–	–	–	–	–	–
		1985	–	140	–	–	–	–	108	–	–	–	–	164
		1986	–	–	–	160	–	–	–	–	–	–	–	–
		1987	–	–	–	–	–	–	–	–	–	–	–	–
		1988	104	–	–	–	–	104	–	–	–	–	–	–
2.	Indus River at Basham Qila	1969	174	–	96	120	110	–	96	112	116	–	130	–
		1971	144	–	144	114	–	94	–	–	106	–	–	–
		1973	–	–	–	230	–	200	240	–	–	–	250	270
		1975	160	152	–	–	102	–	–	116	–	144	–	144
		1981	184	152	208	148	136	792	144	196	120	140	–	172
		1982	260	224	240	188	172	96	116	80	116	124	186	160
		1985	–	–	132	–	–	–	116	112	–	144	148	136
		1986	164	172	136	116	–	92	–	–	–	–	184	–
		1987	–	140	–	–	–	60	164	–	–	–	–	–
		1988	132	–	120	–	–	–	–	–	–	–	–	–
3.	Indus River at Darband	1963	180	168	165	144	–	–	–	160	150	145	165	160
		1965	295	250	250	–	–	186	138	137	270	159	217	179
		1967	184	180	–	160	118	102	114	105	104	–	140	142
4.	Indus River at Khairabad	1981	–	–	–	–	–	–	–	–	–	–	–	216
		1982	312	256	316	204	–	232	–	184	140	200	288	360
		1985	–	–	–	–	–	–	–	–	–	–	–	–
		1986	–	–	–	184	124	96	188	104	96	104	128	144
		1987	236	220	164	–	–	120	248	104	144	–	–	–
		1988	–	–	–	–	–	–	–	–	–	–	–	–
		1989	–	–	–	–	–	–	–	–	–	–	160	132
		1990	232	–	212	184	168	120	–	140	–	200	368	–
		1991	224	–	–	–	–	144	–	–	–	–	–	–
5.	Indus River at Mandori	1965	230	228	218	186	140	370	118	160	192	–	236	250
		1967	–	215	200	164	140	98	116	122	114	168	212	208
		1969	187	191	159	120	1280	80	160	111	104	144	156	170
		1971	–	176	186	134	104	112	92	–	130	–	–	–
		1973	–	196	144	140	122	–	–	–	112	–	170	196
6.	Indus River Near Massan	1973	193	221	158	140	116	86	104	116	161	–	172	190
		1975	194	346	210	146	116	98	128	116	130	128	142	202
		1981	144	216	200	332	160	108	–	756	144	124	–	148
		1982	148	–	224	–	–	106	96	88	140	–	–	172
		1985	184	136	144	184	–	104	–	–	96	–	136	172
		1986	–	–	196	144	136	–	–	132	–	–	184	–
		1987	132	144	–	–	–	276	80	196	120	128	–	140
		1988	160	124	188	124	104	–	124	108	114	100	176	152
		1989	180	–	–	140	164	108	–	–	–	140	–	148
		1990	180	–	132	176	140	128	–	164	–	–	–	–
		1991	236	120	144	–	–	208	–	–	–	–	–	–
7.	Indus River at Taunsa	** 1992	–	–	–	–	170	–	–	–	–	–	–	–
8.	Indus River at Guddu	* 1989	–	–	–	–	–	–	–	–	150	170	280	160
		* 1990	225	260	210	390	130	130	130	140	130	145	180	230
		* 1991	205	240	335	230	–	–	–	–	–	–	–	–

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
9.	Indus River at Sukkur Barrage	1974	–	–	–	275	147	128	122	147	134	156	179	230
		1975	224	237	218	205	211	154	141	179	173	173	186	186
		1976	262	205	198	230	211	156	160	–	–	–	179	–
		1977	–	–	186	294	198	141	186	154	154	129	186	218
		1978	243	192	218	186	186	120	179	192	153	160	211	288
		1979	269	294	166	192	186	173	160	166	154	205	218	198
		1980	230	262	243	204	147	192	204	192	185	204	230	249
		1981	230	237	224	198	160	230	198	186	243	205	224	288
		1982	–	358	371	243	192	154	205	182	179	173	173	256
		1983	263	301	307	–	–	–	–	–	–	–	–	–
		1984	–	–	–	269	141	308	256	154	173	173	182	180
		1985	173	205	204	179	135	173	280	270	138	210	138	220
		1986	240	256	283	–	–	–	–	–	–	–	–	–
		*1989	–	–	–	–	–	–	–	–	–	–	180	150
		*1990	260	190	310	240	235	130	140	135	135	150	180	200
		*1991	185	230	200	200	–	–	–	–	–	–	–	–
10.	Indus River at Dadu Moro Bridge	1981	–	228	208	308	202	196	208	240	212	194	248	288
		1982	302	268	278	362	324	230	210	224	240	–	–	–
		1985	–	272	–	208	200	176	482	–	–	–	–	–
11.	Indus River at Sehwan	1967	–	–	–	–	–	219	160	186	174	152	210	220
		1969	240	280	240	184	174	124	116	140	152	180	174	–
		1971	–	200	–	–	–	–	160	165	–	–	240	–
		1973	260	260	240	290	170	190	190	–	210	180	310	270
		1975	272	272	268	221	224	176	179	172	220	224	281	–
		1981	192	236	220	314	216	222	222	264	186	–	255	256
		1982	270	336	278	344	268	316	236	250	–	–	–	–
		*1989	–	–	–	–	–	–	–	335	255	235	355	555
		*1990	520	380	880	290	255	180	150	160	170	280	375	535
		*1991	445	379	460	510	–	–	–	–	–	–	–	–
12.	Indus River D/S Kotri Barrage	1974	–	–	–	400	280	190	–	–	–	173	245	217
		1975	252	252	295	368	263	168	151	182	210	224	252	252
		1976	260	260	298	296	298	210	227	202	245	242	326	325
		1977	326	304	403	397	298	263	183	182	224	224	226	228
		1978	228	308	273	236	238	182	273	189	306	266	301	266
		1979	228	245	242	273	182	193	140	165	186	139	193	182
		1980	172	305	319	336	273	182	175	189	186	187	235	236
		1981	247	277	350	347	248	168	158	209	195	199	221	239
		1982	249	266	261	350	270	237	181	175	207	162	186	193
		1983	193	221	298	–	–	–	–	–	–	–	–	–
		1984	–	–	–	343	344	337	196	200	224	245	244	241
		1985	224	280	302	280	374	228	183	179	173	168	208	232
		1986	191	249	281	–	–	–	–	–	–	–	–	–
		*1990	–	–	490	210	–	–	150	640	200	170	190	240
13.	Jhelum River at Chinari	1969	–	–	–	–	–	–	–	–	–	–	158	–
		1971	–	156	156	104	96	114	126	150	162	–	168	–
		1973	136	140	116	90	–	–	104	–	–	–	152	–
		1975	136	–	–	–	112	96	100	–	–	172	148	180
		1981	148	148	–	–	84	–	–	–	160	–	–	294
		1982	160	–	–	104	76	108	–	136	–	–	–	–
		1985	–	–	–	–	–	–	144	124	164	148	–	176
		1986	164	136	–	112	96	116	96	116	160	132	156	144
		1987	–	–	–	92	–	268	104	132	160	–	–	160
		1988	–	120	164	–	–	–	–	172	96	–	–	–
		1989	–	–	120	108	148	–	–	–	–	–	–	–
		1990	–	–	–	–	–	–	144	144	–	–	176	160

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
14.	Jhelum River at Azad Pattan	1981	–	148	128	148	–	–	–	824	120	–	144	–
		1982	–	132	–	–	–	92	–	–	–	–	–	–
		1985	–	132	–	96	96	92	–	124	–	148	216	160
		1986	148	144	132	116	104	–	–	–	–	–	–	–
		1987	160	–	200	84	–	–	104	–	124	–	–	136
		1988	–	140	124	96	–	–	–	–	–	116	–	–
		1989	–	168	140	96	144	–	–	–	–	144	–	–
		1990	144	160	–	108	–	–	–	–	–	–	–	–
		1991	–	160	–	–	–	–	–	–	–	–	–	–
15.	Jhelum River at Mangla	1963	184	116	188	180	160	–	–	195	135	155	164	180
		1965	185	166	157	147	102	128	118	138	–	–	–	–
		1967	–	–	–	–	113	99	95	93	130	266	–	113
		1969	121	–	–	–	104	120	32	80	96	102	100	126
		1971	–	–	–	–	–	–	–	–	–	–	–	–
		1973	120	–	–	120	–	–	90	96	–	–	–	–
		1975	–	–	–	–	–	–	–	–	–	–	–	–
		1981	–	–	–	–	–	–	–	–	548	–	–	–
16.	Jhelum River at Rasul	** 1992	–	–	–	–	80	–	–	–	–	–	–	–
17.	Chenab River at Alexandria Bridge	1963	195	–	174	164	170	–	144	230	96	125	244	–
		1965	169	170	143	135	115	148	118	115	300	160	194	211
		1967	150	170	160	145	128	112	108	104	126	168	210	–
18.	Chenab River at Marala	** 1992	–	–	–	–	80	–	–	–	–	–	–	–
19.	Chenab River at Khanki	** 1992	–	–	–	–	70	–	–	–	–	–	–	–
20.	Chenab River at Qadirabad	** 1992	–	–	–	–	80	–	–	–	–	–	–	–
21.	Chenab River at Trimmu	** 1992	–	–	–	–	140	–	–	–	–	–	–	–
22.	Swat River near Kalam	1963	–	60	–	60	110	156	166	100	–	–	72	30
		1965	102	–	–	–	80	83	35	–	–	76	86	83
		1967	96	80	–	75	82	138	90	70	64	66	72	65
		1969	–	60	192	70	48	–	78	52	38	–	66	–
		1971	–	–	68	–	–	–	–	–	59	–	–	–
		1973	–	–	–	84	–	68	64	–	–	–	–	–
		1975	136	120	–	–	54	52	–	58	–	116	60	80
		1981	64	84	96	116	116	72	68	68	76	92	76	–
		1982	136	100	216	148	120	44	–	–	116	56	60	60
		1985	68	104	164	76	76	–	–	–	200	72	76	92
		1986	70	–	–	68	60	64	–	108	60	60	80	64
		1987	76	80	–	80	–	68	84	–	–	–	60	76
		1988	–	–	–	–	–	–	–	–	–	–	–	–
		1989	–	–	–	–	–	–	–	–	–	–	–	–
		1990	–	–	–	–	–	–	–	–	–	–	–	–
23.	Swat River at Chakdara	1963	124	–	140	92	–	1096	165	73	96	110	64	115
		1965	115	176	120	112	89	83	76	76	90	109	131	135
		1967	130	120	110	102	96	68	48	48	64	90	104	100
		1969	118	116	84	72	52	–	46	50	72	–	–	–
		1971	–	–	–	88	50	58	–	–	63	–	96	–
		1973	–	–	78	68	96	64	56	–	–	–	–	116
		1975	120	116	–	–	56	56	–	88	–	60	92	76
		1981	120	116	148	136	116	88	80	80	84	112	–	140
		1982	172	216	200	132	148	48	120	56	76	116	–	132

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		1985	116	100	136	108	64	60	–	–	–	120	124	128
		1986	128	124	–	96	64	104	–	104	88	124	160	116
		1987	92	116	104	96	–	64	60	–	168	–	136	108
		1988	216	428	108	76	56	60	72	84	88	116	–	112
		1989	128	128	72	88	76	–	–	–	–	–	120	104
		1990	116	120	144	100	–	64	108	–	–	56	–	160
		1991	120	116	–	–	–	–	48	–	–	–	–	–
24.	Kabul River at Nowshera	1963	346	260	208	190	196	164	192	155	176	235	250	250
		1965	272	294	221	179	128	150	160	174	163	246	462	384
		1967	314	325	290	167	162	128	140	132	140	212	250	252
		1969	247	256	168	136	–	–	–	90	156	192	–	–
		1971	–	–	220	132	–	–	–	–	102	–	–	–
		1973	220	232	200	158	144	112	116	116	–	176	–	220
		1975	196	–	201	–	136	–	96	–	–	–	244	244
		1981	284	284	300	176	140	204	188	184	232	364	–	172
		1982	272	264	224	188	232	236	104	216	188	216	336	–
		1985	236	–	304	160	132	104	172	148	144	–	124	224
		1986	256	272	248	–	128	104	104	108	164	148	224	212
		1987	224	348	244	160	352	96	204	96	208	160	204	236
		1988	236	172	260	172	110	–	–	148	–	228	224	156
		1989	288	268	260	228	132	–	–	–	–	–	–	184
		1990	244	240	216	–	108	116	140	218	–	272	–	276
		1991	244	–	–	–	–	–	–	–	–	–	–	–
25.	Kurram River at Thal	1967	–	–	–	–	–	–	230	216	216	–	–	1055
		1969	228	248	231	214	260	260	212	226	236	178	250	280
		1971	–	232	208	216	–	–	–	–	–	–	–	–
		1973	–	300	188	180	194	228	232	226	–	–	244	–
		1975	242	240	236	–	188	220	–	–	–	–	236	268
		1981	288	312	248	240	316	276	324	312	368	280	–	336
		1982	344	284	408	292	396	300	316	280	364	328	368	308
		1985	212	204	–	180	208	208	224	284	200	200	220	200
		1986	192	204	224	216	188	188	100	212	380	260	248	252
		1987	260	268	212	276	248	200	372	–	212	244	236	288
		1988	248	136	180	172	168	220	208	252	468	176	228	244
		1989	268	292	228	212	168	184	–	–	–	–	284	368
		1990	324	244	216	200	216	236	356	336	–	336	348	316
		1991	264	256	–	–	–	236	–	–	–	–	–	–
26.	Tochi River at Tangi Post	1963	790	724	872	1070	540	732	964	995	–	840	1000	665
		1965	806	659	619	365	384	618	765	544	947	845	1008	672
		1967	802	734	640	305	550	1010	750	587	955	–	–	664
		1969	552	576	692	746	–	392	–	–	–	–	–	–
		1981	–	632	544	440	544	716	496	746	820	700	–	684
		1982	636	724	428	480	596	604	664	488	1000	604	604	556
		1985	–	–	–	–	852	–	–	–	824	600	868	–
		1986	–	–	–	404	788	784	–	–	–	8520	808	684
		1987	692	808	536	–	888	756	–	549	820	912	856	680
		1988	656	604	732	660	496	436	416	652	712	804	840	736
		1989	620	736	688	376	372	–	–	–	–	–	868	696
		1990	576	484	436	–	420	788	632	404	–	816	748	744
		1991	596	544	–	352	–	468	–	–	–	–	–	–
27.	Gomal River at Khajuri	1963	772	652	912	–	–	–	–	–	–	425	868	845
		1965	910	715	754	592	517	780	608	891	650	900	975	910
		1967	–	898	–	–	615	–	812	–	874	–	–	993
		1981	948	744	–	–	572	864	800	–	920	–	912	852
		1982	920	1220	512	668	684	–	668	664	996	660	724	944
		1985	824	588	832	–	792	800	–	–	800	876	736	732
		1986	–	–	–	488	400	728	–	548	–	828	800	–
		1987	876	824	532	536	–	932	1020	920	724	824	852	868
		1988	880	888	756	660	772	1020	788	304	796	748	756	768

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		1989	824	812	640	512	784	760	–	–	–	–	836	964
		1990	884	648	560	508	456	840	696	688	1060	812	–	840
		1991	884	580	–	–	–	800	–	–	–	–	–	–
28.	Gomal River at Kot Murtaza	1963	784	732	1080	966	676	–	–	–	–	790	885	900
		1965	845	–	791	630	580	910	656	744	965	900	988	897
		1967	–	950	–	–	632	1120	822	662	906	–	–	1030
		1969	820	820	762	804	–	960	856	856	–	928	–	–
		1971	–	856	–	924	620	–	–	–	–	–	–	–
		1973	–	–	–	–	744	880	588	648	482	–	868	830
		1975	850	642	–	–	672	–	–	–	–	–	840	608
		1981	132	760	620	496	476	832	992	896	820	948	800	864
		1982	500	1200	520	628	692	858	–	580	944	600	748	560
		1985	820	500	528	1060	788	808	–	–	804	732	748	756
		1986	–	–	–	500	420	788	812	600	860	820	788	820
		1987	852	884	508	532	–	912	948	868	744	820	808	880
		1988	876	884	760	632	716	1140	860	296	736	780	–	800
		1989	892	688	676	528	796	776	–	–	–	–	–	900
		1990	892	688	556	–	720	864	664	608	664	852	–	836
		1991	880	592	–	–	–	760	–	–	–	–	–	–
29.	Zhob River at Sherik Weir	1973	–	–	870	–	–	–	–	–	1040	–	–	–
		1975	1088	1264	1216	1216	–	1600	640	384	896	1216	830	1495
		1981	388	514	471	501	498	1019	630	744	527	1239	–	–
		1982	655	794	818	702	687	525	614	570	1025	–	–	–
		1985	–	–	536	–	598	451	577	574	473	570	582	561
30.	Haro River near Khanpur	1963	280	–	265	260	320	270	315	240	240	220	227	238
		1965	320	243	250	224	166	198	217	279	254	256	294	230
		1967	288	280	200	210	232	228	260	300	216	254	256	256
		1969	222	234	213	222	208	246	–	243	170	192	236	226
		1971	206	176	172	180	216	219	416	178	178	202	–	228
		1973	200	230	200	192	182	204	164	176	166	202	–	220
		1975	240	228	220	124	180	180	152	208	176	236	240	232
		1981	756	208	208	176	264	208	–	–	212	–	–	192
		1982	204	–	176	–	–	116	184	176	–	–	–	–
		1985	–	–	192	–	–	176	–	–	164	–	–	156
		1986	–	–	–	160	–	–	172	184	–	–	–	–
		1987	–	–	192	132	172	152	96	216	184	–	–	224
		1988	188	176	–	164	–	–	–	196	156	152	–	–
		1989	236	180	168	152	148	–	–	–	–	–	–	176
		1990	216	–	–	–	–	–	–	–	–	–	–	–
31.	Haro River at Ghariala	1969	294	238	256	266	232	236	264	164	200	226	264	–
		1971	–	220	236	232	260	228	176	165	159	258	254	268
		1973	276	270	204	188	208	228	172	164	156	230	–	260
		1975	260	266	268	24	232	208	212	176	184	248	264	272
		1981	–	–	312	200	376	348	–	–	–	–	–	316
		1982	–	–	280	–	–	212	232	168	236	–	–	256
		1985	236	–	236	208	–	256	136	–	212	–	284	272
		1986	–	336	184	212	220	–	216	–	300	236	272	276
		1987	–	324	176	168	–	288	152	232	196	224	–	–
		1988	–	–	168	–	–	–	–	–	–	–	–	232
		1989	–	–	–	264	200	–	–	–	–	–	192	–
		1990	236	–	–	–	–	–	–	–	–	–	–	–
32.	Soan River near Rawalpindi	1963	302	292	277	330	166	–	215	164	200	290	430	307
		1965	326	256	272	173	160	198	224	179	284	307	358	339
		1967	324	325	220	–	276	324	130	140	208	240	282	–
		1969	306	174	212	276	284	320	248	104	144	280	296	–
		1971	–	320	238	344	294	294	118	162	–	–	270	304
		1973	304	290	280	304	398	232	116	144	140	320	–	308
		1975	352	344	288	280	308	348	144	144	108	260	300	324

Total Dissolved Solids (ppm) at various Sites of Indus River and its Tributaries

Sr. No.	Name of River/ Sampling Site	Year	Part Per Million Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		1981	–	272	260	188	204	456	272	488	–	368	–	396
		1982	400	–	256	128	216	348	352	–	216	–	–	–
		1985	–	–	192	292	–	–	-	–	–	364	360	236
		1986	–	288	–	328	352	408	–	204	240	336	280	268
		1987	–	312	284	340	188	216	232	200	368	352	368	372
		1988	396	–	322	384	408	288	–	212	–	236	320	–
		1989	372	208	312	356	408	–	–	–	–	–	–	276
		1990	396	–	–	–	–	–	–	–	–	–	–	–
33.	Soan River near Dhok Pathan	1963	–	–	–	–	–	–	–	–	–	–	–	290
		1965	275	288	243	196	222	256	272	227	263	364	320	291
		1967	295	296	256	328	234	400	269	182	272	222	242	335
		1969	–	309	242	196	260	284	292	164	158	–	176	286
		1971	258	–	–	244	204	173	167	185	185	196	–	240
		1973	268	272	172	192	260	296	176	192	188	206	–	272
		1975	284	294	232	176	228	288	232	220	172	180	244	248
		1981	400	–	408	208	360	372	–	–	336	300	–	328
		1982	396	–	380	232	–	380	540	–	380	–	–	–
		1985	352	–	336	392	–	380	–	–	148	260	256	–
		1986	348	276	316	–	280	–	–	–	–	244	256	256
		1987	352	340	216	184	216	232	236	224	184	256	–	352
		1988	–	392	–	244	288	216	184	176	216	212	288	368
		1989	–	296	364	–	320	364	–	–	–	252	360	384
		1990	–	372	–	–	212	–	216	–	–	–	–	–
		1991	292	388	–	–	–	252	–	–	–	–	–	–
34.	Ravi at Balloki H/W	1965	–	–	–	–	–	–	–	214	–	–	–	–
		** 1992	–	–	–	–	100	–	–	–	–	–	–	–
35.	Ravi at Sidhnai	1965	–	–	–	–	–	–	–	256	–	–	–	–
		** 1992	–	–	–	–	150	–	–	–	–	–	–	–
36.	Sutlej at Sulemanki	1965	–	–	–	–	–	–	–	179	–	–	–	–
		** 1992	–	–	–	–	110	–	–	–	–	–	–	–
37.	Sutlej at Islam	1965	–	–	–	–	–	–	–	227	–	–	–	–
		** 1992	–	–	–	–	110	–	–	–	–	–	–	–
38.	Sutlej at Panjnad	1965	–	–	–	–	–	–	–	198	–	–	–	–
		** 1992	–	–	–	–	130	–	–	–	–	–	–	–

Source: 1) Suface Water Hydrology, WAPDA

2) Data for the Years 1974 to 1986 for River Indus at Sukkur and Kotri obtained from published record of Indus River Commission, IPD, Sindh.

* Right Bank Master Planning Study, 1991

** Observed by DSEA–NDP Consultants

Total Dissolved Solids (ppm) of Drains

Sr. No.	Name of Drain/Location	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1.	Subago Drain US Near O/F 2–L	1991	744	–	–	–	–	692	–	–	–	–	–	–
2.	Lower Raniwah D.S Near O/F Machhar Khadi Drain	1991	5380	–	–	–	–	4760	–	–	–	–	–	
3.	Tarkanawala D.S. Near O/F Jahanian Shah Drain	1991	644	–	–	–	–	284	–	–	–	–	–	–
4.	Mona at Head Reach	1979	–	–	–	–	–	–	–	–	512	640	–	768
		1980	608	608	576	576	512	544	–	518	–	–	–	–
	Mona Near Luk	1979	–	–	–	–	–	–	–	–	832	832	–	1056
		1980	1280	1152	1216	1184	1152	1280	–	1024	–	–	–	–
	Mona at D.S. O/F Bhakhar	1991	1340	–	–	–	–	–	–	–	–		–	–
5.	Sulki D.S. Near O/F Jahanabad	1991	5120	–	–	–	–	3660	–	–	–	–	–	–
6.	Faqirian Sillanwali Drain	1990	–	–	–	–	–	–	–	–	–	–	–	6100
	U.S. Near O/F 7L/FS	1991	–	–	–	–	–	7720	–	–	–	–	–	–
7.	Chot Drain DS Near O/F Amirchand	1990	–	–	–	–	–	–	–	–	–	–	–	568
		1991	–	–	–	–	–	628	–	–	–	–	–	–
8.	Bhikho U.S Near O/F Rairka Drain	1990	–	–	–	–	–	–	–	–	–	–	–	548
		1991	–	–	–	–	–	772	–	–	–	–	–	–
9.	Budhi Nallah D.S Near O/F	1990	–	–	–	–	–	–	–	–	–	–	–	464
	Chak Kamal	1991	–	–	–	–	–	408	–	–	–	–	–	–
10.	Bhera Drain at aquaduct of	1973	–	–	–	–	–	–	480	–	486	–	484	488
	Shahpur Branch	1974	–	540	474	446	474	480	492	480	480	478	–	–
	"	1975	468	580	–	436	472	508	452	488	364	492	488	484
	"	1976	484	516	496	–	484	–	468	–	–	–	–	508
11.	Deg Nallah on Jaranwala	1969	–	–	–	–	–	–	–	–	612	720	286	–
	Sayedwala Road	1970	472	480	496	–	1170	1420	790	456	540	812	–	–
	"	1973	–	–	–	–	–	–	544	–	–	–	–	–
	"	1974	1580	–	–	–	–	–	720	–	–	–	–	–
	"	1975	–	–	–	932	–	872	804	1610	640	1730	488	–
	"	1976	944	432	–	2540	1150	916	–	–	–	800	–	952
	"	1977	2010	–	–	–	–	496	476	–	–	–	–	–
	"	1978	–	748	–	–	–	–	–	–	–	–	–	–
12.	Sarwarwala Drain at Jaranwala	1972	–	–	–	–	–	–	–	868	–	–	–	–
	Lahore Road	1973	–	–	–	–	–	–	540	228	–	–	–	204
	"	1975	–	–	–	–	–	–	–	–	–	–	–	324
	"	1976	–	–	–	–	–	–	–	–	–	–	804	–
	"	1977	–	–	–	–	–	104	–	–	–	–	–	–
13.	Jaranwala Main Drain at RD 102	1969	–	–	–	–	–	–	–	–	1200	1450	442	–
	Burala Branch	1970	774	622	324	–	–	–	1240	384	640	–	–	–
	"	1971	540	–	–	–	–	–	–	–	–	–	–	–
	"	1972	–	–	–	–	–	–	–	–	–	–	–	–
	"	1973	604	–	–	–	–	–	–	–	–	–	–	280
	"	1974	–	–	–	–	–	–	–	–	–	–	–	–
	"	1975	812	1070	–	1120	1750	1100	–	–	–	1440	344	644
	"	1976	–	–	–	–	–	–	–	–	–	1040	554	848
	"	1977	–	–	–	–	–	–	–	–	–	–	–	1090
14.	Machrala Drain at JRW/LHR Road	1972	–	–	–	340		–	–	1010	–	–	–	260
	"	1973	–	–	424	–	–	–	672	372	476	–	354	–
	"	1974	–	–	–	–	–	702	460	–	–	–	–	–

Total Dissolved Solids (ppm) of Drains

Sr. No.	Name of Drain/Location	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
15.	Marh Chiniot Drain on Chiniot Faisalabad Road (RD.98970)	1969	–	–	–	–	–	–	648	–	818	744	–	–
		1970	732	806	702	678	226	536	584	560	680	604	–	–
		1976	–	–	–	–	–	–	–	–	–	844	816	880
		1977	880	1110	1040	–	746	1310	672	–	–	–	–	–
		1978	1050	1120	–	–	–	–	–	–	–	–	–	–
		1987	–	–	–	–	–	743	–	–	–	895	–	–
		1988	–	–	–	–	–	1312	–	–	–	1071	–	–
		1989	–	–	–	–	–	663	–	–	–	1215	–	–
		1990	–	–	–	–	–	–	–	1155	–	–	–	1027
		1991	–	–	–	–	–	1127	–	–	–	–	–	–
	Marh Chiniot Drain on Chiniot Faisalabad Road (RD.79000)	1987	–	–	–	–	–	875	–	–	–	1040	–	–
		1988	–	–	–	–	–	1330	–	–	–	1008	–	–
		1989	–	–	–	–	–	–	897	–	–	1170	–	–
		1990	–	–	–	–	–	–	–	893	–	–	–	990
		1991	–	–	–	–	–	1711	–	–	–	–	–	–
	Marh Chiniot Drain on Chiniot Faisalabad Road (RD.112000)	1988	–	–	–	–	–	1197	–	–	–	1071	–	–
		1989	–	–	–	–	–	–	728	–	–	1110	–	–
		1990	–	–	–	–	–	–	–	749	–	–	963	–
		1991	–	–	–	–	–	1095	–	–	–	–	–	–
16.	Ahmadpur Kot Nikka Drain on Chiniot Pindi Bhattian Road	1969	–	–	–	–	–	–	552	–	548	652	596	–
		1970	632	628	682	672	694	678	756	488	684	648	–	510
		1971	596	262	682	754	1280	–	332	–	540	606	–	526
		1972	–	–	618	632	726	352	–	424	472	596	536	428
		1973	268	344	552	560	540	772	436	456	–	–	600	1260
		1974	590	856	588	–	664	528	–	450	468	470	596	456
		1975	424	548	472	596	720	608	484	312	596	608	548	580
		1976	556	476	512	572	–	–	–	–	–	600	560	336
		1977	480	484	564	–	–	–	–	–	–	–	–	–
17.	Sumandri Main Drain RD 188628	1988	–	–	–	–	–	2688	–	–	–	1728	–	–
		1989	–	–	–	–	–	2243	–	–	–	2050	–	–
		1990	–	–	–	–	–	–	–	1669	–	–	1931	–
		1991	–	–	–	–	–	–	2786	–	–	–	–	–
	Sumandri Main Drain RD 158214	1988	–	–	–	–	–	2470	–	–	–	535	–	1495
		1989	–	–	–	–	–	2496	–	–	–	2030	–	–
		1990	–	–	–	–	–	–	–	2326	–	–	1995	–
		1991	–	–	–	–	–	–	2696	–	–	–	–	–
	Sumandri Main Drain RD 144000	1987	–	–	–	–	–	302	–	–	–	2575	–	–
		1988	–	–	–	–	–	2838	–	–	–	535	–	1530
		1989	–	–	–	–	–	2475	–	–	–	2015	–	–
		1990	–	–	–	–	–	–	–	2120	–	–	1926	–
		1991	–	–	–	–	–	–	2576	–	–	–	–	–
	Samundri Branch Drain RD 0080	1987	–	–	–	–	–	2210	–	–	–	–	1460	–
		1988	–	–	–	–	–	1728	–	–	–	–	204	–
		1989	–	–	–	–	–	1989	–	–	–	1360	–	–
		1990	–	–	–	–	–	1553	–	–	–	–	1916	–
		1991	–	–	–	–	–	–	1926	–	–	–	–	–
	Samundri Branch Drain RD 51000	1987	–	–	–	–	–	2600	–	–	–	6970	–	–
		1988	–	–	–	–	–	1228	–	–	–	–	1339	–
		1989	–	–	–	–	–	–	–	–	–	–	1245	–
		1990	–	–	–	–	–	–	–	2118	–	–	1344	–
		1991	–	–	–	–	–	–	1922	–	–	–	–	–
	Samundri Branch Drain RD 80000	1989	–	–	–	–	–	–	–	–	–	–	1535	–

Total Dissolved Solids (ppm) of Drains

Sr. No.	Name of Drain/Location	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	Maduana Branch Drain RD 28411	1987	–	–	–	–	–	3350	–	–	–	3350	–	–
	"	1988	–	–	–	–	–	–	2990	–	–	–	–	–
	"	1989	–	–	–	–	–	–	3185	–	–	2720	–	–
	"	1990	–	–	–	–	–	–	–	2572	–	–	2765	–
	"	1991	–	–	–	–	–	–	3618	–	–	–	–	–
	Maduana Branch Drain RD 144592	1987	–	–	–	–	–	2730	–	–	–	3890	–	–
	"	1988	–	–	–	–	–	3168	–	–	–	–	2580	–
	"	1989	–	–	–	–	–	8710	–	–	–	–	1340	–
	"	1990	–	–	–	–	–	–	–	2268	–	–	2333	–
	"	1991	–	–	–	–	–	–	2326	–	–	–	–	–
	Maduana Branch Drain RD 203000	1988	–	–	–	–	–	3055	–	–	–	–	2048	–
	"	1989	–	–	–	–	–	–	2197	–	–	–	2785	–
	"	1990	–	–	–	–	–	–	–	2452	–	–	2405	–
	Awagat Branch Drain RD 0096	1987	–	–	–	–	–	2096	–	-	–	1605	–	–
	"	1988	–	–	–	–	–	2145	–	–	–	1495	–	–
	"	1989	–	–	–	–	–	–	–	–	–	1140	–	–
	"	1990	–	–	–	–	–	–	–	1620	–	–	1352	–
	"	1991	–	–	–	–	–	–	1555	–	–	–	–	–
	Awagat Branch Drain RD 60088	1987	–	–	–	–	–	–	–	–	–	1605	–	–
	"	1988	–	–	–	–	–	1664	–	–	–	–	1553	–
	"	1989	–	–	–	–	–	–	1430	–	–	1465	–	–
	"	1990	–	–	–	–	–	–	–	1380	–	–	926	–
	"	1991	–	–	–	–	–	–	1417	–	–	–	–	–
18.	Paharang Main Drain RD 152400	1987	–	–	–	–	–	7350	–	–	–	10050	–	–
	"	1988	–	–	–	–	–	–	–	–	–	3102	–	–
	"	1989	–	–	–	–	–	–	–	–	–	2730	–	–
	"	1990	–	–	–	–	–	–	–	2447	–	–	–	2730
	"	1991	–	–	–	–	–	2210	–	–	–	–	–	–
	Paharang Main Drain RD 108968	1987	–	–	–	–	–	5000	–	–	–	–	–	–
	"	1988	–	–	–	–	–	4680	–	–	–	3484	–	–
	"	1989	–	–	–	–	–	–	2795	–	–	3460	–	–
	"	1990	–	–	–	–	–	–	–	2340	–	–	2747	–
	"	1991	–	–	–	–	–	2508	–	–	–	–	–	–
	Paharang Main Drain RD 181000	1991	–	–	–	–	–	2077	–	–	–	–	–	–
19.	Pandoki Main Drain at Outfall	1978	–	–	–	–	–	755	755	–	–	–	–	–
	"	1979	1210	–	1066	973	1088	800	–	1088	–	1024	1056	–
	"	1980	–	–	1024	–	973	–	832	–	–	–	–	–
20.	Ali pur Main Drain	Averaq	3138	–	–	–	–	–	–	–	–	–	–	–
21.	Hadda main drain		908	–	–	–	–	2500	–	–	–	–	–	–
22.	Lower Budhi Main drain	1991	216	–	–	–	–	384	–	–	–	–	–	–
23.	Kot Adu Main Drain		–	–	–	–	–	–	–	–	–	–	–	–
24.	Ghazi Ghat Main Drain		–	–	–	–	–	–	–	–	–	–	–	–
25.	Sukhrawa Main Drain		–	–	–	–	–	–	–	–	–		–	–
26.	Khairwala Main Drain		–	–	–	–	–	–	–	–	–	–	–	–
27.	Kirthar Canal Before Mixing	1985	–	–	–	–	–	–	–	183	–	–	–	–
		1986	167	–	–	–	–	–	–	176	201	250	189	–
		1987	204	200	320	–	–	188	212	140	176	–	180	192

Total Dissolved Solids (ppm) of Drains

Sr. No.	Name of Drain/Location	Year	Part Per Million											
			Months											
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		1988	152	242	144	–	–	–	–	–	–	–	–	200
		1989	–	–	144	152	344	272	–	–	–	–	–	–
		1990	–	–	204	–	–	–	–	162	180	212	160	–
		1991	448	252	368	–	–	–	–	–	–	–	–	–
28.	Hairdin Pump Station	1986	–	–	–	–	–	–	–	4750	5080	3890	4880	–
		1987		4490	4220	–	5280	5280	2340	2400	2760	–	3310	3570
		1988	8440	4948	4948	–	19248	3440	–	3240	–	6532	–	5940
		1989	–	5280	3082	3240	10040	–	1900	1848	2916	–	–	2460
		1990	3372	–	3572	–	–	–	–	3052	–	2340	1600	2340
		1991	–	2400	–	–	–	2468	–	–	–	–	–	–
29.	Carrier Drain RD–146	1987	–	–	–	–	4160	3990	2270	2210	–	–	–	15160
		1988	11560	29500	28390	–	15180	3400	–	–	–	–	–	6262
		1989	–	9440	2694	–	1608	1660	1600	1532	–	3116	–	–
		1990	3400	–	3568	–	–	–	–	2984	2860	2040	–	4380
		1991	5868	2760	3244	–	–	–	–	–	–	–	–	–
30.	Carrier Drain RD–141	1985	–	–	–	–	–	–	–	5280	–	–	–	–
		1986	7680	–	–	–	–	–	–	–	–	–	–	–
31.	Carrier Drain RD–114/115	1986	6400	2264	2400	–	14480	–	–	8370	–	2970	4680	2470
		1987	–	3640	2930	–	4290	2530	2080	2070	2670	3210	–	–
		1988	7370	2300	–	–	–	–	–	–	–	–	–	6268
		1989	3016	9380	2590	–	1600	–	816	968	–	1780	–	–
		1990	3380	–	12928	3460	–	–	–	3640	2140	2044	–	2628
		1991	–	–	3116	–	–	–	–	–	–	–	–	–
32.	Kirthar Canal RD–142	1985	–	–	–	–	–	–	–	1010	960	–	–	–
		1987	–	–	–	–	–	468	–	–	–	–	–	–
		1988	–	–	–	–	–	–	–	3080	1724	–	–	–
		1989	–	816	–	–	–	–	–	–	368	–	–	156
		1990	708	–	–	–	–	–	–	–		–	–	–
33.	Kirthar Canal RD–214	1985	–	–	–	–	–	–	–	820	1075	–	–	–
		1988	–	–	–	–	–	–	–	1704	–	–	–	–
		1989	–	780	–	–	–	–	–	–	–	–	–	–
34.	Kirthar Canal RD–335	1985	–	–	–	–	–	–	–	936	–	–	–	–
		1988	–	–	–	–	–	–	–	500	1820	–	–	–
		1989	–	756	–	–	–	–	–	–	–	–	–	–

Source: SCARP Monitoring, WAPDA

ANNEX - IV

CANAL COMMAND-WISE DISTRIBUTION OF SOIL TEXTURAL GROUPS

Canal Commandwise Distribution of Soil Textural Groups

Province	Canal Command	Total Acreage	Textural Groups				
			Coarse	Loamy	Somewhat Clayey	Clayey	Misc. Land Type
N.W.F.P	Upper Swat	304908		141351	109044	14533	39980
		%	–	46	36	5	13
	Lower Swat	193092	381	89879	59044	30937	12851
		%	•	46	31	16	7
	Warsak High Level	138676	–	71979	20099	–	46598
		%	–	52	14	–	34
	Kabul River &	146324	501	106968	19844	–	19011
	Jui Sheikh	%	•	73	14	–	13
	Pehur High Level	159105	11099	97701	45793	1933	2579
		%	7	61	29	1	2
	Tarbela Left Bank	4109	122	3955	–	–	32
	(NWFP Portion)	%	3	96	–	–	1
	Katch KOT	136970	4517	25215	96745	2291	8202
		%	3	19	70	2	6
	Landi Dak	39530	–	5120	31910	–	2500
		%	–	13	81	–	6
	Shah Joi	13500	–	796	12366	–	338
		%	–	6	92	–	2
	Paharpur	134000	20632	61076	44667	741	6884
		%	15	46	33	1	5
	Gravity Flow	247000	–	107051	128371	–	11578
	(NWFP Portion)	%	–	43	52	–	5
	Total	1517214	37252	711091	567883	50435	150553
		%	2	47	38	3	10
Punjab	Terbela Left Bank	60891	3080	55358	–	–	2453
	(Punjab Portion)	%	5	91	–	–	4
	Gravity Flow	281000	28075	80151	155298	–	17476
	(Punjab Portion)	%	10	29	55	–	6
	D.G.khan	950000	96053	204662	640880	1312	7093
		%	10	22	67	•	1
	Dajal	320000	24168	22019	272298	–	1515
		%	8	7	85	–	•
	Jalalpur	200000	2808	79588	116389	–	1235
		%	1	40	58	–	1
	Thal	2864354	896474	1833916	98989	858	34117
		%	31	64	4	•	1
	Rangpur	416000	62432	314583	32716	962	5307
		%	15	76	8	•	1
	Muzaffargarh	884000	315234	529790	29135	119	9722
		%	36	60	3	•	1
	Upper Jhelum	700444	44066	500886	120726	7294	27472
		%	6	72	17	1	4
	Lower Jhelum	1700556	271711	1100145	275951	3823	48926
		%	16	65	16	•	3
	Jhang Branch	2048825	204699	1728913	50316	–	64897
		%	10	85	2	–	3
	M.R. Link	206491	132	70164	90598	43150	2447
		%	•	34	44	21	1
	Upper Chenab	1155313	8435	623774	446661	60708	15735
		%	1	54	39	5	1
	Rayya Branch	527304	5695	280606	188237	45208	7558
		%	1	53	36	9	1
	Gugera Branch	1724188	211426	1408601	64692	531	38838
		%	12	82	4	•	2
	Haveli	169567	45791	111930	8226	–	3620
		%	27	66	5	–	2
	Ravi Syphon	703000	348	453238	239372	1636	8406
		%	•	65	34	•	1
	Upper Depalpur	376000	64748	258065	34396	–	18791
		%	17	69	9	–	5

Canal Commandwise Distribution of Soil Textural Groups

Province	Canal Command	Total Acreage	Textural Groups				
			Coarse	Loamy	Somewhat Clayey	Clayey	Misc. Land Type
	Lower Depalpur	650000	71284	553209	19828	–	5679
		%	11	85	3	–	1
	L.B.D.C	1960090	78235	1756001	64866	85	60903
		%	4	90	3	•	3
	Pakpattan	1058330	99210	908382	30331	–	20407
		%	9	86	3	–	2
	Sidhanai	841440	45937	649744	121554	1190	23015
		%	5	77	15	•	3
	Mailsi	1097670	105108	934773	41892	114	15783
		%	10	85	4	•	1
	Fordwah	474000	72608	357076	35412	38	8866
		%	15	75	8	•	2
	Sadiqia	1260000	7154	1132420	100403	412	19611
		%	•	90	8	•	2
	Upper Bahawal Qaim	94236	12841	61915	18550	–	930
		%	13	66	20	–	1
	Lower Bahawal	755764	16914	647101	75509	1268	14972
		%	2	86	10	•	2
	Abbasia	136621	7542	97887	28176	–	3016
		%	6	72	20	–	2
	Panjnad	1501379	252395	999133	221726	535	27590
		%	17	66	15	•	2
	Total	25117363	3054603	17754010	3623127	169243	516380
		%	12	71	14	1	2
Sindh	Pat	46005	660	9082	35923	–	340
(Sind Portion)		%	2	19	78	–	1
	Desert	356963	21964	178214	153538	284	2963
(Sind Portion)		%	6	50	43	•	1
	North West	869642	92234	387579	384161	629	5039
(Sind Portion)		%	11	45	44	•	•
	Begari	1088155	68939	647449	361094	–	10673
		%	6	60	33	–	1
	Rice	604622	24164	270631	287565	–	22262
		%	4	45	47	–	4
	Dadu	577628	12835	276171	267649	10279	10694
		%	2	48	46	2	2
	Ghotki	900000	61551	369071	463299	–	6079
		%	7	41	51	–	1
	Khairpur West	300389	17420	249925	26149	–	6895
		%	6	83	9	–	2
	Khairpur East	505503	88363	306427	88474	154	22085
		%	18	60	18	•	4
	Rohri North	1354103	111552	777840	446012	–	18704
		%	8	58	33	–	1
	Rohri South	1537404	26083	918212	575141	–	17968
		%	2	60	37	–	1
	Nara	2360388	80823	1445480	789221	7504	37360
		%	3	62	33	•	2
	Gaja	104390	1732	79610	20166	–	2882
		%	2	76	19	–	3
	Fuleli	1020762	11362	535849	432983	1020	39556
		%	1	53	42	•	4
	Tando Bagho	507706	3009	201749	236783	167	65998
		%	•	40	47	•	13
	Kalari	687500	8855	370019	259837	5913	42876
		%	1	54	38	1	6
	Pinyari	964810	33493	529986	298558	425	102348
		%	3	55	31	•	11
	Total	13785975	665039	7553294	5126553	26367	414722
		%	5	55	37	•	3

Canal Commandwise Distribution of Soil Textural Groups

Province	Canal Command	Total Acreage	Textural Groups				
			Coarse	Loamy	Somewhat Clayey	Clayey	Misc. Land Type
Balochistan	Pat	602363	8080	95440	496886	–	1957
	(Balochistan Portion)	%	1	16	83	–	*
	Desert	75694	–	27321	47617	–	756
	(Balochistan Portion)	%	–	36	63	–	1
	North West			0			
	(Balochistan Portion)	195234	1692	36294	156547	–	701
		%	1	18	80	–	1
	Total	873291	9772	159055	701050	–	3414
		%	1	18	81	–	*
	Grand Total	41293843	3766666	26177450	10018613	246045	1085069
		%	9	63	24	1	3

* Less than 1%

Source: Planning Division, WAPDA.

ANNEX - V

AGRICULTURE PERFORMANCE

TABLE 1

Pakistan: Wheat Statistics

Year	Area (Mac)	Yield (lb/ac)	Production (Mt)	Year	Area (Mac)	Yield (lb/ac)	Production (Mt)
1947/48	9.8	747	3.3	1970/71	14.8	950	6.4
48/49	10.6	828	4.0	71/72	14.3	1047	6.8
49/50	10.3	828	3.9	72/73	14.8	1096	7.3
				73/74	15.1	1096	7.5
50/51	10.8	804	3.9	74/75	14.4	1161	7.6
51/52	10.1	641	3.0				
52/53	9.4	552	2.4	75/76	15.1	1267	8.5
53/54	10.4	763	3.6	76/77	15.8	1275	9.0
54/55	10.5	658	3.1	77/78	15.7	1172	8.2
				78/79	16.5	1325	9.8
55/56	11.2	658	3.3	79/80	17.1	1397	10.7
56/57	11.6	682	3.6				
57/58	11.3	682	3.5	80/81	17.2	1463	11.3
58/59	11.9	715	3.8	81/82	17.8	1394	11.1
59/60	12.1	706	3.8	82/83	18.3	1495	12.2
				83/84	18.1	1320	10.7
60/61	11.5	723	3.8	84/85	17.9	1436	11.5
61/62	12.2	715	4.0				
62/63	12.4	731	4.1	85/86	18.3	1675	13.7
63/64	12.4	731	4.1	86/87	19.0	1389	11.8
64/65	13.1	731	4.5	87/88	18.1	1544	12.4
				88/89	19.1	1661	14.2
65/66	12.7	666	3.9	89/90	19.4	1626	14.1
66/67	13.2	715	4.3				
67/68	14.8	942	6.3	91/92	NA	NA	NA
68/69	15.2	942	6.5	92/93	NA	NA	NA
69/70	15.4	1031	7.2				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: Production increase 1947/48 to 1989/90:

Overall	:	327%
Area related	:	98%
Yield related	:	118%

TABLE 2

Pakistan: Rice Statistics

Year	Area (Mac)	Yield (lb/ac)	Production (Mt)	Year	Area (Mac)	Yield (lb/ac)	Production (Mt)
1947/48	2.0	771	0.7	1970/71	3.7	1291	2.2
48/49	2.1	780	0.7	71/72	3.6	1364	2.2
49/50	2.3	763	0.8	72/73	3.7	1386	2.3
				73/74	3.7	1429	2.4
50/51	2.4	788	0.9	74/75	4.0	1267	2.3
51/52	2.2	731	0.7				
52/53	2.2	796	0.8	75/76	4.2	1364	2.6
53/54	2.5	796	0.9	76/77	4.3	1394	2.7
54/55	2.4	763	0.8	77/78	4.7	1383	2.9
				78/79	5.0	1483	3.2
55/56	2.4	763	0.8	79/80	5.0	1408	3.2
56/57	2.4	763	0.8				
57/58	2.7	715	0.9	80/81	4.8	1439	3.1
58/59	2.8	755	1.0	81/82	4.9	1546	3.4
59/60	3.0	731	1.0	82/83	4.9	1551	3.4
				83/84	4.9	1488	3.3
60/61	2.9	771	1.0	84/85	4.9	1478	3.3
61/62	3.0	820	1.1				
62/63	2.9	812	1.1	85/86	4.6	1396	2.9
63/64	3.2	812	1.2	86/87	5.1	1503	3.4
64/65	3.4	877	1.3	87/88	4.8	1471	3.2
65/66	3.4	828	1.3	88/89	5.0	1396	3.1
				89/90	5.2	1361	3.2
66/67	3.5	853	1.3				
67/68	3.5	926	1.5	91/92	NA	NA	NA
68/69	3.8	1153	2.0	92/93	NA	NA	NA
69/70	4.0	1307	2.4				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: Production increase 1947/48 to 1989/90:

Overall	:	357%
Area related	:	160%
Yield related	:	77%

TABLE 3

Pakistan: Maize Statistics

Year	Area (Mac)	Yield (lb/ac)	Production (Mt)	Year	Area (Mac)	Yield (lb/ac)	Production (Mt)
1947/48	0.9	869	0.35	1970/71	1.6	991	0.71
48/49	0.9	869	0.37	71/72	1.6	983	0.69
49/50	1.0	893	0.40	72/73	1.6	966	0.70
				73/74	1.6	1064	0.76
50/51	0.9	901	0.38	74/75	1.5	1072	0.74
51/52	1.0	861	0.38				
52/53	1.0	788	0.35	75/76	1.5	1153	0.79
53/54	1.1	836	0.40	76/77	1.5	1090	0.75
54/55	1.1	885	0.43	77/78	1.6	1114	0.81
				78/79	1.6	1094	0.78
55/56	1.1	934	0.45	79/80	1.7	1112	0.86
56/57	1.1	658	0.46				
57/58	1.1	909	0.44	80/81	1.9	1124	0.95
58/59	1.1	942	0.48	81/82	1.8	1130	0.91
59/60	1.2	885	0.48	82/83	2.0	1134	0.99
				83/84	2.0	1131	1.00
60/61	1.2	804	0.43	84/85	2.0	1132	1.01
61/62	1.2	901	0.48				
62/63	1.1	934	0.48	85/86	2.0	1119	0.99
63/64	1.2	926	0.52	86/87	2.0	1212	1.09
64/65	1.2	658	0.52	87/88	2.1	1176	1.11
				88/89	2.1	1239	1.18
65/66	1.3	877	0.53	89/90	2.1	1218	1.16
66/67	1.4	934	0.58				
67/68	1.5	1145	0.78	91/92	NA	NA	NA
68/69	1.5	893	0.62	92/93	NA	NA	NA
69/70	1.6	901	0.66				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: Production increase 1947/48 to 1989/90:

Overall	:	214%
Area related	:	133%
Yield related	:	40%

TABLE 4

Pakistan: Cotton Statistics

Year	Area (Mac)	Yield (lb/ac lint)	Production (Mbales)	Year	Area (Mac)	Yield (lb/ac lint)	Production (Mbales)
1947/48	3.1	138	1.1	1970/71	4.3	277	3.1
48/49	2.6	147	1.0	71/72	4.8	317	4.0
49/50	2.7	179	1.2	72/73	5.0	309	3.9
				73/74	4.6	317	3.7
50/51	3.0	179	1.4	74/75	5.0	277	3.6
51/52	3.3	163	1.4				
52/53	3.4	204	1.8	75/76	4.6	247	3.0
53/54	2.8	195	1.4	76/77	4.6	208	2.6
54/55	3.1	195	1.6	77/78	4.6	278	3.4
				78/79	4.7	223	2.8
55/56	3.5	187	1.7	79/80	5.1	312	4.3
56/57	3.6	187	1.7				
57/58	3.6	187	1.7	80/81	5.2	302	4.2
58/59	3.3	187	1.6	81/82	5.5	301	4.4
59/60	3.3	187	1.6	82/83	5.6	324	4.8
				83/84	5.5	199	2.9
60/61	3.2	204	1.7	84/85	5.5	401	5.9
61/62	3.4	204	1.8				
62/63	3.4	236	2.1	85/86	5.8	459	7.2
63/64	3.6	252	2.4	86/87	6.2	469	7.8
64/65	3.6	228	2.1	87/88	6.3	509	8.6
				88/89	6.5	485	8.4
65/66	3.9	236	2.3	89/90	6.4	499	8.6
66/67	4.0	252	2.6				
67/68	4.4	252	2.9	91/92	NA	NA	NA
68/69	4.3	269	3.0	92/93	NA	NA	NA
69/70	4.3	269	3.0				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: 1 bale = 375 lbs.

Production increase 1947/48 to 1989/90:

Overall	:	682%
Area related	:	106%
Yield related	:	262%

TABLE 5

Pakistan: Sugarcane Statistics

Year	Area (Mac)	Yield (lb '000/ac)	Production (Mt)	Year	Area (Mac)	Yield (lb '000/ac)	Production (Mt)
1947/48	0.5	25.7	5.4	1970/71	1.6	32.1	22.8
48/49	0.5	31.0	6.8	71/72	1.4	31.8	19.6
49/50	0.5	31.5	7.7	72/73	1.3	32.6	19.6
				73/74	1.6	32.6	23.5
50/51	0.5	25.7	5.4	74/75	1.7	27.8	20.9
51/52	0.5	25.0	5.3				
52/53	0.6	25.2	7.2	75/76	1.7	32.5	25.1
53/54	0.7	27.0	8.8	76/77	1.9	33.4	29.0
54/55	0.8	25.6	8.7	77/78	2.0	32.6	29.5
				78/79	1.9	32.3	26.8
55/56	0.7	25.2	8.1	79/80	1.8	34.1	27.0
56/57	0.8	24.7	8.8				
57/58	1.0	25.0	11.1	80/81	2.0	34.9	31.8
58/59	1.1	25.7	12.3	81/82	2.3	34.4	35.9
59/60	1.0	23.6	10.5	82/83	2.3	31.8	32.0
				83/84	2.2	34.0	33.7
60/61	1.0	26.4	11.5	84/85	2.3	31.7	31.6
61/62	1.1	28.4	14.1				
62/63	1.3	30.5	18.1	85/86	1.9	31.8	27.4
63/64	1.2	29.7	15.9	86/87	1.9	35.3	29.4
64/65	1.2	32.6	18.4	87/88	2.1	34.9	32.4
				88/89	2.2	37.6	36.3
65/66	1.5	32.9	22.0	89/90	2.1	37.0	34.9
66/67	1.6	29.8	21.6				
67/68	1.2	32.6	18.4	91/92	NA	NA	NA
68/69	1.3	35.8	21.6	92/93	NA	NA	NA
69/70	1.5	37.4	28.0				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: Production increase 1947/48 to 1989/90:

Overall	:	546%
Area related	:	320%
Yield related	:	40%

TABLE 6

Pakistan: Population

Year	Population (M)	Year	Population (M)
1951 c	33.74	1981 c	84.25
		82	87.76
1961 c	42.88	83	90.48
		84	93.29
1972 c	65.31	85	93.18
73	66.88	86	99.16
74	68.92	87	102.24
75	71.03	88	105.41
76	73.21	89	NA
77	75.44	90	NA
78	77.75	91	NA
79	80.13		
80	82.58		

Source: Pakistan Statistical Year Book, 1989
c : consus figures
Population increase 1951 to 1988 : 212%

TABLE 7

Pakistan: Cultivated Area

Year	Cultivated Area (Mac)	Year	Cultivated Area (Mac)
1947/48	36.30	1970/71	47.47
48/49	37.47	71/72	47.16
49/50	37.03	72/73	NA
		73/74	NA
50/51	37.43	74/75	NA
51/52	37.35		
52/53	37.76	75/76	48.96
53/54	38.39	76/77	48.80
54/55	37.86	77/78	49.65
		78/79	49.35
55/56	38.70	79/80	49.97
56/57	39.56		
57/58	40.14	80/81	50.14
58/59	40.04	81/82	50.44
59/60	40.80	82/83	50.29
		83/84	50.22
60/61	44.76	84/85	50.91
61/62	44.23		
62/63	44.55	85/86	51.08
63/64	45.30	86/87	51.67
64/65	46.26	87/88	51.03
		88/89	NA
65/66	47.54	89/90	NA
66/67	47.60		
67/68	48.00	91/92	NA
68/69	47.67	92/93	NA
69/70	47.53		

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives.
Islamabad
Note: Increase in Cultivated Area 1947/48 to 1987/88 41%

TABLE 8

Pakistan: Irrigation Water Supplies

Year	Surface Water		Tubewells		Total	Rainfall
	Canal Head (MAF)	Farm Gate (MAF)	Public (MAF)	Private (MAF)	Farm Gate (MAF)	(AA, A, BA)
1960/61	NA	56.61	0.55	1.78	58.74	
61/62	NA	58.01	0.67	2.61	61.29	
62/63	NA	60.04	2.29	3.61	65.94	
63/64	NA	62.53	2.53	4.89	69.95	
64/65	NA	62.94	2.79	6.46	72.19	
65/66	NA	63.88	2.91	8.28	75.07	
66/67	NA	67.18	2.03	10.27	79.48	
67/68	NA	66.28	2.31	12.16	80.75	
68/69	NA	68.82	3.33	13.70	85.85	
69/70	NA	69.90	4.17	15.03	89.10	
70/71	NA	61.15	5.10	16.47	82.72	
71/72	NA	60.65	5.27	18.27	84.19	
72/73	NA	70.63	5.56	19.76	95.95	
73/74	NA	67.25	6.35	21.30	94.90	
74/75	NA	61.72	6.84	22.81	91.37	
75/76	98.91	59.42	6.16	19.43	85.01	AA
76/77	97.33	58.40	5.65	20.52	84.57	
77/78	102.69	61.62	6.21	21.61	89.44	AA
78/79	96.63	57.98	6.62	22.79	87.39	
79/80	105.24	63.14	7.03	23.91	94.14	
80/81	104.81	65.21	7.44	25.14	97.79	
81/82	102.83	64.39	8.16	26.31	98.86	
82/83	103.30	65.24	9.96	26.29	101.49	
83/84	100.51	66.47	10.20	27.02	103.69	
84/85	101.08	59.50	10.08	27.75	97.33	
85/86	96.34	60.81	10.74	28.48	100.03	
86/87	105.82	69.69	10.84	29.19	109.72	
87/88	109.07	71.24	11.06	29.96	112.22	
88/89	105.11					
89/90						

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture & Cooperatives, Islamabad

Note: Increase in farm gate supplies 1960/61 to 1987/88 91%
AA Above average
A Average
BA Below average

TABLE 9

Pakistan: Fertilizer Statistics

Year	Imported (Nt'000)	Production (Nt'000)	Consumption (Nt'000)	Year	Imported (Nt'000)	Production (Nt'000)	Consumption (Nt'000)
1947/48	Neg	Neg	Neg	1970/71	151.4	145	283
48/49	Neg	Neg	Neg	71/72	73.0	220	382
49/50	Neg	Neg	Neg	72/73	187.7	283	437
				73/74	335.6	304	403
50/51	Neg	Neg	Neg	74/75	132.1	307	426
51/52	Neg	Neg	Neg				
52/53	2.1	Neg	1.0	75/76	182.7	327	651
53/54	29.3	Neg	14.8	76/77	279.6	321	637
54/55	4.1	Neg	14.1	77/78	548.7	325	679
				78/79	707.7	364	857
55/56	7.6	Neg	6.6	79/80	776.0	439	1071
56/57	18.7	Neg	9.0				
57/58	10.9	1.6	16.4	80/81	574.4	640	1035
58/59	18.3	7.9	18.0	81/82	202.0	753	1070
59/60	10.9	9.0	19.4	82/83	400.1	1072	1217
				83/84	286.8	1107	1232
60/61	36.3	11.4	31.4	84/85	341.9	1119	1265
61/62	40.2	15.0	37.5				
62/63	NA	41.0	40.2	85/86	331.0	1129	1484
63/64	5.1	45.0	68.2	86/87	522.4	1212	1679
64/65	3.2	48.1	87.2	87/88	521.6	1193	1719
				88/89	461.6	1212	1721
65/66	36.1	47.4	70.5	89/90	NA	NA	1834
66/67	123.4	51.5	111.8				
67/68	153.5	52.5	190.4	91/92	NA	NA	NA
68/69	156.4	81.2	214.6	92/93	NA	NA	NA
69/70	303.7	133.4	307.7				

Source: Agricultural Statistics of Pakistan 1975, 1989/90
Ministry of Food, Agriculture and Cooperatives, Islamabad

Note: Nt : Nutrient Metric tonnes
Neg : Negligible

TABLE 10

Pakistan: Pesticide Statistics

Year	Quantity ('000 t)		Year	Quantity ('000 t)	
	imported	utilised		imported	utilised
1950/51	18		1971/72	2,438	
51/52	217		72/73	10,484	
52/53	2		73/74	8,673	
53/54	63		74/75	6,928	
54/55	162		75/76	13,258	
55/56	155		76/77	16,226	
56/57	712		77/78	12,754	
57/58	716		78/79	7,727	
58/59	430		79/80	4,419	897
59/60	1,328		80/81	7,105	1,301
60/61	4,901		81/82	5,481	2,585
61/62	1,949		82/83	8,860	1,924
62/63	4,522		83/84	10,663	2,126
63/64	1,644		84/85	15,889	4,908
64/65	813		85/86	17,499	3,251
65/66	869		86/87	20,648	4,045
66/67	2,405		87/88	15,765	8,066
67/68	5,072		88/89	11,327	12,935
68/69	1,4[illegible]		89/90	10,544	
69/70	1,896		90/91	NA	NA
70/71	2,248		91/92	NA	NA

Source: Agricultural Statistics of Pakistan 1975, 1989/90 – Ministry of Food, Agriculture and Cooperatives, Islamabad

TABLE 11

Factors Increasing Agricultural Production

Factors (Increasing/Improved)	Likely effect on	
	Area	Yield
Population	High	Low
Seed	Low	High
Water	High	High
Other Inputs	Low	High
Mechanization	High	Medium
Drainage	Low	Medium

TABLE 12

Notional Water Demand, 1986

Season	Crop	Water Required(1) at Farm Gate			Cropped Area (2)	Total Water Demand at Mogha
		CU (ft)	Gate (ft)	Mogha (ft)	(Mac)	(MAF)
Kharif	Rice	2.5	3.8	4.2	5.10	21.42
	Cotton	2.1	2.5	2.7	6.20	16.74
	Maize	1.1	1.3	1.4	2.00	2.80
	Millet/Sorghum	1.2	1.4	1.6	2.20	3.52
	Fodder	1.7	2.0	2.2	1.00 (3)	2.20
	Vegetables	1.0 (3)	1.2	1.3	0.50	0.65
Rabi	Wheat	1.4	1.7	1.8	19.00	34.20
	Pulse	0.6	0.7	0.8	3.80	3.04
	Oilseed	0.8	1.0	1.0	1.10	1.10
	Fodder	2.1	2.5	2.7	1.00 (3)	2.70
	Vegetables	1.5 (3)	1.8	2.0	0.50	1.00
Perennial	Sugarcane	6.2	7.4	8.1	1.90	15.39
	Other (fruit)	4.8	5.8	6.2	1.00	6.20
			Total:		45.30 (4)	110.96

Source: (1) Staff Appraisal Report from SCARP VI, World Bank, 1977.
(2) Agricultural Statistics of Pakistan, 1989/90

Note: (3) Estimated
(4) Total cropped is given in (2) as 51.6 Mac. It is assumed that 6.3 Mac is in non irrigated areas.

TABLE 13

Notional Water Demand, 1960/61

Season	Crop	Water Required(1) at Farm Gate			Cropped Area (2)	Total Water Demand at Mogha
		CU (ft)	Gate (ft)	Mogha (ft)	(Mac)	(MAF)
Kharif	Rice	2.5	3.8	4.2	2.9	12.18
	Cotton	2.1	2.5	2.7	3.2	8.64
	Maize	1.1	1.3	1.4	1.2	1.68
	Millet/Sorghum	1.2	1.4	1.6	1.5	2.40
	Fodder	1.7	2.0	2.2	0.5 (3)	1.10
	Vegetables	1.0 (3)	1.2	1.3	0.3	0.39
Rabi	Wheat	1.4	1.7	1.8	11.5	20.70
	Pulse	0.6	0.7	0.8	4.0	3.20
	Oilseed	0.8	1.0	1.0	1.3	1.30
	Fodder	2.1	2.5	2.7	0.5 (3)	1.35
	Vegetables	1.5 (3)	1.8	2.0	0.3	0.60
Perennial	Sugarcane	6.2	7.4	8.1	1.0	8.10
	Other (fruit)	4.8	5.8	6.2	0.7	4.34
			Total:		28.9 (4)	65.98

Source: (1) Staff Appraisal Report from SCARP VI, World Bank, 1977.
(2) Agricultural Statistics of Pakistan, 1989/90

Note: (3) Estimated
(4) Total cropped is given in (2) as 36.72 Mac. It is assumed that 7.82 Mac is in non irrigated areas.